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PREFACE

This book covers key areas of geography, earth science and environment. The contributions by the authors include geohazard mapping, terrestrial lidar, terrestrial laser scanning, global navigation satellite system, digital surface models, geodectic technology, interferometric synthetic aperture radar, uncrewed technology, spectral angle mapper, planetary atmospheres, thermodynamics of gases, planetary balance circumstances, thermodynamic effects, gravitational acceleration force, solar magnetic field, normalized difference vegetation index, blue infrastructure, human vulnerability index, fisher-folk, livelihood dimensional index, economic indicator, climate variability vulnerability index, multivariate cluster analysis, volcanic eruptions, volcanic ash, hierarchical hazard, occupational safety, quarry mining, respiratory protection. This book contains various materials suitable for students, researchers, and academicians in the fields of geography, earth science and environment.

Chapter 1
Print ISBN: 978-93-49729-70-4, eBook ISBN: 978-93-49729-97-1

Water Quality on the Growth Performance of *Clarias gariepinus* in Fish Farms within Fako Division, Cameroon

Benjamin Nchegang [a*], Tabe Maxwell Enow [a], Geneva Ojong Nkongho [b] and Paul Vernyuy Tan [c]

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ABSTRACT

Fish need a decent environment for their living and livelihood. Fish species that have additional organs as a means of breathing, such as catfish, can tolerate fairly extreme water conditions, but in order to develop optimally, water quality parameters need to be within their recommended ranges. Aquaculture in South-West Cameroon has evolved, although not exponentially, due to a qualified workforce. The problem of water quality in fish farms, which could be associated with poor feeding and water quality management, impacts the life of aquatic animals. Fish need a healthy environment for their optimal growth and development (case study of Catfish). The objective of this study was to determine the impact of water quality on the growth performance of *Clarias gariepinus* in some fish farms within Fako Division-South West Region of Cameroon. *Material and methods:* The method used in this research topic is descriptive, with 4 different localities, namely IRAD Batoke (A), Bakingili (B), Cornelius Fish Center (C) and Oumarou fish farm (D), Mile 2 Limbe. A collection of physicochemical and biological data was conducted from the period of March 8th to May 15th 2023, in Fako Division. The data collection was done from Tuesday to Friday, a day each per farm from 7- 8 am. *Results:* The results revealed the highest mean temperature value (28.22±0.94ºC) in IRAD Batoke broodstocks tanks, while Cornelius Fish Center had the highest mean pH, electrical conductivity, and total dissolved solid with values of 8.16±1.25; 0.84±0.35 ms/cm; and 0.38±1.8ppt in the broodstocks tanks and table fish tanks. The highest dissolved oxygen (<8

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mg/l), the highest nitrite level (<0.025 mg/l) and the highest Ammonium (<0.05 mg/l) were noted in three localities except the Bakingili locality. The physicochemical parameters of water had no negative impacts on the growth performance of this species. *Conclusion:* The results of physicochemical parameters such as water temperature, pH, and electrical conductivity, total dissolved solids, NH_4, NO_2 and O_2 obtained during the study experienced a relative variation and were very suitable for the growth and development of *Clarias gariepinus*. The study concluded that although the African catfish can tolerate poor water conditions, optimal water quality is crucial for its optimal growth and development. These findings are relevant to the fish production sector for the sustainable production of African catfish.

Keywords: South West Region; TANKS; physicochemical parameter of water; Clarias gariepinus; growth performance.

1. INTRODUCTION

Fish is an important dietary animal protein in human nutrition and fish provides 60% of animal protein consumption (Mohammad & Haque, 2021). Fish need a decent environment for their living and livelihood. Fish species that have additional organs as a means of breathing, such as catfish, can tolerate fairly extreme water conditions, but in order to develop optimally, water quality parameters need to be within their recommended ranges. Water quality encompasses the water's physical, biological, and chemical parameters. It generally affects the fish growth and welfare (Abdel-Tawwab *et al*., 2019). The physicochemical characteristics of water are important parameters as they may directly or indirectly affect its quality and consequently its suitability for the distribution and production of fish and other aquatic animals (Imam *et al*., 2019). Decreasing water quality will disrupt growth and enhance the proliferation of disease-causing agents that grow and develop in low-quality waters (Nchegang *et al*., 2024). Ayuniar and Hidayat, (2018) stipulated that fish farming activities will be successful if we can manage the water quality of the cultivation media, seeds, and feed provided. Catfish can survive in unfavorable environmental conditions; according to Dauhan and Efendi, (2014), the water quality parameters of water to be used for the culture should be those that can support growth adequately (Nchegang *et al*., 2024). Physical parameters that need to be considered are temperature, pool water depth, turbidity level, TSS, or the presence of dissolved solids (Nchegang *et al*., 2024). Chemical parameters that also need to be controlled are DO, carbon dioxide, water pH value, the amount of Nitrate, Phosphate, and ammonia presence (Yuli *et al*., 2023; Budha *et al*., 2022; Føre *et al*., 2018; Godoy *et al*., 2018).

Physiologically, dissolved oxygen and temperature are essential for sustaining aquatic lives (Abdulraheem *et al*., 2012). Additionally, nutrients enrichment mostly phosphorus and nitrogen stimulate unnecessary primary production, which can reduce oxygen (Nchegang *et al*., 2024).

The changes in physical characteristics like temperature, conductivity, transparency and chemical elements of water (dissolved oxygen, and nitrite)

provide valuable information on the quality of the water (Nchegang *et al*., 2024). According to Valetta, (2013) increase in temperature is known to speed up biochemical activities of metabolism, stimulated by heat energy. Turbidity is also a key water-quality characteristic because its effects on light transmission and water clarity define habitat characteristics (Nchegang *et al*., 2024).

Furthermore, water pH regulates aquatic chemistry, which can affect water use and habitats (Nchegang *et al*., 2024). The health of an aquatic ecosystem is highly dependent on the physicochemical and biological characteristics of the water (Venkateshraju *et al*., 2010; Ramkumar *et al*., 2019). Changes in fish populations can be indicators of aquatic ecosystem health (Moyle, 1994). The existence and richness of fish species can be associated with the physical and chemical properties of water (Deacon & Mize, 1997; Nchegang *et al*., 2024). Reduced water quality is reported to affect fish communities by impacting habitat, food availability, and dissolved oxygen levels, which in turn influence their growth potential and reproductive abilities (Shetty *et al*., 2015). Thus, the present study was designed to determine the impact of water quality on the growth performance of *Clarias gariepinus* in some fish farms in the South West Region, precisely Fako Division in Limbe 1, Limbe 2 (IRAD Batoke) and Bakingili Subdivision (Fig. 1) (Nchegang *et al*., 2024).

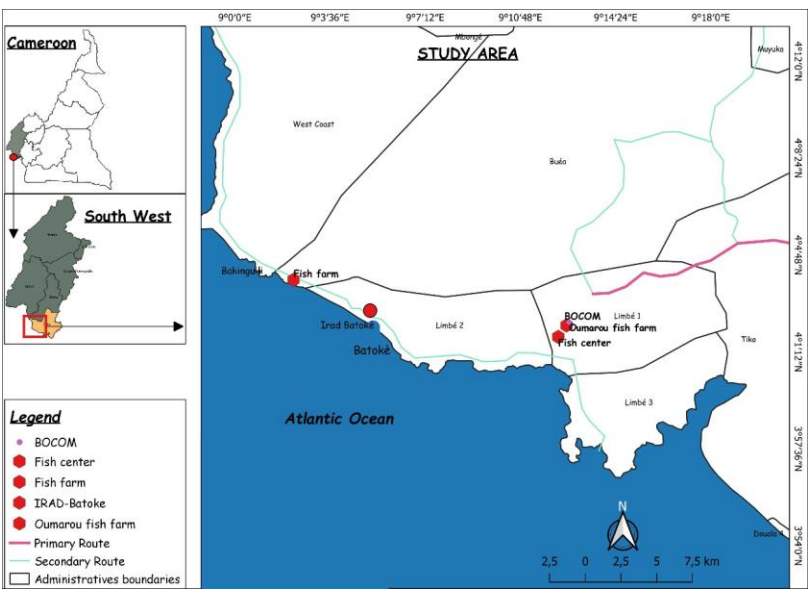


Fig. 1. Location of study Area (Tabe, 2023)

2. MATERIALS AND METHODS

Physical parameters (Temperature, pH, Electrical conductivity, Total dissolved solid) were measured *in-situ* using a Hannah portable meter model HI98129.

Transparency of the water was detected by eye observation while chemical parameters (NH4; NO2; O2 were measured *ex-situ* using a Pro JBL AQUATEST Kit (Nchegang *et al*., 2024). The data collection was done from Tuesday to Friday; a day each per farm from 7- 8 am.

2.1 Experimental Protocol

The Multiparameter was deepen into each fish tank to determine temperature, pH, electrical conductivity and total dissolved solid. The multiparameter was rinsed each time prior to using in another tank (Nchegang *et al*., 2024).

Thereafter, water was collected from each fish farm and transported in an ice chest to the IRAD-Batoke Laboratory for analysis using a Pro JBL AQUATEST KIT (Nchegang *et al*., 2024).

To determine the biological parameters, the tanks were emptied very early in the morning before feeding, and the fish were captured using a deep net and introduced into well-cleaned tanks. Using an electronic scale, the fish in each tank were weighed to determine the total weight (biomass) of fish then a count was made in order to have the total number of fish in the respective tanks (Nchegang *et al*., 2024). Using an ichthyometer, the total length and the standard length were measured for a sample of 10 -20 fish. This was done at the beginning and at the end of the experiment in order to have the initial and final weights (Nchegang *et al*., 2024). In the end, this allowed us to calculate weight gain and Condition factor as follows:

- The initial average weight (Wi) and the final average weight (Wf); which corresponds to the total initial and final weight (biomass) divided by the total initial and total final fish number, respectively (Nchegang *et al*., 2024).
- To calculate the total weight gain, it had been given by the formula:

  Total weight gain = Wf-Wi

At the end of the experiment, each fish was weighed and its length measured. These individual data made it possible to calculate the K-factor using the formula:

$$K=W/L^3$$

W= fish weight (g)
L = total length of fish (cm)

2.2 Statistical Analysis

The data collected was processed using Microsoft Office Excel 2013. One-way analyzes of variance (Anova) and the Kruskal-Wallis test was performed to test for significance $p < 0.05$ using Statistical Package for Social Science (SPSS) (Nchegang *et al*., 2024).

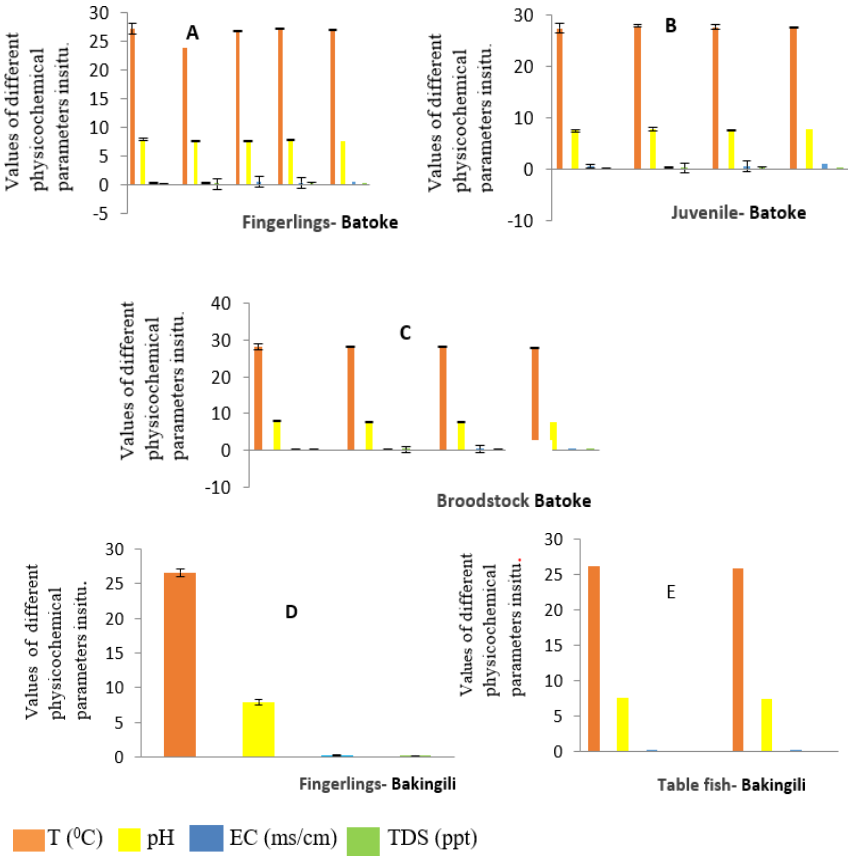
3. RESULTS AND DISCUSSION

3.1 Results

3.1.1 Physicochemical parameters measured in situ

The results for be seen in Fig. 2 and Fig. 3 below.

The values of the different physicochemical parameters in situ in the various tanks, such as T(0C), pH, EC (ms/cm) and TDS (ppt)) in the fish farms of IRAD Batoke (five tanks fingerlings, four tanks juvenile and four tanks broodstock) (Fig. 2 A, B & C) (Nchegang *et al*., 2024); Bakingili (one tank fingerlings and two tanks table fish) (Fig. 2 D & E) (Nchegang *et al*., 2024); Cornelius fish center (two tanks juvenile, one tank table fish and one tank broodsock) (Fig. 2 F, G & H) and Oumarou fish farm (one tank table fish and seven tanks broodstock) (Fig. 2 I & J) were not significantly different (P>0.05) as seen Fig. 2 (Nchegang *et al*., 2024).



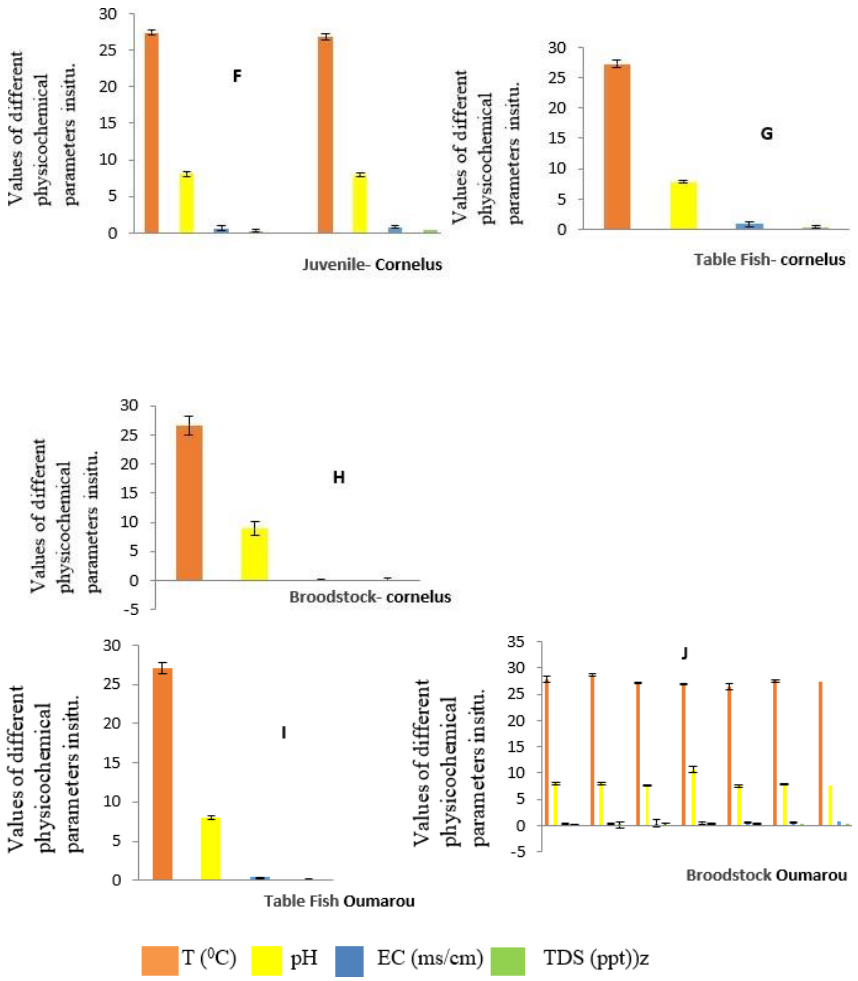


Fig. 2. Physicochemical parameters of water in tanks among four fish farms studied

The comparative studies of physicochemical parameters *in situ* among the four different fish farms, as seen in Fig. 3, such as

Fingerlings tanks Batoke (5t) Compared to fingerlings tanks Bakingili (1t):

As illustrated on Fig. 3A below, the parameters variation for mean temperature was relatively high in Batoke fingerlings, but not significantly different (P= 0.611) from those of the fish farm in Bakingili while the Bakingili Station had a higher pH with P=0.280, which was equally not different from those of the fish farms in IRAD-Batoke (Nchegang *et al*., 2024).

Comparison of juveniles tanks Batoke (4t) / Cornelius Fish Center (2t):

From the Fig. 3B below, the mean temperature was higher in Batoke juveniles tanks, but not significantly different (P=0.201) from those in Cornelius fish center, while the Cornelius Fish Center had a higher pH, EC, and TDS with P=0.172, P=0.395, P=0.000 respectively (Nchegang *et al*., 2024).

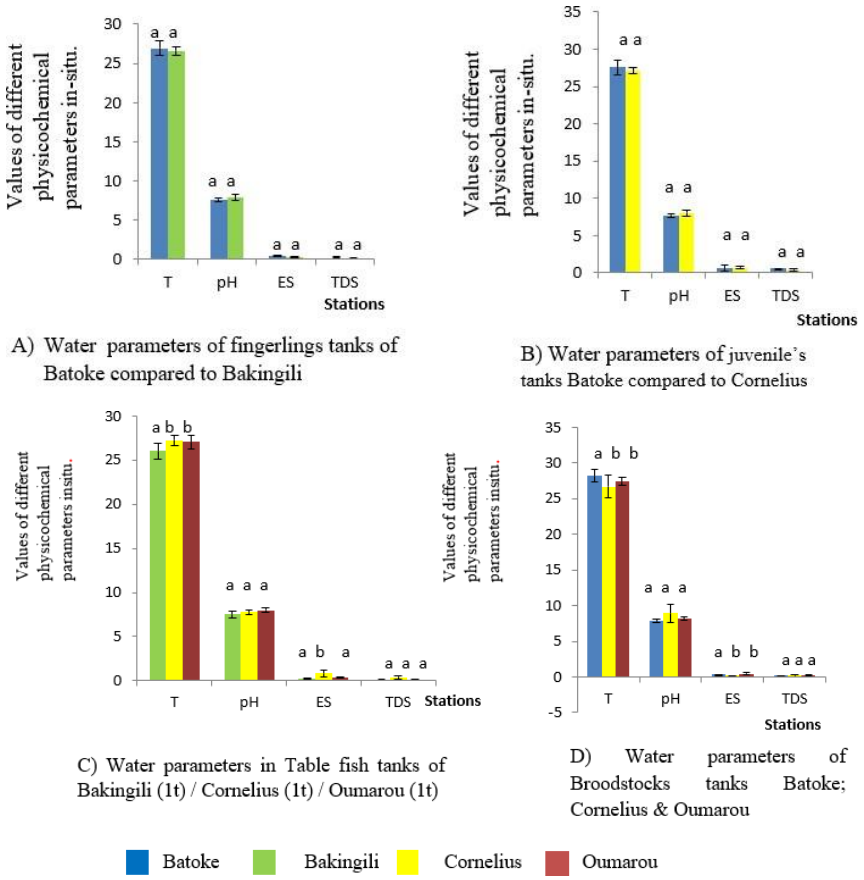


Fig. 3. Comparative analysis of water parameters of tanks in four different localities of Fako Division, South West Region of Cameroon
p < 0. 05, compared with the same parameter (T (0C), pH, EC (ms/cm), TDS (ppt) The values that affect the same letters had no significant difference

Table Fish Tanks Bakingili (1t)/ Cornelius Fish Center (1t) /Dr Oumarou (1t):

From Fig. 3C above, according to the parameter variations of table fish, the mean Temperature and EC were significantly higher with P=0.002 and P=0.000,

respectively, in the Cornelius fish farm, while the station of Oumarou had significantly (P=0.034) (Nchegang *et al*., 2024).

Comparison of Broodstocks Tanks Batoke, Cornelius and Dr Oumarou Fish Farms:

From Fig. 3D above, according to the parameter variations, the mean Temperature was significantly (p=0.000) higher in the Batoke fish farm and but EC was higher, but not significantly different (P=0.472) (Nchegang *et al*., 2024). While the station of Oumarou had a higher pH, which was, however not significantly different with p=0.588 (Nchegang *et al*., 2024).

3.1.2 Physicochemical parameters ex-situ

The results of water quality parameters for physicochemical measurements in situ, such as dissolved oxygen, Nitrite and Ammonium, for catfish broodstock tanks for the various farms are presented in Table 1 (Nchegang *et al*., 2024).

3.1.3 Evolution of biological parameters of African catfish (*Clarias gariepinus)*

Some biological parameters of African catfish were collected during the research period in the four (04) different fish farms and their results obtained, as shown in Table 2 (Nchegang *et al*., 2024). The result showed that growth performance of *Clarias gariepinus* for all the tanks of different stations was favorable. The growth of fingerlings for Batoke fish farm was remarkable greater than that of bakingili fingerlings (Nchegang *et al*., 2024).

3.2 Discussion

Based on the results of water temperature measurements during the study, the results were relatively constant and were very suitable for *Clarias gariepinus* growth and development (Nchegang *et al*., 2024). Water temperature in the IRAD Batoke station was 26.92±0.94ºC, in the Bakingili was 26.58±0.61ºC for the fingerling's tanks. According to the suitable temperature range for *Clarias gariepinus* farming, this varies from 23-30ºC (Teugels, 1986; Nchegang *et al*., 2024). So based on research that has been carried out, the temperature range during the research period on *Clarias gariepinus* farming still meets the feasibility conditions and is good enough for *Clarias gariepinus* farming (Nchegang *et al*., 2024). Too high a temperature will damage the process of growth by preventing enzyme activity in cells. Temperatures less than the ideal range can cause a decrease in processing efficiency, increasing the toxicity of an aquatic organism pollutant (Moyle, 1994). Temperature values at the four different studied fish farms are still classified as supporting the growth rate of *Clarias gariepinus* and do not have an adverse effect on the other parameters (Nchegang *et al*., 2024).

The results of water pH during the period of measurement, with the highest mean pH recorded in the Cornelius Fish Center Farm, had a mean pH value of

8.16±1.25 in the juvenile tanks**.** Hence, the pH range measured in this study is still classified as the optimum pH to support *Clarias gariepinus* life (Nchegang *et al*., 2024). According to a suitable pH for *Clarias gariepinus* farming ranges from 6.5 to 8.5, a pH which is high or above 8.5 can cause increased toxicity in the water, but if the pH is low or below 6.5, it can inhibit the growth rate of *Clarias gariepinus*. Ndubusi *et al*., (2015) stated that microorganisms generally have growth conditions with a pH of 4-9.5. Ammonium compounds that can be ionized are found in waters that have a low pH (Nchegang *et al*., 2024). Ammonium is not toxic, but at high pH conditions, there is more ammonia that is not ionized and is toxic. So, based on research that has been carried out, the pH range during research on *Clarias gariepinus* maintenance in the four stations met the recommendation and is good enough for catfish farming (Nchegang *et al*., 2024).

The highest mean electrical conductivity recorded during the period of experiment was 0.84±035ms/cm in the Cornelius fish farm in table fish tanks (Nchegang *et al*., 2024). Also, the highest mean for total dissolved solids of 0.38±18ppt was recorded in the Cornelius table fish farm.

Dissolved oxygen measurements were within the recommended range reported by Viveen *et al*., (1985) for *Claris gariepinus* in all farms except in the fish farm in Bakingili (Nchegang *et al*., 2024).

The values for nitrite obtained in IRAD Batoke, Cornelius fish center, and Oumarou fish farm were <0.025mg/l, <0.01mg/l and <0.01mg/l, respectively (Nchegang *et al*., 2024). According to Chapman and Kimstach, (1996) mean concentration of 1.8 mgL−1 of nitrite was found to be the upper recommended limit for *Clarias gariepinus* survival and growth. Nitrite concentration in reservoirs ranges from less than 1.0 mgL^{-1} to 5.0 mgL^{-1}.

Ammonium measurements obtained in the broodstock tanks at IRAD were <0.05 mg/l, <0.05 mg/l, and <0.05 mg/l, respectively, with all being within the recommended range as reported by Viveen *et al.*, (1985), Nchegang *et al*., (2024).

Table 1. Physicochemical parameters ex-situ for catfish (*Clarias gariepinus*) tanks

Parameters	IRAD Batoke	Cornelius	Oumarou fish farm
NH4	<0.05 mg/l	<0.05 mg/l	<0.05 mg/l
NO2	<0.025 mg/l	<0.01 mg/l	<0.01 mg/l
O2	<8 mg/l	< 8 mg/l	< 8 mg/l

The biological data collected revealed favorable growth performance as culture conditions in the various farms were within the optimal levels for *C. gariepinus* (Nchegang *et al*., 2024). This was due to the fact that the physicochemical parameters of water were carefully monitored and controlled. However, there was an exceptional case for the growth performance of fingerlings tanks in IRAD Batoke **/** Bakingili; for which the growth performance was more favorable in Batoke fingerlings compared to Bakingili (Nchegang *et al*., 2024). This could be

Table 2. Biological parameters of African catfish

Clarias gariepinus	Stations	Biological Parameters					
		Mean number of fish per tank	Initial mean Weight(g)	Final mean Weight(g)	Total weight Gain(g)	Total length(cm)	Standard length(cm)
Fingerlings	Batoke	6864	4.65	44	39.54	11.23±2.33	9.21±2.14
	Bakingili	2000	3.46	38.50	35.04	9.31±1.01	8.31±1.25
Juveniles	Batoke	1200	94.67	243.11	148.44	24.27±9.41	21.04±8.44
	Cornelius	5000	98.25	260.50	162.25	23.54±8.08	19.62±6.60
Table fish	Bakingili	60	205	1307	1102	34.86±5.82	31.67±6.23
	Cornelius	10000	212	535	323	36.12±9,54	33.07±7.55
	Oumarou	30	245	376	131	37.34±9.78	33.64±7.78
Broodstocks	Batoke	35	1519.5	2048.50	529	75.24±25.53	67.12±25.94
	Cornelius	100	2400	3700	1300	68.26±9.42	59.23±23.59
	Oumarou	145	1134.14	2177.14	1043	78.44±12.79	66.92±13.91

due to the methods of feeding, since IRAD Batoke Station feeds appropriately with high-quality feeds which are extruded compared to Bakingili and the other fish farms (Cornelius and Oumarou), where feeds of variable quality (locally pelleted feeds often used with extruded feeds from time to time) (Nchegang *et al*., 2024). We could also add the factor of human resources.

Batoke fish farm, which is being run by the research station, has skilled personnel who have mastery of the fundamentals of fish production, unlike the other farms which are owned and managed by individuals who have not obtained adequate trainings.

4. CONCLUSION

This study revealed a strong correlation between fish tanks, water quality parameters and the growth performance of the fish in all the studied localities (Nchegang *et al*., 2024). Thus, the a need to maintain healthy water quality through appropriate feeding and water quality management approaches.

The study also revealed that, although the African catfish can tolerate poor water conditions, optimal water quality is crucial for its optimal growth and development. These findings are relevant to the fish production sector for the sustainable production of the African catfish.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc.) and text-to-image generators have been used during the writing or editing of manuscripts.

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COMPETING INTERESTS

Authors have declared that no competing interests exist.

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Chapter 2
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Advancing Geohazard Mapping and Monitoring Using Terrestrial LiDAR: Insights from the British Geological Survey

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ABSTRACT

Geomatics is defined in the ISO/TC 211 series of standards as the discipline concerned with the collection, distribution, storage, analysis, processing and presentation of geographic data or geographic information. It is the discipline of electronically gathering, storing, processing and delivering spatially related digital information; it continues to be one of the fastest expanding global markets, driven by technology. The aim of this chapter is to review the types of work that the BGS carries out using terrestrial and aerial LiDAR systems, and to discuss the merits therein. The British Geological Survey (BGS) geomatics capabilities have been utilized in a variety of scientific studies such as the monitoring of actively growing volcanic lava domes and rapidly retreating glaciers; coastal erosion and platform evolution; inland and coastal landslide modelling; mapping of geological structures and fault boundaries; rock stability and subsidence feature analysis and geo-conservation. In 2000, the BGS became the first organization outside the Mining Industry to use Terrestrial LiDAR Scanning (TLS) as a tool for measuring change; paired with a Global Navigation Satellite System (GNSS), BGS were able to measure, monitor and model geomorphological features of landslides in the United Kingdom (UK) digitally. Many technologies are used by the BGS to monitor the Earth, employed on satellites, aeroplanes, drones and ground-based equipment, in both research and commercial settings to carry out mapping, monitoring and modelling of Earth surfaces and processes. Outside BGS, these technologies are used for close-range, high-accuracy applications such as bridge and dam monitoring, crime and accident scene analysis, forest canopy and biomass measurements and military applications.

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Key methodologies in TLS are discussed under the headings of Mobile Mapping, Monitoring Change, Virtual Outcrop Modelling and Uncrewed Technologies, featuring examples of work for each. This research underscores TLS's transformative potential in geosciences, particularly for advancing disaster resilience, environmental monitoring, and geological conservation efforts. Terrestrial Laser Scanning (TLS) offers high-resolution, three-dimensional data collection capabilities that can significantly enhance the understanding of geological processes and environmental changes.

Keywords: Terrestrial LiDAR; geohazards; mapping; monitoring; modelling.

1. INTRODUCTION

Terrestrial LiDAR has often been used to provide detailed characterization of structural diversity at local scales, but it is largely unknown whether these same structural features are detectable using aerial LiDAR data that are available across larger spatial scales (LaRue et al., 2020; Singh et al., 2024). Geomatics is defined in the ISO/TC 211 series of standards (Anon, 1994) as the discipline concerned with the collection, distribution, storage, analysis, processing and presentation of geographic data or geographic information. It is a relatively new term that has been around in the scientific literature for more than thirty years (Krawczyk, 2022). This broad term applies to both science and technology, and integrates the more specific disciplines and technologies of geodesy, surveying, mapping, positioning, navigation, cartography, remote sensing, photogrammetry and geographic information systems (Jones & Hobbs, 2021). The discipline of geomatics continues to be one of the fastest-expanding global markets (Jones, 2015). Ground-based geomatics is driven by technology and, at present, there is a huge demand for land, hydrographic, and engineering surveyors who are able to utilise this technology fully. This is due to a high media profile, the changing nature of mapping and spatial data management worldwide, the need for faster and more accurate data gathering, and the growth in European Union and national governments' spatial data agendas and legislation. As the underpinning information provider of the land and property lifecycle, geomatics is of fundamental importance to society (Jones & Hobbs, 2021).

The most common geomatics applications in the British Geological Survey (BGS) use a Terrestrial Light Detection and Ranging (LiDAR) Scanner (TLS) or the Pegasus Backpack Mobile Mapping Solution (MMS) and a Global Navigation Satellite System (GNSS) in order to obtain a terrestrial equivalent of an airborne LiDAR survey and create a 3D model. Combining a terrestrial LiDAR scanner with a high-resolution digital camera and a high-precision differential GNSS enables colored point clouds, textured triangulated surfaces, or orthophotos with depth information to be accurately geo-referenced and captured (Jones & Hobbs, 2021). The relative distance, elevation angle and azimuthal angle between the laser and the subject are measured in each scan and once processed, a 3D surface model can be generated. From the 3D models created, a variety of products are derived including Digital Surface Models (DSM), Digital Terrain Models (DTM), Virtual Outcrop Models (VOM), Cross-Sections, Area & Volume

Calculations, 3D Photo-realistic Video 'Fly-throughs', Discontinuity Maps, Stratigraphic Facies Profiles, Attributed 3D Reservoir Models, Fossil Assemblage Maps, Surface Deformation Feature Recognition Layers, Soil Erosion Maps, Cave Surveys and Change Models (Jones & Hobbs, 2021). TLS allows geological outcrops to be digitally captured with unprecedented resolution and accuracy (Buckley et al., 2007). Structure from Motion photogrammetry (SfM) has emerged as an alternative viable technology, whether deployed on an Unmanned Aerial Vehicle (UAV) or otherwise. It is lighter, more compact, cheaper to buy and repair and has lower power requirements than TLS. The BGS has six multi-rotor and two fixed-wing small (<20 kg) UAVs, which it uses to collect data for ground surface and geological investigations (Jones & Hobbs, 2021).

The BGS geomatics capabilities have been utilized in a variety of scientific studies (Jones, 2014) such as the monitoring of actively growing volcanic lava domes and rapidly retreating glaciers; coastal erosion and platform evolution; inland and coastal landslide modelling; mapping of geological structures and fault boundaries; rock stability and subsidence feature analysis; geo-conservation and public understanding of science (Jones et al., 2020). These capabilities can be utilized alongside airborne LiDAR and other Remote Sensing techniques in order to provide a range of complementary applications or information (Jordan and Napier, 2015). It is clear that LiDAR scanning technology is playing an increasingly important role in the acquisition of 3D spatial environmental data worldwide (Jones & Hobbs, 2021). This technology and its applications are certain to develop further as the rapidity and accuracy of data acquisition, and associated monitoring and modelling capabilities, become more widely recognized. Outside the BGS, ground-based LiDAR systems are used for close-range, high-accuracy applications such as bridge and dam monitoring (Artese and Zinno, 2020), architectural restoration, crime and accident scene analysis (Kamnik et al., 2020), landslide and erosion mapping, forest canopy and biomass measurements (Moskal and Zheng, 2012, Ariza et al., 2024) mobile mapping systems, adaptive cruise control systems, manufacturing, and military applications (Pejanović et al., 2012, Nafisah et al., 2021, Jones & Hobbs, 2021).

2. GEODECTIC TECHNOLOGY

In 2000, the BGS became the first organization outside the Mining Industry to use TLS as a tool for measuring change; paired with a GNSS, BGS were able to measure, monitor and model geomorphological features of landslides in the United Kingdom (UK) digitally. Since the acquisition of the first TLS, this technology has proved to be essential for the monitoring of geohazards in both the UK and overseas (Jones & Hobbs, 2021). The authors have ~20 years of experience in engineering surveying techniques, using TLS, GNSS and MMS. They have carried out terrestrial scanning and terrain modelling of inland and coastal landslides, eroding coastlines, actively growing volcanic lava domes, retreating glaciers, rock stability and subsidence features, soil erosion and geo-conservation. Modern Terrestrial LiDAR Scanners have significantly increased the level of resolution and accuracy achievable, and the speed of data acquisition

has dramatically increased, from 4 pts/sec in the early days to around 2M pts/sec latterly (Jones & Hobbs, 2021). A combination of airborne and terrestrial LiDAR can create high-resolution digital terrain models (DTM) and provide realistic 3D models of built infrastructure with accurate rooftops and realistic building facades. Combining these factors, with the increase in point density and accuracy achievable, means that the ability to capture and measure temporal changes in geological features via repeat surveys has revolutionized ground-based geomatics research (Jones, 2019).

UAVs are becoming increasingly common for civilian use, with the Chinese manufacturer DJI dominating the market (Jones & Hobbs, 2021). They are used in such fields as surveillance (West and Bowman, 2016), filmmaking (Karakostas et al., 2018), scientific research, surveying, disaster relief (Nguyen et al., 2019) law enforcement, archaeology (Tapete et al., 2017), forestry and agriculture, as well as recreation. The BGS, like many other organizations, are able to utilize the usability and portability of these aircraft to create photogrammetric images and point clouds using SfM techniques, at a fraction of the cost of using a system mounted on an aeroplane (Jones & Hobbs, 2021). The resultant data, though not as accurate as that obtained from TLS, should still be considered more than adequate for most uses, and gives the ability to create both 2D and 3D full color models. In general, UAV results appear to underestimate the volumes of smaller features, most likely due to the reduced line of sight (Glendell et al., 2017).

Radar Interferometry is another technique used in geodesy and remote sensing to generate maps of surface deformation or digital elevation (Jones & Hobbs, 2021). The technique can potentially measure millimeter-scale changes in deformation over spans of days to years. It has applications for geophysical monitoring of natural hazards, for example earthquakes, volcanoes and landslides, and in structural engineering, in particular monitoring of subsidence and structural stability. It can be deployed in satellites, as Interferometric Synthetic Aperture Radar (InSAR), as the European Space Agency (ESA) launched Sentinel-1A and Sentinel-1B 1-B, which provide global scale coverage on a 6-day repeat cycle. Alternatively, it can be ground-based, as Terrestrial Radar Interferometry (TRI), used primarily to remotely measure the stability of natural and engineered slopes, in near real-time (Jones & Hobbs, 2021). It can be used for a wide variety of applications, including surface mining operations, quarrying, landslides, rockfalls, glacial retreat, volcano growth and coastal retreat, but at considerable cost.

Mobile mapping is the process of collecting geospatial data from a mobile vehicle or backpack, typically fitted with a range of GNSS, Photographic, Radar or LiDAR systems. These are combined with time synchronized navigation and imaging sensors. This gives the ability to survey an area as fast as you can drive, or walk.

Many publications have been written about these techniques and their relative merits and pitfalls. A comparison of the above techniques is given in Table 1, along with some useful references detailing their typical uses (Jones, 2019, Glendell et al., 2017, Giordan et al., 2020, Xiaoye, 2008, Gaong et al., 2025,

Noori et al., 2024, Takada and Motono, 2020, Toschi et al., 2015). The table presents a synopsis of the pros and cons of each of the techniques, the times taken for data collection in the field and data processing back in the office, and some typical costs of both the hardware (purchase or hire) and software (full price and/or yearly licence) that are required (Jones & Hobbs, 2021).

Table 1 shows that LiDAR is the best technique to use in order to generate digital models of the earth's surface and accurate topographic maps for planning, construction, engineering design and environmental management. It is also best suited for measuring distances, heights, areas, volumes and quantities. However, it does have its limitations (Montoya-Sánchez et al., 2024). LiDAR data is not colored and relies on RGB overlays from photography. The point clouds it generates are often sparse (dependent on point density) compared to photogrammetry, giving a general shape rather than contextual detail. Terrestrial LiDAR can also suffer from gaps (or voids) in the data, primarily where there are shadow areas caused by sight-line issues or inaccessible sections due to adverse terrain.

Table 1. Pros and cons of geodetic techniques

System	Terrestrial LiDAR	Airborne LiDAR	Aerial SfM	UAV SfM
Pros	Long range	Very long range	Good accuracy	Good accuracy
	High accuracy, high precision	Downward scanning	Full colour	Full colour
	High speed data acquisition	Cover large areas	2D & 3D models	2D & 3D models
	Built-in camera	Good accuracy	Downward looking	Down & side looking
Cons	Shadow' areas			Pilot required
	Dense vegetation	Very expensive	Dense vegetation	Dense vegetation
	Severe weather	Lower accuracy	Expensive	Severe weather
	Cloud & fog	Downward scanning	Downward looking	Strong wind
Data Collection	Active Sensing	Active Sensing	Passive Sensing	Passive Sensing
	Survey: 1 hour - 5+ days	Survey: 4 hours - 5+ days	Survey: 4 hours - 5+ days	Survey: 1 hour - 5+ days
	Data: 1 day - 5+ days	Data: 1 day - 5+ days	Data: 1 day - 5+ days	Data: 1 day - 5+ days
Typical Cost	Hardware: £130k	Hardware: £20k (rental)	Hardware: £20k (rental)	Hardware: £5k
	Software: ~£2.5k p.a.	Software: ~£10k p.a.	Software: £3500	Software: ~£3k p.a.
Reference	13	20, 21	22, 23	18, 19
System	Terrestrial Radar	Satellite Radar	Backpack MMS	Vehicle MMS
Pros	Extremely long range	Extremely long range	Scan as fast as can walk	Scan at up to 30mph
	Very high accuracy	High accuracy	High accuracy	High accuracy
	Real time, continuous monitoring	Cover large areas	High speed data acquisition	High speed data acquisition
	Works in cloud & fog	Works in any weather	Works inside & outside	Built-in cameras
Cons	Dense vegetation	Dense vegetation		Heavy
	Severe weather	Expensive	Only scan where walk	Only scan where drive
	Strong wind	Downward looking	Point density	Expensive
	Points must be coincident	Points must be coincident	Bad weather	Severe weather
	Heavy			
Data Collection	Active Sensing	Active Sensing	Active Sensing	Active Sensing
	Survey: 1 hour - 5+ days	Survey: Multiple days	Survey: 1 hour - 5+ days	Survey: 1 hour - 5+ days
	Data: 1 day - 5+ days	Data: 1 day - 5+ days	Data: 1 day - 5+ days	Data: 1 day - 5+ days
Typical Cost	Hardware: £140k	Hardware: Free (data)	Hardware: £230k	Hardware: £350k
	Software: ~£2.5k p.a.	Software: ~£10k p.a.	Software: ~£16k p.a.	Software: ~£16k p.a.
Reference	24, 25	26	27	28

3. TLS OUTSIDE BGS

In the UK, the Office of Science and Innovation identified LiDAR technology within one of eight key clusters of emerging science and technology with "the potential to...transform the delivery of public services, challenge society and/or affect wealth creation" (Jones & Hobbs, 2021). Following this, LiDAR: net was set up to bring together a network of interested parties from academia, industry and NERC. Participants include the Universities of Leicester, Salford, Nottingham, Aberystwyth, Glamorgan, Loughborough and Durham, Queen's University Belfast, Infotera, 3D Laser Mapping, Halo-Photonics, Willis Research Network, Halcrow Group Ltd., Ordnance Survey and BGS. Deliverables included the running of dedicated workshops and a scoping report to input into both the technical Earth Observation development and the NERC Technologies program.

Other major exponents of terrestrial LiDAR scanning in the UK include Geomatics Group (Environment Agency), Merrett Survey Partnership, Digital Surveys, Bluesky International Ltd., Terrain Surveys, Manchester University and University College London (Jones & Hobbs, 2021).

Outside the UK, the USGS employs terrestrial LiDAR technology for a variety of projects, including shoreline mapping, digital elevation and terrain mapping, dam monitoring, soil erosion, snow measurement, ground deformation & landslide mapping, flood hazard mapping & levee failure analysis, habitat mapping and volcano monitoring. It has setup the Center for LiDAR Information Coordination and Knowledge (CLICK) in order to facilitate data access, user coordination and education. UNAVCO maintains a pool of TLS instruments and associated peripherals, digital photography equipment, software and ancillary equipment optimized to support earth science investigators (Jones & Hobbs, 2021). Projects include landside deformation studies in Puerto Rico, wildfire extent surveys in Colorado, volcano conduit process modelling in Utah and biomass measurements in Florida.

European universities with ground-based geomatics departments include Z-GIS at the University of Salzburg (Austria), Delft University of Technology (Netherlands), University of Bergen (Norway) and Institut Cartografic de Catalunya (Spain). Other organizations working in terrestrial LiDAR include Laserdata (Austria), IMAO (France), SPECIM (Finland), IGI TopScan (Germany), Sineco (Italy), Artescan 3D (Portugal), Dielmo 3D (Spain), Galileo Group Inc. (USA) and Fugro (Jones & Hobbs, 2021).

The BGS uses Riegl and Faro terrestrial LiDAR scanners. Besides these, the major manufacturers of terrestrial LiDAR systems are Teledyne Optech (Canada), Z+F (Germany), Topcon (Japan), Leica (Switzerland), MDL (UK) and Trimble (USA). The major software companies with programs able to download and process LiDAR data include Virtalis, Golden Software, Maptek, Applied Imagery, Qcoherent, Pointools, Innovmetric, Blue Marble, Erdas and Esri (Jones & Hobbs, 2021). Open source and free viewers are available to download,

including Cloud Compare, Quick Terrain Reader, Fugro Viewer, Mars Viewer, MeshLab and LASTools.

Over the past two years, the use of TLS has increased dramatically:

- In the gaming industry, it is used to quickly and precisely render whole cities, or to recreate an accurate reproduction of a racetrack, including every undulation, etc. (Jones & Hobbs, 2021).
- The Police can record the scene of a car accident within a few minutes, enabling the emergency services to clear the scene, reducing traffic jams as well as preserving the evidence digitally.
- The automotive industry uses TLS as a guidance system for autonomous vehicles. Near real-time processing allows the device controlling the vehicle to detect obstacles and to update its route almost instantly (Jones & Hobbs, 2021).
- TLS gives archaeologists the ability to create high-resolution DEMs of archaeological sites that can reveal micro-topography that may otherwise be hidden by vegetation.
- LiDAR is used in a variety of ways in meteorology including the studies of atmospheric composition, structure, clouds, and aerosols.
- In astronomy, a worldwide network of observatories uses LiDAR to measure the distance to reflectors placed on the moon, allowing the moon's position to be measured with mm precision.
- LiDAR allows conservation research scientists to not only measure the height of previously unmapped trees but to determine the biodiversity of the forest (Jones & Hobbs, 2021).

In geology and seismology, a combination of aircraft-based LIDAR and GPS have evolved into an important tool for detecting faults and measuring uplift. This combination was used most famously to find the location of the Seattle Fault in Washington, USA. The USGS is using TLS more and more for land surface analysis, including the monitoring of sand dune movement and soil microtopography and erosion.

4. MOBILE MAPPING

In January 2018, the BGS became one of the first companies in the UK to purchase the Leica Pegasus Backpack, a wearable reality-capture sensor platform, combining five cameras with two LiDAR profilers and a triple band GNSS and Inertial Measurement Unit (IMU), weighing less than 13 kg (Jones & Hobbs, 2021). It is designed to be deployed quickly and easily and can be used for indoor, outdoor or underground surveys, offering speed and portability advantages over existing terrestrial LIDAR systems, which can weigh in excess of 30 kg and are extremely bulky to carry. This enables the BGS to carry out 3D mapping of long coastal (or inland) sections, landslides and other unstable surfaces, ice sheets and glaciers, railway lines, roads and other linear features (Jones & Hobbs, 2021). The backpack can also be vehicle-mounted, for fast mobile mapping, or pole-mounted, for lowering into voids or sinkholes.

Mapping underground structures has always proved to be quite difficult and complex, especially related to the issues of positioning and visibility. The Pegasus Backpack allows the user to move from an area of good GNSS coverage to one with limited or even no coverage. The BGS used this system when mapping the show caverns (old mineshafts) associated with the Heights of Abraham Park (Fig. 1), with a 3D section of part of a cavern shown in Fig. 2 (Jones & Hobbs, 2021). Use of the Leica SLAM (Simultaneous Localization and Mapping) technology and the high precision IMU ensured the achievement of accurate positioning, even during GNSS outages, and with the addition of the external light-source, precise scanning of the caverns was possible (Jones & Hobbs, 2021).

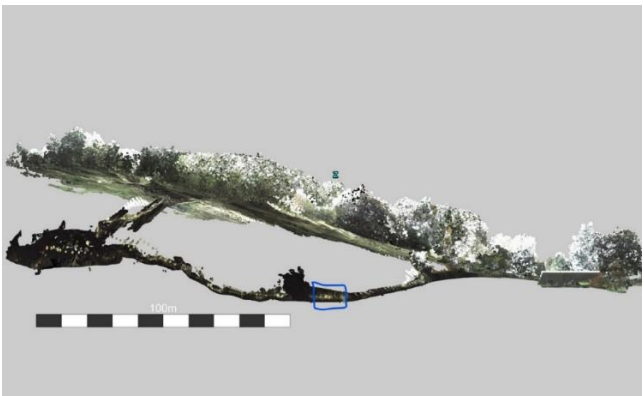


Fig. 1. Elevation view of the Masson Cavern at the Heights of Abraham, Derbyshire (BGS, 2018)

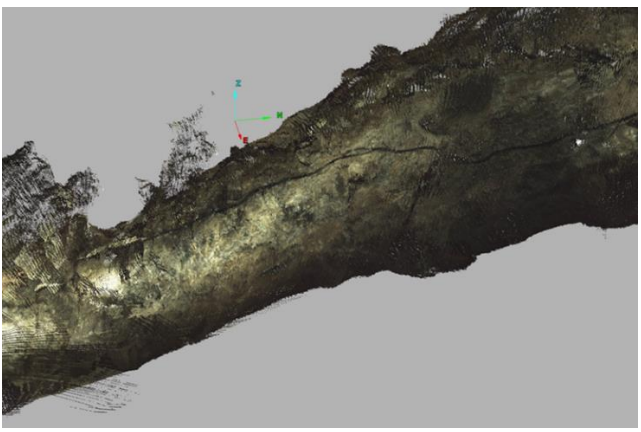


Fig. 2. Close-up view of a scanned section of the Heights of Abraham, Derbyshire (BGS, 2018)

The MMS technology has been used by BGS for many current and emerging developments in laser scanning surveys. The Pegasus Backpack has proven invaluable in many projects, some of which are shown in Table 2; where one Pegasus Walk can cover the same area as multiple TLS set-ups. The resolution of both surveys will be equivalent, but the point density of the Pegasus will not be as great as that of the TLS survey (Jones & Hobbs, 2021).

Table 2. Pegasus Backpack Surveys

Project	Use	Reason	Figure
Bacton Sandscaping Project	Initial geological modelling	4 km long cliff section	
Hollin Hill Landslide Observatory	Inland landslide monitoring	1 Pegasus walk = 17 TLS setups	
Aldbrough Coastal Observatory	Coastal landslide monitoring	Extend coastal section, beach and foreland	
Incipient Block Failure, Dover	3D survey & change anaylysis	Difficult ground, requiring multiple TLS setups	
Nottingham Castle Pre-design	Internal & external facade scans	Impossible with any other form of laser scanner	3 & 4
BGS Geological Walkway	Incorporation into Minecraft	2 Pegasus walks = 12 TLS setups	
Tenby Beach E-trike Survey	Tidal sand volume calculations	8 km on E-trike only possible with Pegasus	5



Fig. 3. Plan view of Nottingham Castle and grounds, Nottingham (BGS, 2018)



Fig. 4. Pegasus scan from inside the Long Gallery of Nottingham Castle (BGS, 2018)

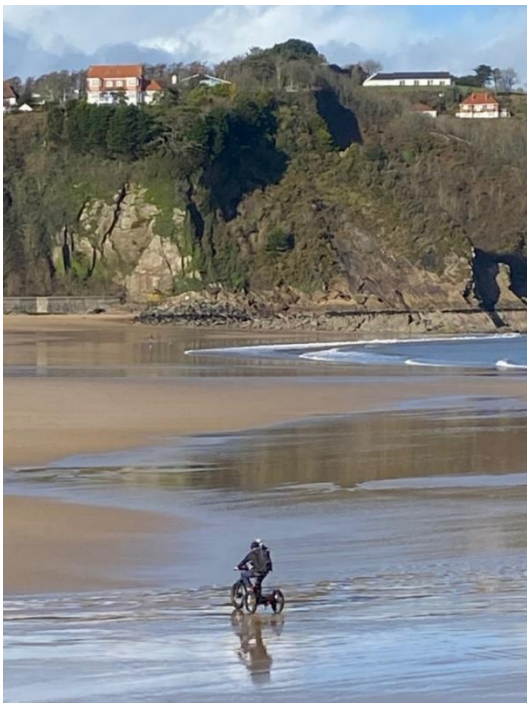


Fig. 5. Pegasus Backpack mounted on an e-trike at Tenby, Wales (BGS, 2020)

5. MONITORING CHANGE

BGS uses its laser scanning systems to monitor actively eroding sections of coast around the UK. The techniques developed are ideal for this application as TLS is particularly suited to measuring vertical cliff sections (Jones & Hobbs, 2021). Due to the orientation of these sections, measurements from a plane or satellite may only be able to measure the top edge of the cliff and the platform below, whereas TLS may struggle with data voids on long, non-linear sections. These voids are gaps, or holes, in the data, usually caused by shadow areas or temporary obstructions. UAVs may offer a compromise solution; they are able to survey both the top and face of the section, albeit at a lower resolution, without any data losses.

Measuring these features in the field is often very dangerous, due to tidal restrictions and slope access (Jones & Hobbs, 2021). These techniques enable the measurement of change in the whole cliff section, from a safe distance, and can be used to model how internal processes within the cliff slope affect coastal erosion.

The BGS Coastal Landslide Observatory, at Aldbrough, East Riding of Yorkshire, has been monitored since 2001, using a variety of GNSS, TLS and MMS techniques (Hobbs et al., 2019). The cliff faces northeast, is low (17m) and of regular height. It consists of glacial till and is actively receding, both by toppling and rotational mechanisms (Fig. 6) (Jones & Hobbs, 2021). This work has shown that the cliff is eroding at a rate of up to 3m per year; both landslides and the direct action of the sea crashing against it cause this erosion (Hobbs et al., 2019).



Fig. 6. Rotational landsliding is the primary mechanism at Aldbrough (BGS, 2017)

Terrestrial LiDAR surveys have been carried out at the Aldbrough site over a 19-year period from 2001 to 2020 (47 surveys in total), centered on the same section of cliff and platform (Jones & Hobbs, 2021). The scans have varied in the area covered, and this has necessitated clipping to an arbitrarily defined datum in order for DTM's, change models and volume calculations to be produced from them. Since 2012 the utilization of the Riegl VZ-1000, with its improved scan range of 1400 m, accuracy of ± 8mm, measurement rate of 62000 points/second and high-resolution digital camera, the quality and resolution of the scans and therefore the surfaces created from them has improved considerably (Fig. 7). Added to this is the improvement in the acquisition and pre-processing software (RiSCANPro, Infinity), the post-processing software (I-Site Studio, Cloud Compare) and the visualization software (QT Modeler, Global Mapper, GeoVisionary) leading to improvements in the ability to visualize and quantify change more accurately and at better scales (Fig. 8) (Jones & Hobbs, 2021). The histogram in Fig. 8 shows the colors associated with gain (blue) and loss (red) and also the quantity of each of these values, and where they occur within the model.

The BGS operate an observatory site at Virkisjökull in southeast Iceland, studying the evolution of the glacier and the surrounding landscape and their responses to regional climate change. Sensors at the site were constantly

collecting climate and seismic data, on an hourly basis, over a ten-year period, from 2009 to 2018 (Jones & Hobbs, 2021). Repeated high resolution (25-40 mm accuracy) surveys were carried out annually, over the same time period, and studied how both the glacier and land surface, and the deposits beneath, changed over time. Cutting-edge technologies, not used in such a combination anywhere else in the world, were used to give unique insights into processes of landscape formation and responses of glacial systems to climate forcing. Virkisjökull and its neighbor Falljökull are retreating rapidly, like most glaciers in Iceland (Anon, 2018). Since 1996, the glacier margin has retreated nearly 500m, and it appears that this rate has accelerated over the last five years (Fig. 9) (Jones & Hobbs, 2021).

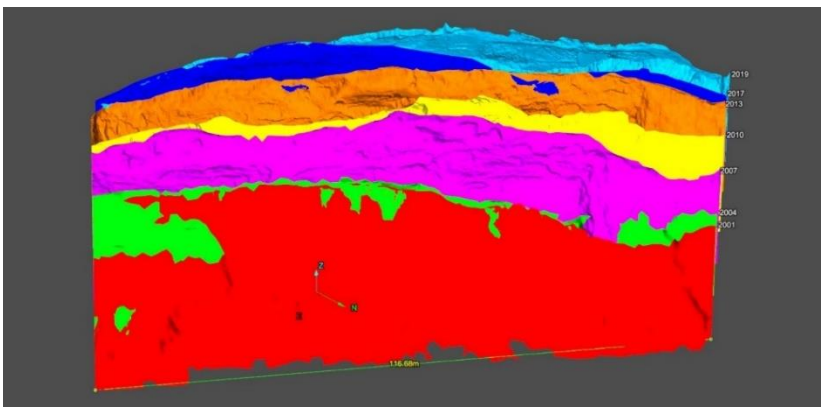


Fig. 7. 3D models of the retreating cliff line at Aldbrough, 2001 to 2017 (BGS, 2021 (Jones & Hobbs, 2021))

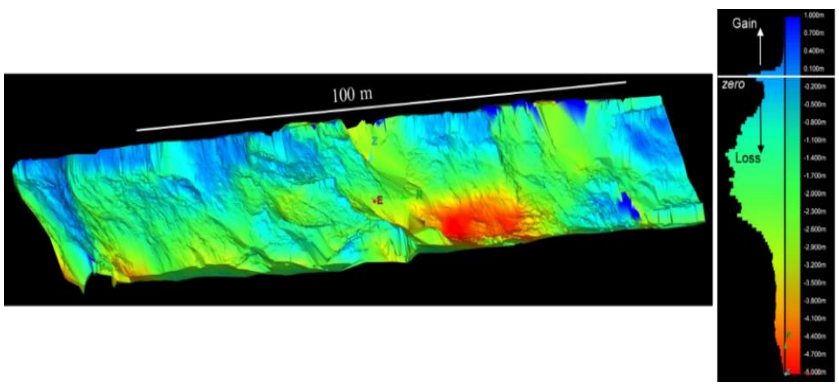


Fig. 8. Aldbrough change model, colored by elevation change, Feb 2016 – Mar 2017 (BGS, 2021(Jones & Hobbs, 2021))

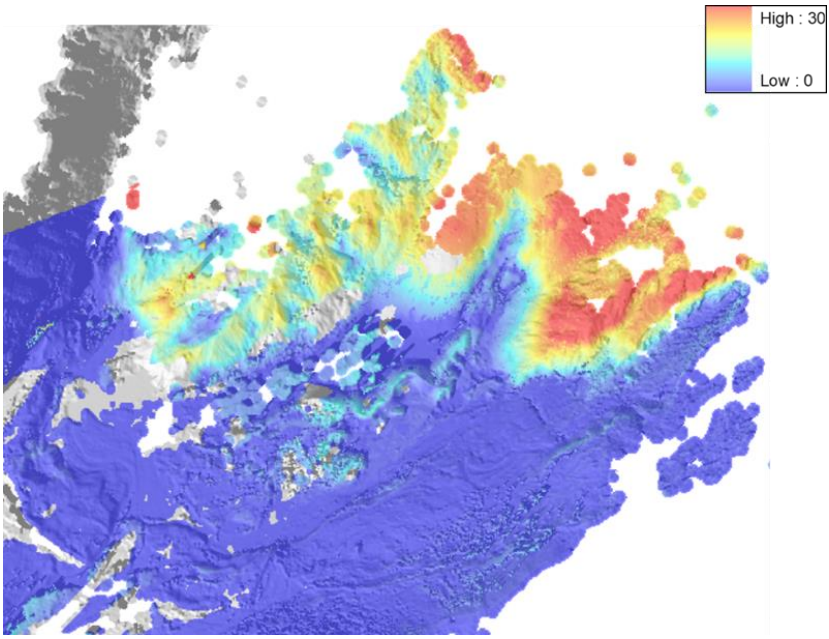


Fig. 9. Falljökull glacial margin; change detection model: 2009 to 2012 (BGS, 2021 (Jones & Hobbs, 2021)

By using a combination of repeat TLS surveys, on-ice based GNSS and Ground Penetrating Radar (GPR), the BGS have been able to see the full picture of glacial retreat for the first time (Fig. 10) (Jones & Hobbs, 2021). This unique approach allowed the TLS surveys to show changes in the glacier, glacial ice margin and glacial foreland, whilst the GNSS stations indicated the speed and direction of glacial flow. This work has shown that the margin of the Falljökull glacier has ceased moving and is now undergoing stagnation (Philips et al., 2013). However, field and photographic evidence show that the icefall remains active, feeding ice from the accumulation zone on Öraefajökull to the lower reaches of the glacier. To accommodate this continued forward motion, the upper section of the glacier below the icefall is undergoing intense deformation (folding and thrusting) and, as a result, is being thrust over the lower, immobile section of Falljökull (Philips et al., 2014). This type of behavior has never been described before and could have implications for how other steep, mountain glaciers around the world are responding to changes in the climate (Dochartaigh et al., 2019; Jones & Hobbs, 2021). Because of this Falljökull glacier has been called the ‘zombie’ glacier (Anon, Zombie Glacier Surprises Scientists, 2014).

There can be few more hazardous situations than that of monitoring a volcanic andesite lava dome for signs of an impending collapse. Partial collapse of a lava dome generates hot, fast-moving pyroclastic density currents (Jones & Hobbs,

2021). Monitoring in such circumstances requires that measurements be taken from a distance that minimizes the threat from eruption and from asphyxiation by volcanic gases. The method also needs to be rapid to minimize the time spent by the monitoring team in the hazardous zone (Jones, 2006). We have used TLS techniques to monitor the growth of the lava dome of the Soufrière Hills Volcano, on the Caribbean Island of Montserrat (Fig. 11). In May 2006, our measurements helped to identify the changes in lava dome growth rate that directly preceded a total lava dome collapse (Ryan et al., 2010). The technique utilizes multiple passes of the scanner, as the presence of volcanic gas and ash (from occasional rock falls) clouds can lead to a reduction of 10-20% in the number of scan points returned (Jones & Hobbs, 2021). Multiple passes increase the chance of 'seeing' the dome through these drifting clouds.

6. VIRTUAL OUTCROP MODELLING

A Virtual Outcrop Model (VOM) is a digital 3D representation of the outcrop surface, mostly in the form of a photorealistic textured polygon mesh or a point cloud with associated RGB data.

The BGS has operated a landslide field observatory site at Hollin Hill, in North Yorkshire, since 2004 (Fig. 12) (Jones & Hobbs, 2021). It brings together a combination of field surveying, geomorphological, geophysical, engineering and hydrogeological expertise at BGS. The landslide is several hundred meters wide and extends some 200 m down slope. It is located on the south-facing side of a degraded Devensian ice-margin drainage channel; the slope has an angle of approximately 12° (Jones & Hobbs, 2021).

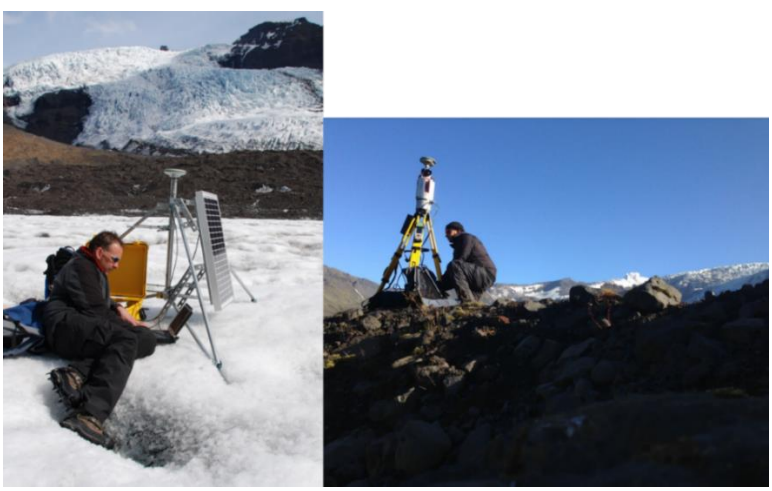


Fig. 10. Data download of on-ice GNSS unit. Terrestrial LiDAR Scanning of glacial foreland, Falljökull Glacier, Iceland (BGS, 2014)



Fig. 11. Scanning the active lava dome of the Soufrière Hills Volcano, Montserrat (BGS, 2014)



Fig. 12. Aerial photograph of the Hollin Hill landslide (BGS, 2011)

The 'complex' landslide is largely caused by the movement of the Whitby Mudstone Formation, which is highly prone to landsliding (Jones & Hobbs, 2021). The landslide is characterized by shallow rotational failures at the top of the slope in the Whitby Mudstone, which move through an area of translational landslide movement and "cascade" over the *in situ* Staithes Sandstone Formation feeding into four larger-scale slow-moving (i.e. tens of centimeters per year) lobes of slumped material at the bottom of the slope (Chambers et al., 2011). Relatively little is known about the hydrogeology of the site. However, a spring line extends across the base of the slope where the Staithes Sandstone overlies the underlying Redcar Mudstone Formation (Fig. 13). Fig. 14 shows a stylized model (not to scale) of the solid geology and slip surface of the landslide (Jones & Hobbs, 2021).

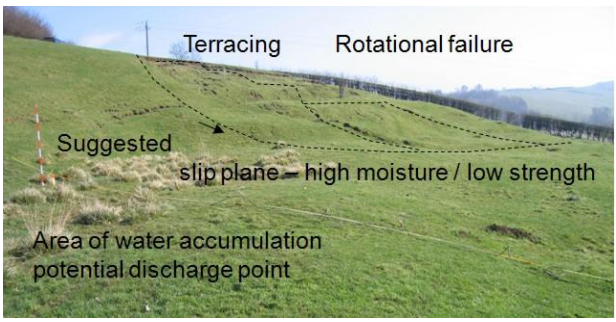


Fig. 13. Annotated photograph of the Hollin Hill landslide (BGS, 2011; Chambers et al., 2011)

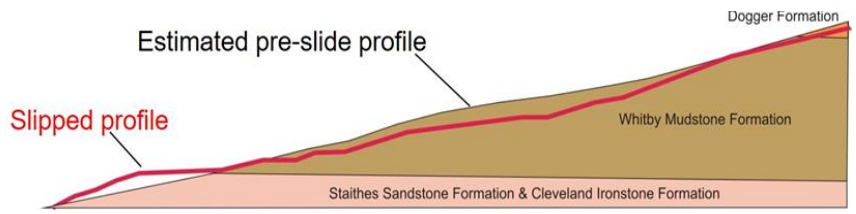


Fig. 14. Solid Geology model of the Hollin Hill landslide – not to scale (BGS, 2011, 39. Chambers et al., 2021)

Both aerial and terrestrial LiDAR surveys have been carried out at this site (Boyd et al., 2021). These have provided geologists at BGS with the opportunity to study the different technologies for monitoring an active slow-moving landslide site in the UK. The initial method of survey is long-range laser scanning. The first (baseline) survey records the ground surface of the landslide and surrounding area in three dimensions (Fig. 15) (Jones & Hobbs, 2021). Subsequent surveys of this active landslide record the movement of the slope. Surveys are carried out at six-monthly intervals, the data collected in the field by LiDAR, and GPS are entered into computer-modelling packages. The results are processed to provide

data to model the way in which the landslide is moving. Information obtained by the Terrestrial LiDAR survey will allow us to monitor the changing shape of the surface and create change models between visits (Fig. 16) (Jones & Hobbs, 2021).

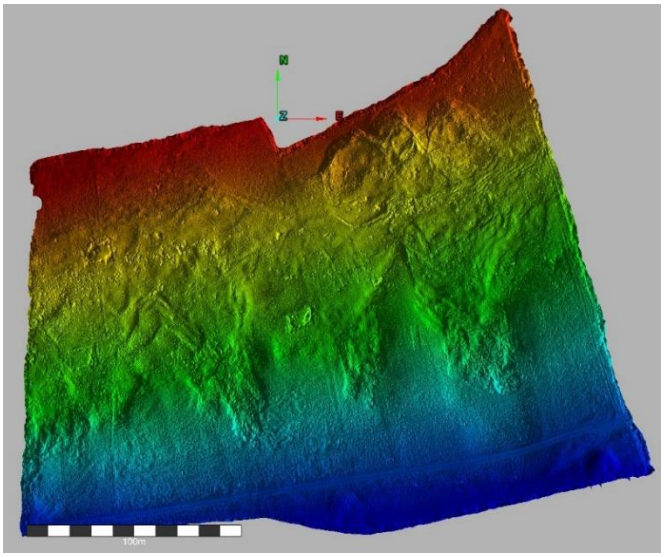


Fig. 15. Elevation model of initial TLS survey of the Hollin Hill landslide (BGS, 2021)

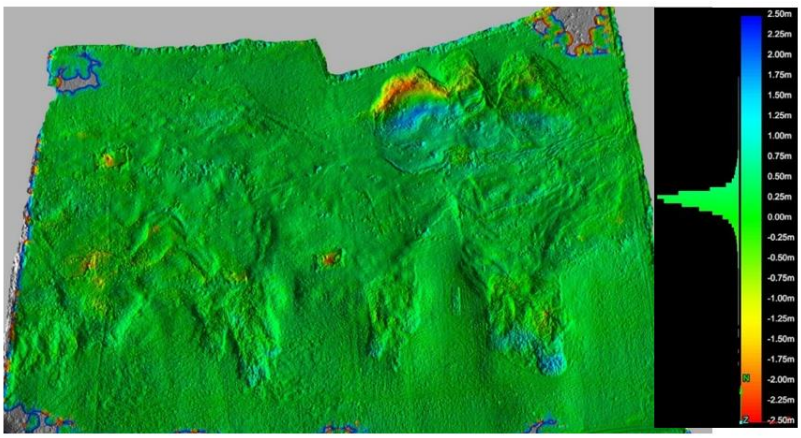


Fig. 16. Hollin Hill landslide Change model (2008 – 2018). Red -2.5m, Blue +2.5m (BGS, 2021)

Since 2018, the landslide has been monitored using the Leica Pegasus backpack mobile mapping solution, enabling digital terrain and surface models (DTM, DSM) to be created in a shorter time, and changes to be assessed (Fig. 17) (Jones & Hobbs, 2021). The use of MMS on this slope means that the shadow areas that are potentially present when using TLS are eliminated (Jones & Hobbs, 2021). However, the point density, and therefore the overall resolution, is lower and issues do arise where the vegetation becomes dense.

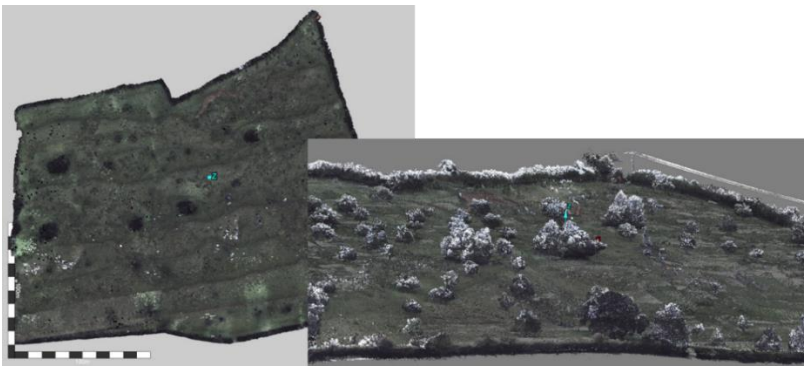


Fig. 17. Digital terrain (DTM) and surface (DSM) models of the Hollin Hill landslide (BGS, 2021)



Fig. 18. Seven Sisters cliff, East Sussex – 4 km long x 75 m high (BGS, 2018)

The Seven Sisters, in East Sussex, is a 4 km long and approximately 75 m high series of eight cliffs between Seaford and Eastbourne (Fig. 18) (Jones & Hobbs, 2021). As part of a larger BGS research project into the distribution of flint beds and their relative positions across the Chalk, TLS was carried out along the full

length of the Seven Sisters site. TLS provided the optimum solution to the problem of measuring the extent and distance between the Flint bands in such a large cliff, which cannot be reached directly. Due to the tidal range and the need to work at a safe distance from the cliffs, the Riegl VZ-1000 TLS was used to carry out the survey (Jones & Hobbs, 2021). The work took 3 days to carry out and was able to provide geologists with extremely detailed (25 mm accuracy) scans of the full section, in both true RGB color and intensity (Fig. 19), the latter of which provides a more noticeable way to view the much darker flint beds and therefore map them more easily (Jones & Hobbs, 2021).

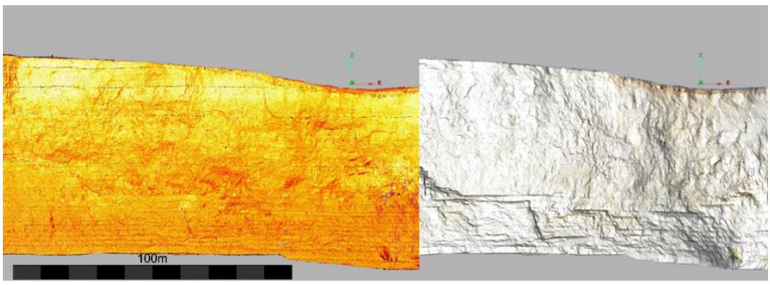


Fig. 19. Intensity and RGB color scans of a section of the Seven Sisters cliff (BGS, 2018)

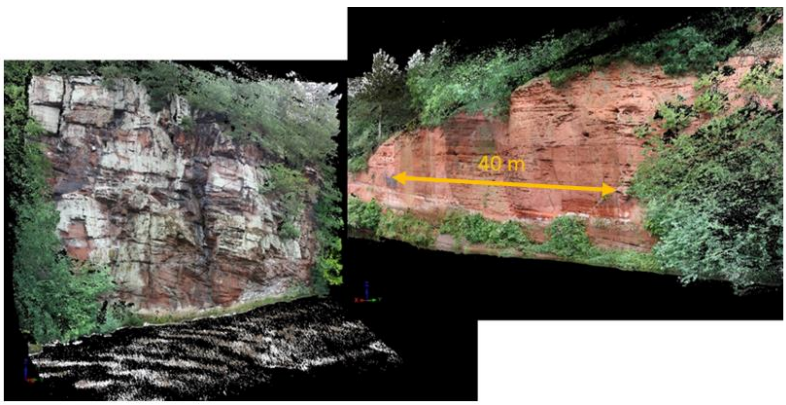


Fig. 20. Virtual Outcrop Models of Eden Valley outcrop (BGS, 2010)

The main objectives of the BGS Permian Sandstone study, in the Eden Valley, are to undertake discontinuity analysis using LiDAR scans and to provide high-quality 3D images for further stratigraphical study (particularly where outcrops are inaccessible) (Jones & Hobbs, 2021). The study also links with the Permo-Trias Cross-Cutting Project, to provide detailed geological information to assess the suitability of Permian and Triassic strata for carbon sequestration and to

create 3d property models for flow modelling of CO_2. Locations of potentially suitable sandstone outcrops were identified by interrogating several BGS data sets, and following consultation with hydrogeological and mineralogical colleagues, six outcrops were scanned in the Eden Valley (Jones & Hobbs, 2021).

The RGB colored LiDAR scans were used in order to create DTM's and virtual outcrop models (VOM's) of the outcrops (Fig. 20). In turn, these models were used to create 3D S-Grid facies models (in SKUA-GOCAD) to be used for CO_2 flow modelling, as part of research into carbon capture and storage in sandstone reservoirs (Fig. 21) (Jones & Hobbs, 2021). These models provide important sub-seismic scale information about geological heterogeneity for flow simulations.

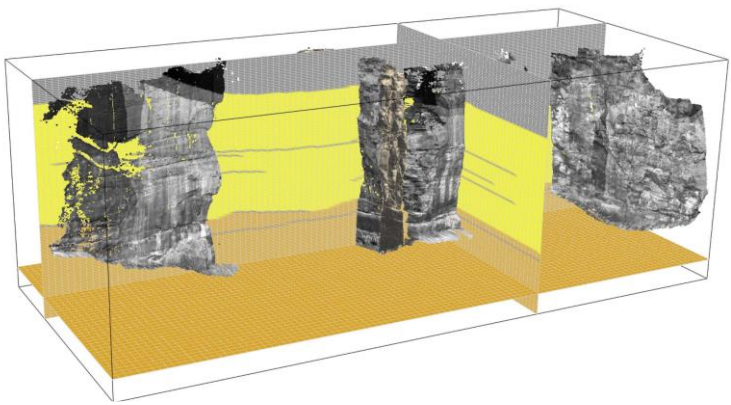


Fig. 21. 3D attributed S-Grid generated from VOM (BGS, 2010)

7. UNCREWED TECHNOLOGY

The British Geological Survey (BGS) uses Unmanned Aerial Systems (UAS), or drones, primarily to collect data for ground surface and geological investigations, mapping, and surveying (Jones & Hobbs, 2021). Drones are a fantastic platform for rapidly capturing data, which can be used to understand and monitor geological processes. The most common deliverables created from the data collected by UAS include Digital Terrain Model (DTM & Digital Surface Model (DSM) generation using Structure-from-Motion (SfM) or Light Detection and Ranging (LIDAR) technologies. RGB photos & videos, 3D maps, ortho mosaics, 2D & 3D meshes & point clouds, and actionable reports. Some of these are shown below. BGS undertake UAS operations under the Civil Aviation Authority approval given in the Uncrewed Aircraft - Operational Authorization granted by the Civil Aviation Authority Permission for Commercial Operations (PfCO) Operator registration number: GBR-OP-NX2PQN84K2VB, CAA reference UAS 2700. All BGS pilots are registered, have a General Visual Line of Sight Certificate (GVC) and are fully insured to fly in accordance with the BGS Flight Operations Manual.

BGS owns and operates a suite of sub-25kg rotary UAS for aerial data collection associated with filming, aerial photography, multispectral imaging, thermal imaging, LiDAR and/or gas detecting conducted in surveying, monitoring and/or mapping modes flown under VLOS (Visual Line of Sight) permissible limits for operation. BGS conduct research using:

- DJI Matrice 210-RTK: Red Green Blue (RGB, Zenmuse X5) and Thermal Infrared (TIR, FLIR XT2)
- DJI Inspire-1 RAW: RGB (Zenmuse X5S) and TIR (FLIR XT)
- DJI Inspire-2: RGB (Zenmuse X5), TIR (FLIR XT2) and Rededge Multispectral (Micasense)
- DJI Matrice 300-RTK: RGB (Zenmuse X5), TIR (FLIR XT2) and Light Detection and Ranging (LiDAR, Zenmuse L1) and Carbon Dioxide and Methane gas detection (GasFinder2)
- DJI Phantom-3 Professional: RGB
- DJI Mavic Air (x2): RGB
- DJI Mini 3 Pro: RGB
- DJI Mavic 3: RGB

UAS technology has been used by BGS for many current and emerging developments in laser scanning, photogrammetric, thermal and gas detection surveys. The Matrice UAS have proven invaluable in many projects, some of which are shown in Table 3; where a UAS LiDAR flight can cover the same area as a large Pegasus Walk or multiple TLS set-ups, in a much shorter time. The resolution and accuracy of these surveys are similar, although the point density of a TLS survey would be greater.

Table 3. UAS Surveys

Project	Type of Flight	Use	Figure
Kilbourne Hole Volcano	LiDAR & Photogrammetric	3D Model	22 & 23
Croft Quarry	LiDAR	3D Geological Model	24 & 25
Selwick Bay	LiDAR & Photogrammetric	3D Map	26 & 27
Slaithwaite Reservoir	LiDAR	DEM	28 &29
Los Humeros Cladera	Thermal & Photogrammetric	Thermal DEM	30 & 31
Gresham Marshes	LiDAR & Photogrammetric	3D Model	32 & 33

Kilbourne Hole Volcano (KBH) is a maar volcanic crater that is between 24,000 and 100,000 years old and forms part of the Potrillo volcanic field, in New Mexico (Fig. 22). This area is part of the Rio Grande Rift, where rising magma interacts with groundwater-saturated sediments. The elliptical crater is approximately 3.2

km long x 2.2km wide. BGS were invited by the NASA Goddard Instrument flight team to join their field excursion to KBH. The core objectives of this trip were:

- To learn more about the history of phreatomagmatic eruptions at the KBH maar volcano.
- To compare terrestrial and aerial LiDAR measurements of surge-bed cross-bedding foresets from within the tuff rings (Fig. 23).
- To contribute to the RISE EVA exercises.



Fig. 22. Kilbourne Hole volcanic crater (BGS, 2021)

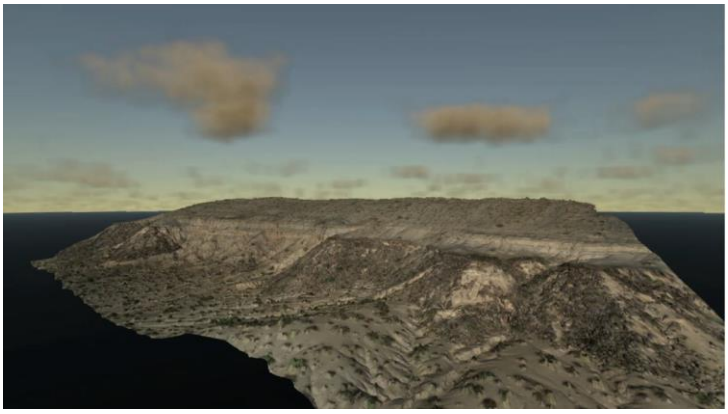


Fig. 23. Photogrammetry results from the SE rim of Kilbourne Hole (BGS, 2021)

UAS flights were defined in DJI Pilot as pre-programmed surveys. Altitude was set at c.50m, with sensor view angles set at NADIR to collect both RGB and LiDAR surveys with flight paths set to enable front-back and side-side overlap. Output pixel resolution of <1.5 cm. OBLIQUE surveys were flown free hand, capturing multiple RGB images that overlapped, to allow for photogrammetry of the crater walls. Output pixel resolution was <1cm. Flights took approximately 40 minutes each to perform, and there were eight in total.

Croft Quarry SSSI forms part of a group of igneous rock bodies that are exposed at, or near the surface, often as upstanding hills in south Leicestershire, known as the South Leicestershire Diorite Suite. Protruding above, and surrounded by, the red Triassic strata, the quartz-diorite and tonalite rocks at Croft have been exploited since Roman times; the current quarry has a depth of -136 m OD, representing one of the deepest quarries in Europe (Fig. 24).



Fig. 24. Aerial view of Croft Quarry, Leics (BGS, 2022)

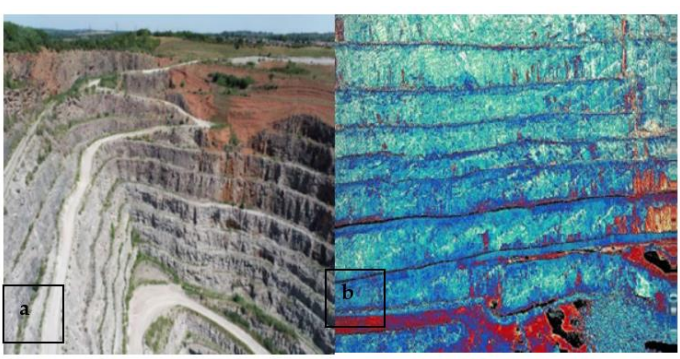


Fig. 25. UAS Image looking ENE showing buried topography (a). TLS image enhanced with multi-spectral filters looking East (b) (BGS, 2022)

The quarry was surveyed using both Terrestrial LiDAR Scanning (TLS) and UAS, from 9 TLS positions, including the upper quarry edge and from within the quarry (10 scans in total). UAS flights were defined in Pilot, and manually, as a nadir and at 45° and 90° using the Zenmuse L1 to perform LiDAR and matching RGB imagery. The nadir altitude was set at 50 m to provide a point cloud density of 283 points/m2 and a photo output pixel resolution of ~1.5 cm, with a side overlap of 70%, the scanning mode set to repetitive, a flight speed of 10 m/s, the number of returns set to three, and calibration enabled. The UAS was flown from 5 positions, both on the upper edge and from within the quarry (10 flights in total, 162 minutes flight time); the data was used to create both a 2D and 3D ortho-rectified photogrammetric image (Fig. 25a) and a 3D RGB colored point cloud (Fig. 25b).

Selwick's Bay at Flamborough Head on the East Yorkshire coast of England has a well-exposed Chalk wave-cut platform. Behind the gravel beach are chalk cliffs (Fig. 26).



Fig. 26. Aerial view of Selwick Bay, Yorkshire. Looking North (BGS, 2022)

Mapping the three-dimensional orientation of geological structures. The structures (faults) are well exposed in the wave-cut platform and also on the cliff at the back of the beach. Having exposures in two planes (horizontal platform and vertical cliff) allows for the full orientation of the structures to be mapped. The objective was to assess the integration of nadir photography and vertical (cliff) photography into a single model using geolocation through RTK-enabled UAS (Fig. 27).

The DJI M300 was used due to the high quality RGB imagery acquired by the L1 sensor and the RTK GNSS positioning meaning that no external ground control data was required.

The M300 UAS was carried down to the beach at approximately 4pm, once the beach was empty of the public. Take off was completed from the shingle beach.

A nadir survey of the bay was programmed in DJI pilot software with a 50m elevation and 70% forward and 50% side overlap. This survey was flown, once completed the pilot took manual control of the M300 and the camera adjusted to be side looking.

The pilot then flew the UAS at approximately 10m elevation and 30m distance from the cliffs. RGB imagery were manually acquired, the UAS was flown slowly, and frequency of photography was such to ensure sufficient overlap. This required the pilot to keep a careful eye on both the controller screen and UAS, it was therefore essential to have a spotter to ensure safety.

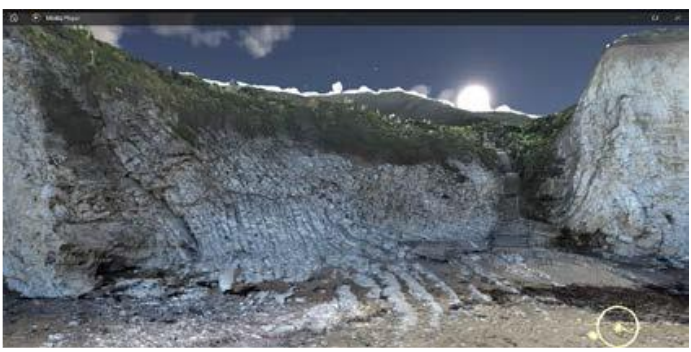


Fig. 27. 3D model showing structure in the cliff face at Selwick Bay, Yorkshire (BGS, 2022)



Fig. 28. Aerial view of Slaithwaite Reservoir, Yorkshire. Looking North (BGS, 2021)

Slaithwaite Reservoir (Fig. 28) was constructed in 1797 to supply the Huddersfield Canal and impounds the waters of Merry Dale Clough, a tributary of the River Colne. The reservoir has a surface area of 46,000 m2 and an estimated volume of 175,000 m3. The dam is oriented roughly north-south and is built at the east end of the reservoir. It is a 17 m high earth fill dam with a reported puddle clay core. The Canal and River Trust had commissioned SOCOTEC Monitoring LTD to implement geophysical monitoring in collaboration with the BGS to investigate suspected leakage issues of the core, embankment dam, and tunnel using a combination of geophysical monitoring, terrestrial LiDAR, and UAV surveys.

UAS flights were defined in Pix4D as pre-programmed single-grid surveys. The RGB survey was performed with sensor view angle set at NADIR and flight paths set to enable front-back and side-side overlap of 70 %. Altitude was set at 50 m to provide output pixel resolution of <1.5 cm. Imagery was recorded in JPG format with ground control targets set out across the study site, providing accurate geo-positioning using Network RTK GNSS. Flights took approximately 20 minutes each to perform, and there were four in total. Terrestrial LiDAR surveys and geophysical surveys were also carried out during this time to help produce a Digital Elevation Model (DEM) of the site (Fig. 29).

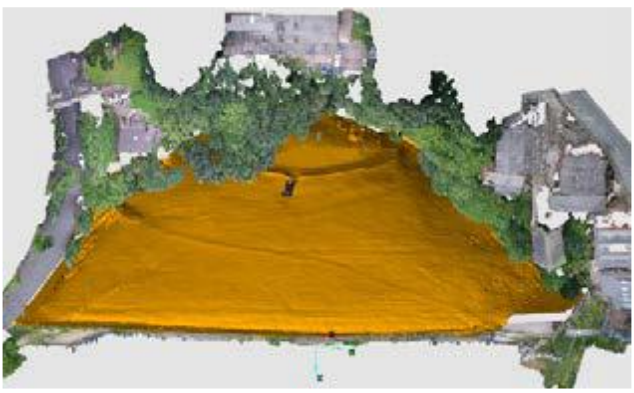


Fig. 29. DEM of Slaithwaite Dam and Tunnel (BGS, 2021)

Los Humeros is a 5-8km wide oval caldera situated 200km East of Mexico City (Mexico) at an elevation of approximately. 2800±50m (Fig. 30). The caldera is bounded by rim faults with steep cliffs and has an active geothermal system that is exploited for energy production. The topography within the caldera is gently sloping, with in-caldera fault escarpments evident in the central and southern areas. The caldera contains evidence of mineral alteration, high temperature anomalies and fault structures indicative of an active geothermal system.

Assessing the relationship between thermal UAS data and tectonic structures, surface geothermal manifestations and soil gas monitoring to better understand

the tectonic, sedimentary and geochronological evolution of a geothermal system at Los Humeros caldera, Mexico, where geothermal features occur both with and without surface expression.



Fig. 30. Los Humeros caldera, Mexico (BGS, 2018)

UAS flights were defined in Pix4D as pre-programmed single-grid surveys. Both RGB and thermal surveys were performed with sensor view angle set at NADIR and flight paths set to enable front-back and side-side overlap of 70% for RGB and 90% for LWIR Thermal. Altitude was set at 80m to provide output pixel resolution of 2cm (RGB) and 15cm (LWIR Thermal). Imagery was recorded in JPG (RGB) and RJPG (LWIR Thermal) formats with ground control targets set out across the study site, providing both accurate geopositioning using GPS and temperature calibration using FLIR T650sc. Flights were repeated pre-dawn and post solar heating maximum to enable assessment of temperature amplitude indicative of volumetric variation in ground surface (Fig. 31).

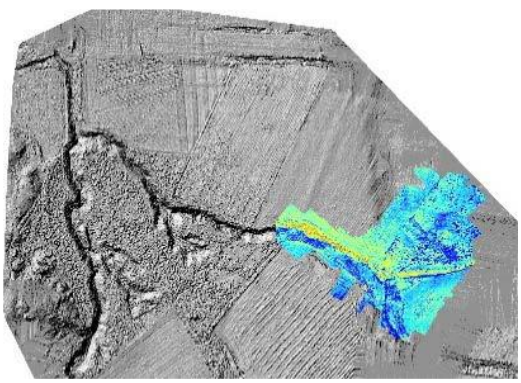


Fig. 31. OrthoLWIR on DEM of Los Humeros caldera (BGS, 2018)

Gresham Marshes is an Environment Agency owned small area of land. It is a local wildlife site and wildlife reserve located off Gresham Park Road and Wilford Lane in West Bridgford, Nottingham (Fig. 32). The ponds were dug in December 2018 by EMEC. There wasn't a report or official design for the ponds, they were just created by the chaps in the diggers whilst onsite, so there are no maps showing where they are. The Environment Agency, working closely with the Wildlife Trust, are currently undertaking eco-surveys, footpath clearance and hedge laying activities at Gresham Marshes. The EA contacted the BGS to provide an aerial survey of the pond locations within the site.



Fig. 32. Gresham Marshes, Nottinghamshire (BGS, 2021)

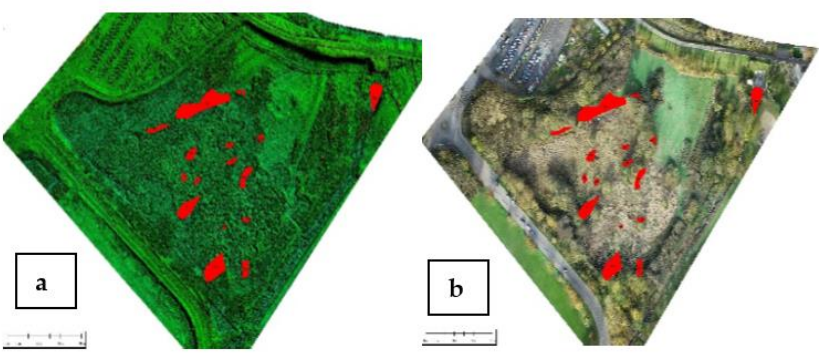


Fig. 33. DEM of marsh area with ponds highlighted in red (a). Colored point cloud of marsh area with ponds highlighted (b) (BGS, 2021)

UAS flights were defined in DJI Pilot as a series of one nadir and four matching oblique (45°) flights. The LiDAR and the matching RGB imagery were performed

using the Zenmuse L1 sensor with a side overlap of 70%, the scanning mode set to repetitive, a flight speed of 10 m/s, the number of returns set to three, and calibration enabled. The altitude was set at 50 m to keep within regulations and to provide a point cloud density of 283 points/m2 and a photo output pixel resolution of ~1.5 cm. Imagery was recorded in JPG format with ground control targets set out across the study site, providing accurate geo-positioning using Network RTK GNSS. The five flights took a total of 60 minutes. The outputs were a DEM from LiDAR (Fig. 33a) and orthophotography and models from RGB (Fig. 33b).

8. DISCUSSION

The BGS is currently at the forefront of TLS, MMS and UAV-based research for geohazard mapping, monitoring and modelling, with over 500 surveys from ~230 sites, both in the UK and overseas, carried out over a 20-year period. During this time, the BGS has utilized these technologies for widespread applications in earth sciences, engineering geology, geohazards and climate change, including terrain modelling of inland and coastal landslides, eroding coastlines, actively growing volcanic lava domes, retreating glaciers, rock stability and subsidence features, soil erosion and geo-conservation. The BGS also has a long history of overseas work in geodetic monitoring, using satellite and airborne Radar and LiDAR and, more recently, using both LIDAR and photogrammetric techniques on UAVs (Jones & Hobbs, 2021).

Examples of these projects can be found at: https://www.bgs.ac.uk/geology-projects/geomatic-surveys/. Besides these, we believe that TLS and other associated techniques can be applied to the following applications:

- Use of TLS to generate digital models of historic buildings, geological outcrops or other structures for geo-conservation, geo-tourism and distance learning.
- Combine TLS with multi-spectral imaging cameras in order to create virtual outcrop models (VOM) of geological structures.
- Use TLS to detect cracks and wet spots in the clay-lined caps of Landfill sites, particularly with respect to ‘closed’ landfills.
- Use TLS to monitor long-term ground movement due to shrink–swell clays, compressible soils or Karst as a ground-truthing procedure for InSAR techniques (Jones & Hobbs, 2021).
- Use TLS to ‘Join-up’ fractures and faults in outcrops. The structures of one side of the outcrop can be measured and then the other side (inside, if a cave, etc.) and a virtual model can be created of the two parts.
- Use TLS to measure the loss of glacial volume. This could be compared to the volume of water flowing out of the glacial lake (during the same time interval). This would give us the total runoff volume that was accounted for by the net change in glacier snout storage over time.

- Use of TLS for 3D mapping of temporary exposures, made by excavations, blasting, mass movements (landslides, etc.) or even trial pits (Jones & Hobbs, 2021).
- Use of TLS to study coastal evolution, in order to determine the changes in beach thickness over time.
- Use of TLS to monitor precursor displacement failures in cuttings or embankments.
- Use of TLS to complement and calibrate the outputs of digital aerial photogrammetry.
- Use of MMS (Pegasus Backpack) allows Building Information Modelling (BIM) using both imagery and point cloud data to document outdoor, indoor and even underground areas to be carried out (Jones & Hobbs, 2021).
- Use of InSAR to calculate long-term and future effects of coastal retreat in low-lying areas, or glacial retreat, on a local, national or global scale.
- Use of InSAR to calculate potential subsidence rates due to natural subsidence.
- Use of UAVs in tandem (or more) to cover medium-large areas of land simultaneously, making surveys quicker.
- Use of UAV incorporating Blue/Green LiDAR to look at near-shore and inland water sources to determine depth and structure of bed.
- Use of UAV as a carrier for geophysical sensors for magnetic and electromagnetic surveys, ground penetrating Radar, seismic surveys and gravity (Jones & Hobbs, 2021).

9. CONCLUSIONS

The developments in, and applications of, terrestrial LiDAR at BGS in the last two decades have demonstrated the wide capability of this, and related techniques, in the field; some examples of which have been described in this paper. Advances in technology have meant that the applications of terrestrial LiDAR in the field of engineering geology have expanded dramatically over this time period. For example, the use of the Pegasus Backpack has demonstrated that disaster responders will be able to capture data in 3D, on foot, in danger zone areas, and, in combination with SLAM and IMU, achieve accurate positioning even during GNSS outages (Jones & Hobbs, 2021). With the addition of the external light source, precise scanning of tunnels and cave systems is also possible. The Pegasus Backpack can also be vehicle-mounted, for mobile mapping of large areas, or pole-mounted, for lowering into voids or sinkholes. Current GNSS units operate to sub-centimeter accuracy, and can be used to acquire accurate surveys of non-linear features, including watercourses where the depth to the bed cannot be recorded as easily as other features. The ability to combine different technologies (LiDAR, GNSS, UAV, InSAR) to create a system greater than the sum of its parts has resulted in a powerful tool for research and industry (Jones & Hobbs, 2021). The BGS is in the process of developing a Low-Cost GNSS system as a rugged ‘disposable’ solution for monitoring the movement of a landslide, the flow velocity of a glacier, volcanic deformation,

coastal evolution or differential ground subsidence. Also, UAVs can now be fitted with digital cameras, thermal detectors, multispectral cameras, LiDAR scanners and geophysics sensors to provide highly accurate data, with full-coverage 2D and 3D models.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc.) and text-to-image generators have been used during the writing or editing of this manuscript.

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COMPETING INTERESTS

Author has declared that no competing interests exist.

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Chapter 3
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Human Vulnerability Index: Adaptation and Alleviation to Climate Change among Fishing Communities in India's Bay of Bengal Region

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ABSTRACT

Climate change involves complex interactions and changing likelihoods of diverse impacts. Climate change may alter the frequency and intensity of extreme events like floods, droughts, heat waves, and hurricanes, but difficult to attribute to anthropogenic climate change. Evidence of climate change impact is the strongest and most comprehensive for natural systems. Impacts on human systems have also been attributed to climate change, with a major or minor contribution of climate change distinguishable from other influences. The objective of the present study is to identify the intensity of the human vulnerability to climate change in a multi-dimensional context. The study area is located along the coastline 20-30 m above sea level on the southeast coast of the Indian Peninsula, bordered by the Bay of Bengal and severely prone to cyclonic occurrences, namely Hudhud, Phailin, and Vardah. A track of cyclonic disturbances, cyclones, and severe cyclones over the Bay of Bengal (BOB) was observed from 1994 to 2015. The study was commissioned as a part of the national project, National Innovations on Climate Resilient Agriculture (NICRA), Indian Council of Agricultural Research (ICAR) during the year 2017. The impact of climate change on the livelihood of the fishing community was measured through an integrated Human Vulnerability Index (HVI) approach. The domain-specific HVI was found to be high for livelihood followed by climate variability, economic, socio-demographic, and health dimensions. In the present study of assessment of the vulnerability index of fisher communities, vulnerability was found to be high for the livelihood dimension followed by climate variability, economic, socio-demographic, and health dimensions.

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Keywords: Climate change; fisher-folk; Bay of Bengal; human vulnerability index; hushed cyclone; adaptation and mitigation.

1. INTRODUCTION

Climate change is a long-lasting change in the weather that affects the tropics to poles. It is a global threat that has embarked on to put stress on various sectors (Abbass et al., 2022). It involves complex interactions and changing likelihoods of diverse impacts (IPCC, 2018). A remarkable aspect of climate change is that seemingly modest amounts of global warming are associated with severe outcomes: those that have already occurred and, especially, those expected to occur in the future. Global temperatures have risen slightly more than 1°C since the late 19th Century and are expected to increase, depending on human emissions, by another 1.5–4.5°C by the end of this century (Robinson, 2021; Abhijeet et al., 2023). Climate change may alter the frequency and intensity of extreme events like floods, droughts, heat waves, and hurricanes, but difficult to attribute to anthropogenic climate change (Brander, 2010). The environmental fluctuations and extreme climate change events exert aggravated effects on global food production, and the growth rate begins to decrease in some regions, influencing the stability of the future global food supply and demand (FAO, 2015; IPCC, 2018). Extreme weather events such as cyclones and floods may further intensify these impacts by disrupting fishing operations and land-based infrastructure (Islam, 2014). The land-based assets can also be deteriorated by sea-level rise, land erosion, and variations in temperature and rainfall (Sumankalyani et al., 2023). These impacts may result in vulnerability of fishery-dependent livelihoods (Islam, 2014; ResearchGate, 2016)**.**

Vulnerability is a function of the character, magnitude, and rate of climate change, which encompasses a variety of concepts and elements, including sensitivity or susceptibility to harm and lack of capacity to cope and adapt (ResearchGate, 2016). In addition to synthesizing the climate science, there is also a need to analyze the multi-dimensional climatic projections in the context of human eco-systems, livelihood patterns, socio-demographic profiles, and underlying health factors in order to realize the complex network of issues due to the impact of climate change (Lutz, 2017; Sumankalyani et al., 2023). Quantitative projections of climate change by combining models of varying complexity and observations and expressed in terms of probabilities can measure current uncertainties (Lutz, 2017; Colburn and Jepson, 2016). These projections illustrate the robust concept of vulnerability and the capacity of individuals and social groups to respond, to cope with, recover, adapt to external disastrous events placed on their livelihoods and well-being (Collins and Chandler, 2012; Allison, 2009; Sumankalyani et al., 2023).

Ecosystem-based coastal adaptation approaches are still emerging, which require an improved understanding of key ecosystem services of coastal communities in order to assess risk, aid coastal development planning and build decision support systems (Luers, 2005; Das & Srivastava, 2016). The present paper sheds insight into delineating the issues and challenges in the context of vulnerability to climate change. The objective of the present study is to assess the multi-dimensional vulnerability index of fishing communities and to identify

adaptation and mitigation measures in relation to climate change, and propose strategies & policies to protect the livelihood of fishing communities in the context of climate change.

2. MATERIALS AND METHODS

The locale of the study is East Godavari District in the state of Andhra Pradesh in India. The coastal belt of Andhra Pradesh extends up to a length of 972 km and is the third longest among the states of India. The Coastline of Andhra Pradesh is located on the southeast coast of the Indian Peninsula, bordered by the Bay of Bengal (Sumankalyani et al., 2023). The coastal corridor consists of several ports, harbors, and industries, and forms the major platform for exports and marketing. As per the Revenue disaster management of Andhra Pradesh, A.P. is exposed to cyclones, storm surges, floods, and droughts. A moderate to severe intensity cyclone can be expected to make landfall every two to three years. About 44 percent of the state is vulnerable to tropical storms and related hazards (Sumankalyani et al., 2023). In India, the cyclones develop in the pre-monsoon (April to May) and post-monsoon seasons (October to December), but most of them tend to form in the month of November (Weatherdon, 2016; Jacob, 2015)**.**

The implications of vulnerability are an interactive phenomenon involving various social and environmental factors in climate change and the ability to adapt to future challenges (Sharma, 2009; Hahn et al., 2009; Michel, 2017). Generally, poorer and less empowered countries and individuals are more vulnerable to climate impacts, and the vulnerability of fisheries is likely to be higher where they already suffer from overexploitation or overcapacity (Daw, 2009; Sumankalyani et al., 2023). Vulnerability is approached as a complex, constantly evolving and changing phenomenon that needs to be situated within interactions between biophysical, socio-economic, and livelihood elements (Adger, 2006; Jurgilevich, 2017; Luers, 2005).

The study area is located along the coastline, and situated 22-28 m above sea level, ranging from Latitude: 17.04 N-17.26 N and Longitude: 82.27 E to 82.51 E (Sumankalyani et al., 2023). The area is climate sensitive and affected severely by cyclones viz. Hudhud, Phailin, etc. in the district and lies in the coastal belt with a number of hamlets. Hudhud is the very severe cyclonic storm that crossed Visakhapatnam coast during 07-14 October, 2014 after 1985 which affected East Godavari, Visakhapatnam, Vizianagaram & Srikakulam districts of Andhra Pradesh and Ganjam, Gajapati, Koraput & Malkangiri districts of Odisha with a wind speed of 180 kmph and gusting wind speed of 195 kmph affecting about 4484 villages and 20,93,508 families. Cyclonic track of Hudhud and Phailin is illustrated in (Fig. 1) (Sumankalyani et al., 2023). A representative sample of 200 families from Konapapa peta, Moolapeta, Chinthakayalapeta, Perumallapuram, Hukhumpeta villages from U. Kothapalli and Thondangi mandals of East Godavari district, A.P, India were selected based on purposive and stratified random sampling. All the respondents from selected villages have their major occupation as fishing (Sumankalyani et al., 2023). The data was elicited by using different methods from the focus group discussions, open forum feedback

sessions and interviews from the key informants within the framework of the fishing community (Johnson, 2016).

Vulnerability is a function of the character, magnitude, and rate of climate change and variation to which a system is exposed, the sensitivity and adaptive capacity of that system (Saxena, 2013; Sumankalyani et al., 2023). Indicators for assessment of vulnerability include food security, infrastructure sensitivity, ecosystem sensitivity, human health sensitivity, water resource sensitivity and economic capacity, human and civic resources, and environmental capacity (Bahinipati, 2014; Pandey, 2012)**.** Interpreted vulnerability in terms of integration of three components: exposure, sensitivity, and adaptive capacity (Sumankalyani et al., 2023). Indicators were classified as climate, water, and land resources, and adaptive capacities were classified as human capital, social capital, financial capital, and physical capital. As per the data retrieved from the Regional Specialized Meteorological Centre (RSMC), New Delhi, a track of cyclonic disturbances, cyclones, and severe cyclones over the Bay of Bengal (BOB) was observed from 1994 to 2015 (Sumankalyani et al., 2023). Recently there is an increasing trend in overall cyclones as shown in (Fig. 2) with an annual frequency of cyclonic disturbances (maximum wind speed of 17 knots or more), cyclones (34 knots or more) and severe cyclones (48 knots or more) over the Bay of Bengal (BOB) (Sumankalyani et al., 2023). Hence, the vulnerability studies of coastal communities attain major importance with respect to the increasing trend in cyclones.

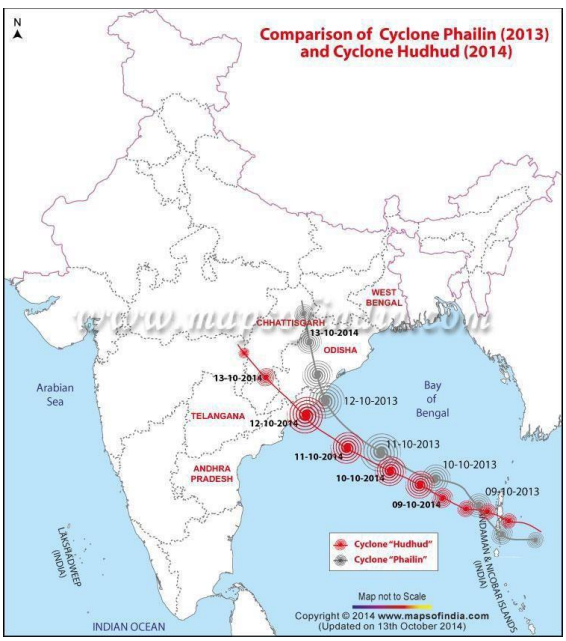


Fig. 1. Cyclonic track of Hudhud and Phailin

2.1 Measuring and Assessing Vulnerability

Vulnerability index describes the indicators and sub-indicators used to compute the vulnerability of fisherwomen to climate change factors (Sumankalyani et al., 2023). The main goal of the indicators is to map the relative vulnerability of the fishing community to various components during the Hudhud cyclone. Livelihood, socio-demographic, economic, health, and climate variability are the different indicators used to assess the vulnerability index of fishing communities (Sumankalyani et al., 2023). The indicator scores are determined by summing the scores of sub-indicators by the number of sub-indicators. The vulnerability index was developed by using the procedure developed by IPCC, by including several factors in the frame work.

$$\text{Vulenerability index of each of the indicator } (DI_j) = \frac{(\text{Actual } X_i - \text{Minimum } X_i)}{(\text{Maximum } X_i - \text{Minimum } X_i)} \quad (1)$$

$$\text{Actual } X_i = \frac{(\text{Sub-indicator}(X)_1 + \ldots\ldots + \text{Sub-indicator}(X)_i)}{\text{Number of Sub-indicators (m)}} \quad (2)$$

$$\text{For example, Livelihood indicator (Actual } X_i) = \frac{(\text{Shelter loss} + \text{Fuel shortage} + \text{Drinking water shortage} + \text{Loss of assets} + \text{Food shortage})}{\text{Number of Sub-indicators (5)}} \quad (3)$$

X_i = Sub-Indicator
m=Number of Sub-Indicators
Actual X_i= Average of percentage of population affected by the sub-indicators
Minimum X_i =percentage of population affected by all the sub-indicators
Maximum X_i = percentage of population affected by at least one sub-indicator

The HVI is calculated by summing the indicator scores using the following equation:

$$\text{Human Vulnerability index (HVI)} = \frac{(\text{Indicator}(DI)_1 + \ldots\ldots + \text{Indicator}(DI)_j)}{\text{Number of indicators (n)}} \quad (4)$$

$$\text{Human Vulnerability index (HVI)} = \frac{\sum_{j=1}^{n} \text{Indicator}(DI)_j}{n} \quad (5)$$

$$\text{For example, Human Vulnerability index (HVI)} = \frac{\text{Livelihood } (DI_L) + \text{Sociodemographic } (DI_{Sd}) + \text{Economic}(DI_E) + \text{Health}(DI_H) + \text{Climate variability}(DI_{CV})}{\text{Number of indicators (5)}}$$

Each indicator was subdivided into three to five sub-indicators, and the vulnerability dimensional index of fishing communities was computed and categorized as least vulnerable (<0.25), moderately vulnerable (0.25-0.5), and highly vulnerable (>0.5) as shown in Table 1 (Sumankalyani et al., 2023).

Livelihood indicator (DI_L) includes five sub-indicators viz., 1. Loss of shelter 2. Shortage of fuel 3. Shortage of drinking water 4. Loss of assets, inputs, and production of capacity 5. Shortage of food (Sumankalyani et al., 2023). The number of family members affected and the percentage of families affected by the cyclone with regard to primary needs was calculated as reported by at least

one family member in the family. The livelihood dimensional index (DI_L) was calculated by using equation 1. Socio demographic indicator (DI_{SD}) includes four sub-indicators, viz., 1. Loss of public facilities, public utilities 2. Terminating education (school dropouts) 3. Teenage/ early marriage 4. Loss of infrastructure. The number of family members affected and the percentage of families affected by the cyclone were calculated based on the respondents' statements about the disaster in a social perspective, as reported by at least one family member in the family (Sumankalyani et al., 2023). The socio-demographic dimensional index (DI_{SD}) was calculated by using equation 1.

Economic indicator (DI_E) includes four sub-indicators 1. Interruption of market access and unemployment 2. Reduced income and increased debts 3. Depletion of savings. 4. Loss of occupation and reduced fish catch (Sumankalyani et al., 2023). The number of family members affected and the percentage of families affected by the cyclone with regard to economic indicators was calculated based on loss of economic sources, which in turn affect the purchasing power of families (Sumankalyani et al., 2023). The economic indicator dimensional index (DI_L) was calculated by using equation 1. Health indicator (DI_H) includes four sub-indicators 1. Spread of epidemics 2. Ill-health due to contaminated aquatic species and zoonotic diseases 3. Nutritional deficiencies and post-traumatic stress 4. Lack of health facilities. The percentage of households that report at least one family member with chronic illness, prolonged stress was calculated based on the respondents' statements about the disaster in a health perspective. The health indicator dimensional index (DI_H) was calculated by using equation 1 (Sumankalyani et al., 2023).

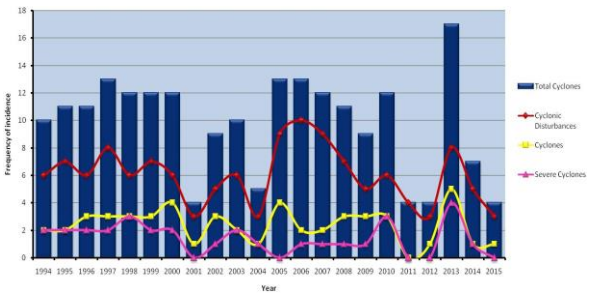


Fig. 2. Annual cyclones over Bay of Bengal

Climate variability index (DI_{CV}) includes three sub-indicators 1. Solidarity issues and conflicts 2. Injury or death of a family member 3. Damage to the ecosystem. The percentage of households that reported having death, injury, conflicts regarding fishing borders, ecosystem damage, and contamination was calculated based on the respondents' statements about the disaster in a climate change perspective (Sumankalyani et al., 2023). The climate variability dimensional index (DI_{CV}) was calculated by using equation 1. The human vulnerability index (HVI) was calculated by taking the average of five dimensional indicators, as shown in equation 3.

Table 1. The consequences & vulnerability index of Hudhud on the fishing community (n=200 families)

Indicators	Sub-Indicators	Families Affected (%)	Conceptual explanation	Dimensional index (DI)	Degree of vulnerability	Human Vulnerability Index (HVI)
Livelihood	Shelter loss	96	Crisis in primary needs in coastal area viz. food, shelter, water & fuel classified under livelihood indicator as reported by at least one family member	0.60*	highly vulnerable	0.51803* (highly vulnerable)
	Shortage of fuel	83				
	Shortage of drinking water	96				
	Loss of assets, inputs and production capacity	87				
	Shortage of food (98%)	98				
Socio demographic	Loss of facilities, public utilities	82	Calculated based on the respondents' statement to the disaster in a social perspective	0.49	moderately vulnerable	
	Education (school dropout rate)	10				
	Early marriages	4				
	Loss of fishing infrastructure	74				
Economic	Interruption of market access and unemployment	97	Loss of economic sources which in turn effect the purchase power of the families	0.52*	highly vulnerable	
	Reduced income & increased debts	85				
	Depletion of savings	89				

Indicators	Sub-Indicators	Families Affected (%)	Conceptual explanation	Dimensional index (DI)	Degree of vulnerability	Human Vulnerability Index (HVI)
	Loss of occupation & reduced fish catch	94	due to disaster			
Health	Spread of epidemics	58	Percentage of households that report at least one family member with chronic illness, prolonged stress and lack of health facilities	0.42	moderately vulnerable	
	Ill-health due to contaminated aquatic species & Zoonotic diseases	24				
	Nutritional deficiencies & post-traumatic stress	86				
	Lack of health facilities	32				
Climate variability	Solidarity issues and conflicts	52	Percentage of households that reported having death, injury, conflicts and eco-system damage and contamination	0.56*	highly vulnerable	
	Injury or death of family member	5.5				
	Damage to the eco system	75				

(Note: > 0.5: highly vulnerable; 0.25-0.5: moderately vulnerable; < 0.25: least vulnerable.)*

3. RESULTS AND DISCUSSION

Results of the paper were discussed and presented under the following indicators: a) livelihood vulnerability index, b) socio-demographic vulnerability index, c) economic vulnerability index, d) health vulnerability index, and e) climate variability vulnerability index. Each indicator was subdivided into three to five sub-indicators, and the vulnerability dimensional index of fishing communities was computed and categorized as least vulnerable (<0.25), moderately vulnerable (0.25-0.5), and highly vulnerable (>0.5) (Sumankalyani et al., 2023). Frihy and Sayed (Patnaik, 2009) classified vulnerability variables/sub-indicators on similar grounds depending on the geographical application of the coastal vulnerability index. As climate change is causing uncertainties and as a result, there is depletion in fish catch, affecting their livelihood and making them vulnerable to the cyclones as the fishermen are forced to venture deeper into the sea by risking their lives (Sumankalyani et al., 2023).

3.1 Vulnerability Index

The domain-specific vulnerability indicators were chalked out and depicted in Table 1 (Sumankalyani et al., 2023). The human vulnerability index was calculated in order to identify the critical domain. Dimension-wise vulnerability indices appear to differ considerably among different habitations (Frihy & El-Sayed, 2013)**.** The vulnerability index with different spatial patterns provides an important basis for policy makers to develop appropriate adaptation strategies to minimize the risk of the fisheries sector to climate variability (Saxena, 2013; Yuan, 2017). Deb and Haque revealed that fishing households cope with climatic adversities by applying a host of strategies based on the opportunities and capabilities they have, constraints they face, and the changing social relations dictated by external and internal forces (Jurgilevich, 2017). The vulnerability index was computed by using the Parameter, Attribute, Resilience indicator, and Score (PARS) method which is useful for planning suitable remedial measures to mitigate the adverse effects of climate change (Sumankalyani et al., 2023). Multivariate cluster analysis is an alternative approach for identifying hotspots and is useful in capturing complex interactions with respect to hazard, exposure, and impact, which has greater policy relevance (Deb & Haque, 2017; Patwardhan, 2006).

Fishing communities are primary victims and are more prone to climate change consequences due to the fact that they live very close to nature and their habitat and livelihood are affected by natural calamities (Sumankalyani et al., 2023). Crisis in primary needs viz. Food, shelter, water and fuel create a struggle for existence which demands the sustenance of life. Vulnerable regions face multiple stresses that affect their exposure and sensitivity as well as their capacity to adapt (Sumankalyani et al., 2023). These stresses arise from current climate hazards, poverty, and unequal access to food availability, food stability, utilization, and access to food (Sharma, 2008; FAO, 2015).

Vulnerability index was also assessed by Hahn for estimating the climate change vulnerability (Blasiak & Spijkers, 2017; Malakar, 2017). The results of the study were quite useful in drawing conclusions, calculating the composite index, and

comparing differential vulnerabilities. Results suggest that some areas are more vulnerable in terms of exposure and sensitivity, while some areas are more vulnerable in terms of adaptive capacities with respect to the climate vulnerability index (Sumankalyani et al., 2023). Allison explains that vulnerability of the fishing population by building adaptive capacities among countries that have weak economies and poor governance is a necessary response, and climate change may bring improved fishing opportunities (Huang, 2012).

3.2 Livelihood Vulnerability Index (DI_L)

The dimensional index of livelihood indicator was found to be 0.6, which is classified as highly vulnerable because loss of shelter, lack of access to food and water are the primary basic needs of the families and which are deprived during natural disasters (Sumankalyani et al., 2023).

The fishing infrastructure of the fishing communities in coastal areas includes fishing gears, fishing vessels, cold storage, fuel tanks, safety devices, processing areas, etc., which are landed openly on the shores and are prone to damage and loss during cyclones (Sumankalyani et al., 2023). These non-climate stresses can increase vulnerability to climate change by reducing resilience and adaptive capacity because of resource deployment to competing needs. The rising value of infrastructure in coastal areas increases vulnerability to climate variability and future climate change, with losses projected to increase if the intensity of tropical storms increases (Ratner & Asgard, 2014). The vulnerability of communities depends upon social capability, infrastructural facilities, and financial resources (Rufat, 2015; ADB, 2016). The intensity of vulnerability depends on the loss of livelihood index but has benefited little from economic development (Sumankalyani et al., 2023)**.**

3.3 Socio-demographic Vulnerability Index (DI_{SD})

The dimensional index of socio demographic indicator was found to be 0.49, which is classified as moderately vulnerable due to the impact of loss of public facilities, fishing infrastructure, increased conflicts, early marriages and increased number of school dropouts (Sumankalyani et al., 2023). The results are in coherence with Hahn and Colburn in terms of vulnerability index, indicating that socio-demographic factors are equally important in the estimation of vulnerability index (Ahamed, 2013; Chiu et al., 2017).

In order to get rid of major responsibilities and family burdens, early marriages among girl children are a common phenomenon before they reach adolescence due to financial crisis. The school dropout ratio was found to be more after the occurrence of disasters due to the reason that the families engage the children in economic activities to increase the family income (Sumankalyani et al., 2023). Lack of employment, loss of infrastructure, reduced income source, and declined fish catch demand them to engage in social conflicts. The present study indicated that fishing communities lack social capacities to utilize opportunities for adaptation to potential impacts. In other words, the rate of resilience to climate

change is low compared to urban communities (Sumankalyani et al., 2023). Fisherfolk are subjected to violence resulting from disputes over resource access, and the rate of child labor enormously increases during disasters and calamities.

3.4 Economic Vulnerability Index (DI_E)

The economic variability index of fishing communities was found to be 0.52, which is categorized as a highly vulnerable dimension because of the reason that the economic position of the fisherfolk folk considerably reduced to the disastrous event of Hudhud in this area (Sumankalyani et al., 2023).

The economic factors, such as loss of livelihood and unemployment, have a great impact on the communities living in absolute poverty as they may not be able to cope with the challenges posed by climate change. Vulnerability is often reflected in the condition of the economic system as well as the socioeconomic characteristics of the population living in that system (Sumankalyani et al., 2023). Patnaik and Narayan attempted to construct a picture of the socioeconomic context of vulnerability by focusing on indicators that measure both the state of development of the region as well as its capacity to progress further (Malakar, 2017). The demographic characteristics, socioeconomic status, and health as the leading empirical drivers of social vulnerability to damaging flood events. Fisheries exist at the nexus of food security, income generation, and natural resource management, and earmarking of climate finance to enhance adaptive capacity of various countries depending on their economies (Sumankalyani et al., 2023).

3.5 Health Vulnerability Index (DI_H)

Deteriorating environmental conditions are the major contributing factors to poor health and poor quality of life. Prolonged stress and health problems are the major reasons for deterioration of health and post-traumatic stress (Sumankalyani et al., 2023). The dimensional index for health is found to be 0.42, which is categorized as moderately vulnerable.

Mismanagement of natural resources, excessive waste production, industrial pollution, open defecation, and slurry & sewage runoff from toxic resources in this Uppada region have affected the health of the communities by posing a major challenge to climate change. Diseases such as diarrhea, dysentery, cholera, skin infections, and respiratory problems were found to be high among these communities soon after the incident (Sumankalyani et al., 2023). Loss of a parent during disasters results in an increased number of orphans, early marriages in young girls, leading to high dropout rates. Vulnerable groups face multiple stresses that arise from climate hazards, food insecurity, deficiency disorders, and the incidence of diseases such as HIV/AIDS. Climate-sensitive vulnerabilities on health include vector borne diseases, infectious waterborne diseases, and diseases related to extreme weather conditions such as floods, droughts, windstorms, fires, and heat (Sumankalyani et al., 2023). Lack of

resources, such as poor infrastructure, poor water and environmental management, and high poverty rates, reduces their ability to adapt.

3.6 Climate Variability Vulnerability Index (DI_{CV})

The climate change impact was ranked high for fishery attributes followed by economy, environment, social, and development drivers. Climate variability index was found to be 0.6, which is categorized as a highly vulnerable dimension because of the reason that the deaths and injuries for people and damage to the ecosystem occurred during the disastrous event of Hudhud in this area (Sumankalyani et al., 2023).

The impact of climate change can be assessed by drawing some explicit linkages between the objectives of vulnerability and adaptation assessment**.** Due to the death and migration of spouses, the number of female-headed households is increasing and women are significantly overburdened with heavy workloads significantly during natural disasters (Sumankalyani et al., 2023). Socio-ecological safety of coasts is an urgent need which cannot be sacrificed for short term economic gains (Chiu et al., 2017). Hopper felt that significant correlations exist for all vulnerability (exposure/surge) and storm damage combinations. Hence, coastal protective features ranging from storm surge barriers, mangrove plantations and levees to natural features have to be promoted to decrease future flood risk to communities in coastal areas around the world as they protect coastal areas from erosion, storm surge, tsunamis, and are efficient at dissipating wave energy [(Sumankalyani et al., 2023). The climate change impact on various sectors of forced migration, mortality, morbidity, and tropical storms continues to be negative after the year 2050**.**

The results of the major vulnerability dimensions are presented collectively in a spider diagram depicted in (Fig. 3). The scale of the diagram ranges from 0 (less vulnerable) at the center of the web, increasing to 0.8 (more vulnerable) at the outside edge in 0.2unit increments (Sumankalyani et al., 2023). It is evident from the figure that the fishing communities are more vulnerable in the livelihood dimension, followed by climate variability, economic, socio-demographic, and health aspects. The figure portrays that all the dimensions are above 0.4, indicating a higher intensity in terms of vulnerability (Sumankalyani et al., 2023).

3.7 Mitigation and Adaptation to Vulnerability

According to IPCC, adaptation measures are seldom undertaken in response to climate change alone but can be integrated with risk-reduction strategies such as water resource management and coastal defense (Sumankalyani et al., 2023). The ability to cope with threats includes the ability to absorb impacts by guarding against or adapting to them. It includes provisions made in advance to pay for potential damage, such as contingency reserves and insurance repayments. Adaptation is the adjustment in natural or human systems in response to actual or expected climatic stimuli or their effects, which moderates harm or exploits beneficial opportunities (Sumankalyani et al., 2023). To mitigate climate change

impacts, the interventions should start at the micro-level by considering the needs, problems, and priorities of the coastal communities.

Adaptation to climate impacts includes reactive or anticipatory actions by individuals or public institutions, which range from finding suitable alternative occupations, provision of insurance, and improving warning systems (Sumankalyani et al., 2023). The adaptive capacity to tropical cyclones depends on various attributes such as environmental, social, psychological and physiological attributes**.** The low adaptive capacity of the coastal ecosystem is due to the exploitation of the natural resources and inadequate biodiversity conservation. Adaptation and mitigation options for different sectors of agriculture, water bodies, forests, and energy can make the community resilient to climate change consequences (Sumankalyani et al., 2023)**.**

The adaptation goal is to reduce vulnerability to the harmful effects of climate change, like sea-level encroachment, more intense extreme weather events, or food insecurity (Sumankalyani et al., 2023). The resistance and resilience of coastal ecosystems for human livelihoods is necessary to understand the processes to integrate knowledge with the decision-making process and permeate social networks, which are crucial to develop the social capital, political, legal and cultural frameworks that enhance resilience (Chiu et al., 2017; Vanderlinden, 2017).

3.8 Strategies for Adaptation and Mitigation

Spatial and temporal vulnerability analysis through modelling, satellite imaging and mapping as tools for decision support systems during risk hazard analysis, food economy analysis, contingency planning, monitoring, and evaluation. Key policy actions were illustrated by Cinner to reduce vulnerability at different spatial and temporal scales (Vanderlinden, 2017).

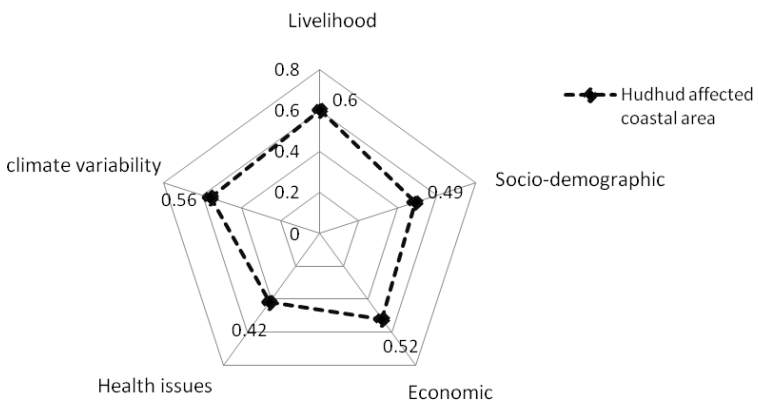


Fig. 3. Spider diagram of vulnerability index

Development of a multi-level natural hazard framework and thematic maps suitable for micro zones will be an effective tool for mitigation and adaptation strategies for coastal areas (Sumankalyani et al., 2023).

The vulnerability assessment in terms of indicators has to be quantified in order to prepare micro-level action plans.

As a long term goals providing livelihood diversified activities viz. seafood value addition, sea weed and sea shell processing; fish preservation and fish vending units for livelihood dimension, cooperative marketing, provision of transport & packaging, logistics, fish byproduct utilization units will pave a long way for improvement of socio-economic status of fishing communities (Sumankalyani et al., 2023).

Information and communication technology approach by conservation planning, coastal infrastructural development, social networks, early warning systems & planned evacuation of victims from shores and improved navigation systems.

Emergency medical services, improved climate-sensitive disease surveillance and control, safe drinking water, personal hygiene & improved sanitation, and promotion of health education are the important adaptation and mitigation options under the health sector (Sumankalyani et al., 2023).

Community activities, viz. construction of seawalls and storm surge barriers, dune reinforcement, creation of marshlands/wetlands as buffer against sea level rise, and protection of existing natural barriers and provision of safety devices, are the important mitigation options for coastal zones through strong local governance.

Involving the victims by creating awareness, training and sensitizing the people, providing alternative livelihood activities and strengthening them as climate change agents in rescue operations, food for work programmers, multi-hazard disaster preparedness and management plans, mangrove plantation & biodiversity conservation programs and climate change resilient programers (Sumankalyani et al., 2023).

Strengthening disaster management policy regimes and devising a holistic mitigation plan with the sustainable development strategies by creating linkages between researchers, policy makers and stakeholders (Martinez, 2017).

4. CONCLUSION

This HVI method is supportive of IPCC methodology for assessing vulnerability to climate change impacts. The degree of intensity of each dimension has been identified by the HVI method. This approach provides a detailed depiction of factors responsible for human vulnerability in a particular region at a particular point in time (Sumankalyani et al., 2023). These types of vulnerability studies are highly helpful in identifying critical components, allocating resources, and providing emergency services during natural disasters. Government and governmental policies & programs need to be aimed at addressing the issues of climate change based on livelihood, socio-demographic, economic, and health dimensions of vulnerability (Sumankalyani et al., 2023).

The present study of assessment of vulnerability index of fisher communities, vulnerability was found to be high for the livelihood dimension, followed by climate variability, economic, socio-demographic and health dimensions.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc.) and text-to-image generators have been used during the writing or editing of this manuscript.

COMPETING INTERESTS

Authors have declared that no competing interests exist.

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Chapter 4
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Balance Circumstances of Planetary Atmospheres and Causes of Gaseous Escape and Collapse of Earth's Ozone Layer

Armen Ohan [a++*]

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ABSTRACT

Understanding the mechanism of the planetary atmosphere and the forces involved in balance circumstances may help to know how gaseous escape occurs and guide the ways to reduce its rate via preserving the most important factors of Earth's ecosystem, where the balance circumstances of the planetary atmosphere can be defined as the balance among a group of dynamic and static forces that determines the volume and pressure of gases of the planetary atmosphere. The first force is the gravitational force of the planet that attracts gases from space to confine them within the range of the planet's gravitational field, and the value of this force varies among the planets of the solar system due to the differences in their masses. The second main force is the thermodynamic effects that generated as a result of the gaseous expansion that depends on the value of their temperature due to the increase in the kinetic energy resulted of the atomic and molecule collision of the gaseous elements of the atmosphere and their values change according to the value of the planet's temperature that results from their differences in their distances from the Sun, all planetary atmospheres have a specific state of equilibrium within the different circumferences, physic characteristics of elements and amount of gases in the atmosphere, the total result of these forces is determines the specific feature of the balance circumstances of each planetary atmosphere according to the value of the planet's gravitational force and the value of generated dynamic forces due to the expansion of gases according to the temperature of the planet as well as the physical characteristics of atmospheric elements such (boiling, liquefaction

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and solidification temperature) that differs according to the physical properties of elements that are strongly affected by their mass weight.

Keywords: Planetary atmospheres; balance circumstances; gaseous escape; system of the planetary atmospheres.

1. INTRODUCTION

Studying the physical properties of gases under different conditions and values of temperatures and pressures according to the thermodynamics laws (Atkins, n.d.) guides us in understanding the mechanism of balance circumstances in the atmosphere of the planets of the solar system, which is in a state of balance between the magnitude of the planet's gravitational force and the extent of the thermodynamic effect of atmospheric gases, which is strongly affected by the rates of the temperature of the planet (NASA, n.d.-a), which is determined by the planet's distance from the Sun. Emphasis should be placed on secondary factors such as the effect of the atomic gravitational field of gaseous elements, as an extreme drop in temperature in planets far from the Sun leads to decreased rates of motion and activity of atoms. This leads to the activation of the force of the atomic gravitational field to attract atoms to each other, reducing the distances among them and increasing their density, causing the formation of high-density masses of gases attracted forcedly by the gravity force of the planet, greater than the effects of the same gases on planets to have a higher temperature. that raises the atomic kinetic activity, which leads to a decrease in their density, therefore, planets that are far from the Sun, such as (Saturn, Jupiter, Uranus and Neptune, and Pluto) have a thick atmosphere, while the planets that are close to the Sun, such as (Mercury, Venus, Earth, and Mars), their temperature rates rise and suffer higher rates of gaseous escape, due to the increase in the thermodynamic activity of gases at high rates of temperature, causing a high-pressure force that overcame the planet's gravitational force.

The rising pressure in the lower atmospheric layers, such as (Troposphere) is affected by increasing temperature, and a reduction of density and an increase in the volume of gases in the Troposphere lead to the rise of these expanded gases to higher altitudes, such as the (Stratosphere) which leads to rebalancing the atmospheric pressure according to the value of the gravity force of the planet. The value of the generated pressure due to this expansion raises the pressure in the Troposphere layer, which pushes gases toward the upper layers in the Mesosphere and Thermosphere, causing rushing gases of these layers towards space in atomic diffusion under pressure, which causes gaseous escape from Earth's atmosphere to space. The effect of atomic diffusion leads to an acceleration of the spread of atmospheric elements in the lower layers (Troposphere and Stratosphere) towards the upper layers (Mesosphere and Thermosphere) as a result of a decrease in their concentration in the higher layers after the gaseous escape in large quantities from (Mesosphere and Thermosphere) to the space, due to the effect inertia feature of materials, also due to the acceleration of the speed of the Earth in space in its orbit around the Sun, where the elliptical orbit of Earth results changes in Earth's velocity (NASA,

n.d.-b; NASA, n.d.-c). Because Earth's speed gets to its minimum rate when the Earth is at the farthest distance from the Sun (152 million km) and equals 29.29 km/sec, after that point, the speed begins to accelerate gradually, where the speed reaches the highest value when the distance gets the closest value to the Sun at 147 million km and equals 30.29 km/sec, leads to the failure of catching gases of the atmosphere up with the Earth leaving them behind its path in the space, causing the phenomenon of gaseous escape, in the meantime, the mechanical effect of the bombardment of solar energy particles on the upper layers of the atmosphere (Mesosphere and Thermosphere) pushes gases elements away, especially in the limits of the apogee of the Earth when the distance between the Earth and the Sun is as at closest value (147 km million), where the density of solar particles increases in the space to its higher rates. Planets that have close distances to the Sun such as (Mercury, Venus, Earth, and Mars) suffer the greatest rates of gaseous escape from the planetary atmosphere, and escaped gases are often from the group of light elements (low mass weight) in the periodic table, so any increase in temperature may increase the value of generated pressure and volumetric expansion of the gases of the atmosphere, forms a pressure upon the upper layers of the atmosphere, which causes the displacement of gases in the upper layers to areas in the higher altitudes (lower gravity forces) leads them to be rushed into the space resulting the gaseous escape, where the time factor which represents the age of the planet, the temperature of the surface of the planet, which is affected forcedly by their distances from the Sun, the time range of exposure of the surface of the planet to Sun radiation (number of hours of the day and night) resulting from the speed of spinning of the planet, and the amount of heat gain and heat loss, and speed of rushing in their orbit around the Sun, because of the balanced circumstances of the planetary environment based upon two main factors, the first is the effect of the gravity force of planets that works to confine gases within its gravitational field range, and the second is the effect of the thermodynamic processes of gases at high rates of temperature that pushes the gases of the atmosphere to expand and overcome the effects of gravity force and to be escaped to space. The relationship between these two forces on the value of atmospheric pressure and the rate of planetary gaseous escape, the rushing of the solar system in space in its orbit around the centre of the galaxy Milky Way, collides with clouds of gases, which mostly consist of Hydrogen and its isotopes (Deuterium and Tritium), Helium and some other elements. A high percentage of these materials gets attracted to the gravitational field of the Sun due to its immense gravity force to be consumed in its core to generate the heat by nuclear fusion, some percentage of them collides with the gravitational field of the planets in the solar system (Saturn, Jupiter, Uranus, Neptune, and Pluto).

2. THE FACTORS THAT HAVE INFLUENCED UPON STATE OF BALANCE CIRCUMSTANCES OF THE PLANETARY ATMOSPHERE

There are many physical factors, material properties, and thermodynamic effects, as well as the planetary gravitational force, as shown in Illustration 1, where the effects of the planetary gravitational force and the thermodynamic effect of

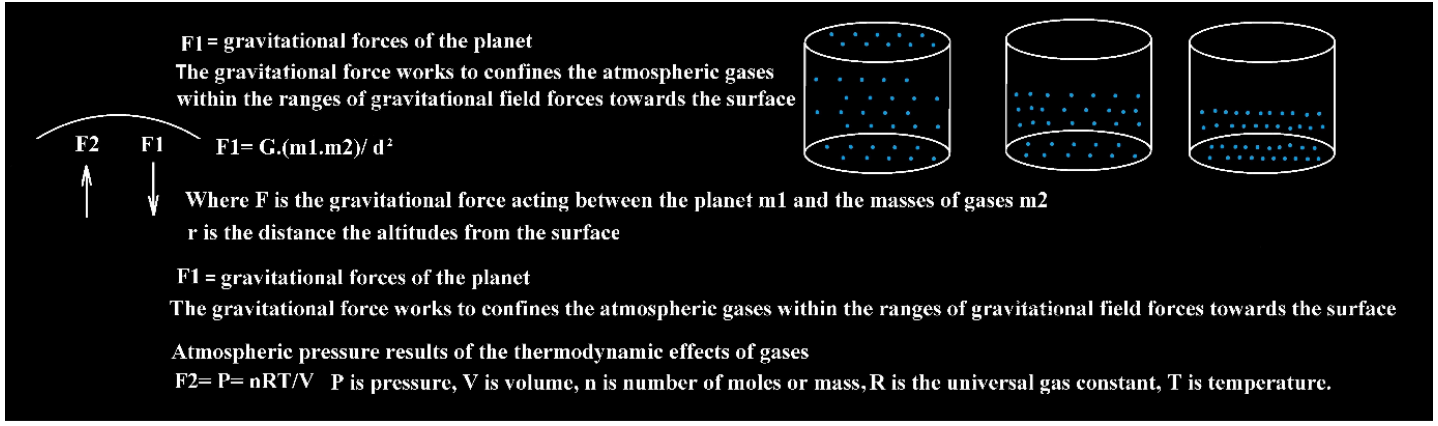


Illustration 1. The effects of the two forces of the planetary gravitational force and the thermodynamic effect of atmospheric gases on the value of atmospheric pressure

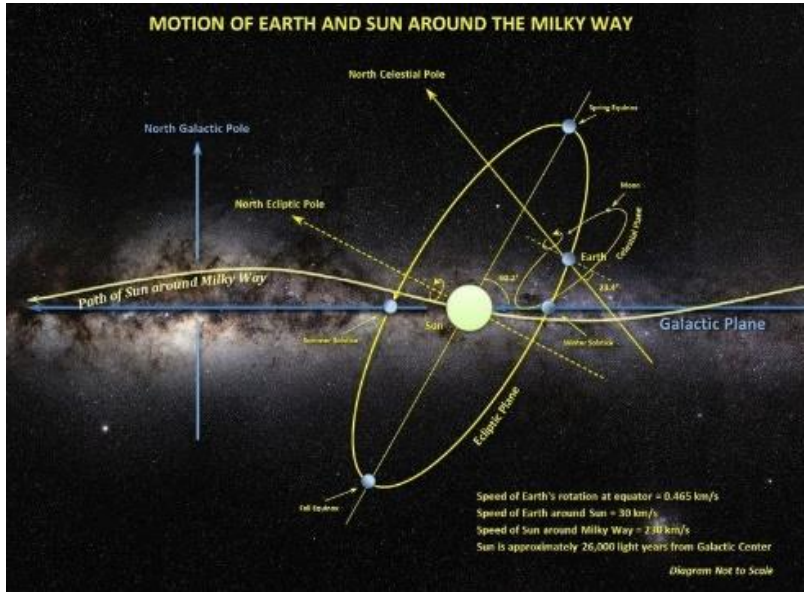


Illustration 2. The motion of solar system around the centre of Milky Way. [7]

atmospheric gases upon the value of atmospheric pressure collectively participate in balanced circumstances of the planetary atmosphere. These effects can be summarized in a set of points as follows:

2.1 The Effects of the Gravity Force upon the Value of Atmospheric Pressure

The greater mass of the planet creates the greater gravitational force that pulls and confine gases from the space in greater pressure towards the surface reducing the possibility of escaping gases from the atmosphere, according to the equation for universal gravitation theory of Newton (Sfetcu, 2019) the greater mass creates a greater gravitational force in space, which attracts gases from space with a greater force toward the surface, according to the equation for the universal gravitation theory of Newton:

$$F= G.(m1.m2)/ d^2 \quad \text{.....1}$$

Where F is the gravitational force acting from the planet to the material of gases, m1 = mass of the planet, m2 = mass of atmospheric gases, and d is the distance between the center of the planet to the center of different elevations of the atmosphere and G is the gravitational constant = 6.674×10^{-11} $N\cdot m^2/kg^2$.

2.2 The Effects of the Thermodynamics of Gases upon Atmospheric Pressure

The first group of the solar system planets (Mercury, Venus, Earth, and Mars) that are close to the Sun, due to the increase of their distances from the Sun, decreases received heat from Sun that has the thermodynamic effects upon gases of the atmosphere to expand at the limits of gravity fields to generate a specific value of atmospheric pressure, according to the universe ideal gas equation (Mortimer, 2008):

$$PV=nRT \quad \text{.....2}$$

Environmental factors that represent variables and constant in the ideal gas equation PV = nRT where P is pressure, V is volume, n is the amount of substance (n amount of gases in the atmosphere), T is the temperature, and R is a constant = 8.31446261815324 $J\cdot K^{-1}\cdot mol^{-1}$, as shown in Illustration 1, the effects of the planetary gravitational force and the thermodynamic effect of atmospheric gases on the value of atmospheric pressure.

2.3 The Effect of Rates Temperature

The nature and state of balance of the planet's atmosphere play the main role in sort of the planet's atmosphere and then the nature of the conditions on the surface, because the value of their pressure determined by the rates of the temperature of the environment of Earth due to their thermodynamic properties,

which may be calculated via the ideal gas equation PV = nRT, where P is pressure, V is volume, n is the amount of substance, T is the temperature and R is a constant (8.31446261815324 $J \cdot K^{-1} \cdot mol^{-1}$). Therefore, the significant increase in temperature due to the planet's decrease in the approaching distance to the Sun or the effects of global warming leads to an increase in the volume and pressure of gases in the Troposphere, leading them to be raised from their original layers in the lower layers, (Troposphere) to be rush towards higher altitudes, such as the (Stratosphere) to restore a state of balance between the magnitude of the planet's gravity force and the thermodynamic effect of the atmospheric gases to restore the standard value of the atmospheric pressure to the stable value that equals 101.325 kPa. The temperature factor plays an effective role in determining the amount of planetary atmospheric pressure in addition to the rest of the factors, such as the values of the gravitational force of the planet and the number of available gases in the atmosphere. This is the reason why the atmospheric pressure on Mars drops to 4% of its value compared to Earth's atmospheric pressure, because the low temperatures, in Mars's winter, around -153 °C, lead the atmospheric gases to shrink and reduce their generated pressure upon the surface. So the phase transformation of atmospheric gases guides them to be transferred (some gases, such as carbon dioxide) into the solid phase (Carbon frost), getting its effect completely out of the rest of the effect of rest gases in the Mars atmosphere in addition, the low value of the gravity force of Mars, which is (3.721 m/s^2), which is unable to attract some relatively low weight elements gases such as Oxygen (atomic mass 15.999 u) and Nitrogen (atomic mass14.0067 u), which are abundantly available in the Earth's atmosphere due to the relatively high value of the Earth's gravity force, which equals 9.807 m/s^2, which is able to attract these two gases forcedly to confine them within the Earth's atmosphere elements. There is a proportional relationship between the temperature rate of the planet and the rate of gaseous escape from the atmosphere, as the higher temperature rate of the planet (because of its distance from its star and the eruption rate of the star), according to the proportional relationships among the volume, the temperature, the pressure, and the amount of a gas can be combined into the ideal gas equation PV = nRT.

2.4 The Effect of the Amount of Gases

Increasing the amount of gases disturbs the balance state between the effects of gravity force and the thermodynamic effects, for instance, a large quantity of emissions of high molecular weight industrial gases into the atmosphere disturbs the state of balance and raises the pressure, which leads to pushing light gases such as Oxygen and Nitrogen into the higher layers to restore the value of the atmospheric pressure and restore the state of balance between the two mentioned forces. The specific conditions of balance circumstances of the planetary atmosphere of each planet lead to the formation of different environmental conditions of the values of atmospheric pressure and creating specific environment conditions for each planet and different rates of gaseous escape, where the rates of the gaseous escape in the Earth's atmosphere at moderate rates due to the moderate rates of gaseous expansion which results

from the thermodynamic effects of gases at the limit of the temperature of Earth (average 15 °C) this expansion overcoming the value of the Earth's gravitational force that equal 9.807 m/s² and pushes gases towards space at moderate rates comparing with the rates gaseous escape from the atmosphere of Mercury, Venus, and Mars.

2.5 The Range of the Active Gravity Field

The gravitational force of the planet that attracts gases from space confines it within the range of the planet's gravitational field, which represents the distance or d in the law of universal gravitation of Isaac Newton and (the volume or v) in the ideal gas equation. Therefore, atmospheric gases cannot be considered merely free gases in space, but much as they can be likened to the pressure vessel limited by the planetary gravitational active field, and any causes that disturb the balance factors lead to the failure of the system of the pressure vessel of the planetary atmosphere.

2.6 Physical Characteristics of Gases

The atmospheres of the planets are at state of balance between the value of the gravity force that pull gases into the surface of the planet and the value of the thermodynamic forces of their atmospheric gases generated within their various circumstances, where the thermodynamic effects are affected by the physical properties of the elements of the atmosphere and the amount of gases that present in the atmosphere, and the final result of these forces determines the specific character of the equilibrium conditions for each planetary atmosphere, the rates of the generated thermodynamic forces are the result of the expansion of gases according to the planet's temperature, as well as the physical properties of their elements such as (boiling, liquefaction, and solidification temperature), which differ according to their atomic mass, and this allows the presence of certain elements in the atmosphere according to the value of the gravity force and the limits of thermodynamic forces generated according to their temperature rates.

The similarities and differences between the gravity force and the thermodynamic action that form the planetary balance circumstances:

Table 1. The common and different factors between the two effects the gravity and the thermodynamic processes upon the planetary atmosphere

	Effects of the gravity force.	**Effects of thermodynamic processes**
1.	The value is stable at any point in the gravity field according to its altitude and value mass of the planet.	The value is variable and depends upon variables mentioned in the ideal gas relationship such as T temperature, n is the amount of substance.

	Effects of the gravity force.	**Effects of thermodynamic processes**
2.	Mathematical relationship is Newton's law of universal gravitation equation: $F= G.(m1.m2)/ d^2$ where m1 is the mass of Earth, m2 is the mass of atmospheric gases, d is the distance between the center of	The mathematical relationship is the ideal gas equation: PV=nRT Where P is pressure, V is volume, n is the number of moles or mass R is the universal gas constant, T is the temperature, and R is a constant = 8.31446261815324 $J{\cdot}K^{-1}{\cdot}mol^{-1}$.
3.	Earth and the middle of the atmospheric layer, and G is the gravitational constant = 6.674×10^{-11} $N{\cdot}m^2/kg^2$	The value is affected by the distance of the planet from the Sun due to the changes in the intensity of radiation and heat in each distance from the Sun.
4.	Its value is not affected by the distance of the planet from the Sun.	The elements have a greater thermal expansion coefficient are more affected.
5.	The elements with higher molecular weight are more affected, due to their greater influence by the gravitational force.	Directly proportional to the thermal expansion coefficient of gases.
6.	Directly proportional to the atomic (molecular) weight of gases.	Mass of atmospheric gases represents (n) the amount of substance in the equation PV=nRT
7.	Mass of atmospheric gases represents (m2) in the equation $F= G.(m1.m2)/ d^2$	Affected by the temperature and the physical characteristics of gases.
8.	Affected by the gravity of the planet and the molecular weight of gases.	It cannot exist without the effect of the gravity force of the planet that confines gases, therefore, it's the reaction of the effects of gravity force upon the masses of gases.
9.	Can exist without gases and their thermodynamic effects.	An extreme increase in its value leads to the collapse of the equilibrium system as it happened the planet Mercury.
10.	An extreme increase in its value leads to the confinement the atmospheric gases in a greater density and in lower ranges in the space of the planet as in the planet Jupiter.	The direction effect is toward in three dimensions in the atmosphere (3D).

	Effects of the gravity force.	**Effects of thermodynamic processes**
	The direction effect is towards the surface of the planet in one dimension (1D)	

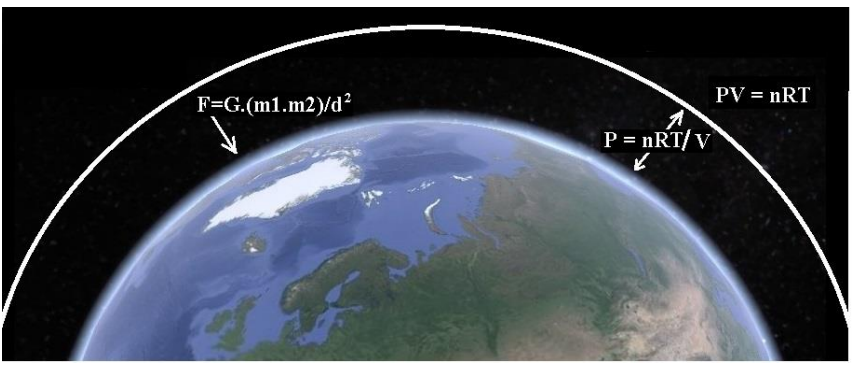


Illustration 3. The balance of circumstances between the two main forces in Earth's atmosphere, the gravity force of Earth's mass to the gases and thermodynamic processes to raise the pressure and the volume of gases

According to the second law of thermodynamics states that a spontaneous process will increase the entropy of the universe, because entropy is typically defined as either the level of randomness of a system and measure of the energy dispersal of the molecules in the system according to Ludwig Boltzmann theory (Viegas, n.d.) the entropy for the gases depend on the kinetic energy at a particular thermodynamic state to expand and the resulting increase in the number of possible arrangements is an increase in the randomness of the system and increases the entropy, this could be calculated via Boltzmann equation:

$$S = k.\ln W \qquad \text{……..3}$$

Where k is the Boltzmann constant ($k = 1.380\ 649 \times 10^{-23}$ J K^{-1}) and ln is the natural logarithm function W is the number of real microstates corresponding to the gas's microstate. Therefore, any change that results in a higher temperature, more molecules, or a larger volume yields an increase in entropy. As a substance is heated, it gains kinetic energy, increasing molecular motion and a broader distribution of molecular speeds. This increases the number of microstates possible for the system increasing the number of molecules in the system also increases the number of microstates to be more arrangements possible of the molecules as well, increasing the volume of a substance increases the number of positions where each molecule could increase the number of microstates, therefore any changes that result in the higher rates of

temperature or more molecules or a larger volume increases entropy. Thus we understand that increasing in the entropy of the atmospheric gases as a result of the rising rates of temperature leads to an increase in their total volume to be extended above their original altitudes in the atmosphere, restoring the value of pressure and the balance circumstances between the value of the gravity force and the thermodynamic effects of the atmospheric gases. For this reason, we find that the value of atmospheric pressure on the Earth at sea level is relatively stable around the limit of 101.325 kPa despite the increase and decrease in the Earth's temperatures during the four seasonal changes. On Earth`s atmosphere the water (H_2O) is the only material that changes its phases among the gas, liquid, and solid phases, but on other planets, this transformation occurs with the rest of the elements, such as Nitrogen, Oxygen, Argon, and carbon dioxide, where the phase transformation of these elements occurs in the atmospheres of the planets fared by the Sun, such as (Saturn, Jupiter, Uranus, Neptune, and Pluto), where the temperature drops to an extreme range within the limits of their liquefaction temperatures. The effect of the sharp rise in the temperature on the rate of gaseous escape from the Earth's atmosphere into space is evident. For the first four planets of the solar system (Mercury, Venus, Earth, and Mars), Illustration 3. Shows how the interaction between the two forces (the planetary gravitational force and the thermodynamic effect of atmospheric gases) determines the state of the planetary atmosphere.

3. THE TOTAL VALUE OF THE ATMOSPHERIC PRESSURE ON THE SURFACE OF THE EARTH IS THE RESULT OF THE EFFECTS OF THE GRAVITY FORCE UPON THE MASS OF ALL LAYERS OF THE ATMOSPHERE AND THE THERMODYNAMIC EFFECTS OF ATMOSPHERIC GASES

The atmospheric pressure of the Earth's atmosphere is the reaction of the gravity force of Earth to the mass gases of the atmosphere towards the surface, the rest of the affecting factors upon the value of atmospheric pressure are physics characteristics (coefficient of thermal expansion and contraction of atmospheric gases), which generate pressure and expansion as a result of thermodynamic processes increasing the atomic collision of the elements and gases in the atmosphere, and the temperature value that determines the Earth's distance from the Sun, and according to the ideal gas equation $PV = nRT$, where P is pressure, V is volume, n is the amount of substance, R is the universal gas constant, and T is temperature). When temperatures rise in warm seasons, the pressure and volume raise gases, raising the higher altitudes of the atmospheric layer and pushing towards the Stratosphere or Mesosphere and even to the Thermosphere, causing gaseous escape from the atmosphere of the planet towards space, overcoming the gravitational force of the planet, because it has become at a fared distances (altitudes) from the surface in which the values of the gravitational force decrease. Because the value of the force of the atmospheric pressure on the surface of the Earth is the result of the attraction force of the Earth to the masses of gases of the atmosphere, the values of the atmospheric pressure at the level of the surface are the summation of the attraction forces of the Earth to the materials of all layers the atmosphere

(Troposphere, Stratosphere, Mesosphere, Thermosphere), therefore to calculate the value of the Earth's attractive force to the material of each layer of the atmosphere, first, the volume of the spherical layer should be calculated, where the density of each layer through the relationship density = masses/volume.

$$D = M/V \qquad \text{.....4}$$

Where D is density, M is mass, and V is volume, and the density is commonly expressed in units of grams per cubic centimeter.

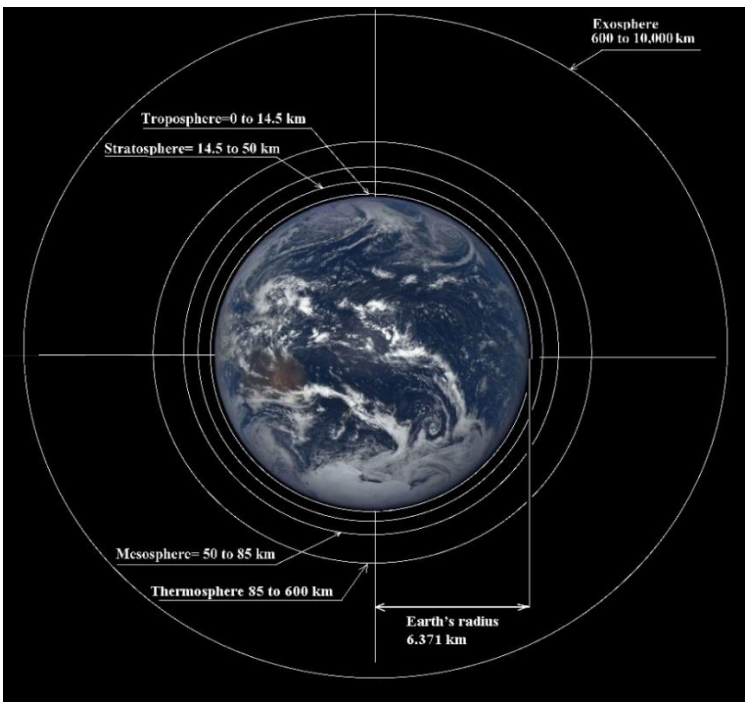


Illustration 4. The actual dimensions of the thicknesses of Earth's atmospheric layers, Troposphere, Stratosphere, Mesosphere, Thermosphere, and Exosphere, compared to the Earth's radius

We calculate the mass of each layer, and then it is possible to calculate the Earth's attractive force to each layer, where the summation is equal to atmospheric pressure at sea level. The pressure of the atmospheric gases on the surface of the planets is a reaction force of gravity force that attracts gases toward the surface, and because gases are compressible fluids, the values of atmospheric pressure change according to their distances from the surface of the planet due to decreasing the decreasing quantities of gases in the higher layers which get attracted towards the surface, until the atmospheric pressure reaches

zero pressure in space afar from the surface, when there are no quantities of gases attracted via the influence of gravity to the surface, and because of the effects of the thermodynamic factor of the gases, their volume and pressure decrease with the decreasing of their temperatures, which leads to a decrease in the atmospheric pressure on the surface. On the cold planets, the thickness of the atmosphere layers decreases to consist of dense and cold gases due to a decrease in the thermodynamic effects of gases at low temperatures, which compress more easily due to the effect of the gravity force of the planet. When the temperature rises (in the summer), the volumes of the atmospheric gases expand. Earth's atmosphere is not a firmly closed pressure vessel, and it's merely confined due to the value of the gravitational force. Therefore, it's not possible to raise the values of pressure absolutely, due to the thermodynamic effects of atmospheric gases. Illustration 4 represents the actual dimensions of the thicknesses of Earth's atmospheric layers: Troposphere = 0 to 14.5 km, Stratosphere = 14.5 to 50 km, Mesosphere = 50 to 85 km, Thermosphere = 85 to 600 km, Exosphere =600 to 10,000 compared to the Earth's radius = 6,371 km. (NASA Science Editorial Team, 2019):

Calculating the volumes of atmospheric gases according to thermodynamic theories: Generally, Earth's atmosphere has a stable value of pressure at sea level despite the differences in the values of their volumes due to the differences in the rates of expansion in the different ranges of temperatures due to their differences in their distance from the equator latitude and the Arctic and Antarctic regions, and if any changes occur in the values of their volumes as a result of changing the temperatures during the four season cycles, which leads gases to expand and contract to occupy a different volume and cause a temporary imbalance of gaseous pressure in the atmosphere, that will be equalized by moving air currents between these areas of different pressures generating wide storms and hurricanes, the movement of air currents on the water surfaces leads to a decrease in the temperatures of air as a result of the phase transformation of the water from the liquid phase to the gaseous phase, which absorbs heat to achieve this phase transformation of water in the nature, because this phase transformation of the water occurring in the limits lower than the boiling point temperature of water under the influence of atomic or molecular diffusion. Therefore, this process leads to the absorption of a large amount of heat from the atmosphere to complete the requirements of these phase transformations. This leads to a further decrease in atmospheric temperatures, more contractions in the volumes of atmospheric gases, and more imbalances in the values of atmospheric pressure, so these currents gradually grow to form extreme changes in the values of pressure in the atmosphere, leading to the great masses of air to move, creating destructive hurricanes. note that the origin of hurricanes is the surfaces of the oceans, where the phase transformations of the water from the liquid phase to the gaseous phase occur at low rates of temperature, which cause a decrease in temperatures, volume, and pressure of atmospheric gases, and thus the phase transformations of the water in the low rates of temperature the main resource of the energy supplies to form and grow hurricanes.

Steps to calculate the summation of the Earth's gravitational force on the layers of the Earth's atmosphere:

1. Determine the rate of elevation of each atmospheric layer.
2. Determining the average density of each atmospheric layer.
3. Calculating the volume of each atmospheric layer through the relationship of calculating the Volume of a spherical shell (Stack Exchange, n.d.):

$$\text{Volume of spherical shell} = 4/3\pi(R^3 - r^3) \quad \ldots\ldots 5$$

R is the radius of the outer layer, and r is the radius of the inner layer.

4. Calculate the mass of each atmospheric layer via the relationship:

   Density = Mass / Volume or Mass = Density × Volume

5. Calculating the Earth's gravitational force for the masses of each atmospheric layer via Newton's law of universal gravitation:

$$F = G.(m1.m2)/d^2$$

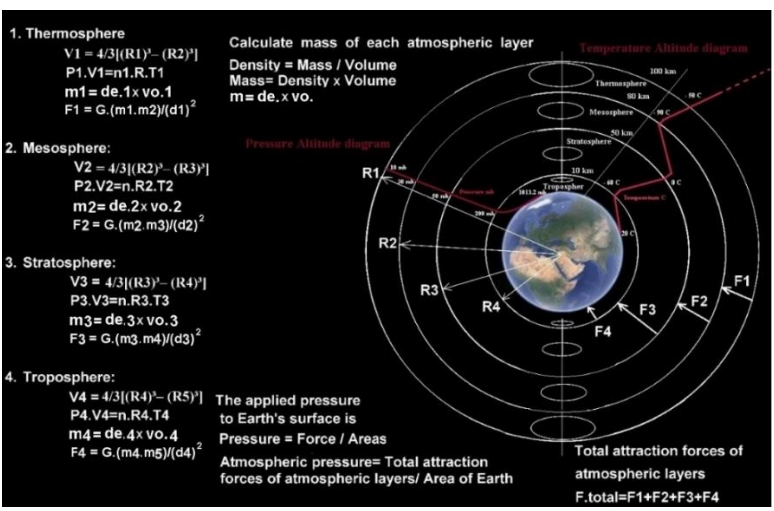


Illustration 5. The layers of Earth's atmosphere with required data to calculate the summation of the Earth's gravitational force on the layers of Earth's atmosphere

Where F is the gravitational force acting between Earth to the masses of each atmospheric layer

d is the distance between the centers Earth and the middle of the elevation of each atmospheric layer, G is the gravity constant = 6.674×10^{-11} $N\cdot m^2/kg^2$

6. Calculate the summation of the Earth's gravitational force for all atmospheric layers.
7. Calculate the applied pressure to Earth's surface through the relationship:

Pressure = Force/Area

Considering the areas the spherical areas of the Earth's surface and the result of this calculation represents the atmospheric pressure at sea level.

Volume of spherical shell: $V1 = 4/3[(R1)^3 – (R2)^3]$

The mass of the atmospheric layer m1 = de.× V

The thermodynamic effects P1. V1 = n1. R.T1

F. total is the summation of gravity forces between the masses of atmospheric layers and Earth.

F. total = F1 + F2 + F3 + F4 ……6

Atmospheric pressure = Total weight of all atmospheric layers/ Area of the surface of Earth.

Atmospheric pressure = F. total / Area of the surface of Earth (The area of Earth is 510.1 million km²)

= atmospheric pressure at the level of sea

Table 2. The main equations could be considered to calculate the summation of the Earth's gravitational force to the layers of the Earth's atmosphere

	The thermodynamic effects of atmospheric gases: P.V = n.R.T	**The gravitational effects upon the atmospheric layer: F= G. (m1.m2)/ d^2**	**Calculate the volume and mass of each atmospheric layer: V1 = 4/3[(R1)³– (R2)³] Mass = Density ×Volume**
1.	Thermosphere P1.V1=n1.R.T1	F1= G. (m1.m2)/ $(d1)^2$	$V1 = 4/3[(R1)^3 – (R2)^3]$ m1=de.1× V1
2.	Mesosphere P2.V2=n2.R.T2	F2= G. (m1.m3)/ $(d2)^2$	$V2 = 4/3[(R2)^3 – (R3)^3]$ m2= de.2× V2
3.	Stratosphere P3.V3=n3.R.T3	F3= G. (m1.m4)/ $(d3)^2$	$V3 = 4/3[(R3)^3 – (R4)^3]$ m3= de.3×vo.3
4.	Troposphere P4.V4=n4.R.T4	F4= G. (m1.m5)/ $(d4)^2$	$V4 = 4/3[(R4)^3 – (R5)^3]$ m4= de.4× V4

Therefore, when the planetary atmospheric pressure increases, it pushes the gases toward higher altitudes, the thermodynamic effect overcomes the effects of the gravitational force of the planet to rebalance the atmospheric pressure that is

determined by the value of the gravitational force of the planet and the value of the thermodynamic effect of each layer of Earth`s atmosphere.

Table 3 symbols of the calculations in the Table 2

P1 is pressure in Thermosphere layer.	m1 is the mass of Earth.	V1 is the volume of Thermosphere layer.
P2 is pressure in Mesosphere layer.	m2 is the mass of the Thermosphere layer.	V2 is the volume of Mesosphere layer.
P3 is pressure in Stratosphere layer.	m3 is the mass of the Mesosphere layer.	V3 is the volume of Stratosphere layer.
P4 is pressure in Troposphere layer.	m4 is the mass of the Stratosphere layer.	V4 is the volume of Troposphere layer.
	m5 is the mass of the Troposphere layer.	
V1 is volume of the Thermosphere layer.		R1 is altitude of Thermosphere
V2 is volume of the Mesosphere layer.	d1 is the distance between surface of Earth and Thermosphere layer.	R2 is altitude of Mesosphere
V3 is volume of the Stratosphere layer.	d2 is the distance between surface of Earth and Mesosphere layer.	R3 is altitude of Stratosphere
V4 is volume of the Troposphere layer.	d3 is the distance between surface of Earth and Stratosphere layer.	R4 is altitude of Troposphere
n1 is amount of substance in Thermosphere layer.	d4 is the distance between surface of Earth and Troposphere layer.	R5 is radius of Earth = 6378 km.
n2 is amount of substance in Mesosphere layer.	d5 is the radius of Earth.	
n3 is amount of substance in Stratosphere layer.		
n4 is amount of substance in Troposphere layer.	F1 is the gravity force between the mass of the Thermosphere and Earth.	
T1 is temperature rate at Thermosphere.	F2 is the gravity force between the mass of the Mesosphere and Earth.	
T2 is temperature rate at Mesosphere.	F3 is the gravity force between the mass of the Stratosphere and Earth.	
T3 is temperature rate at Stratosphere.	F4 is the gravity force between the mass of the Troposphere and Earth.	
T4 is temperature rate at Troposphere.	G is the gravitational constant = 6.674×10^{-11} $N\cdot m^2/kg^2$	
R is a constant 8.31446261815324 $J\cdot\underline{K}^{-1}\cdot mol^{-1}$	M is the mass of Earth = 5.9722×10^{24} kg.	

4. COMPARISON AMONG THE ATMOSPHERIC BALANCE CIRCUMSTANCES OF THE ATMOSPHERES OF THE PLANETS MERCURY, VENUS, EARTH, AND MARS WHICH HAVE CLOSE TO THE SUN

The planet Earth has mass (5.973×10^{24} kg) and the gravity acceleration force (9.807 m/s²) has an effective dynamic atmosphere resulting from the gaseous interference between the surface and the atmosphere (steam) as a result of the phase transportation due to atomic diffusion (Reed & Ehrlich, n.d.) at a rate of temperature lower than the boiling temperature of water, Earth's atmosphere is composed generally of Oxygen (20.95%), Nitrogen (78.08%), and some other elements at low rates, such as argon (0.93%) and carbon dioxide (0.04%). In the meantime, the air also contains a variable amount of (the steam) vapor of water, on average around 1% at sea level. the effects of the two forces of the planetary gravitational force and the thermodynamic effect of atmospheric gases on the value of atmospheric pressure as in Illustration 5 in the limits of the temperature of the environment of Earth, we find that the physics properties of Oxygen and Nitrogen guide them to be in a gaseous phase in the rates of the temperature of the environment of Earth, composing a pressure that represents the atmospheric pressure of Earth around 1013.25 millibars as the effect of the gravitational force of the Earth on the masses of these gases. The distance between the Earth and the Sun is around 152.09 million km, which prevents the temperature of the Earth from reaching the high rates as on the planet Venus. Therefore, the atmospheric pressure on the Earth, 101.325 kPa is lower than the atmospheric pressure on the planet Venus, which is equal (9100 kPa), despite the higher gravity force on the surface of the Earth, which around 9.807 m/s². The condition of the specific balance on the Earth remains for the elements that have the specific physical properties that allow them to expand and contract according to the available thermal ranges on the Earth, such as Oxygen and Nitrogen.

In the Illustration 6 Oxygen phase diagram (Engineering ToolBox, n.d.-a) shows that Oxygen is in the state of superfluid and starts at a temperature of -118.57 °C, where the rate of temperature of Earth`s atmosphere is around 14 °C. Oxygen is on the state of supper fluid and starts at temperature -118.57 °C where the rate of temperature of Earth`s atmosphere is around 14 °C, the molecules weight of Oxygen is 31.9988 g/mol and the molecules weight of Nitrogen is 28.02 g/mol, and the Illustration 7 which shows the Nitrogen phase diagram, we find that state Nitrogen of supper fluid that starts at temperature -146.9 °C

The equilibrium state of the planetary atmosphere depends on the value of the gravitational force of the planet and the average temperature of the planet as well as the type of gases in the atmosphere, which have suitable thermodynamic properties to the limits of the temperature of the planet and the gravitational force, the state of qualitative equilibrium in the Earth's atmosphere, to the elements that have physical properties that allow them to expand and contract according to the available thermal ranges on the Earth, such as Oxygen and

Nitrogen according to the current rates of temperature and gravity on the Earth to specify the atmospheric pressure as a balance between the gravity force and the thermodynamic properties and physics circumstances, we find the order of the first group of planets in the Illustration 8 of the solar system (Mercury, Venus, and Earth) their masses increase according to their distance from the Sun. Illustration 8 shows the values of gravitational force and atmospheric pressure on planets Mercury, Venus, Earth, and Mars, which have close distances to the Sun, as shown in Illustration 8 and its data on planets Mercury, Venus, Earth, and Mars, which are relatively close distances to the Sun.

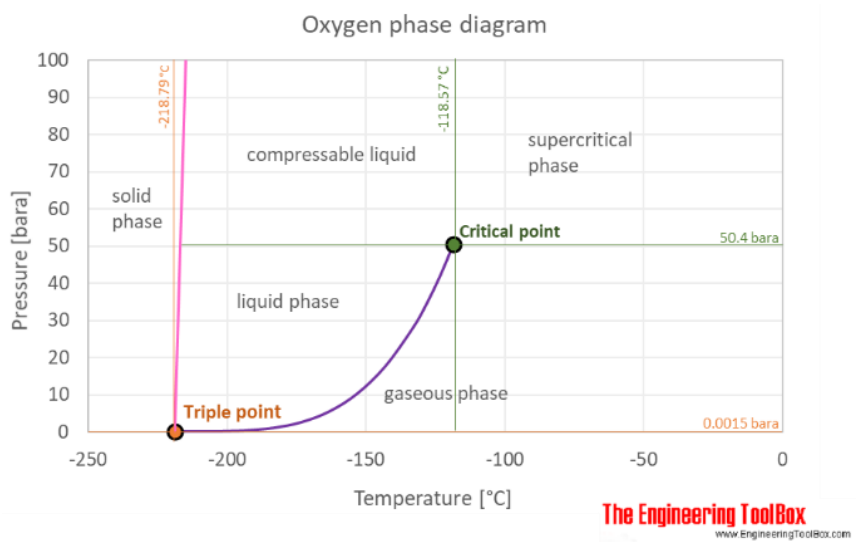


Illustration 6. The Oxygen phase diagram (Engineering ToolBox, n.d.-a)

Mercury: the collapse of the balance of circumstances due to high temperatures (the thermodynamic effect) and the decrease in the value of the planetary gravity force, which is unable to confine gases within the limits of the planetary gravity field.

Venus: the sharp increase in the value of atmospheric pressure due to high rates of temperature (the thermodynamic effect) within the presence of a moderate value of the planetary gravity force able to confine gases within the limits of the gravity field.

Earth: the state of a unique balance between the two forces (the planetary gravity force and the thermodynamic effect of atmospheric gases) that supports and sustains life on Earth.

Mars: the reduction in the atmospheric pressure due to the sharp drop in temperature (low thermodynamic effect) and low value of the planetary gravity force.

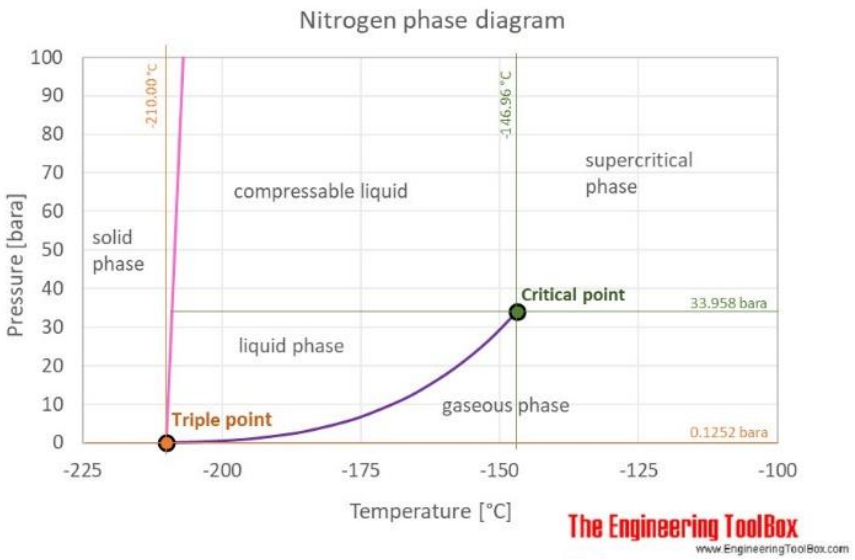


Illustration 7. Nitrogen phase diagram (Engineering ToolBox, n.d.-b)

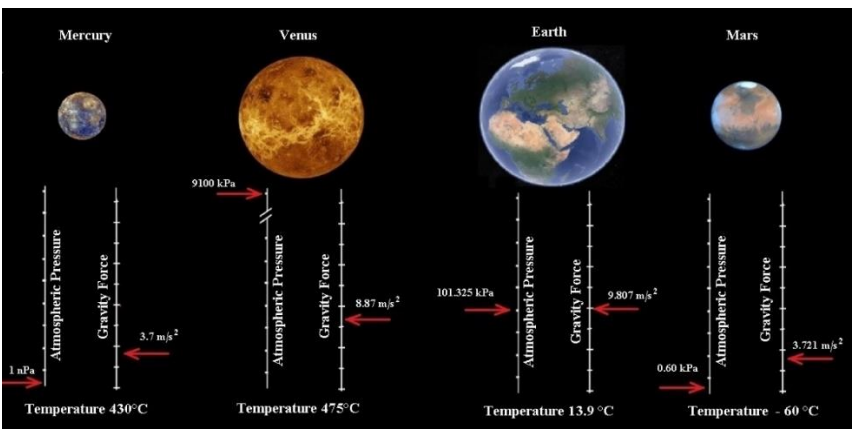


Illustration 8. The values of gravitational force and atmospheric pressure on planets Mercury, Venus, Earth, and Mars, which have close to the Sun

But the reason for the raising of the atmospheric pressure of planet Venus to 9100 kPa where it`s about 48 times above the value of the atmospheric pressure

of the Earth's atmosphere despite its low value gravity force of 8.87 m/s² than what is on the Earth 9.807 m/s² is due to rise in temperature rate to 475°C that raise thermodynamic effects to expand gases and raise the pressure of the atmospheric gases in its middle limit gravity force because that value is able of confining those expanded gases in limits the gravity field of planet Venus.

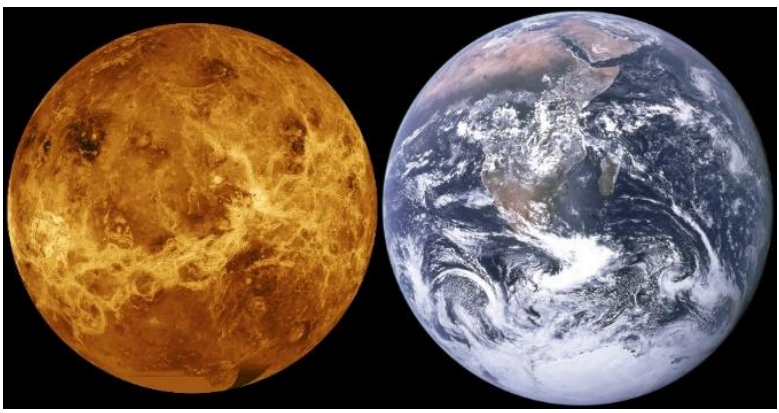


Image 1. The relation of the size of Venus to the size of Earth (Wikimedia Commons, n.d.-b)

Table 4. Data comparison between Venus and Earth (Wikimedia Commons, n.d.-b)

		Venus	**Earth**
1.	Mass	4.867 $\times 10^{24}$kg	5.973 x10^{24} kg
2.	Gravity	8.87 m/s²	9.807 m/s²
3.	Temperature	464°C	13.9 °C
4.	atmospheric pressure	9100 kPa	101.325 kPa

The distance between Venus and the Sun is around 108.18 million km leading to increasing temperatures on the planet Venus, which is around 464°C, leading to the expansion of atmospheric gases on the planet Venus due to the thermodynamic effects of atmospheric gases. Where the atmospheric pressure on the planet Venus is 9100 kPa and greater than the atmospheric pressure on the planet Earth 92 times, the mass of Venus is less than the mass of the Earth, where the mass of Venus is 4.867 $\times 10^{24}$ kg and gravity force on the surface is 8.87 m/s², it's the effect of the thermodynamic factor of the gases. The gravity force factor of Venus upon these gases determines their volume within the gravitational field zone so that it is more like a pressure container for these gases affected by the gravitational field zone to prevent these gases from expanding their volumes, leading to a rise in atmospheric pressure due to the effect of the thermodynamic of gases. Illustration 9 shows the pressure and temperature axes of the Carbon dioxide (CO_2) phase diagram (Wikimedia Commons, n.d.-b), Carbon dioxide is in the state of supper fluid a starting temperature is 31.1°C, where the rate of temperature of Venus`s atmosphere is around 474 °C.

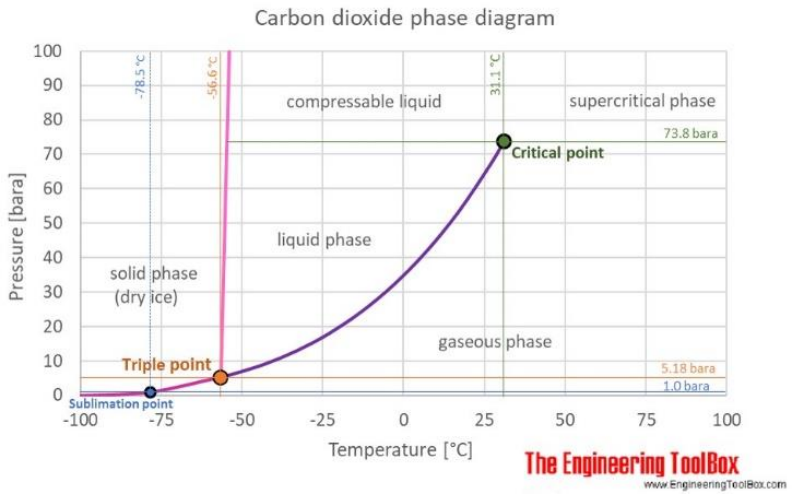


Illustration 9. Carbon dioxide (CO_2) phase diagram (Engineering ToolBox, n.d.-c)

Illustration 10 shows the Hydrogen sulfide H_2S diagram **(Engineering ToolBox, n.d.-c)**, where hydrogen sulfide is in the state of superfluid and starts at a temperature of 99.95 °C where the rate of temperature of Venus`s atmosphere is around 475 °C.

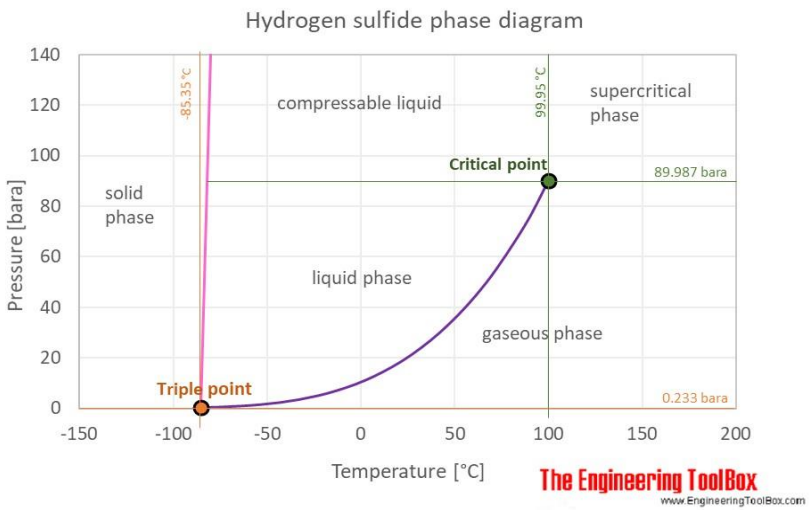


Illustration 10. Hydrogen sulfide phase diagram (Engineering ToolBox, n.d.-d)

The planet Mars has an atmosphere composes (Engineering ToolBox, n.d.-d)of 3% Nitrogen molecular (N_2), 1.6% Argon (Ar), 0.16% molecular Oxygen (O_2), and 0.06% Carbon monoxide (CO), which is generally composes of 95% Carbon dioxide (CO_2) which has molecular weight 44.01 g/mol, and low rates of Oxygen (O_2) molecular weight Oxygen gas is 15.999 u and Nitrogen (N_2), 14.0067 u this is due to its mass (6.39×10^{23} kg) and the gravity acceleration (3.721 m/s²), therefore there isn't a capability to attract Nitrogen (N_2) and Oxygen (O_2) forcedly as it occurs on the Earth which composes its atmosphere of Nitrogen (N_2) 80% and Oxygen (O_2) 20 % due to the relatively high gravity force of Earth, where the mass of Mars is about % 64 of Earth`s, leads to generate gravity force less than the Earth's gravity force prevent it to attract these two gases forcedly, therefore these two gases represents as a light gases and raises upward of the atmosphere of Mars and gets escaped to the space, and this drop in the value of the gravity force of Mars lead to a significant drop in the atmospheric pressure of the Mars to reach (0.60 kPa) which represents 1% of Earth's atmosphere raising the gaseous escapability of the atmospheric gases of Mars to the space at very high rates.



Image 2. The difference in the masses of the planets Mars, Earth and Venus

The thermodynamic factor that works to reduce the value of the atmospheric pressure of Mars (PV = nRT, P is pressure, V is volume, n is the amount of substance, R is the universal gas constant, T is temperature) is the effect of the reduction of the temperature of Mars to be around -60°C due to its far distance from the Sun, around 208.22 million km, checking the Carbon dioxide (CO_2) phase diagram in Illustration 9 the environment circumstances of the Mars is in the triple point to combine three phases of this material (solid phase, liquid phase and gaseous phase), when gases getting to the solid phase or the liquid phase due to the dropping temperature will drop its effect extremely on the atmosphere pressure, it descend the atmospheric pressure to (0.60 kPa) or 1% of Earth's atmosphere, causes the high rates of gaseous escape from the atmosphere towards space, as it happens in the planet Mercury which has mass (3.285×10^{23} kg), and gravitational acceleration force (3.7 m/s²), thus the effects of thermodynamics of the atmospheric gases overcoming the effect of gravitational force is the reason behind the collapse of the balance circumstance of the atmosphere of the planet Mercury.

Table 5. Comparison of the data between Earth and Mars in their orbit around the Sun

	Factors	Earth	Mars
1.	Distance from Sun	152.09 million km	208.22 million km
2.	Average of temperature °C	13.9 degrees °C	Minus 60 °C
3.	Atmosphere composition	78% Nitrogen, 21% Oxygen, and one percent other gases.	Carbon Dioxide 95.3%, Nitrogen.
4.	Atmospheric pressure	Pressure at sea level 1013.25 mill bars.	Pressure at surface is 6.518 mill bars
5.	Mass of the planet.	5.9736×10^{24} kg	6.39×10^{23} kg
6.	Gravitational acceleration on the surface.	9.807 m/s²	3.721 m/s²
7.	Speed of rushing around Sun	107,000 km/h	86,871 km per hour

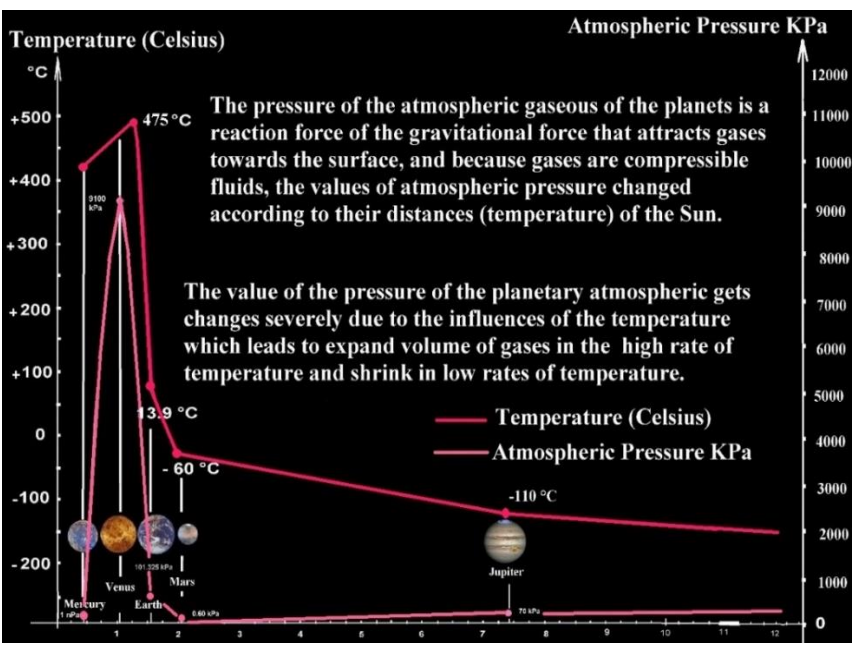


Illustration 11. The relationship among the distance, mass, and pressure of the first group of planets of the solar system

The temperature of the atmosphere is affected by the distance of the planet from the star and also the amount of mass and value of the activity of the star, where the Illustration 11 shows the relationship among the distance, mass, and pressure of the first group of planets of the solar system, therefore, in the solar system, the farthest planets of the solar system, their temperatures drop at extreme rates. The closest planets to the Sun (Mercury, Venus, Earth, and Mars), raise their temperature, leading to an increase in the volume of their atmospheric gases, which leads them to rise to the height altitudes of the atmosphere, due to the effects of the thermodynamics of gases, where the value of the planet's gravity decreases on the mass of these gases, so the possibility of their escape from the atmosphere to space increases. Therefore, we find the absence of Oxygen and Nitrogen from the atmospheric compositions of (Saturn, Jupiter, Uranus, Neptune, and Pluto), which are often composed of Hydrogen and Helium, which have a very low liquefaction temperature, while Oxygen and Nitrogen are abundantly available in the Earth's atmosphere, because of their liquefaction or solidification at the surface of these extremely cold planets. Hence, it is unavailable in the gaseous phase in the atmosphere of those planets, and possible to find it in a liquid or solid state on the surface. The gaseous escape from the atmosphere for a long time leads to the formation of a new balance circumstance of the planetary atmosphere that is composed of higher atomic mass elements and compositions that are released from the materials of the crust due to extremely high temperatures, as occurs with the planet Venus. Where the atmosphere of Venus is composed of Carbon dioxide (CO_2) molecules weight and Sulfuric acid (H_2SO_4) molecules weight is 98.079 g/mol. Note that the weight of the Carbon dioxide molecule is 44.01 g/mol, which often gets released from the rocks of the surface due to the extremely high temperature as well as the volcanic emissions, because the planetary gravitational force is relatively high (due to the relatively large masses) and the thermodynamic effect of atmospheric gases is also high (due to the high temperature), the state of balance circumstances between the two forces leads to the confinement of the molecules of the atmospheric gases in the gaseous phase in the limits of gravitational force, leading to the rising of the atmospheric pressure on the surface of the planet and moderate rates of gaseous escape from the atmosphere into space.

Table 6. Boiling and freezing of temperature of gases [(Poling, Prausnitz, & O'Connell, n.d.), Oxygen, Nitrogen, Carbo dioxide CO_2, Methane CH_4, Hydrogen H_2, Argon Ar

	Gas	Boiling temperature	Freezing temperature
1.	Oxygen O_2	-183 °C	-218.8 °C
2.	Nitrogen N_2	-195.8 °C	-210 °C
3.	Carbon dioxide CO_2	-78.46 °C	-78 °C
4.	Methane CH_4	-161.6 °C	-182 °C
5.	Hydrogen H_2	-252.87 °C	-252.9 °C
6.	Argon Ar	-185.8 °C	-189.4 °C

The atomic diffusion (Setser, 2013) occurs due to moving atoms and molecules of materials from a high density zone to the lower density zone therefore, atomic

diffusion occurs strongly from the Troposphere and Stratosphere to the Exosphere, Thermosphere, and Mesosphere, after the gaseous escape from the Thermosphere and Mesosphere) to space, due to inertia effects when the speed of Earth accelerates, the mass and momentum of the gases prevent them from accelerating at the speed of the Earth and increasing in the percentage of radiation and solar energetic particles when the Earth approaches the Sun, which causes friction of these particles with the gases, and pushes them toward space, the lack of gases disturbs the balance circumstances in these high layers, where this atomic and molecule diffusion under pressure occurs as a reaction to restore the value of atmospheric pressure in (Exosphere, Thermosphere, and Mesosphere).

Remember the atomic diffusion theory is the movement of molecules or atoms from the region of high concentration to the region of low concentration as a result of the random motion of the molecules or atoms where diffusion is driven by the gradient in the chemical potential of the diffusing species, affected by factors such as gradient, value, concentration, pressure, temperature, and distance, where the gradient is the change in the value of the concentration, the change in pressure over distance is called pressure gradient, and the change in temperature over distance is called temperature gradient. This fully applies to the concept of diffusion under pressure, which claims (if the diffusion takes place in a closed vessel, the pressure difference drives the heavy gas into the light gas region at a faster rate) (Walther, n.d.).

The atmospheric of Mars generally composes of 95% Carbon dioxide (CO_2) which has molecular weight 44.01 g/mol, 2.6% molecular Nitrogen (N_2) molecular weight 14.0067 u, 1.9% Argon (Ar), 0.16% Oxygen (O_2) molecular weight is 15.999 u, and 0.06% Carbon monoxide (CO) (Greeley & Batson, 2001). Because the mass of Mars is around 6.39×10^{23} kg the gravity acceleration (3.721 m/s²), therefore it hasn't the capability to attract Nitrogen (N_2) and Oxygen (O_2) forcedly as it occurs on the Earth which composes its atmosphere of Nitrogen (N_2) 80% and Oxygen (O_2) 20 % due to the relatively high gravity force of Earth 9.807 m/s², therefore Nitrogen (N_2) and Oxygen (O_2) represent light gases on Mars, and rise upward high altitude in the atmosphere of Mars and escape to space gradually, the low gravity force of Mars leads to a significant drop in the atmospheric pressure to reach (0.60 kPa), which represents 1% of Earth's atmosphere, raising the gaseous escape of the atmosphere of Mars to space at very high rates, in the meantime the thermodynamic factor works to reduce the value atmospheric pressure of Mars affected by the reduction of the temperature of Mars to be around - 60 °C due to its fared distance from the Sun around 208.22 million km.

Checking the phase diagram of Carbon dioxide (CO_2) in illustration 9, we find environmental circumstances of Mars are in the triple point to combine three phases of this material (solid phase, liquid phase, and gaseous phase). When gases of the atmosphere get to the solid phase or the liquid phase due to the temperature reduction, it will drop their physics effect extremely upon the value of

pressure of the atmosphere, leading the atmospheric pressure to drop extremely to reach (0.60 kPa) around 1% of Earth's atmosphere.

Table 7. Data comparison between Mercury and Earth

		Mercury	Earth
1.	Mass	3.285×10^{23} kg	5.973×10^{24} kg
2.	Gravity	3.7 m/s²	9.807 m/s²
3.	Temperature	430°C	13.9 °C
4.	atmospheric pressure	1 nPa	101.325 kPa

5. FACTORS CAUSING GASEOUS ESCAPE FROM THE EARTH'S ATMOSPHERE

The effect of the type of orbit of the planet upon the gaseous escape: If the orbit of the planet is circular around the Sun, it leads to the stability of the planetary atmosphere, but each planet's orbit around the Sun is elliptical. According Kepler's Second Law implies that a planet moves fastest when it is at perihelion and slowest at aphelion; therefore, the planet's velocity changes. Where the planet's speed is at its lowest value (perigee) at the farthest distance from the Sun, after that point, the speed of planets starts to be accelerated gradually and the speed reaches the highest value in (apogee) (NASA, 2008), where the distance is the closest distance to the Sun, these change in speed confuses the stability state of the gases of the atmosphere of the planet, especially when the speed of the planet starts accelerating, as shown in Illustration 16 because the mass of gases generates momentum that obstructs the planet's acceleration, thus a quantities of atmospheric gases fail to catch up with the planet to remain behind the planet in space, this related to the inertia phenomenon (Zimba, 2009), where the momentum of masses of the atmospheric gases resists changes in the speed motion of Earth during the acceleration period, causing the phenomenon of gaseous escape from the atmosphere to the space, when the speed of the Earth starts to increase from the lowest value (perigee) in Earth's farthest distance from the Sun, where after that point Earth`s speed begins to be accelerated gradually to reach the highest value in (apogee) where the closest distance to the Sun, thus masses of gases in the highest altitudes prevents them to be accelerated with the same speed of the Earth, losing a big quantities of gases lag in space behind the direction of rushing of the Earth in its orbits around Sun to be missed in space, where we can speculate the shape of the aerobic bubble of the Earth's atmosphere layers; the thickness of the Earth's atmosphere is the shortest distance possible on the front side of the Earth's direction in the orbit and increases gradually towards the back of the rushing direction, where the longest distance is on the back side of the direction in orbit around the Sun.

The effect of inertia (Zimba, 2009)**:** The phenomenon related to inertia where the momentum of masses of the atmospheric gases resists changes in the speed motion of Earth during the acceleration period, these procedures occur in the planets that have elliptical orbits, because when the speed of the Earth starts to

increase from its lowest value (perigee) when the Earth is at the farthest distance from the Sun. The illustration 12 shows Earth's orbit around the Sun is elliptical and how the velocity of the rushing of Earth changes, where Earth's speed is at a lower value (Perigee) when the planet is at the farthest distance from the Sun (152 million km/s), where the speed is 29.29 km/s, after that point Earth's speed begins to accelerate gradually and reaches the highest value (Perigee) when the distance is as close as possible to the Sun (14.7 million km), where the speed is 30.29 km/s. These speed changes confuse the stability state of the gases in the atmosphere of Earth. Especially at the point where the speed of the Earth starts to accelerate, where the mass of gases generates momentum that obstructs them from being accelerated with the acceleration of Earth, is called the inertia phenomenon. This leads to the missing a large quantity of atmospheric gases that remain backward, causing gases to escape from the atmosphere in space in the direction of Earth around the Sun. Causing the phenomenon of gaseous escape from Earth's atmosphere, especially of the high layers of the Mesosphere, the Thermosphere, and the Exosphere, where the value of Earth's gravity force is relatively lower.

Stages of gaseous escape from the Earth's atmosphere into space: the stage of gaseous escape from the upper layers (Thermosphere and Mesosphere) to space due to inertia effects when the speed of Earth gets accelerated, the mass and momentum of the gases prevent them from accelerating at the speed of the Earth and increase in the percentage of radiation and solar energetic particles when the Earth approaches the Sun, which causes friction of these particles with the gases and pushes them towards space. The thicknesses decrease in the front of the rushing direction of the Earth, and they increase on the opposite side because air is a compressible fluid and has relatively low mechanical resistance; we could expect the real shape of the layers of Earth as shown in Illustration 13.

After the losses of quantities of gases in the upper layers (Mesosphere and Thermosphere) due to the acceleration of Earth in its elliptical or the collision of the energetic solar particle with Earth's atmosphere, this leads to an imbalance in the state of gaseous equilibrium between the upper and lower layers, which guides to spread atoms and molecules from high density to low-density zones causes the activation of atomic diffusion from the lower layers (Troposphere, Stratosphere) towards the higher layers (Mesosphere and Thermosphere) to restore the state of equilibrium in the upper layers (Mesosphere and Thermosphere) as a result the atomic and molecules diffusion from the high density zone of the lower layer (Troposphere, Stratosphere) to low-density zones in the (Mesosphere and Thermosphere), this leads losses a great amount of gases such as Oxygen and Nitrogen from (Troposphere, Stratosphere), note that the loss of gases from the upper layers to (Mesosphere and Thermosphere) occurs in every location of Earth in its orbit around the Sun, but the peak of its activity in the phases is the point of starting the acceleration and the shortest distance to the Sun.

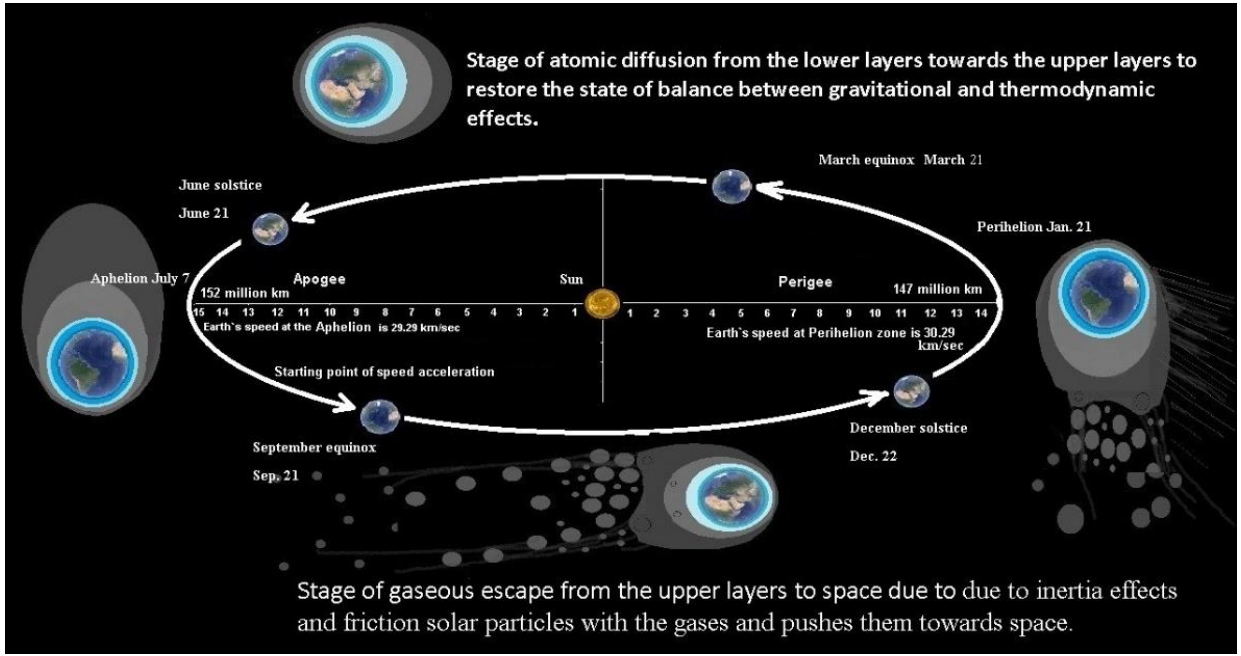


Illustration 12. The two stages of gaseous escape from Earth's atmosphere start with the inertia effects and the collision with the solar particles, then the stage of the atomic diffusion from the low layers to the upper layers

The effects of industrial gases and volcanic emissions upon the balance of circumstances in Earth's atmosphere:

These industrial gases have a second effect in increasing the temperature, which leads to the activate the variable T in the relationship (PV=nRT) due to the effect of global warming, this leads to increasing the volume of air in the Troposphere, which accelerates the process of displacement of small molecular weight gases into the Stratosphere, for example, HFC-23 is a by-product of refrigerant manufacture have around 14,700 times as powerful as carbon dioxide at warming. When the high molecules weight industrial gases such as Carbon dioxide CO_2, Methane (CH_4), Sulfur hexafluoride (SF_6), Nitrogen trifluoride (NF_3), the high molecular weight gases are emitted to the atmosphere in large quantities and will confuse the effect (n) of the relationship PV=nRT, from what leads to an increase in the magnitude of (PV) which will overcome the magnitude of the gravitational force to the masses of atmospheric gases in the Troposphere layer, for what causes confusion in the state of balance between the gravitational force and the thermodynamic effects, and because the magnitude of the Earth's gravitational force is the constant that attracts the gaseous masses towards the surface and that results a specific value of pressure in the Troposphere layer, this will rush of the lighter weight elements in the layer (Nitrogen and Oxygen) to the higher layers (Stratosphere and Mesosphere), in order to restore stability to the state of balance between the two mentioned forces in the Troposphere layer, and this is the main reason behind the phenomenon of gaseous escape of light elements such as Oxygen and Nitrogen and the collapse of the ozone layer in the Earth's atmosphere. When huge quantities of relatively heavy molecular weight industrial gases and volcanic emissions (such as Carbon dioxide, molecular weight 44.01 g/mol and sulfuric dioxide, molecular weight 64.066 g/mol) are emitted and are forcedly affected by the gravity force due to their relatively higher molecular weight into the atmosphere. This will lead to an increase in the amount of matter of gases in the atmosphere, which is represents (n) in the relationship of ideal gases, the relationship PV=nRT which has a direct relationship with pressure and the volume (PV), this will lead to a rising value of air pressure and volume in the Troposphere more than its original value, which is restricted according to the value of the gravitational force to the masses of gases and the thermodynamic effects of the atmospheric gases in the equilibrium state of the mentioned forces. These industrial gases with large molecular weight displace light gases such as Oxygen and Nitrogen from the Troposphere to the Stratosphere (to restore the balance state between the gravitational force and thermodynamic effect of gases in the Troposphere and maintain the standard value of atmospheric pressure in the Troposphere around 101,325 Pa).

This will lead the lighter gases molecular weight such as Oxygen molecular weight of 15,999 u and Nitrogen molecular weight of 14.0067 u which are less affected by the gravity force due to their lower mass weight (lower mass), to be rushed to the (Stratosphere and Mesosphere), this causes damage to the Ozone layer, which is formed from triple Oxygen or O_3 and due to the rise of the triple molecular Oxygen to higher altitudes where atmospheric pressure is lower and different physics circumstance that required to formation of triple Oxygen (Ozone

O_3) to be form, note that Ozone gas exposures to the ultraviolet rays of the Sun prevented them from reaching the surface of Earth, protecting life on Earth and its absence poses a great danger to life on Earth.

This is the main physics factor causes gap in the Ozone layer in the Earth's atmosphere, in the meantime the process of gaseous escape from the Earth's atmosphere is the result of the expansion of the gases of the Troposphere and to be rushing towards the upper layers to Stratosphere, and pushing the gases of the upper layers to escape to Stratosphere and Mesosphere towards space, of the triple Oxygen O_3 molecules (molecular weight 48 g/mol) in the first layers to the higher altitudes where devoid the conditions of forming the triple Ozone molecule, so the triple Oxygen molecule decomposes into the binary molecule or the singular atoms as found at high altitudes Stratosphere and Mesosphere, leaving a gap in the Ozone layer of the atmosphere of the Earth in addition to the chemical factor of the chemical reactions of the active industrial gases and even due to collision the energetic solar particle with Earth's atmosphere. Therefore, the erosion of the Ozone layer in the Earth's atmosphere is not considered an escape of gases as much as a disturbance in the qualitative balance system of gases in the Earth's atmosphere. The Illustration 14 shows how the high molecule weight such as Carbon dioxide CO_2, Methane (CH_4), Sulfur hexafluoride (SF_6), Nitrogen trifluoride (NF_3), Nitrous oxide (N_2O) etc., that are highly affected by the gravity force of the Earth due their high molecules weight (molar mass), that represents (n) in the relationship PV=nRT leads to change in value PV, the pressure and volume to increase the value of the atmospheric pressure in the Troposphere, is what causes the imbalance in the state of the natural balance circumstances between the two effected forces (the gravity force of the Earth and the effects of the thermodynamic processes of the gases), and this replacement leads to pushing the elements of lighter molar masses, such as Oxygen (O_2) = 32.0 g/mol and Nitrogen (N_2) = 28.014 g/mol) to move from the troposphere to the higher layers, the stratosphere and Mesosphere to restore the state of balance in these mentioned forces in the Troposphere, to return the value of the atmospheric pressure in the Troposphere to its original value around 101.325 kPa.

Table 8. High molar mass gases of the industrial emissions in Earth`s atmosphere (NASA, n.d.-d)

	Gases of the industrial emissions	**Molecules weight (molar mass) *g/mol***
1.	Methane (CH_4)	16.04246 g/mol
2.	Carbon dioxide (CO_2)	44.01g/mol
3.	Nitrous oxide (N_2O)	44.013 g/mol
4.	Nitrogen trifluoride (NF_3)	71.002 g/mol
5.	Sulfur hexafluoride (SF_6)	146.06 g/mol

High percentages of industrial residues are in the gas phase, and it is clearly seen in the emissions of petroleum refineries, car smoking, and heavy machines global greenhouse gas emissions, which add around 50 billion tonnes annually of gases to the Earth's atmosphere. Since the middle of the 20th century, the

annual emissions from burning fossil fuels have increased every decade from 11 billion tons of carbon dioxide per year in the 1960s to 36.6 billion tons in 2022 according to the Global Carbon Budget 2022 (World Meteorological Organization, n.d.).

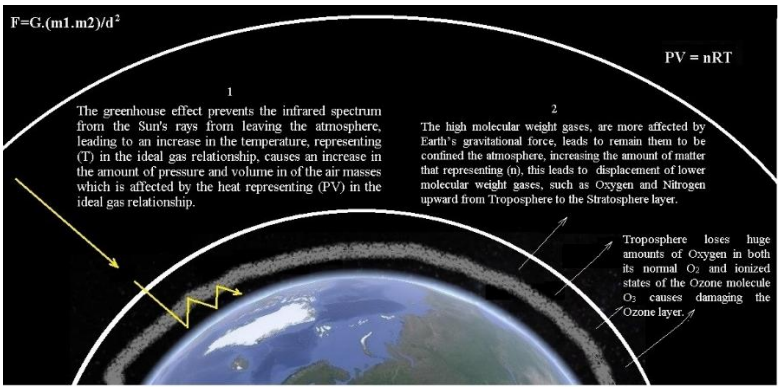


Illustration 13. The effects of the greenhouse and the displacement of lower molecular weight gases to higher altitudes, such as Oxygen and Nitrogen upward, restore the value of atmospheric pressure to the value that is equivalent to the effect of the Earth's gravitational force upon the mass of confined gases in the Troposphere layer, restoring the state of balance in the Troposphere layer, where Troposphere carries the state of the ozone molecule (O_3), this causes damages in the Ozone layer, which protects living organisms from the fatal effects of the ultraviolet of Sun rays spectrums

Table 9. The gas emissions to the atmosphere yearly by the industry

	Sector	Gas emissions
1.	Energy	% 73.2
2.	Agriculture, forestry, land use	% 18.4
3.	Industry	% 5.2
4.	Waste	% 3.2

Some quoted data with respect of rates of greenhouse gas emission and ozone depletion in the last twenty years since the beginning of the 21st century:

1. Green House Gas (GHG) emissions of all world countries (EDGAR, n.d.): China, the United States, India, the EU27, Russia and Brazil were the world's largest GHG emitters in 2023. Together they account for 49.8% of the global population, 63.2% of global gross domestic product, 64.2% of global fossil fuel consumption and 62.7% of global GHG emissions. Among these top emitters, in 2023, China, India, Russia and Brazil increased their emissions compared to

2022, with India having the largest increase in relative terms (+ 6.1%) and China the largest absolute increase by 784 Mt CO_2. With only two exceptions, 2009 (global financial crisis) and 2020 (COVID-19), global GHG emissions have grown steadily since the beginning of the 21st century, mainly due to the increase in fossil CO_2 emissions by China, India, and other emerging economies (EDGAR, n.d.).

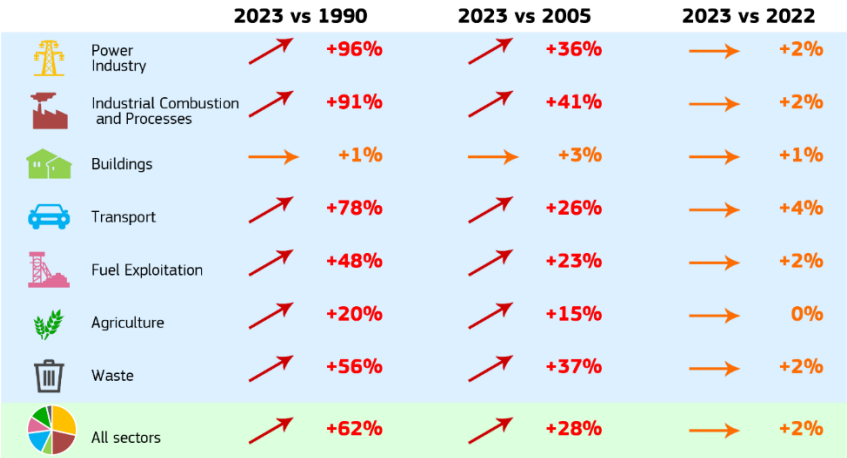


Illustration 14. Global emission trends by sector and years) (EDGAR, n.d.)

In 2023, the majority of GHG emissions consisted of fossil CO_2, accounting for 73.7% of total emissions, while CH_4 contributed 18.9% to the total, N_2O for 4.7% and F-gases 2.7%. Global fossil CO_2 emissions increased by 72.1% since 1990. The increases in CH_4 and N_2O emissions have followed a somewhat slower pace: CH_4 increased by 28.2% and N_2O by 32.4% between 1990 and 2023, while F-gases have seen a four-fold increase (+294%) in the same period. As far EU27 is concerned, its GHG emissions, excluding LULUCF, were 33.9% lower than in 1990, reaching 3.22 Gt CO_2. In 2023, EU27 emissions decreased by 7.5% (-261 Mt CO_2) and the EU27's share of global emissions decreased from 6.8% in 2022 to 6.1% in 2023. In 2023, all EU27 countries except Croatia and Cyprus experienced a decrease in their emission levels compared to the previous year. In terms of contribution to the EU27's GHG emissions in 2023, Germany remained the largest emitter, followed by France, Italy, Poland and Spain. In the EU27, all sectors experienced a decrease in their GHG emissions in 2023 compared to 2022. The largest relative drop was observed in the power industry sector, in which emissions decreased by 20.1%. The industrial combustion and processes showed the second-highest decrease, falling by 8.1% below the 2022 levels. The data are presented in the table for key years, to allow easy comparison of different countries. Data for individual GHG (fossil CO_2, CH_4, N_2O and F-gases) are available in the excel file in the download section (data sheet GHG by sector and country) (EDGAR, n.d.).

2. Related data in respect of the Antarctic ozone hole via the European Centre for Medium Range Weather Forecasts (ECMWF) operates services from the EU's Copernicus Earth observation (Copernicus, 2024): The development of the Antarctic ozone hole in 2024 started later than average, mainly due to the disruption of the polar vortex following two sudden stratospheric warming episodes in July. Illustrations 16 and 17 shows the time series of the daily ozone hole area between 1 July and 8 December 2024. As the polar vortex became established through August, the chemical depletion of stratospheric ozone over Antarctica was able to take effect, as it does every year. The area of the ozone hole kept increasing throughout September, closely following the 1979-2021 average, reaching its maximum size of 22 million km^2 by the end of the month. Not only is this smaller than the area in 2023 and 2022, when the maximum area was around 25 million km^2, but the maximum occurred later than in 2023 and is in line with the historical average. While some of the recent year's ozone holes were notable for closing later in the year on average, the 2024 closure has been more typical. The ozone hole area steadily decreased in size throughout October, in line with the average, but stabilized through November with an approximate area of 10 million km^2 per day during the month. The polar vortex broke down in the first week of December, quickly bringing the 2024 ozone hole closure in line with the 1979-2021 average closure date. Once the ozone depletion started to be significant, the ozone hole quickly reached an area of 15 million km^2, the area of Antarctica, in the first days of September. It reached its peak extent of about 22 km^2 by the end of the month, also slightly later than in 2023, in line with the average, and it was smaller than in 2023 and 2022 when the area reached approximately 25 km^2 Global warming and the greenhouse effect tend to cool the stratosphere favoring the formation of Polar Stratospheric Clouds and ozone depletion. In Illustration 17 the ozone hole above the Antarctic is in line with the 1979-2024 average date (ECMWF, 2024) (Copernicus, 2024).

3. The double effect of the industrial gases upon the ozone layer: The illustration 14 shows the double effect of the presence of industrial gases in the atmosphere upon the Ozone layer, where the first is (the greenhouse effect), as it prevents the infrared spectrum of Sun's rays from leaving the atmosphere, thus it keep rising the temperature of gases, this representing (T), and this causes an increase in the value of pressure and volume (PV) in the ideal gas relationship in PV=nRT.

The second is that these industrial emission gases have a high molecular weight, so they are more affected by the effect of the Earth's gravitational force, so they remain confined within the atmosphere, which increases the amount of matter, represented (n) in the ideal gas relationship. This leads to the displacement of lower molecular weight gases, such as Oxygen and Nitrogen upward layers such as the Stratosphere, restoring the state of balance in the Troposphere layer, and restoring the value of atmospheric pressure to the value equivalent to the effect of the Earth's gravitational force upon the mass of confined gases in the altitude of the Troposphere layer. Thus, the troposphere loses huge amounts of Oxygen in both its normal (O_2) and excited and ionized states of the ozone molecule (O_3),

ultimately damaging the Ozone layer, which protects living organisms from the fatal effects of the ultraviolet of Sun rays spectrums.

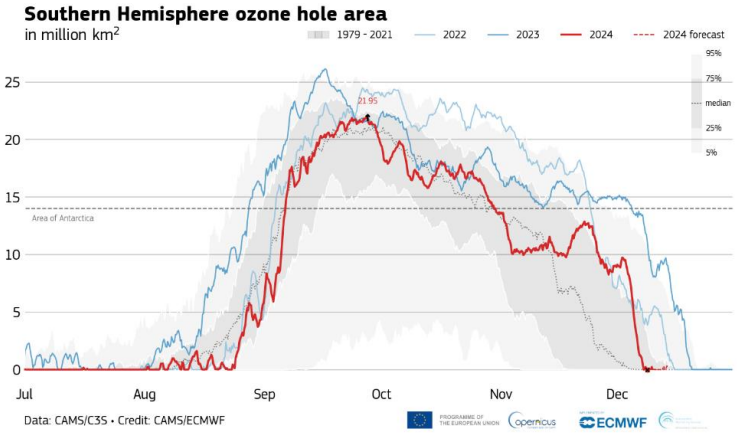


Illustration 15. The time series of daily ozone hole area between 1 July and 8 December, where the red line shows the data for 2024 and the dotted line the forecast for the following 5 days. The thicker, darker blue line represents the evolution of the ozone hole during 2023, and the lighter blue line represents 2022. The grey colours represent the percentile values and the median between 1979 and 2021 (Credit: CAMS/ECMWF) (Copernicus, 2024)

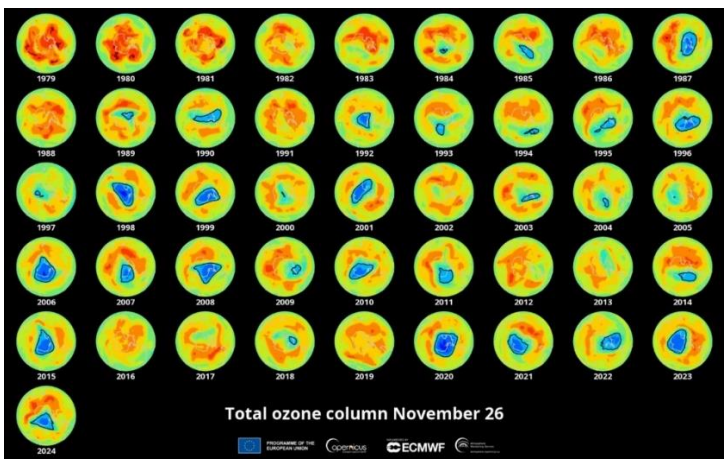


Illustration 16. Data on the ozone hole above the Antarctic are in line with the 1979-2024 average date. Atmosphere Monitoring Service, the Copernicus Programme (Copernicus, 2024)

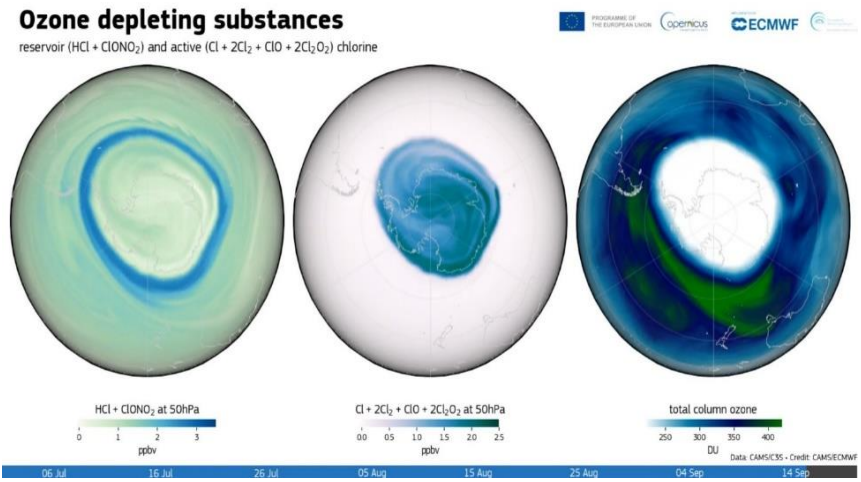


Illustration 17. CAMS suite provides forecasts and analysis of ozone and several ozone depleting substances, allowing it to represent the ozone hole depletion process. Credit: CAMS (Copernicus, 2024)



Image 3. The part of the emissions of power plants that are fuelled by fossil fuels

Emissions of industrial (Phillips, 2021) high-weight molecules such as Sulfur hexafluoride (SF_6) = 146.06 g/mol, Nitrogen trifluoride (NF_3) = 71.002 g/mol, Nitrous oxide (N_2O) = 44.013 g/mol and Carbon dioxide (CO_2) = 44.01 g/mol into the atmosphere gases composition leads to a disturbance in the state of balance between the amount of the Earth's gravity force and the thermodynamics effects at the Troposphere at the height (0-14.5 km), pushing a large percentage of low

molecules weight gases, such as Nitrogen (N_2) = 28.02 g/mol and Oxygen (O_2) = 31.999 *g*/mol towards the Stratosphere at the height 14.5 to 50 km, the value of atmospheric pressure (101.325 kPa) in Troposphere at the height 0 to 14.5 km, so that the state of balance between the effects of gravity and gases thermodynamics effects at the Earth's temperature average 14 °C which is determined by the Earth's distance from the Sun. The relatively high temperature of the Earth causes the physical activity of the atoms of gases and expands in volume of the atmosphere, this causes a reduction in the effect of atomic gravity on the atoms of the gases so that the volume of one cubic meter in space around the Earth contains a lower value of gas matter. Here we can count that the nature of the planetary atmosphere is one of the aspects of the age of the planet's formation, as the greater the amount of heavy elements in the planet's atmosphere indicates the older age of the planet's formation, not only in terms of traditional chronological age, but also due to thermal effects that accelerate the chemical and physical processes to accelerate the passage of the atmosphere to its older formation ages.

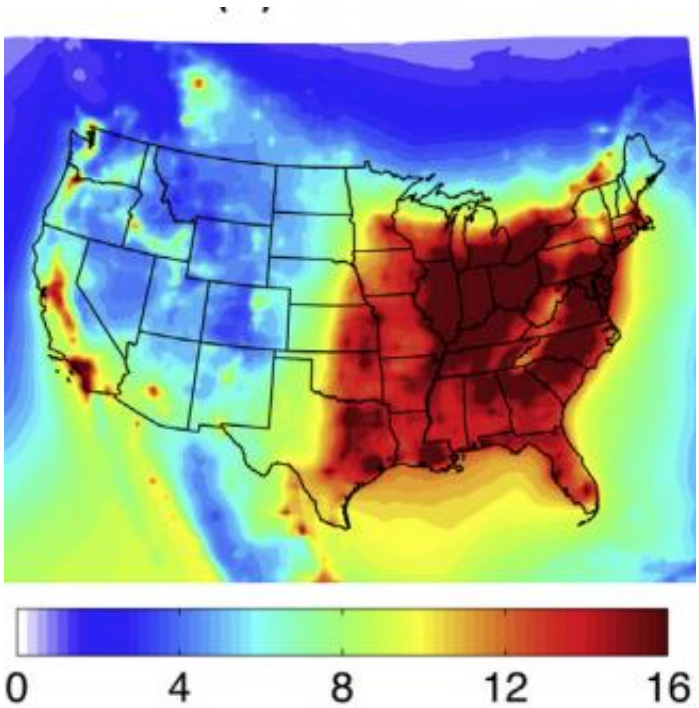


Illustration 18. The map of the America where all the red represents the quality of the air in that area and how harmful it is to show air pollution all across the world from industrial companies used to produce toxic air to enter the atmosphere (Phillips, 2021)

Therefore, we find that the atmosphere of Venus is filled with heavy gases due to the high temperature of the planet, which led to the release of these gases from the crustal rocks into the atmosphere, displacing light gases such as Oxygen and Nitrogen into space, causing gaseous escape. This will also happen with the Earth's atmosphere after a few billion years when the Earth is in the advanced stages of formation. Gas escape is a natural path during the planet's formation ages, which is forcibly affected by the rate of temperature and availability of those gases as a result of geological activities or industrial emissions. The volume of one cubic meter of any gas in space contains a larger amount of the substance of that gas due to the physical inertness of atoms, leading to the activation of the atomic gravitational field, which causes the attraction of their atoms, therefore, we find low atomic weight elements such as Hydrogen and Helium are available in abundance in distant planets such as Uranus and Neptune, despite their moderate gravitational force value and closeness to the value of the Earth's gravitational field, while these two components are absent in the Earth's atmosphere. In the meantime, the volume of one cubic meter of any gas in space around the Earth contains a lower amount of the substance of that gas represented (n) in the ideal gas relationship, due to the relatively high temperature increasing the physical activities of atoms and the expansion of their volume, which leads to the reduction of the effects of their atomic gravitational field due to the increase in the distances among their atoms, therefore the low atomic weight elements, such as Hydrogen and Helium are unavailable in the Earth's atmosphere.

4. The stages of the transfer of triple oxygen from the O_3 phase into dioxygen O_2 or the atomic oxygen O causing decomposition of the ozone layer: The effect of the heavy industrial gases high molecules weight industrial gases such as Carbon dioxide CO_2, Methane (CH_4), Sulfur hexafluoride (SF_6), Nitrogen trifluoride (NF_3), released into the atmosphere on the displacement of low molecular weight gases (O_3, O_2 and N_2) and moving from the Troposphere towards the higher layers of the atmosphere, such as the Stratosphere and the Exosphere. Where Archimedes' principle applies, the volume of the sinking object is equal to the volume of the displaced fluid, but here the volume of the sinking part represents the volume of heavy industrial gases (high molecular weight) released towards the Troposphere and the volume of displaced fluid represents the displacement of low molecular weight gases towards the Stratosphere. This transformation of gases from the Troposphere to the Stratosphere occurs to restore the balance in the Troposphere between the value of the Earth's gravitational force and the thermodynamic effect of gases in the altitude of the Troposphere layer because rushing these great quantities of high molecular weight gases towards the Troposphere leads to an increase in the value of pressure in the pressure container of the Earth's atmosphere, therefore the displacement of low molecular weight gases (O_3, O_2 and N_2) towards the Stratosphere returns the value of the atmospheric pressure in the Troposphere to its original value that is limited by the amount of the Earth's gravitational force at the altitude of Troposphere layer and the thermodynamic effects of the existent gases in that layer. This is in addition to the chemical effects of these industrial gases upon the existent gases in the Troposphere, which causes chemical

reactions due to their higher chemical activity of the industrial gases, as well as the effect of global warming of heating atmospheric gases, raising the temperature of the atmosphere and the expansion of the volume of gases in the Troposphere, causing low molecular weight gases (O_3, O_2 and N_2) upward to the Stratosphere to restore the original value of the atmospheric pressure at the Troposphere layer. Where the physical conditions that lead to the formation of triple Oxygen (Ozone) are absent, this leads to the decomposition of triple Oxygen into binary or atomic Oxygen, thus causing the dissolution of the Ozone layer of the Earth's atmosphere.

5. Volcanic activities: Most of the gases in the Earth's atmosphere are caused by the gaseous escaping from the hot rocks towards the Earth's atmosphere during its formation about 4.543 billion years ago and this process has been continuous until nowadays and it appears the volcanic activities where most of these gases are (Carbon monoxide, Carbon dioxide, Sulfuric monoxide, Sulfuric dioxide, Oxygen, Nitrogen, Argon, and water vapour (USGS, n.d.), where the chemical activity of the emissions led to the decomposition complex components of these emissions such Carbon monoxide, Carbon dioxide, Sulfuric monoxide, Sulfuric dioxide upon the surface of the Earth and the deposition of the heavy elements such as Sulfur and Carbon upon the surface, and the release of Oxygen and Nitrogen into the atmosphere. Volcanic activities where most of the gases in the Earth's atmosphere are caused by the gaseous escaping from the hot rocks towards the Earth's atmosphere during its formation about 4.543 billion years ago and this process is continuous until nowadays and it appears during the volcanic activities where most of these gases are (Carbon monoxide, Carbon dioxide, Sulfuric monoxide, Sulfuric dioxide, Oxygen, Nitrogen, Argon, and water vapour, where the chemical activity of the emissions led to the decomposition complex components of these emissions such Carbon monoxide, Carbon dioxide, Sulfuric monoxide, Sulfuric dioxide upon the surface of the Earth and the deposition of the heavy elements such as Sulfur and Carbon upon the surface, and the release of Oxygen and Nitrogen into the atmosphere. There are about 1,350 potentially active volcanoes worldwide, but we will choose a specific sample in this research, such as the volcano Cumbre Vieja on La Palma Island, Spain (USGS, n.d.), which lasted from 19 September to 13 December 2021 for 85 days, and the most dangerous volcano, Hole of Mount Semeru Balkh volcano, is 3676 km in Indonesia (Smithsonian Institution, National Museum of Natural History, n.d.), as well as the Tonga island blast (National Geographic, n.d.), which is estimated to be a 10 megaton explosion. Through our long observations of several volcano eruptions around the world, we recognized several important notes that could help us to understand the mechanism of how volcanoes erupt, in which the fundamental issue depends upon immense amounts of gases released from the rocks in the materials lying underneath due to exposure to the heat of the magma to form a gap of gases in the tunnels of melted rocks.

The pressure of gases in these gaps increases due to the increasing amount of released gases and also due to the increase in the value of their heat. Therefore, once the pressure of gases in these gaps, gets to tremendous values, it will be able to fail the toughness of the rocky crust that covers volcanoes and to get out

carrying multiple sorts of particles from the magma, the hot rocks, and the cool rocks, which lie in the upper layers in the tunnels of volcanoes. These gaps of gases are not linking with each other directly, where some of them are in the upper crust and some of them are behind the melted rocks in these tunnels, pushing the melted rocks to the upper side after the explosion occurs in the gate of the tunnels via the first group of gaps of gases in the upper side. This is why sometimes we find some volcano eruptions are merely explosions that occur in the crust and release gases; these melted rocks will stay in their places underneath, while sometimes we find other volcano eruptions are explosions that occur in the crust with a pushing out of melted rocks into their places underneath, while sometimes we find other volcano eruptions are explosions that occur in the crust with a pushing out of melted rocks into their gates because of the existence of gaps of gases behind the melted rocks in the materials that lie underneath, pushing these melted rocks due to their high value of pressure. Therefore, the duration of the volcano eruptions determines the resources of volcanic ashes that come out of volcanoes, where as much as the eruption takes a longer time, as much as these ash resources are from deeper layers of the materials lying underneath.



Image 4. Mount Semeru, East Java is the highest volcano in Java Island Indonesia with an altitude 3,676 meters above sea level (Smithsonian Institution, National Museum of Natural History, n.d.)

6. FACTORS THAT LEAD TO THE FORM OF AIRGLOW

There are some explanations for this phenomenon, which state that airglow occurs when the atmospheres of planets, including Earth and Mars, as sunlight interacts with atoms and molecules within the planetary atmospheres, and lights

up as airglow, scientist claim that this phenomenon, There are some explanations for this phenomenon, which state that airglow occurs when the atmospheres of planets, including Earth and Mars, as sunlight interacts with atoms and molecules within the planetary atmospheres. Scientists claim that this phenomenon occurs due to glowing Oxygen when energetic electrons from the Sun hit the upper atmosphere, from excited oxygen atoms radiating light at a particular wavelength of light (557.7 nanometers) green radiation emitted giving polar auroras a characteristic green hue (Royal Belgian Institute for Space Aeronomy, n.d.). The Image 5 shows Airglow in Earth's atmosphere observed from the International Space Station on Earth in the atmosphere of Earth

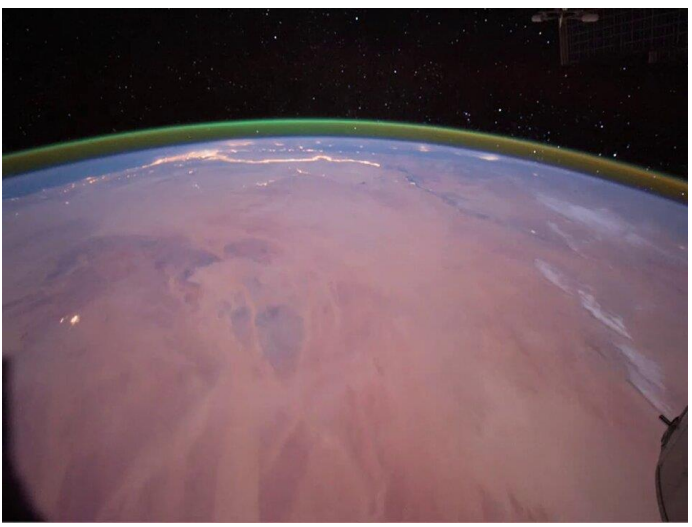


Image 5. This optical phenomenon of airglow was captured with cameras on board the International Space Station (Royal Belgian Institute for Space Aeronomy, n.d.)

7. THE EFFECT OF THE TOTAL INTERNAL REFLECTION TO FORM THE AIRGLOW IN EARTH'S ATMOSPHERE

The airglow known that occurs due to glowing Oxygen is produced during polar auroras when energetic electrons from interplanetary space hit the upper atmosphere. The aurora, however, is just one way in which planetary atmospheres light up. There is another factor that causes this phenomenon, as all conditions are available for the total internal reflection (Axelrod, 2022) due to the difference in density of the atmospheric layers between the Troposphere and Stratosphere, where the sun's rays coming from the horizon collide with the layer between the Troposphere and Stratosphere) and due to the reduction of the density of the Stratosphere than the Troposphere leads to the occurrence of the phenomenon of total internal reflection due to the light reflection of the layer

between (Troposphere and Stratosphere), so leads the area between them to be illuminated to appear as airglow around the Earth. This is why the intensity of the airglow increases during the day and decreases during the night because the source of the light is from the Sun rays coming from beyond the horizon therefore, we could call this the atmosphere lensing and its colour changes in the atmosphere of Earth according to the observation angle due to the differences in the angles of light reflection according to its frequency energy, when we return to the basic concepts of the properties of light, we find that one of these properties is the refraction of light during the transition from the highly dense medium to the least dense medium.

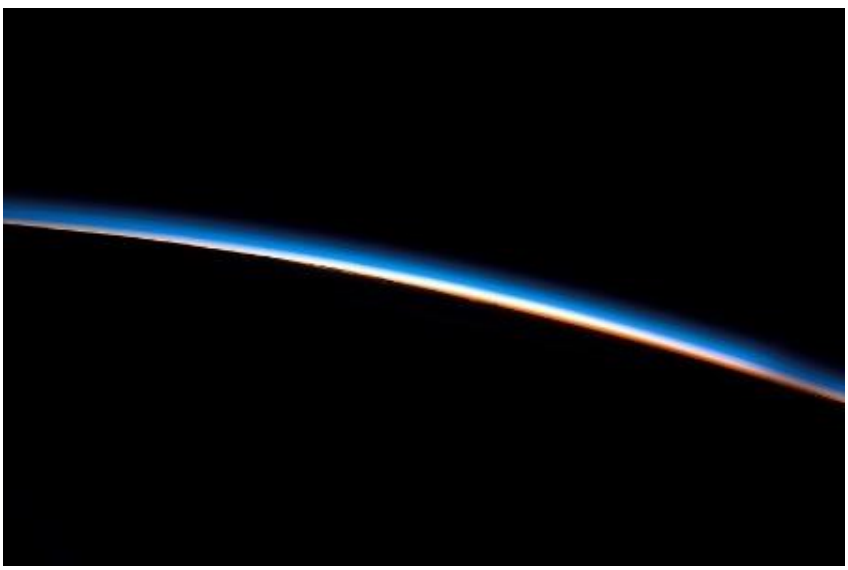


Image 6. The blue airglow on Earth

The effect of the total internal reflection to creates airglow in Earth's atmospheric layers between the Troposphere and Stratosphere:

$$n1.\text{Sin}(\theta 1) = n2\ \text{Sin}(\theta 2) \qquad \ldots\ldots\ldots\ldots\ldots 8$$

Could described as Snell's low

n1= the reflection index of the first medium (Troposphere layer).
n2= the reflection index of the second medium (Stratosphere layer).
$\theta 1$= angel of the incident.
$\theta 2$= angel of the reflection.

The total reflection occurs when the angle of the incident $\theta 1$ is greater than the reflection angle $\theta 2$, where Sin ($\theta 1$) = n1/ n2.



Image 7. The yellow airglow on Earth



Image 8. The green airglow on Earth



Image 9. The distortion of the airglow on the Earth is due to the effect of two factors: first, the collision of cosmic particles with the atmosphere, and second, the effect of light reflection

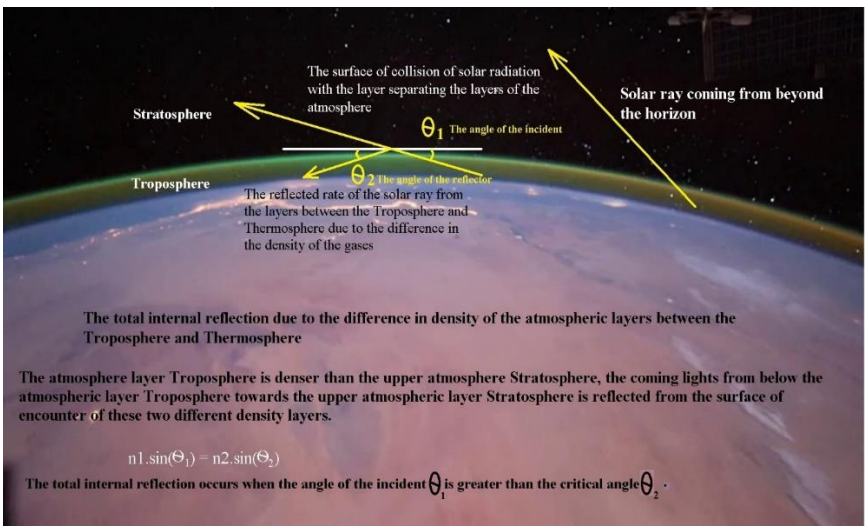


Illustration 19. Shows how the total internal reflection creates airglow in Earth's atmospheric layers between the Troposphere and Stratosphere

Because the atmosphere layer Troposphere is denser than the upper atmosphere Stratosphere, the coming lights from below the atmospheric layer Troposphere towards the upper atmospheric layer Stratosphere are reflected from the surface of encounter of these two different density layers, this is why the airglow of Earth's atmosphere could be visible only at specific altitudes where the reflected rays are at the same angle in the observation direction, In the meantime, the reflected light analyzes according to its frequency energies at different angles. The evidence is that the intensity of the luminosity of the airglow dims at night because it often depends upon the Sun's light, while its luminosity increases during the day due to the presence of high proportions of the Sun's light. Note that this phenomenon occurs clearly when we see the sky from under the water, where we see the surface is such a mirror reflecting the ground image of the swimming pool, but we do not see the sky except when the angle of sight is zero or exactly in a straight line. The atmospheric glow reflects the value of the gravity force of the planet, whereas the value of gravity in the atmosphere is at a higher rate, leading to generating the higher value of the gravity force of the planet because it will attract larger quantities of gases from space and increase the thickness of the atmosphere, generating visible light below the 20 km altitude level to have a good probability of producing airglow. This is why we find the airglow around the Earth in several colors. If it was only in a single color characteristic green hue, it means that it was only due to the effect of solar particles, but we witness this phenomenon happen in multiple colors, which means that the direction angles of the photograph were different so that it would be directed towards the angle of the analyzed frequency of the light after it was reflected because the reflected light analyzes according to its frequency at different angles. Thus, we conclude that the airglow on the Earth occurs due to the effect of two factors: first, energetic electrons hitting the upper atmosphere, and second, the effect of light reflection. This is why the airglow of Earth's atmosphere could be visible only at specific altitudes where the reflected rays are at the same angle in the observation direction.

8. THE RESEMBLANCE OF THE EMERGENCE OF A COMET'S TAIL WITH A PLANETARY GASEOUS ESCAPE PHENOMENON ACCORDING TO THE THERMODYNAMIC CONCEPT OF GASES

The fireball spots during the impact of Comets or Meteors: Fireball spots in Earth's atmosphere during the impact of Comets or Meteors with the highest density of the atmosphere layer (Troposphere) which represents the elevation of the atmosphere glow of Earth, where before their impact with Troposphere they merely burning quietly in the upper layers (Stratosphere) which has a lower density of gases. Often, rocky objects usually burn up between 15 to 20 kilometers above the ground, creating fireballs at that altitude (PW Only IAS, n.d.). Meteors light up almost as soon as they hit Earth's atmosphere when we see a meteor it means look at a piece of dust burning bright about 80 to 120 km in altitude above Earth's surface due to the friction the materials of a meteor with the gases of the atmosphere at the high altitudes, but when this meteor gets closer to the Earth will impact with the higher density gases layers of the atmosphere, the style motion of meteors will be changed from friction to the

collision with the high-density gases of the atmosphere which results from totally crash of meteor which often results from an explosion and some damages on the surface of the Earth due to the expanding gases via effects of the high temperature created during the collision.

Comets: solid body the nucleus is solid body around tens km made of ice, rock, dust, and frozen gases as they fracture and disintegrate leaves a trail millions of km of solid debris and gases.

Asteroids: small rocky iron or ice debris flying in space around 1 to 100 km.

Meteoroid: are small asteroids from 1 micron to 1 meter.

Meteorite: fragments of a meteoroid or an asteroid of a few grams to several dozen that survive passage through the atmosphere and hit the ground.

The gaseous escapes from Comets and asteroids: Looking closely at Image 10 Comet Hyakutake (NASA, n.d.-e) of an asteroid with two tails allows us to understand the reason behind the presence of two tails instead of one tail, as the first tail is caused by the phase transformation of frozen gases on the surface of the asteroid, such as Methane and Ammonia, precipitated on the surface into the gaseous phase “sublimation phenomenon” when the temperature of the asteroid rises as it approaches the Sun, whereas the first tail which consists of gases, has a low momentum as a result its low mass weight of than the relatively high mass weight of sand (silicates) of which the second tail is composed, thus the tail that consists of sand particles are greatly affected by the solar particles bombardment, and be pushes forcedly away more than the gases particles of the first tail, so the second tail takes sharper angle than the first. The effects of extreme reduction of the temperature upon the physics the environment feature of fared planets and Comets due to their a fared distances of the Sun, where Comets and asteroids are completely frozen when they are far from the Sun, and most are just a few kilometers across and appearance changes as it approaches the Sun, when solar heat warms the comet, while gets completely frozen when they are far from the Sun, and most are just a few kilometers across, because the comet's nucleus is the snowball that is essentially the entire comet when it is frozen and far from the Sun, the crust called (coma) is that is relatively large where dusty atmosphere surrounding the nucleus, formed of sublimated gases and dust where the tail consists of gas and dust that can extend hundreds of millions of kilometers away from the coma. Most comets have two tails, the first is a plasma tail made of ionized gas, and the second is a dust tail made of small solid particles, their tails point away from the Sun and expansions of the coma away from the Sun, regardless of the direction in which the comet is traveling, forming two tails is due to escaping gases and dust are influenced by the Sun in slightly different ways, and point in slightly different directions. Gaseous escape from the comet are ionized by the ultraviolet photons from the Sun, the solar wind carries them straight outward away from the Sun, these gases form the plasma tail, where dust size particles that escape from the comet experience a much weaker push from the Sun caused by the pressure of sunlight itself (called

radiation pressure), and the charged particles of the solar wind while the dust tail also points generally away from the Sun has a slight curve back in the direction the comet came from. Most comets remain frozen balls of ice in the outer reaches of our solar system and some of them warm up as they get closer to the Sun and grow the long tails that we characteristically see as they flash through the sky.

Often asteroids move in an elliptical orbit around the sun, having different distances to the Sun, where they get their closest distance to the Sun then begin to move away gradually to very distant regions around Jupiter or Saturn zones, where the zones are rich of gaseous elements due to the effects of their gravitational fields or they may reach farther distances in the final limits of the solar system where temperatures drops to a several hundred degrees under zero, asteroids quickly lose their heat gained from the Sun due to their small mass, and they turn into best the deposition zone for light elements such as Methane, which have a low physics activities due to the extremely low temperatures, thus these light elements starts to deposit upon the surface of comets due to the effects of the force of the gravitational field of the asteroids, forming an ice crust above these asteroids of frozen gases of Hydrogen, Helium and Methane around the asteroids. When the asteroids approach closer distances to the sun during their orbit around the sun, their temperatures begin rise gradually, and the ice crust the frozen gases around the asteroids begins to heat up, and transfer into the gaseous phase via sublimation influences, where due to the small masses of these asteroids, their gravitational field will not be able to confine them within their gravitational field zones. These gases begin to escape from the range of their gravitational fields, forming a tail behind them to appear as comets, and what increases the state of the escape of gases from these asteroids is their speeds acceleration during their rushing in their elliptical orbit towards the Sun, where the momentum of the masses of the materials of gases prevents them of accelerating their speeds with the speed acceleration of the asteroids, this is exactly what happens during the gaseous escape from the atmosphere of the planets but at slower speeds due to the huge values of masses of the planets, which generate high values of gravitational force able to confine gases within their gravitational field zone, similar to the known pressure vessels used in our industrial purposes. But the limits of the pressure vessels of the atmosphere of planets are not confined via the metal cover used in our industrial pressure vessels, but via the gravity field forces of those planets, and gaseous escape of the planet's atmosphere also occurs, but at low rates. We conclude from these results that the gaseous escape from the planetary atmospheres occurs naturally, rebalancing the effects of gravitational field forces of the planet with the atmospheric pressure already generated as a result of the attraction of the gas elements towards the surface maintaining the atmospheric pressure values to their natural limits, resulting from the effect of the values of the gravitational field forces upon the mass of the materials of gases, where the value of the atmospheric pressure gets within the limits of the equilibrium system of the atmosphere to determine within a certain range according to the available energy (heat and wind) in the material of the Earth, in the meantime the rates of atomic energy also has a major role in controlling the rate of atomic spread in the

material of the environment, therefore any increase in the rates of atomic (molecules) energy guides to increase the speed of water molecules in the water surfaces to increase the rates of the atomic diffusion of the water to the air results in confusion in the value of atmospheric pressure in specific regions, where quickly equalized by generating winds and storms and hurricanes among those regions in order lead the energy to stability on the materials of the surface of the planets and the equalize of atmospheric pressure values in the restricted value which eventually resulted via the effects of the gravitational force of the mass planets upon the mass of gases towards the surface of the planets.

Fireball spotting is due to the impact of the comet with the high density of the atmosphere layer Troposphere, which creates the air low where it was merely burning in the upper low-density layers, while rocky objects usually burn up between 15 to 20 kilometers above the ground, fireball occurred at an altitude of 46 kilometers. The fireball advanced in a southwest direction until it extinguished at an altitude of 69 kilometers above sea level. Meteors light up almost as soon as they hit Earth's atmosphere. On average, when you see a meteor, you're looking at a piece of dust burning bright about 50 to 75 miles (80 to 120 km) in altitude above the Earth's surface, generating visible light below the 20 km (12 miles) altitude level to have a good probability of producing a meteorite fall.



Image 10. Comet Hyakutake came as close as 9.3 million miles of Earth in March 1996 (LASP, n.d.)

9. THE INFLUENCES OF THE VARIATION IN THE VALUES OF GRAVITY FORCES AND THERMODYNAMICS PROCESSES OF ATMOSPHERIC GASES UPON THE EQUILIBRIUM STATE AND THE RATES OF GASEOUS ESCAPE OF THE PLANETARY ATMOSPHERES

There are three probabilities of the diversity of the values of gravity forces and thermodynamic processes of the planetary atmospheres:

1. When the effects of the thermodynamic process of the atmospheric gases are greater than the effects of the gravitational acceleration force of the planet (F1 < F2): When the gravitational acceleration force F1 of the planet is at very low and less than the thermodynamic effects F2 of the atmospheric gases, this will certainly collapse the atmosphere of the planet due to increasing the gaseous escape, because the gravitational acceleration force will not be able to confine gases of the atmosphere to the surface, as it occurs on the mass Mercury planet 3.3011×10^{23} kg generate low gravity force around 3.7 m/s² in the meantime temperature rises to very high rates around 430°C which exiting the thermodynamic processes which guides to generate great value of pressure to expand atmospheric gases above the ranges of the gravitational fields, and its gravitational value (3.7 m/s²) is devoid of atmosphere, which means that the rate of gaseous escape is very high due to the weakness of the value of the gravitational field generated by its small mass resulting collapse of the balance of the atmosphere of the planet when the planetary gravitational force is low (due to small mass of planet) and the thermodynamic effect of atmospheric gases is (high temperature), the thermodynamic effect of the atmospheric gases will overcome the effect of the planetary gravitational force, which leads to the generation of high pressure that pushes the atmospheric gases towards the upper layers of the atmosphere.

In group of planets that are close to the Sun (Mercury 430 °C, Venus 475 °C, Earth 14 °C, and Mars -60 °C) (NASA, 2022) the temperatures rise relatively in their environments leading to the expansion of the volume to generate higher rates of pressure increases able to overcome their influence upon the gravitational fields of the planets, this causes the presence of heavier elements such as Oxygen, Nitrogen, Argon and Carbon dioxide, which has high atomic or molecular weight and are affected forcedly by the gravitational force of the planets, where their attraction towards the surface with relatively higher force leads to the displacement of lighter elements (Hydrogen and Helium) towards higher altitudes to be escaped from the atmospheres of the planets (Mercury, Venus, Earth, and Mars). The gaseous escape of Mercury is Sodium tails (Phillips, T., 2009), as shown in Image 12, where the observation of the atmosphere of Mercury indicates the existence of a high concentration of Sodium in the exosphere of planet Mercury. As the solar wind encounters Mercury, it slows down, piles up, and flows around the planet (grey ball). This figure shows the density of protons from the solar wind, as calculated by modeling the planet's magnetic sheath, or magnetosphere. The highest density, indicated by red, is on the side facing the Sun, yellow indicates a lower density, and dark blue is the

lowest. The tiny planet endures endless assaults by intense sunlight, powerful solar wind, and high-speed miniature meteoroids called micrometeoroids. The planet's flimsy covering, the exosphere, nearly blends in with the vacuum of space, making it too thin to offer protection. Because of this, it's tempting to think of Mercury's exosphere as just the battered remains of the ancient atmosphere. The exosphere is constantly changing and being renewed with sodium, potassium, calcium, magnesium, and other species that are liberated from Mercury's soil by barrages of particles. Both these particles and Mercury's surface materials respond to sunlight, the solar wind, Mercury's magnetic sheath (the magnetosphere), and other dynamic forces. The exosphere may not look the same from one observation to the next. Far from being dead, Mercury's exosphere is a place of amazing activity that can tell astronomers a lot about the planet's surface and environment.

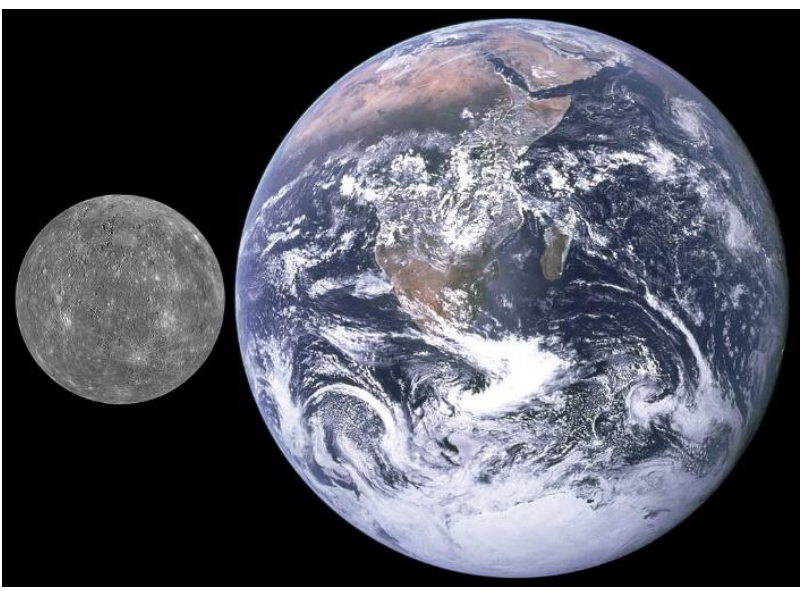


Image 11. The relation of the size of Mercury to the size of the Earth (Wikimedia Commons, n.d.-c)

Image 12 was taken from an animation showing the seasonal effects of sodium in Mercury's exosphere. (For the animation, a simple, uniform source of sodium is assumed.) Going their separate ways shows the large whitish-reddish mass is the sodium in the exosphere around Mercury (dark blue circle). In the upper left corner, Mercury's position (purple ball) in its orbit (light blue circle) around the sun (yellow ball) is shown. (TAA stands for true anomaly angle, the angle between Mercury, the sun, and the closest point to the sun along the orbit, which is known as the perihelion point); as Mercury orbits, the distance between the planet and the sun changes, and so does the speed at which Mercury is moving

relative to the sun. These changes, in turn, affect both the brightness of the sodium emission and the amount of radiation pressure a sodium atom experiences. These factors yield the greatest amount of sodium in the exosphere when Mercury is at a middle distance from the Sun. The atoms and molecules in Earth's atmosphere bounce around and collide all the time, but this doesn't happen much in Mercury's exosphere. Instead, the species tend to follow their paths and are more likely to collide with the planet's surface than with each other. Therefore we may assume that the absence of the atmosphere on the planet Mercury is due to the collapse of the balance circumstance between the gravity force and the thermodynamic effect of atmospheric gases due to the excessive rising in temperatures on the surface of the planet Mercury that led to expanding the volume of gases to great values to rush those gases towards the space overcoming the force of gravity the planet Mercury, due to increasing its temperature extremely around (427 °C) while the gravity force is at a very low limit (3.7 m/s^2) due to its low value of mass (3.285×10^{23} kg), thus the thermodynamic effect of the atmospheric gases overcame the value of the gravitational force of the planet, which finally led of gaseous escape extremely in the first ages of the formation of the planet Mercury.

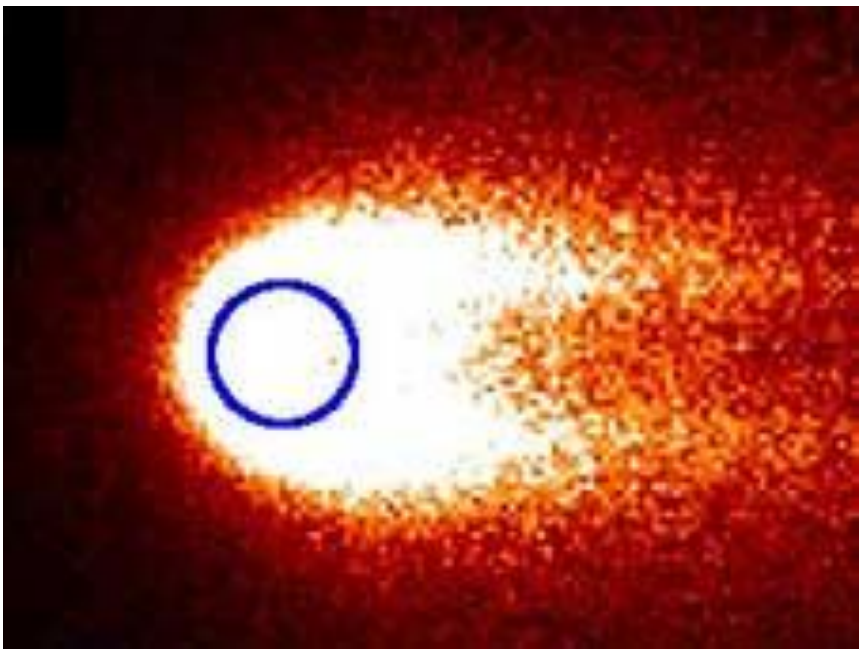


Image 12. The large whitish-reddish mass is the sodium in the exosphere around Mercury (Cassidy et al., n.d.). Image Credit: NASA/GSFC/Mehdi Benna Pity (Editors of EarthSky, 2011)

2. When the effects of the gravitational acceleration force of the planet are greater than the effects of the thermodynamic processes of the atmospheric gases (F1 > F2): If the effects of the gravitational acceleration force F1 of the planet are greater than the effects of the thermodynamic processes of the atmosphere gases F2, the gravitational acceleration force will be able to confine the atmosphere gases in the low altitudes of the low ranges of the planetary gravitational field and will not have any gaseous escape, exactly what occurs on the Jupiter planet, which has a great gravitational acceleration force around 24.79 m/s²). In the meantime, the low temperature is around -110°C, reducing the thermodynamic effects, which lead gases to shrink. This low limit of temperature on Jupiter will guide the atmospheric gases to shrink to liquidize and reduce the pressure due to the thermodynamic processes, therefore, we find that the atmospheric pressure on Jupiter planet is 70 kPa lower than the atmospheric pressure on Earth (at sea level), despite the great gravitational acceleration force that confines gases toward the surface of the planet. The farthest planets, which have very distances from the Sun (Jupiter -140 °C, Uranus -195 °C, Neptune -200 °C, and Pluto -232°C) (NASA, 2022), the elements of their atmosphere elements liquefy or even freeze due to the effect of the reduction of the temperature and separate from the gaseous atmosphere to form liquid or solid layers of these elements on the surface of the planet and stop their influence on creating atmospheric pressure and their role gets cancel in the formation of planetary atmospheric, checking the liquefaction and solidification temperature of the gaseous elements in the periodic table of elements.

We can predict the type of elements that form regions surrounding the planets after knowing the temperatures on their surfaces, it is logical to expect the presence of meteors consisting of solid Nitrogen and Oxygen that strike the surfaces of very distant planets such as Neptune and Pluto, a solid-state phase even when the pressure is zero, and during their collision with the surface of those fared planets, will form layers of ice (solid state) of these elements, although the gravitational force of those planets is low, because at extremely low temperatures, these elements are attracted to each other in distant space due to the effects of the atomic gravity fields of these elements and approach their atoms to each other more when they are at the gaseous phase.

Planets of the first orbits of the solar system (Jupiter -140°C, Uranus -195°C, Neptune -200°C, and Pluto -232°C) get higher percentages of gases to confine them with the composition of their atmospheric gases. What encourages this is the extreme decrease in temperatures in the planets in the first orbits of the solar system, which leads to a significant reduction in the volume of these gases and an increase in their density, causing a decrease in their influence on the gravitational fields of the planets. Planet Jupiter has a mass (1.898×10^{27} kg) and an amount of gravity (24.79 m/s²), which is 2.4 times that of Earth. The planetary gravitational force is very high (the masses of the planet are large), and the thermodynamic effect of atmospheric gases is low (the temperature is low). This leads to a decrease in the value of atmospheric pressure for the following reasons:

A. Gases shrink due to the low temperature, which leads to a decrease in the value of pressure generated due to thermodynamic effects.
B. The pressure is distributed over a wide area, which leads to a decrease in the value of the generating force, as in the relationship:

Pressure = Force / Area ……9

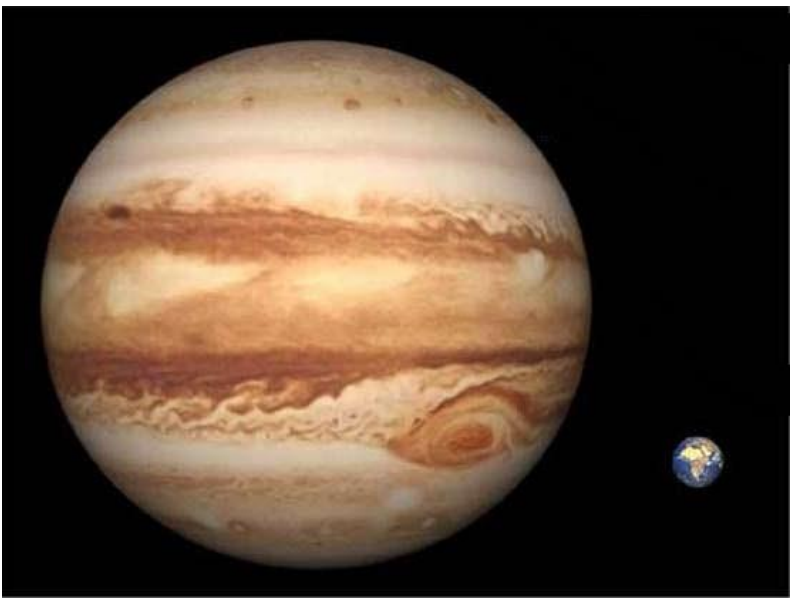


Image 13. The relation of the size of Jupiter to size Earth (Lea, 2022)

Therefore the gaseous escape is non-existent in the planet Jupiter's atmosphere, and the evidence is the presence of the elements Hydrogen and Helium at a very dense rate in the atmosphere, confined due to the influence of the massive gravitational force, where these elements, Hydrogen and Helium does not exist in the atmosphere of the planets Mercury, Venus, Earth, and Mars, which have lower values of mass and gravity than Jupiter.

Therefore in Jupiter`s atmosphere, the effect of the gravitational force of the planet is greater than the effect of thermodynamic gases, especially the temperature of the surface of Jupiter within the limits of (-110°C) where the physical activities of the atoms decrease extremely, leading to an increase in their density, volume and pressure, occupying small volumes being more affected by the influence of the gravitational force of its gravitational field of Jupiter, because the thermodynamic processes related to the phase transformations of gases from the liquid phase to the gas phase are closely related to the atomic diffusion, supported by the atomic dynamic forces of the properties of the elements in the conditions of those transformations, so the

atomic diffusion occurs with extreme violence. When the effects of the gravity force of the planet F1 are greater than the effects of the thermodynamic processes of the atmosphere gases F2, the gravitational force will be able to confine the atmosphere gases in the low altitudes of the ranges of the planetary gravity field and will never get any losses of gases to the space, preventing the gaseous escape exactly what occurs on the Jupiter planet, which has a great gravity force around 24.79 m/s^2, while the temperature drops around -110 °C preventing the thermodynamic processes that lead gases to expand, this low limit of temperature in Jupiter's planet will guide the atmospheric gases to shrink and even to liquidize and reduce the generated pressure due to the thermodynamic processes therefore, we find that the atmospheric pressure in the Jupiter's planet is 70 kPa, slightly less than atmospheric pressure on Earth's sea level, despite the great gravity force, which confines gases toward the surface of the planet forcibly.

Table 10. Data comparison between Jupiter and Earth

		Jupiter	**Earth**
1.	Mass	1.898×10^{27}kg	5.973×10^{24} kg
2.	Gravity	24.79 m/s^2	9.807 m/s^2
3.	Temperature	-110 °C	13.9 °C
4.	atmospheric pressure	70 kPa	101.325 kPa

Note that Jupiter has mass (1.898×10^{27} kg) and gravitational acceleration (24.79 m/s^2) while the value of atmospheric pressure is (70 kPa) less than the atmospheric pressure on the planet Earth about (101.325 kPa) dropping temperatures on the surface of Jupiter to the limits (-110 °C) of liquefaction of Oxygen the gas of critical temperature (-90°C) and Nitrogen (-147°C) leads to the formation of liquid surfaces of these two elements upon the surface of Jupiter, exactly like the water surface on Earth, which is considered the best solvent area for gases in the atmosphere of Jupiter (Hydrogen and Helium) it plays an important role in dissolving these two elements (Hydrogen and Helium) of the atmosphere in those liquid surfaces, which causes the decrease in atmospheric pressure of Jupiter. These temperatures are within the limits of the temperatures of liquefaction, freezing, and evaporation of Oxygen and Nitrogen gases, therefore, it is likely that these swirling shapes in the atmosphere of Jupiter are the result of the explosions accompanying the processes of phase transformation among solid, liquid, and gas phases of Oxygen and Nitrogen, as occurs in our laboratories when observing these phase transitions of these two gases because these temperature rates on Jupiter are within the limits of the temperatures of freezing, liquefaction, and evaporation of Oxygen and Nitrogen gases, therefore the influence of the planetary gravitational force overcomes the thermodynamic effects of the elements of the atmosphere, leading to preventing gaseous escape from the atmosphere into space.

Note that -110 °C is the average temperature on Jupiter, as 14°C is the average temperature on Earth, therefore we should expect many regions on Jupiter with extremely lower than -110 °C, where the minimum temperature in Jupiter is

minus -160 °C, in the meantime, gravity force of Jupiter which is greater than gravity force on Earth 2.4 times able to confine Hydrogen 1.00784 u and Helium atomic weight 4.002602 u (where these two elements are the highest average of Jupiter atmosphere composition) will be able to attract Oxygen molecule weight 15.999 u and Nitrogen molecule weight 14.0067 u, 16 times harder than the attraction force to Hydrogen and 4 times harder than the attraction force to Helium to the surface, thus we could expect of formation of pools of liquid Oxygen and Nitrogen on many regions on the surface of Jupiter. This is why these two elements are absent from the atmospheric composition of Jupiter, where it's mostly Hydrogen and Helium, and visible clouds of Ammonia.

When the planetary gravitational force is high (due to its huge mass), and the thermodynamic effect of atmospheric gases is very low (temperature is very low), the effect of the gravitational force will overcome the thermodynamic effect, preventing the gaseous escape. In the meantime, dropping the temperature extremely will lead to the freezing of some elements and prevent the chances of the formation of a liquid surface for these gases on the surface, preventing the solubility of gases on the surface of the planet, which leads to the sharp increase in atmospheric pressure as it happens on Neptune, preventing the dissolution of atmospheric gases in any liquid surface, which leads to the sharp increase in atmospheric pressure. The planet Neptune, which has a mass of (1.024×10^{26} kg), a gravitational value 11.15 m/s² and a temperature rate of around (-200°C) that's means around (-180°C to -240°C). Note that Oxygen freezing temperature of (-218.8 °C) and freezing temperature of Nitrogen (-210 °C), therefore it's not possible to find these two elements as liquid phase on the surface of planet Neptune, this absences of the liquid surfaces and the absence of the opportunity of dissolve atmospheric gases (Hydrogen and Helium) to in those liquid surfaces, leads to increase the value of the atmospheric pressure to tremendous values to reach around 10 GPa or about 100,000 times of Earth's atmosphere, this is why Neptune atmosphere is made up mostly of Hydrogen and Helium with just a little bit of methane, note that the location of the Neptune is at the second orbit around the Sun, grant it the biggest shares of the attracted gases from the space during rushing the solar system in the space in the orbit around the center of Milky Way galaxy to increase the variable (n) the amount of substance in the equation PV=nRT which has proportional relationship with the value of generated pressure, P is pressure, V is volume, n is amount of substance, R is ideal gas constant and T is temperature.

Table 11. Data comparison between Neptune and Earth

		Neptune	**Earth**
1.	Mass	1.024×10^{26}kg	5.973×10^{24} kg
2.	Gravity	11.15 m/s²	9.807 m/s²
3.	Temperature	-200 °C	13.9 °C
4.	atmospheric pressure	10 GPa or about 100,000 times of Earth's atmosphere	101.325 kPa

The reason behind the extremely high atmospheric pressure of the planet Neptune, which is equal to 10 GPa, is due to the presence of a large amount of elements in its atmosphere, which causes a double effect:

A. It increases the thermodynamic effect in the atmosphere and represents the amount of substance (n) in the ideal gas equation: PV=nRT.
B. Increases the effect of the gravity field where it represents (m2) Newton's law of universal gravitation: F= G. (m1.m2)/ d^2.

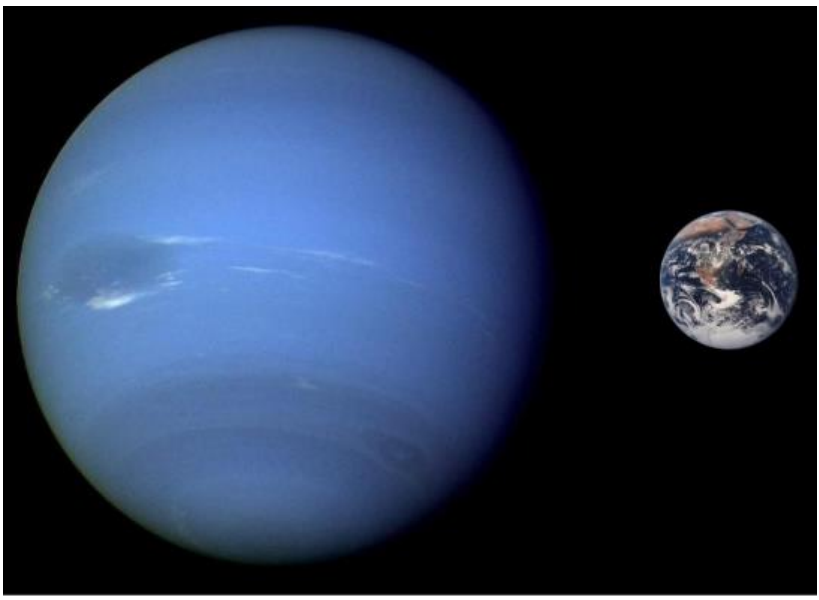


Image 14. The relation of the size of Neptune to size Earth

The reason for the existence of a huge amount of elements in the atmosphere of the planet Neptune is because of the ability of the value of the planet's gravitational pull, which is equal to 11.15 m/s² to attract those elements that are in a low thermodynamic activity due to the decrease in entropy in extremely low temperature. The second reason is that the planet Neptune's location occupies the second orbit of the planets of the solar system. Where elements coming from space pass through the gravity field of the planet Neptune before passing through the gravity field of the rest of the planets. This gives a great opportunity to the planet Neptune to attract a great amount of these elements and higher than the shares of attracted elements of the rest of the planets, where the high gravitational force of the Sun in the low altitudes leads to attracting these gases to the Sun before they are attracted to the planets (Mercury 46.098 million km, Venus 108.18 million km, Earth is 152.09 million km and Mars is 208.22 million km), note that these elements are just gaseous clouds in space that collide with

the gravitational field of the solar system during the traveling within its trajectory around the galactic center of the Milky Way galaxy at an average speed of 230 km/s, as shown in Illustration 2. But this does not happen with the planet Pluto, which occupies the first orbit of the solar system, due to the meager value of the gravitational force of the planet Pluto (0.62 m/s^2), which prevents it from attracting a large quantity of these gases from space to its gravitational range, leading to the value of the atmospheric pressure of the planet Pluto remaining low, equal to 10 μbar.

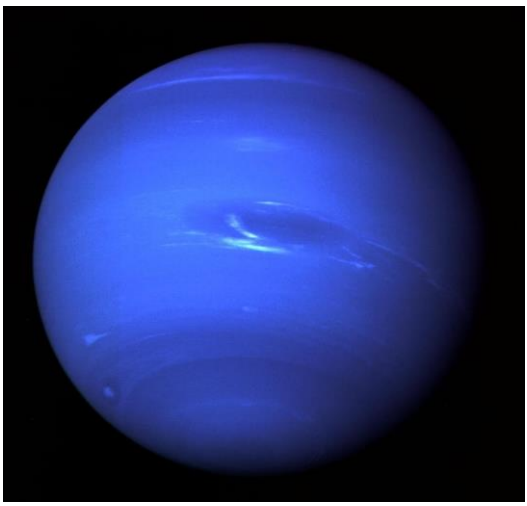


Image 15. Neptune planet was taken by Voyager 2 less than five days before the probe's closest approach to the planet on Aug. 25, 1989. The picture shows what's called the Great Dark Spot — a storm in Neptune's atmosphere and the bright, light blue smudge of clouds that accompanies the storm. (NASA/JPL-Caltech) (Greicius, 2019)

10. THE PRESENCE OF ELEMENTS HYDROGEN AND HELIUM IN THE ATMOSPHERE OF PLANETS WHICH ARE VERY FAR DISTANCES FROM THE SUN, SUCH AS NEPTUNE

It's possible to predict the presence of the elements Hydrogen and Helium in the atmosphere of planets that are far from the sun, such as Neptune, because the low thermodynamic influences of gases lead them to condense to the liquid phase due to the extremely low rates of temperature, creating highly intense masses of these two gases in the space, this allows the attractive force of planets to confine them in the limes of the planetary gravitational field, and these conditions do not exist in the circumstances of rest of the planets, which are close distances to the Sun, which have relatively high temperatures rates, which leads to increase these gases to expand to at high rates and granting them its extremely low density, which makes it difficult for the planetary gravitational

forces to have any influence upon them, as we find in the Earth's atmosphere the Hydrogen and Helium ascending up (in balloons) toward very high altitudes and escaping to the outer limits of the atmosphere, where the thermodynamic effects increases and leads them to expand extremely, due to the relatively high rates of temperature and reducing their density, preventing the planetary gravitational force from doing their gravitational effects upon the materials of these gases and causing the failure to confine them within the limits of the gravitational field of these relatively hot planets.

Note that this thermodynamic effect of gases at extremely low temperatures occurs in the dense gas clouds of the early stages of star formation, where the clouds begin to shrink as a result of the effect of the atomic gravitational field of the Hydrogen element, which is available abundantly in those clouds, leading it to shrink, forming the first nuclei of the formed stars. These Hydrogen clouds are in cold, deep space within the limits of absolute zero, where the atoms are at their lowest kinetic activity, so the force of the atomic gravitational field of the Hydrogen overcomes the force of the propulsion force of the collisions among the atoms, this guides the Hydrogen clouds to contract. But these conditions are not available in the atmosphere circumstances of the Earth nor the rest and close planets to the Sun, where temperatures rise to several handed above absolute zero. For what leads to an increase in the kinetic activity of Hydrogen atoms and an increase in their atomic collisions, which leads to an increase in the volume of the atmospheric gases at high rates of temperature, what makes the problem more complex is the presence of heavy elements (high mass weight) in the planetary atmosphere such as Oxygen, Nitrogen, and Argon, which are affected harder by the force of the planetary gravitational field due to their relatively high mass weights. These relatively high mass weights gases (Oxygen, Nitrogen, and Argon) descend towards the surface of the planet with a greater force and leads to the displacement of light gases (relatively high mass weights) such as Hydrogen and Helium towards the upper zones of the atmosphere, thus it's possible to apply the Archimedes rule "the volume of the submerged part of the object is equal to the volume of the floating part of the object". Here, the submerged body represents the relatively high mass weight gases in the atmosphere (such as Oxygen, Nitrogen, and Argon), and the volume of the floating body is the light gases (relatively low mass weights) such as Hydrogen and Helium, and this causes the displacement of the light gases towards the upper zones of the atmosphere and occupying very high zones of the atmosphere (where the planetary gravity force is lower), especially if the proportions of the heavy gases such as (Oxygen, Nitrogen, and Argon) are at high rates as it is in the atmosphere of the Earth, which eventually causes the escaping of the light gases relatively low mass weights such as (Hydrogen and Helium) from the atmosphere towards the space.

11. THE COMPARISON BETWEEN THE PLANETARY ATMOSPHERE CIRCUMSTANCES OF THE EARTH AND PLUTO

When the planetary gravitational force is very low 0.62 m/s² (due to its very small mass of 1.303×10^{22} kg) and the thermodynamic effect of atmospheric gases is

also very low (due to the extreme reduction of the temperature to -232 °C), the effects of the gravitational force F1 and the thermodynamic effect F2 work on decreasing its atmospheric pressure, which leads to generating very low atmospheric pressure (about 1 Pa or 10 μbar) about 1/100,000 of Earth's atmospheric pressure.

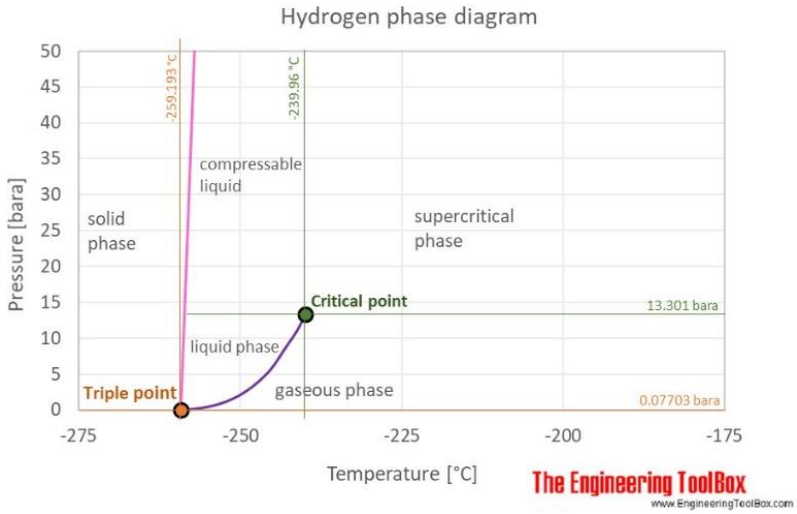


Illustration 20. The Hydrogen phase diagram (Engineering ToolBox, n.d.-e)

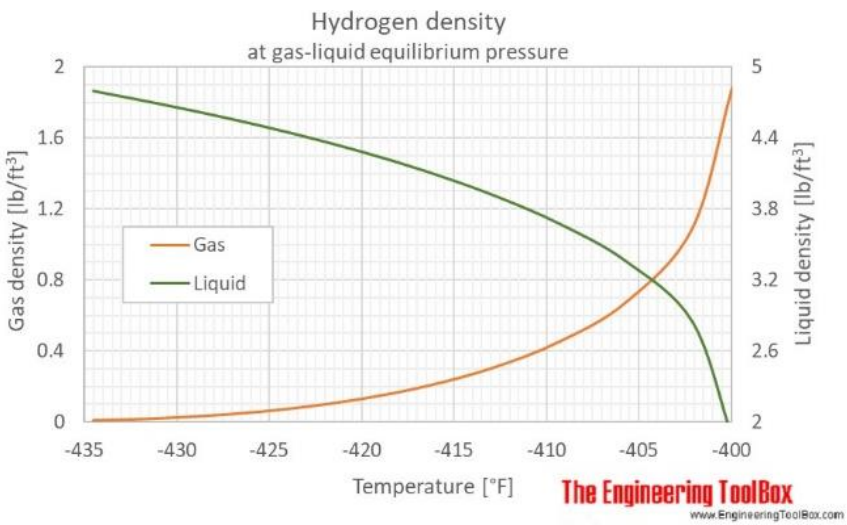


Illustration 21. The Hydrogen - Density and Specific Weight vs Temperature and Pressure (Engineering ToolBox, n.d.-f)

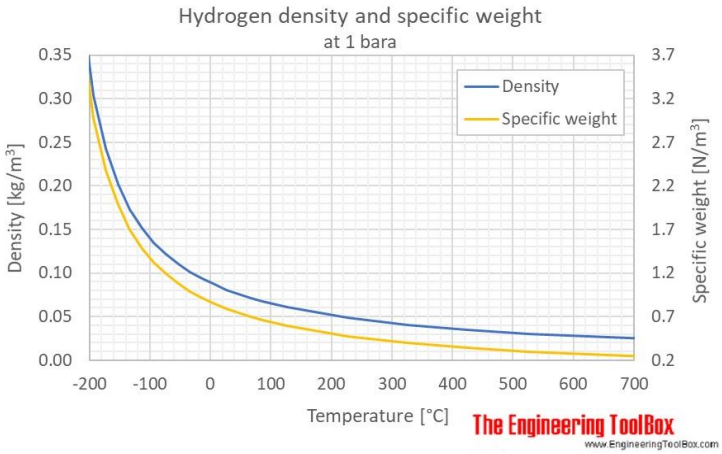


Illustration 22. The Hydrogen - Density and Specific Weight vs. Temperature and Pressure (Engineering Tool Box) (Engineering ToolBox, n.d.-g)



Image 16. The relation of the size of Pluto to size Earth (Talbert, 2015)

Table 12. Data comparison between Pluto and Earth

		Pluto	Earth
1.	Mass	1.303×10^{22} kg	5.973×10^{24} kg
2.	Gravity	0.62 m/s²	9.807 m/s²
3.	Temperature	-232 °C	13.9 °C
4.	atmospheric pressure	10 µbar, roughly 1/100,000 of Earth's atmospheric pressure.	101.325 kPa



Image 17. NASA's New Horizons looked back after passing Pluto to photograph the haze surrounding the dwarf planet. The blue tint is thought to come from the scattering of sunlight by fine soot-like particles that form in Pluto's atmosphere (New York Times, 2015)

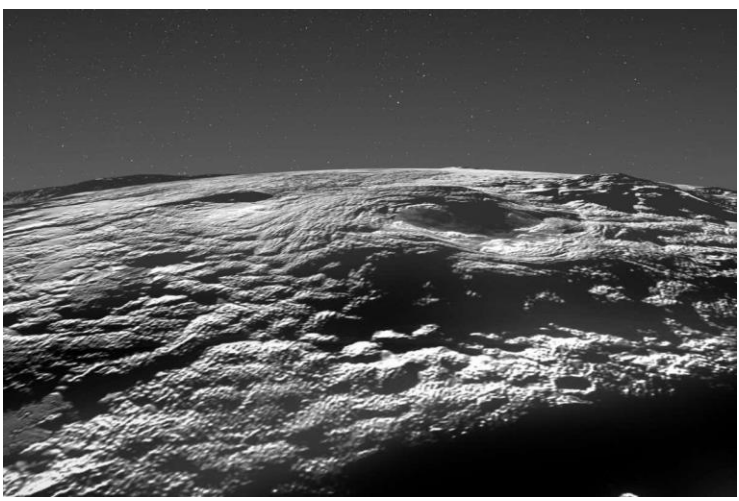


Image 18. This image, released by NASA / Johns Hopkins University Applied Physics Laboratory / Southwest Research Institute, shows a perspective view of Pluto's icy volcanic region, March 29, 2022. (AFP Photo / NASA / Johns Hopkins University Applied Physics Laboratory / Southwest Research Institute / Isaac Herrera / Kelsi Singer) (Daily Sabah, 2022)

The freezing temperature of Oxygen at −218.79 °C and Nitrogen at -210 °C causes of forming icebergs on the surface of planet Pluto at -232 °C, therefore the low value of gravity of the planet Pluto at 0.62 m/s² cannot attract Hydrogen and Helium gases to confine them in the range of their gravity fields, what leads to the gaseous escapes of Hydrogen and Helium reach high rates towards space. Pluto has huge ice volcanoes that may still be active to this day. The comprehensive analysis of data from NASA's New Horizons spacecraft, which flew past Pluto in 2015, has revealed that a large area of its surface, at least 180,000 square kilometers, is made up of ice that seeped out from underground via volcanism relatively recently. Note that velocity distribution data for gas of rubidium atoms confirms the existence of the new phase of matter, the Bose-Einstein condensate, (ScienceDirect, n.d.) in the esteem low temperature with new physic characteristics such as the superconductivity effect or Meissner effect, and this may represent one of the fundamental features of material in the environment of deep space.

3. When the effects of the gravitational force of the planet are equal to the effects of the thermodynamic process of the atmosphere gases (F1 = F2): When the effects of the gravitational force of the planet F1 are equal to the effects of the thermodynamic process of the planetary atmosphere gases F2, the gravitational acceleration force will be able to confine the atmosphere gases of the planetary gravitational field strongly where the thermodynamic effects at low rates of gaseous escape as it occurs on the Earth and Venus which the gravitational acceleration force of Earth is 9.807 m/s² while the temperature rate reaches 14 °C, and the gravitational acceleration force of Venus is 8.87 m/s² while the temperature rate reaches 475 °C, this grants the opportunity to the thermodynamic processes to create high rates of pressure due to the expansion of atmospheric gases in the high gravity force range to be confined in their gravity fields. The tail of Venus is evidence of the gaseous escape from the atmosphere. When the solar wind nearly breaks off Venus, the ionosphere layer of Venus expands into space, and the sheath of electrons and ions enveloping Venus at a height of 150 to 300 kilometers can expand into space like a tail. This exceptional deformation occurs on the planet's night side, when the solar wind, the flow of charged particles from the Sun, nearly comes to a stop, data obtained by instruments on board ESA's spacecraft Venus Express. Venus is surrounded by an ionosphere, a sheath consisting of electrons and ions, this is referred to as plasma, where the charged particles are created on the planets' day sides when ultraviolet light with extremely short wavelengths and X-rays enter the outer layers of the atmosphere, on Earth, the particles are trapped by the strong magnetic field forcing them to rotate together with the Earth. The electrons and ions in the ionosphere flow from the day side to the night side by the high plasma pressure on the day side, such as compressed gas escaping from a pressure vessel and the plasma travels from the region with high pressure to one with lower pressure. The new measurements prove that the night-side ionosphere protrudes approximately 15000 kilometers into space, where the tail could be much longer. It might measure up to millions of kilometers.

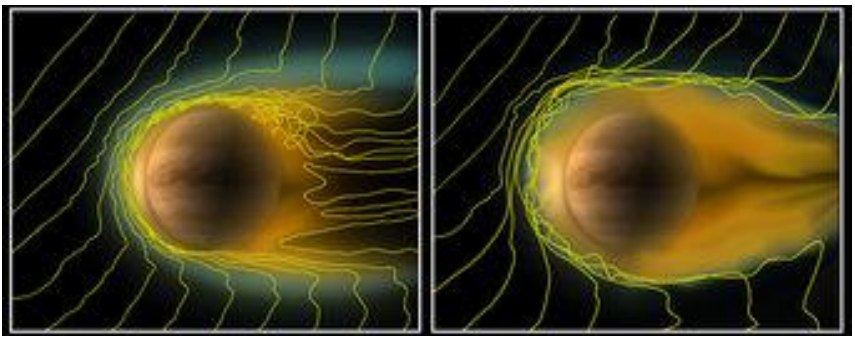


Image 19. The change in ionosphere of Venus during normal solar wind conditions (left) and reduced solar wind activity (right), as observed by ESA's Venus Express spacecraft in August 2010. The yellow lines show a projection of the solar magnetic field lines as they interact with the ionosphere. Venus Express follows an elliptical orbit around the planet once every 24 hours, passing within 250 km of the North Pole and 66,000 km over the South Pole. The observations were made on the night side of the planet, when Venus Express was within 15,000 km of the centre of the planet (SciTechDaily, n.d.)

12. THE EFFECT OF MASS WEIGHT OF THE ELEMENTS OF GASES OF THE ATMOSPHERE

Elements that have a low mass weight in the periodic table, such as (Hydrogen and Helium) are less affected by the force of the planet's gravity field therefore, they occupy regions at high altitudes in the atmosphere and therefore are more candidates for escape from the atmosphere, as well as the rest of the elements occupy regions in the atmosphere according to their mass weight. Concerning the Earth`s atmosphere, the mass number of Nitrogen is 14 and the mass number of Oxygen is 16 therefore, we find Nitrogen gas covering the higher layers of Earth's atmosphere and above the Oxygen gas therefore, the escape ability of Nitrogen to the space is at a higher value than the value of the escape ability of Oxygen from Earth's atmosphere to the space, if the atmosphere loses a big ratio of gases for a long term next several million years, that could not change the value of atmosphere pressure, because any reduction in the value of the atmosphere pressure of the Earth will have the effect of reducing the boiling temperature of the water in Earth's oceans and water surfaces to evaporate and transfer to steam, substituting the position of the lost gases. Also, reducing atmosphere pressure will urge the rest of the dissolved gases either in the oceans or even in the soft ground to be released into the atmosphere. In the meantime, the ultraviolet rays of the sun rays able to break water molecules in the atmosphere into Hydrogen and Oxygen which may also substitute the value of lost Oxygen in the atmosphere. Therefore, when we see the violet colour is above the green colour in the aurora, because the violet colour is the effect of the collision of cosmic particles with Oxygen which occupies higher altitudes, while

the green colour is the effect of its collision with Nitrogen, which occupies lower altitudes in the atmosphere.

This is why (Hydrogen and Helium) represent the main shares of the atmospheric gases of (Jupiter -140 °C, Uranus -195 °C, Neptune -200 °C, and Pluto -232 °C), due to the low temperature that reduces the thermodynamic effects to be shrank to extremely small volumes and reduce their generated pressure, causes of their attraction towards the surface with a relatively higher forces, leads to the displacement of light elements (Hydrogen and Helium) towards the higher altitudes therefore and do not form a proportion of the components of the atmospheric gases of the planets (Mercury 430 °C, Venus 475 °C, Earth 14 °C, and Mars -60 °C). We find that the density of this element is very high at low temperatures, and its density is very low at average temperatures among the planets close to Sun. The Illustration 23. Shows the thermodynamic effects, mass of the planet and entropy of atoms upon the balance circumstances of planetary atmospheres, the relationship between the temperature and pressure according to the universal ideal gases theory of planets of solar system when the planetary gravitational force is high (large mass) and the thermodynamic effect of atmospheric gases is very low (temperature is very low), the effect of the gravitational force will overcome the thermodynamic effect, preventing the gaseous escape, dropping the temperature extremely will lead to the freezing of light elements, this prevent the chances of the formation of liquid surface for these gases on the surface, and preventing the solubility of atmospheric gases in any liquid surface, which leads to the sharp increase in atmospheric pressure, as happens on the planet Neptune, which has a mass of (1.024×10^{26} kg) and a gravitational value (11.15 m/s² and its temperature rate around (-200 °C) that's mean around (-180°C to -240°C), note that the freezing temperature of Oxygen (-218.8 °C) and Nitrogen (-210 °C) therefore it's not possible to find these two elements in the liquid phase on the surface of planet Neptune.

This absence of liquid surfaces and the absence of the opportunity to dissolve atmospheric gases in those liquid surfaces leads to an increase in the value of the atmospheric pressure to tremendous values to reach around 10 GPa or about 100,000 times Earth's atmosphere, note that the location of the Neptune is at the second orbit around the Sun, grant it the biggest shares of the attracted gases from the space during rushing the solar system in the space in the orbit around the Center of Milky Way galaxy to increase the variable (n) the amount of substance in the equation PV=nRT which has proportional relationship with the value of generated pressure, P is pressure, V is volume, n is amount of substance, R is ideal gas constant, T is temperature.

Dropping temperature extremely decreases the kinetic activity of atoms and molecules of gas reduces the distance among their atoms, activating the effect of the atomic gravity fields forming masses of gases that are attracted by the gravity force of the planet to form a thick atmosphere as the planet Neptune knows the atmospheric pressure (10 GPa) despite the decrease in the thermodynamic activities of gases at low temperatures (-200 °C) due to the effect of the gravity force of planet Neptune to confine those gas masses toward the surface forcedly.

Tablet 13. General data upon planets of solar system distance from Sun, temperature, and atmosphere composition of Pluto, atmosphere pressure, mass of the planet, gravitational acceleration and orbital speed. [From different sources on Wikipedia]

	Planet	Distance from Sun	Temp. °C	Composition	Pressure	Mass	Gravity	Speed
1.	Mercury	46.098 million km	-180, 430 Celsius	Oxygen, Sodium and Hydrogen	1 nPa	3.285×10^{23} kg	3.7 m/s²	47 km/ sec
2.	Venus	108.18 million km	-41, 464 Celsius	Carbon dioxide, and Sulfuric acid.	9100 kPa	4.867×10^{24} kg	8.87 m/s²	10.36 km/sec
3.	Earth	152.09 million km	-89, 56.7 Celsius	78% Nitrogen, 21% Oxygen.	101.325 kPa	5.973×10^{24} kg	9.807 m/s²	30 km/sec
4.	Mars	208.22 million km	-153, 20 Celsius	Carbon Dioxide 95.3%, Nitrogen	0.60 kPa	6.39×10^{23} kg	3.721 m/s²	24.07 km/*sec*
5.	Jupiter	742.46 million km	-110, -160 Celsius	Hydrogen, Helium and Methane	70 kPa	1.898×10^{27} kg	24.79 m/s²	13.07 km/*sec*
6.	Saturn	1.4754 billion km	-140, 11 Celsius	Hydrogen, Helium, Methane & Ammonia.	140 kPa	5.683×10^{26} kg	10.44 m/s²	9.69 km/*sec*
7.	Uranus	2.9455 billion km	-195, -200 Celsius	Hydrogen83%, Helium15% and Methane 2%	130 *kPa*	8.681×10^{25} kg	8.87 m/s²	6.80 *km*/*sec*
8.	Neptune	4.4742 billion km	-200, -225 Celsius	Hydrogen 80% and Helium 18.5%	10 GPa	1.024×10^{26} kg	11.15 m/s²	5.43 *km*/*sec*
9.	Pluto	4.5 to 7.3 billion km	-233, -223 Celsius	Nitrogen, Methane and Carbon Monoxide	10 μbar	1.303×10^{22} kg	0.62 m/s²	4.67 *km*/*sec*



Image 20. This image is a composite of observations made of Saturn in early 2018 in the optical and of the auroras on Saturn's North Pole region, made in 2017 (ESA Hubble, n.d.)

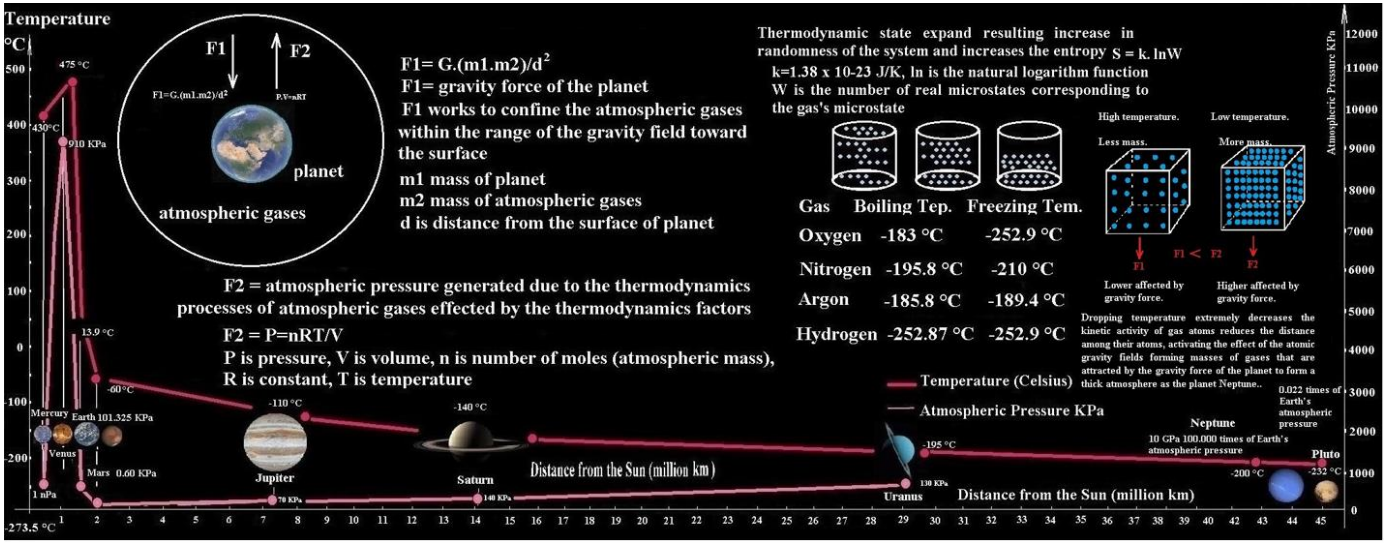


Illustration 23. The effect of temperature, mass of the planet and entropy of atoms upon the balance circumstances of planetary atmospheres

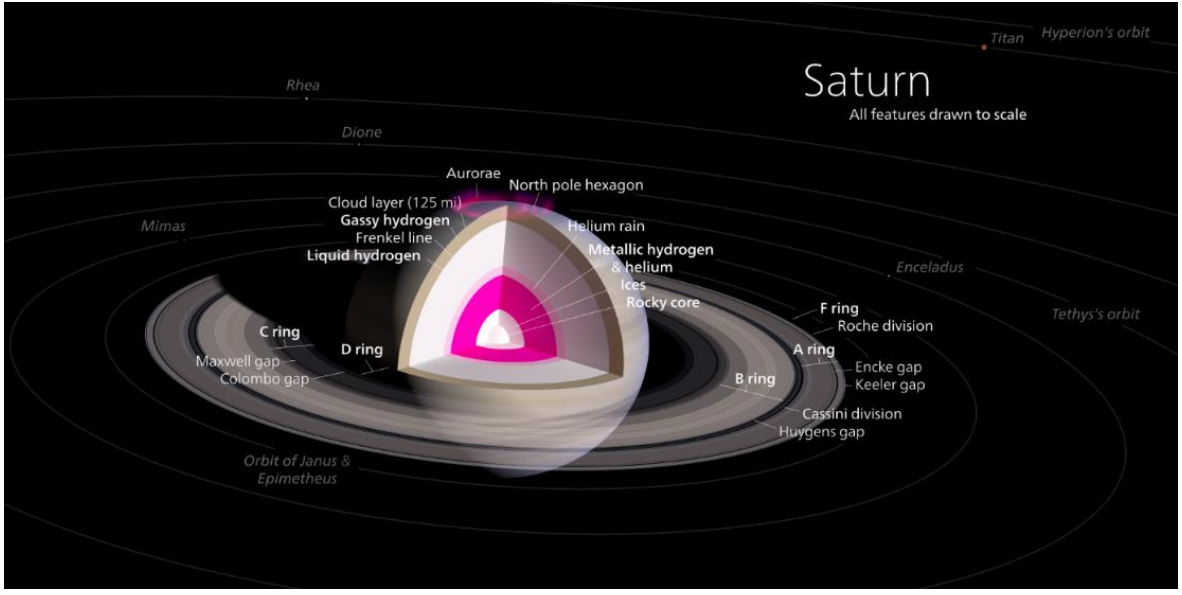


Illustration 24. The atmospheric layers of Saturn (Williams, 2015)

For a conventional substance (inset), there exists a triple point (TP), where solid, liquid, and gas coexist. Helium lacks such a point. Where Bose-Einstein condensation is predicted to occur for an ideal gas, because the thermodynamic effects of gases decrease with the increase in planetary distances from the Sun, due to the decrease in the received heat from the Sun, gives the planetary gravitational force the possibility to overcome the effects of thermodynamics to confine atmospheric gases to narrower limits in the planetary gravitational fields and then lead these gases to liquefy and freeze due to the reduction of atomic entropy of atmospheric gases, to be out of the forces acting in the atmosphere of the planet. Where in the fared planets from the Sun (Jupiter 742.46 million km, Saturn 1.4754 billion km, Uranus 2.9455 billion km, Neptune 4.4742 billion km, and Pluto 7.3 billion km), the reduction of temperature leads to a shrinking volume of atmospheric gases, reducing the thermodynamic effect upon the value of atmospheric pressure on the surface of the planet, as long as these gases are in the gaseous state, where the effects of the gravity force of the planet keep their impact upon the masses of atmospheric gases to confine them in the range of the gravity field of the planet, and when the temperature drops to extremely low rates and results phase transformation of some elements such as Oxygen −183 °C and Nitrogen -195.8°C from the gas phase to the liquid phase, leads to form swamps of liquidized gases of Oxygen and Nitrogen on the surface of the fared planets where atmospheric pressure drops due to the reduction of the thermodynamic effects of many gases from the total thermodynamic effects of the rest of atmospheric the elements such Hydrogen and Helium may be found in the gaseous phase in the atmospheres of these cold planets have a low liquefaction temperature, such as Oxygen −183 °C and Nitrogen -195.8 °C remain in the gaseous state in the atmosphere within the limits of the extremely low temperature of the planet Jupiter around -110 °C, where its high gravity force of 24.79 m/s² can attract and confine them within the range of its gravitational field. Note that the liquefaction temperatures of Oxygen −183 °C and Nitrogen -195.8 °C in the standard laboratory conditions or at Earth`s circumstances (temperature 20 °C, pressure 1 atm, 101.325 kPa), where their values change according to the surrounding conditions. The huge size of Saturn (at around 10 times the diameter of Earth), while Saturn's gravity is only 1.08 times the gravity of Earth, because the solid part of Saturn is nearly the same mass as Earth, but the extremely low temperature causes condensed gases of the atmosphere to compose thick clouds, resulting in a huge size of Saturn due to the thermodynamic laws due to effects of temperature upon the material characteristics of gases.

Increasing the amount of gases in the atmosphere causes increasing atmospheric pressure because of its effect upon both relationships $F= G.(m1.m2)/d^2$ in the meantime in the relationship PV=nRT because, in the first relationship, it represents m2. In contrast, the second relationship represents the amount of substance n, this is also evident in the extreme increase in the atmospheric pressure of the planet Neptune (10 GPa and about 100,000 times that of Earth's atmosphere), as a result of the increase in the amount of gases in the atmosphere, which increases the effect of the gravity force to confine the downward at the range of the gravity field, it increases its effect on the

thermodynamic factor, which leads to an increase in pressure as a result of the increase in the amount of gases. The thermodynamic action of gases at high rates of temperature guides to raise the value of pressure due to the expansion of gases pushing gases in the opposite directions of the gravity force and this force can overcome the value of the planetary gravity force therefore, the different values of the gravitational forces of the planets, which stem from the difference in the values of their masses and the differences in their temperature rates, which result from the differences in their distances from the Sun, cause the different conditions of the equilibrium state in the atmosphere of each planet, which leads to the formation of different environmental conditions of the values of atmospheric pressure and creating specific alien environment conditions for each planet. Therefore raising the temperature leads to an increase in the volume of gases, which leads to the rise of the region of gases occupied in the atmosphere to higher altitudes, where the value of the planet's gravity decreases on the mass of these gases, so the possibility of the gaseous escape increases. Therefore, a drop in temperature leads to atomic vibration; the randomness of molecules also decreases, and this leads to a decrease in entropy. As mentioned in the 3rd law of thermodynamics states, the value of entropy of a purely crystalline substance is zero at absolute zero temperature. If we have a pure crystalline solid substance and its temperature is absolute zero, then its entropy will be zero. Whereas molecules move very fast throughout the container or the atmosphere, it has more randomness, which means they have more entropy. Note that the entropy of the universe keeps on increasing for all the processes occurring on their own, absolute zero is so much cold temperature that all the atoms and molecules of a substance get frozen up; the atoms and molecules of any substance will not be able to move, vibrate, or oscillate inside the substance.

Therefore, the physical characteristics of (Hydrogen and Helium) get changed according to their temperature and their distance from the Sun, therefore, they have different physical characteristics on each planet of the solar system:

1. The elements and compounds that form the atmosphere of Venus (Carbon dioxide and Sulfuric acid) have a high molecular weight that is more affected by the gravitational force of the planet.
2. The amount of gases that form the atmosphere is large which represents (n) in the equation ideal gas relationship PV = nRT where P is pressure, V is volume, n is the amount of substance, R is the universal gas constant, and T is temperature. Therefore, the sharp rise in the temperatures of the atmosphere of Venus led to an extreme increase in the atmospheric pressure, as the high molecular weight prevented those gases from rising to high altitude as a result of their influence by the gravitational force of Venus, which confined these hot gases within their gravitational zone at high pressure.

13. SOLUBILITY OF GASES AT EXTREMELY LOW TEMPERATURE

Dropping temperatures lead to increasing in the solubility of gases in liquids, which leads to decreased proportions of these gases in the atmosphere in their gaseous state due to their soluble in the existing liquids on the surface either the

water surfaces in the liquid state or liquidized gases due to the extremely low temperatures, this causes the melting of other gases which have liquefaction temperature is higher than the liquefaction temperature of the solvent elements, to obtain a mixture of liquids for the gaseous elements at low rates temperatures, and this is seen in Saturn (gravitational acceleration force 10.44 m/s^2 temperature -140°C) and Jupiter (gravitational acceleration force 24.79 m/s^2, temperature -110°C), their high gravitational acceleration forces and the low temperature rates lead to the formation of surfaces of liquid elements that are similar to the water surfaces on Earth but consist of relatively light elements such as Oxygen and Nitrogen, it also causes the presence of Hydrogen gas (Atomic mass 1.00784 u) in abundance confined in the high gravitational field of the Jupiter planet at relatively low altitudes from the surface. As for the fared planets of the solar system such as Uranus (gravitational acceleration force 8.87 m/s^2, temperature -195°C), Neptune (gravitational acceleration force 11.15 m/s^2, temperature -200°C), and Pluto (gravitational acceleration force 0.62 m/s^2, temperature -232°C), we find that these elements are in the solid state and represent thick ice on the surface and to be part of the crust materials as a result of the extreme drop in temperatures, due to the effects of the thermodynamic factor of the gases in those physical conditions which changes the physical properties of those elements and finally leave their effects completely from the rest of effects of the rest of gases in the planet's atmospheric pressure. Therefore, the gas giant planets have circumstances that prevent gas from escaping, such as huge amounts of masses and high gravity force, as well as extremely low temperatures that drop due to the extremely long distances of their stars. This leads to the reduction in the volume of gases, which guides the formation of a dense atmosphere of cold gases at very high altitudes from the solid surface of the planet. Although the rates of the temperature of gases in the upper layers are low, the rates of expansion of the gases in the first layer are at high rates and push the expanding gases forcedly toward the upper layers.

14. THE FACTOR OF CHEMICAL ACTIVITY OF ELEMENYS:

In the first group of planets of the solar system that have close distances to the Sun (Mercury, Venus, Earth, and Mars), the temperatures rise to the limits of the chemical activity of materials, because this heat provides the electrons of the atoms with the energy needed to jump among the orbits of the atom, which urges the chemical activation of the substance. For this reason, we find that most of the elements on Earth (metals and non-metals) are in the form of oxides and chemical compounds only chemically inert elements such as gold can be found free (pure) and the elements of the atmosphere are in continuous chemical interaction with Earth's substance in the formation of Nitrogen compounds that move from the atmosphere to Earth's substance, which are absorbed by plants or the respiration process of creatures, which leads to the transfer of atmospheric Oxygen to the biological building of creatures and many other examples, this transition of elements from the gaseous state of the atmosphere to other phases in the surface leads to their exit from the atmosphere zone, then the absence of their effects upon the amount of atmospheric pressure, but in the planets that are fared distances from the Sun (Jupiter, Uranus, Neptune, and Pluto), where their

temperatures drop at extreme rates, leading to inactivity of the chemical properties of the elements of the atmosphere and the elements on the surface of the planet, the chemical reactions do not occur, so these elements remain present in the atmosphere, and this is also one of the reasons for the extremely high atmospheric pressure on Neptune, which has a temperature rate (-200 °C) and the atmospheric pressure (10 GPa) or about 100,000 times of Earth's atmosphere.

15. THE EFFECT OF THE AMOUNT OF HEAT GAINED AND HEAT LOST (HEAT CONTENT) OF THE SURFACE OF THE PLANET UPON THE STABILITY OF THE ATMOSPHERE

The thermal content of the planet has an effect upon the duration while the surfaces of the planet are at the limits of temperature that obtained from the solar radiation, to activate the thermodynamic processes of gases leading to gaseous expansion and increase pressure (temporarily) that causes the movements of air masses to lead winds and storms, and even gases expansion that leads to the rush of gases from Thermosphere layer towards Troposphere, Stratosphere, Mesosphere and finally causes the gas escape to the space, therefore as much as mass of the planet are greater will lead to increase in the heat content and prolonging the period of the while the surface is hot the effect of thermodynamic processes will increase and vice versa when the masses of the planet are smaller, that will lead to shortening the period while the surface is hot thus the effect of thermodynamic processes will be reduced upon the atmospheric gaseous, the amount of energy required to change the state of a sample of matter depends on three things depends upon what the substance is, how much substance is undergoing the state change, and the phase change that is occurring. The mathematics related to the heat gain and heat loss could be calculated via the following equation:

$$Q = m.C.\Delta T \qquad \ldots\ldots\ldots\ldots 10$$

Q = the quantity of heat transferred to or from the object, m is the mass of the object, C is the specific heat capacity of the material the object is composed of ΔT is the resulting temperature change of the object. Note that during the heat gain and heat loss on the surface of the planet, many thermodynamic processes occur with the sorts of elements on the surface of the planet and the atmosphere. In the case of melting, boiling, and sublimation, the energy would have to be added to the sample of matter to cause the change of phase. Such phase changes are referred to as being endothermic, where the freezing, condensation, and deposition are exothermic, energy is released by the sample of matter when these changes in the phase transformations occur.

These mentioned processes represent the main energy resources of the thermodynamic processes in the atmosphere, in the meantime the spinning speed of the planet has a role in the time period of heat gain and heat loss, because the increase in spinning speed has an effect upon creating turbulences in the air masses guides an atmosphere of instability supplies gases to the atmosphere the kinetic energy that activates the thermodynamic processes such

the atomic diffusion which accelerate the phase transfers from the liquid phase to the gas phase at lower limits of temperature than the boiling temperature of the liquids exactly what happens in Earth's nature where all the water transfers from the liquid phase to the gas phase in lower limits of temperature the boiling temperature of the water (100 °C) or even the vice versa, in the phase transformation from the gaseous phase to the liquid phase as a result of the collision of air currents loaded with clouds with heights mountainous, which gives an opportunity for vapor particles to collide with each other to form larger heavy grains that cannot resist gravity of the Earth so they begin to descend down as rain falls, the same these examples occur on all planets, but with another compounds and elements have diverse phase transition temperatures and their activities interact with the conditions of the environment of those planets in the same concept of the known thermodynamic laws or gases.

The effect of the heat content (heat gain) factor on the thermodynamic processes of atmospheric gases leads to gaseous expansion and pressure generation that causes the movement of winds and hurricanes, as well as the duration of the age of the geological activities of planets. The thermal content of the planet which come of heat gain obtained from solar radiation has an effect upon the length of time for the surface of the planet to remains in the high rates temperature, therefore, the thermodynamic processes lead to the gaseous expansion and rise the pressure that causes the movement of air masses to neutralization the energy generates winds and hurricanes, and even expansion that leads to the rush of gases towards the top causing the gaseous escape to the space, because increase the value of mass of the planet leads to an increase in the heat content prolonging the heating period of the surface of the planet and prolonging the period of the thermodynamic processes for longer time and vice versa, when the mass of the planet is lower, that leads shorten the period when the surface of the plane is hot and shorten the period of the thermodynamic processes for shorter limits of the gases of the atmosphere which leading to gaseous expansion and rise the pressure, the movement of winds and hurricanes as well as the expansion of gases towards the higher altitudes causing the gaseous escape from the atmosphere towards the space, note that the effect of thermal content has also an effect upon the length of the age of the geological activities of the planet, because increase in the mass of the planet increase the value of the thermal content which already came during the formation of the planet leading planet's core to remain hot for longer time and prolonging the period of geological activities of the planet's age.

16. DISCUSSION

The balance circumstances of the planetary atmosphere depend upon the value of the gravity force of the planet and the thermodynamic effects of elements, which mainly depend upon the value of temperatures of the planet and physic properties of elements, where the thermodynamic laws and effects cause the expansion and contraction of air masses in the planetary atmosphere, leading to the movement of air masses to restore the balance circumstances between the gravity and the thermodynamic effects when any temporary changes occur in the

value of the temperature, generating winds and storms in the range of the balance circumstances between the these two mentioned forces, for example: The elements Oxygen and Nitrogen are in the range of the balance circumstances of the planet Earth, due to the value of the moderate value of gravity force and temperature, where carbon dioxide is for the same mentioned circumstances for the planet Venus and carbon dioxide and Argos for planet Mars and element Sodium for the planet Mercury, which already sufferers of the collapse of the balance circumstances in its atmosphere due to the wide differences between the low gravity force of the planet and the high thermodynamic effects of gases due to the extremely high temperature of Mercury that comes from its close distance from the Sun. There is a reversal relationship between the mass of the planet and the weight number of elements of the atmosphere, where the low and middle masses planets generate moderate values of gravity force, especially the have moderate distance from the Sun (Mercury, Venus, Earth and Mars) that have relatively high temperature, are unable to attract low weight elements such as Hydrogen and Helium in the limits of their gravity fields, but we find these two elements in the great mass of planets such Jupiter which have great gravity force which is able to confine them in the limits of its gravity field, in the meantime the fared planets from the Sun such Uranus, Neptune and Pluto where their temperature reduces extremely the thermodynamics effects of gases reduces extremely in their atmospheres and results contraction of the volume of gases extremely, descending the value of generated pressure to be liquidized in the extremely low temperature of the planet, to be confined in their middle value for gravity force, forming clouds in their atmosphere.

The effect of the inertia phenomenon where the momentum of masses of the atmospheric gases resists changes in the speed motion of Earth during the acceleration period, causing the phenomenon of escaping gaseous from the atmosphere when the speed of the Earth starts to increase from lowest value (perigee) in Earths the farthest distance from the Sun, where after that point Earths speed begins to be accelerated gradually to reach the highest value in (apogee) when the distance is the closest distance to the Sun, the mass of gases in the highest altitudes prevents them from being accelerated with the same acceleration rates of the Earth, a big quantity of compressible gases losses in the space from the upper layers (Mesosphere and Thermosphere) due to the acceleration of Earth in its elliptical. We can speculate the shape of the aerobic bubble of the Earth's atmosphere layers, the thickness of Earth's atmosphere is the shortest distance possible on the front side of the Earth's the direction in orbit, and increases gradually towards the back of the of rushing direction where the longest distance is on the back side of the direction in orbit around the Sun.

When the effect of the thermodynamic process of the atmospheric gases is greater than the effect of the gravitational acceleration force of the planet (F1 < F2): When the gravitational acceleration force of the planet is at very low rates and less than the effects of the thermodynamic process of the gases of the atmosphere this will certainly collapse the atmosphere of the planet due to increasing the gaseous escape, because the gravitational acceleration force will

not be able to confine gases of the atmosphere to the surface, as it occurs on the mass Mercury planet 3.3011×10^{23} kg generate low gravity force around 3.7 m/s² in the meantime temperature rises to very high rates around 430°C which exiting the thermodynamic processes which guides to generate great value of pressure to expand atmospheric gases above the ranges of the gravitational fields, and its gravitational value (3.7 m/s²) is devoid of atmosphere, means that the rate of gaseous escape is very high due to the weakness of the value of gravitational field generated by its small mass results collapse the balance of the atmosphere of the planet.

When the effect of the gravitational acceleration force of the planet is greater than the effects of the thermodynamic processes of the atmospheres gases (F1 > F2): If the impact of the gravitational acceleration force of the planet is greater than the effects of the thermodynamic processes of the atmospheres gases, the gravitational acceleration force will be able to confine the atmospheres gases in the low altitudes of the ranges of the planetary gravitational field and will never get any losses of gases to the space preventing the gaseous escape exactly what occurs on the Jupiter planet which has a great gravitational acceleration force around 24.79 m/s²), while the temperature drops around -110 °C which preventing the thermodynamic processes which leads gases to expand, and this low limit of temperature in the Jupiter planet will guides the atmospheric gases such as Oxygen and Nitrogen their liquidation temperature are as following Oxygen -183 °C, Nitrogen -195 °C to shrink and even to liquidize under its immense value of gravity force and reduces the generated pressure due to the thermodynamic processes, therefore we find that the atmospheric pressure in the Jupiter planet is 70 kPa slightly less than the average atmospheric pressure on Earth at sea level, despite of the great gravitational acceleration force which confine gases toward the surface of the planet.

When the effect of the gravitational force of the planet is equal to the impact of the thermodynamic process of the atmosphere gases (F1= F2): If the effects of the gravitational force of the planet are equal to the effects of the thermodynamic process of the gases of the atmosphere, the gravitational acceleration force will be able to confine the gases of the atmosphere in several layers and altitudes in the ranges of the planetary gravitational field and that will guides to partially gaseous escape as it occurs on the Earth which has middle gravitational acceleration force around 9.807 m/s² while the temperature rises around 13.9 °C which granting the opportunity to the thermodynamic processes to expand atmospheric gases regularly during the season revolution and due to the global warming. Where the planets of the first orbits of the solar system get a large percentage of gases to confine them with the composition of their atmospheric gases, what encourages this is the extreme decrease in temperatures in the planets in the first orbits of the solar system, leading to a significant reduction in the volume of these gases and an increase in their density, which leads to their influence in the gravitational field to overcome upon the thermodynamic effects. The balanced circumstances of the planetary atmosphere permit certain elements in the range of their gravity due to them

according to their gravitational force and the value of the dynamic forces generates the expansion of gases as a result of the sharp rise in temperature, in the state of failure of this balance that leads a gaseous escape to space, other sorts of elements get to have existed in their atmosphere, that have greater mass weights because they are more affected by the influence of the gravitational force, creates a new system in the planetary atmosphere has different balance circumstances, such changes in the value of the generated dynamic forces of the gases (due to changes in temperature or an increase in emitted gases) as occurs in the Earth's atmosphere due to global warming and an increase in industrial gases emitted into the atmosphere. This leads to the expansion of the volume of the atmosphere of the first layer (Troposphere), leading to the spread of atmospheric elements to the second and third layers (Stratosphere and Mesosphere) gradually leading to the gaseous escape and the collapse of the Ozone layer.

There is another factor that causes the airglow, because all conditions are available for the total internal reflection due to the differences in density of the atmospheric layers between the Troposphere and Stratosphere, where Sun's rays coming from the horizon collide with the layer between the Troposphere and Stratosphere) and due to the lower density of the Stratosphere than the Troposphere leads to the occurrence of the phenomenon of total internal reflection, what causes the light to reflect of the layer separating (Troposphere and Stratosphere), so the area between them is illuminated to be shown as airglow around the Earth, we could call this the atmosphere lensing and its colour changes in the atmosphere of Earth according to the observation angle due to the differences in the angles of light reflection according to its frequency energy, the basic concept of the properties of light is the refraction of light during the transition from the high dense medium to the least dense medium. This is why the airglow of Earth's atmosphere could be visible only at specific altitudes where the reflected rays are at the same angle in the observation direction. Thus we conclude that the airglow on the Earth occurs due to the effect of two factors: first, energetic electrons hitting the upper atmosphere, and second, the effect of light reflection. This is why the airglow of Earth's atmosphere could be visible only at specific altitudes where the reflected rays are at the same angle in the observation direction. Because if it was only in a single color characteristic green hue, if it was only due to the effect of solar particles, there should only be in one color, but we witness this phenomenon happen in multiple colors due to the different direction angles of the observation being different so that it would be directed towards the angle of the analyzed frequency of the light after it was reflected because the reflected light analyzes according to its frequency at different angles.

17. CONCLUSION

Gaseous escape is a natural phenomenon in most planets, moons, and asteroids which have moderate amount of mass, where their gravitational force is at a medium rates and unable to overcome the effects of the thermodynamic processes of atmospheric gases, it is a phenomenon associated with the history of the formation of the planets related to the gases which got released from the

materials of the planets during their formation especially during the hot decades of ancient ages of the formation the planet, gaseous escape phenomenon continues as long as the factors causing it exist, such as the atomic diffusion, temperature and high rates of atmospheric pressure as well as the instability of the value of the velocity of the planet in the elliptical orbit, where it could be considers to estimate the age of the formation of the planet via evaluating the thickness of the atmospheric layers and the value of the atmospheric pressure to determine the amount of gases which got escaped in during their ancient ages of the history of the formation of the planet.

Factors that result in the gaseous escape from the Earth's atmosphere into space are:

- **A.** The factor of the increase in the rates of radiation and solar energetic particles when the Earth approaches the Sun, causes friction of these particles with the gases and pushes them towards space, and increasing the speed of the Earth's orbit around the Sun leads to the lag of quantities of gases from the upper layers (Exosphere, Thermosphere, and Mesosphere) in space behind the direction of the Earth's launch due to the mass and momentum of the gases that prevent them from accelerating their speed with the acceleration of the Earth, this leads to an imbalance in the amount of pressure in (Exosphere, Thermosphere and Mesosphere), which activates the process of atomic diffusion from (Troposphere and Stratosphere) to the (Exosphere, Thermosphere and Mesosphere) to restore the amount of atmospheric pressure at those heights.
- **B.** Terrestrial factors such as the release of industrial and volcanic gases of high molecular weight into the atmosphere disturb the state of the natural balance between the gravity force and the amount of atmospheric pressure, which leads to moving gases of less weight, such as Oxygen and Nitrogen towards the higher layers to return the natural value of atmospheric pressure in the lower layers (Troposphere and Stratosphere).
- **C.** The factor of global warming that leads to the expansion of gases in the lower layers (Troposphere and Stratosphere) to create pressure leads to a rush of relatively lower-weight gases towards the higher layers upwards which causes the collapse of the Ozone layer. Global warming has obvious effects on increasing the rates of gaseous escape from Earth's atmosphere, and this can be clarified by applying the ideal gas relationship in calculating the rates of the expansion of Oxygen and Nitrogen gases within the increasing temperature limits to calculate the increase the expansion rates of their volume, where during the expansion of the aerobic bubble of the lower layers of Earth's atmosphere, will create pressure upon the gases in the upper layers, which will push them to higher altitudes, reaching a range of low values of the Earth's gravitational force, and escape gradually into space in the phenomenon of gaseous escape from the atmosphere, although the upper layers have lower temperatures, the expansion that occurs in the lower layers will exert direct mechanical pressure on the gases of the upper layers, this expansion in the volume of gases of the atmosphere can be calculated in

the hottest regions in the world such as the Sahara desert in African, that represents the hottest area in the world, the atmospheric air that passes from these regions is exposed to extreme heating and expansion. This also happens in other areas of the world, but at different rates according to their rates of temperatures and the type of material on the surface, whether it is dry or water because this affects the values of these materials' thermal content (heat gain) and the extent of their impact by solar radiation, we can predict the outer shape of the Earth's atmosphere by estimating the momentum generated for each layer, where it should be an elliptical shape, with decreasing radius of the elliptical shape in the front side of the direction of the Earth's rushing in its orbit around the Sun, and increasing in the opposite side the forming a long tail of the light gases which are the most likely to escape and exactly like comets long tail, the number of meteors that hit the earth is about 17,000 a year, their components are added to the components of Earth`s atmosphere.

D. The effect of the type of orbit: When the speed of planets increases, their velocity in their elliptical orbit around their stars. Will urge gaseous escape from their atmospheres, and for this reason, the age of the planet and the decrease in the values of its masses as well as the value of their gravity forces, play a major role in determining the thickness of the atmosphere of these planets. We also find the decay of the atmosphere of Mars which has mass 6.39×10^{23} kg and gravitational values 3.721 m/s², and that is 0.375 that on Earth, its atmosphere is thin layer and about 0.6% of the pressure of the atmosphere Earth =101.3 kPa and consists of 95% by volume of carbon dioxide CO_2, 2.6% molecular nitrogen N_2, 1.9% argon Ar, 0.16% molecular Oxygen O_2, and 0.06% carbon monoxide CO, where the other elements that are available abundance in the Earth's atmosphere were escaped into the space since the first eras of the formation of the planet Mars, this is because the mass of the Earth is 5.973×10^{24} kg and relatively larger and its gravity 9.807 m/s² can attract relatively light gases such as (Oxygen and Nitrogen) therefore these two important elements are available abundance in the Earth's atmosphere, but the factors that we mentioned for the gases escape are also available in Earth physics circumstances, therefore this will also happens with the Earth's atmosphere, but over the next billions of years. After the losses quantities of gases in the upper layers (Mesosphere and Thermosphere) due to the acceleration of Earth in its elliptical or the collision of the energetic solar particle with Earth's atmosphere, this leads to an imbalance in the state of gaseous equilibrium between the upper and lower layers, which guides to spread atoms and molecules from high density to low density zones causes the activation of atomic diffusion from the lower layers (Troposphere, Stratosphere) towards the higher layers (Mesosphere and Thermosphere) to restore the state of equilibrium in the upper layers (mesosphere, thermosphere) which guides to spread atoms and molecules from high density zone the lower layer (Troposphere, Stratosphere) to low density zones the higher layers (Mesosphere and Thermosphere), note that the loss of gases from the upper layers due to (Mesosphere and Thermosphere) occurs in every location of Earth in its

orbit around the Sun, but the peak of its activity in the phases the point of starting the acceleration and the shortest distance to the Sun. The circular orbit around the star gives a stable speed for planet rushing and grants to the stability of the atmosphere, but if the orbit is elliptical, the planet's velocity changes, whereas the planet's speed is less value when the planet is at the farthest distance from the star which revolves around, after that point, the planet speed begins to accelerate gradually with constant acceleration value, and the planet's speed reaches the highest value when the distance is as closest to the star that revolves around it, this speed changes confuses the stability state of the gases in the atmosphere of the planet, especially at the point where the planet begins speed to accelerate, where the mass of gases generates momentum that obstruct the planets acceleration, this leads to the failure of a quantity of atmospheric gases to catch up with the planet and remain behind in the orbit, causing the phenomenon of escaping gaseous from the atmosphere.

E. The effects of industrial gases and volcanic emissions upon the balance of Earth's atmosphere: When huge quantities of relatively heavy molecular weight industrial gases and volcanic emissions (such as carbon dioxide molecular weight 44.01 g/mol and sulfuric dioxide molecular weight 64.066 g/mol) are emitted and are forcefully affected by the gravity force due to their relatively higher mass to the atmosphere. This will lead to an increase in the amount of matter of gases in the atmosphere, which represents (n) in the relationship of ideal gases relationship PV=nRT which has a direct relationship with pressure and the volume (PV), this will lead to rise value of air pressure and volume in the Troposphere more than its original value which restricted according to the value of the gravitational force to the masses of gases and the thermodynamic effects of the atmospheric gases in the equilibrium state of the mentioned forces, this will lead the lighter gases molecular weight such as Oxygen molecular weight of 15,999 u and Nitrogen molecular weight of 14.0067 u which are less affected by the force of gravity due to their lower mass weight (lower mass) to be rushed to the (Stratosphere, Mesosphere), which causes damage to the Ozone layer, which is formed from triple atoms of Oxygen, due to rises the triple molecular Oxygen to high altitudes in which the atmospheric pressure is less than the physical conditions for the formation of triple Oxygen (O_3) to form Ozone gas despite its exposure to the ultraviolet rays of the sun, this is the main physics factor for the results gap in the Ozone layer in the Earth's atmosphere, we conclude that the process of gaseous escape from the Earth's atmosphere is the result of the expansion of the gases of the Thermosphere and their rushing towards the upper layers to Troposphere, and pushing the gases of the upper layers to escape to Stratosphere, Mesosphere and towards space, of the triple Oxygen molecules (molecular weight 48 g/mol) in the first layers to the higher altitudes where devoid the conditions of forming the triple Ozone molecule, so the triple Oxygen molecule decomposes into the binary molecule or the singular atoms as it found at high altitudes Stratosphere, Mesosphere leaving a gap in the Ozone layer of the atmosphere of the Earth in addition to the

chemical factor of the chemical reactions of the active industrial gases and even due to collision the energetic solar particle with Earth's atmosphere. These industrial gases with large molecular weight displace light gases such as Oxygen and Nitrogen from the Troposphere to the Stratosphere (to restore the balance stat between the gravitational force and thermodynamic effect of gases in the Troposphere and maintain the standard value of atmospheric pressure in the Troposphere around 101,325 Pa), these industrial gases have a second effect in increasing the temperature, which leads to the activate the variable T in the relationship (PV=nRT) due to the effect of the global warming, this leads to increasing the volume of air in the Troposphere, this accelerates the process of displacement of small molecular weight gases into the Stratosphere, for example, HFC-23 is a by-product of refrigerant manufacture have around 14,700 times as powerful as carbon dioxide at warming the globe, this accelerates the proses of forming a gap in the Ozone layer.

Resources that supply gases to the atmosphere of planets:

1. **Industrial emissions:** High percentages of industrial residues are in the gas phase, and it is seen in the emissions of petroleum refineries, car smoking, and heavy machines global greenhouse gas emissions, which add around 50 billion tons annually of gases to the Earth's atmosphere, high percentages of industrial residues are in the gas phase, and it is seen in the emissions of petroleum refineries, car smokes, and heavy machines global greenhouse gas emissions add around 50 billion tonnes annually of gases to the Earth's atmosphere.
2. **Volcanic activities:** Most of the gases in the Earth's atmosphere are caused by the gaseous escaping from the hot rocks towards the Earth's atmosphere during its formation about 4.543 billion years ago and this process has been continuous until nowadays and it appeared during the volcanic activities where most of these gases are (Carbon monoxide, Carbon dioxide, Sulfuric monoxide, Sulfuric dioxide, Oxygen, Nitrogen, Argon and water vapor, where the chemical activity of the emissions led to the decomposition complex components of these emissions such Carbon monoxide, Carbon dioxide, Sulfuric monoxide, Sulfuric dioxide upon the surface of the Earth and the deposition of the heavy elements such as Sulfur and Carbon upon the surface, and the release of Oxygen and Nitrogen into the atmosphere. Volcanic activities where most of the gases in the Earth's atmosphere are caused by the gaseous escaping from the hot rocks towards the Earth's atmosphere during its formation about 4.543 billion years ago and this process is continuous until nowadays and it appears during the volcanic activities where most of these gases are (Carbon monoxide, Carbon dioxide, Sulfuric monoxide, Sulfuric dioxide, Oxygen, Nitrogen, Argon, and water vapor, where the chemical activity of the emissions led to the decomposition complex components of these emissions such Carbon monoxide, Carbon dioxide, Sulfuric monoxide, Sulfuric dioxide upon the surface of the Earth and the deposition of the heavy elements such as Sulfur and Carbon upon the surface, and the release of Oxygen and Nitrogen into the atmosphere.

3. **The effect of the planetary gravitational field:** The planet's rotation around the Sun in its elliptical orbit leads to a change in the locations of the planet in space, and its distance from the Sun grants planets opportunities to pass through regions of high-density gas clouds of gases and be affected by their force of the gravitational field to be confined into the limits of the active planetary gravity forces to be added to the composition of the gases of the planetary atmosphere. The combustion of meteors during their penetration into the atmosphere generates another amount of gases that are also added to the composition of the gases in the planetary atmosphere. The occurrence of tails behind meteors is one of the forms of gaseous escape that becomes clear when the thermodynamic effect of gases overcomes the gravity force, where the thermodynamic effect of gases increases when these comets approach close distances to the sun and while their temperature increases, which results in phase transformation of materials (methane, ammonia, etc.) from the solid state to the gaseous phase. These materials attracted to the meteors when they were at far distances from the sun. The temperature drops to extremely low rates, in the meantime the combustion of meteors during their penetration into the atmosphere generates another amount of gases that are also added to the composition of the gases of the planetary atmosphere. The reason behind the explosion of meteors in the atmosphere of the Earth is that these meteors enter the zone of the high-density atmosphere layers Stratosphere and Troposphere that are closer to the surface of the Earth, which leads to changing the state of these meteors from the friction to the state of collision, which accuses an explosion at depends on the mass of the meteors and the speed collusion with the high density of gases. There is another factor that causes the airglow, because all conditions are available for the total internal reflection due to the differences in density of the atmospheric layers between the Troposphere and Stratosphere, where Sun's rays coming from the horizon collide with the layer between the Troposphere and Stratosphere) and due to the lower density of the Stratosphere than the Troposphere leads to the occurrence of the phenomenon of total internal reflection, what causes the light to reflect of the layer separating (Troposphere and Stratosphere), so the area between them is illuminated to be shown as airglow around the Earth, we could call this the atmosphere lensing and its color changes in the atmosphere of Earth according to the observation angle due to the differences in the angles of light reflection according to its frequency energy, the basic concept of the properties of light is the refraction of light during the transition from the high dense medium to the least dense medium.

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Author hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

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Author has declared that no competing interests exist.

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Chapter 5
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Evaluation of Spatial Expansion of Rift Valley Lakes Using Satellite Data: A Brief Study

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ABSTRACT

Assessing and studying lake expansion can reveal important information about climate change and regional responses; normally, a lake recedes during the dry season and floods during the wet season. In recent years, Lake Nakuru has had variations between dry and wet seasons' water levels, suspected to be caused by increasing watershed land conversion to intensive crop production and urbanization, both reduce the capacity of soils to absorb water, recharge groundwater and thus increase seasonal flooding. The present work assessed the expansion and fluctuation of Lake Nakuru in Kenya by using satellite data and information. This study was carried out in the Lake Nakuru area in Nakuru County using two methods. In the first method, data was obtained from the USGS website, whereby Landsat 7 and 8 images were used, though images collected after May 31, 2003, had data gaps when the SLC failed, the images had about 78% of their pixels missing. In the second method shapefiles were used, to find out how the lake surface area had changed, it was done by using different methods. Surface water magnitude was measured from optical sensors, such as Landsat. ENVI software was used to process and analyze data from the satellite images. The data was then used to create a shapefile to get the area of the lake only. The shapefiles were classified using both Supervised and Unsupervised classification, and the area of the lake was obtained in hectares. The obtained area in hectares was recorded in a table and graphs were plotted to show the trend of the lake in the years 1972-2019. Furthermore, correlation was done by assuming the area of the shapefile before any classification is more accurate, therefore, it was compared with the other results obtained by using different methods. Maximum likelihood gave the best correlation values. For R2, it gave 0.8627, and R was 0.9312. The perfect correlation coefficient is 1, the

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graphs for correlation gave 0.91 for NDVI, 0.83 for K-MEANS, 0.93 for Maximum Likelihood and 0.92 for Spectral Angle Mapper. In conclusion, the change was seen using a line graph and that the Lake Nakuru area is expanding from the years 2010 to 2018.

Keywords: Remote sensing; Landsat; soil erosion; supervised classification.

1. INTRODUCTION

The normal status of water ecosystems, particularly the surface water and wetland systems, is increasingly threatened in many countries due to rapid population growth, climate change, and land degradation. For example, the freshwater ecosystems in the Ethiopian Central Rift Valley (CRV) that include marshes, multiple lakes and river channels, and wetlands with significant economic, ecological, and social importance are endangered by the same factors (Tola et al., 2024). Despite the changing lake morphology, there are a lot of disorganized and even contradicting publications on the status of water level fluctuation and the spatiotemporal pattern of the Rift Valley lakes' morphology (Gebreegziabher et al., 2024). Numerous natural and man-made variables are now having an impact on water resource systems all around the world (Gownaris et al., 2018). Depletion of groundwater supplies, increased water demand from a growing human population, and changes in climate and land-use conditions continue to be major concerns for water managers (du Preez & van Huyssteen, 2020). Several lakes in different parts of the world are undergoing expansion mainly due to background climate oscillations, such as the El Niño–Southern Oscillation and anthropogenic controls. Again, several other lakes, including endorheic lakes in arid and semi-arid climates, were noted to have experienced severe water loss and level decline in the past few decades, mainly due to variations in the long-term climatic conditions and human impact (Gebremichael et al., 2022). Water resource systems are crucial to local economies in sub-Saharan Africa because they provide water for home and agricultural utilities, fisheries, recreation, and the promotion of local tourism (Cowx & Ogutu-Owhayo, 2019). The Rift Valley Lakes in Eastern Africa are among the Global 200 Freshwater Ecoregions of today because they support ecoregions with significant biodiversity. However, the Rift Valley Lakes have seen increases in their lake water levels, areas, and volumes, impacting not just the area's unique biodiversity but also the local populations' infrastructure and way of life (Mulama & Ondieki, 2023). Given the prolonged above-average rainfall during the extended rainy season, the situation has worsened more recently (Archer et al., 2020).

Lake Nakuru is positioned at the base of the Kenyan Rift Valley. The lake is within Lake Nakuru National Park and is served by five main rivers: Njoro, Makalia, Nderit, Larmudia and Ngosur, Ngosur being a permanent river and the other four being seasonal rivers. The lake is at an altitude of 1754 m (5755 ft) above sea level with it having no outlets (Mulama & Ondieki, 2023). It has a surface area of about 45 km2 within latitude −0.3562˚N and longitude 36.1002˚E. Being in the Rift Valley, where volcanic activities as well as climatic changes have been very notable, the lake experienced radical changes; the lake's water

level dropped vividly in the early 1990s but has since generally recovered (Mulama & Ondieki, 2023). In 2013, the lake experienced a startling upsurge in water levels, which led to the relocation of flamingoes (Makenzi, 2019). The water levels in Lake Nakuru have been rising progressively since 2012 at an average of three meters (Koskey et al., 2021). The changes in Lake Nakuru have been documented by S.M. Onywere, from 31.8 Km2 in January 2010 to a high of 54.7 km² in Sept 2013, an increase of 22.9 Km^2 (71.9%) (Onywere et al., 2012). The rise in water levels has caused diverse effects on the neighboring community and Nakuru County at large (Mulama & Ondieki, 2023). The neighboring community in Mwariki has been displaced and the farms in Barut submerged. The wild animals in Lake Nakuru National Park have also been displaced from their habitats, causing human-wildlife conflict which leads to the destruction of farms. Economically, the County government has been affected by the drop in tourism as the floods have caused the flamingoes to migrate to other regions. The submerged offices of the Kenya Wildlife Services have also posed huge losses. The floods have also affected the sewage and, in turn, led to a decrease in the quality of the water in Nakuru County. Due to the mixing of the sewage and sediments, the pH of the water has been altered, leading to the death of fish (Mulama & Ondieki, 2023).

Recent years have seen some scientific considerations of the rising water levels in the lakes of Kenya's Rift Valley. The phenomenon of rising lake levels was initially described by Onywere et al. (2012), who also provided thorough descriptions of the affected lake attributes. Using these findings as a foundation, Obando et al. (2016) demonstrated the profoundly detrimental effects of flooding brought on by water level rises on the local people residing in riparian zones (Mulama & Ondieki, 2023). Similar to (Gichuru & Waithaka, 2015) concentrated their analysis on the patterns of Lake Nakuru water surface fluctuations between 1984 and 2013, coming to the conclusion that there was no discernible relationship be- tween rainfall and changes in the lake surface area, (Gichuru & Waithaka, 2015) from the Jomo Kenyatta University of Agriculture and Technology (JKUAT) did a study where they tried to link the water level increase to rainfall changes; he however established that there was no correlation between the two (Mulama & Ondieki, 2023). Other hypotheses that have been formulated include the tectonic plate's movement, Mau Forest degradation, sedimentation and siltation. However, little work has been done to prove these hypotheses.

In this paper, we examined siltation as a factor affecting the rise in the water level of the Rift Valley Lakes. This was achieved through the selected epochs of 1990, 2000, 2010 and 2020 (Mulama & Ondieki, 2023). The investigation employed remote sensing techniques through Google Earth Engine Cloud Platform. Landsat Images were acquired, and Land classification was carried out for the four epochs using a suitable classification algorithm. Four classes (vegetation, water, bare land and built-up areas) were used, and the corresponding areas computed to enable the determination of a correlation (Mulama & Ondieki, 2023). NDVI analysis was also carried out, and this information, together with the DEM, Rainfall data and Soil data, was used to model the risk of soil erosion in the study area.

2. MATERIALS AND METHODS

2.1 Description of the Study Area

Lake Nakuru is positioned at the base of the Kenyan Rift Valley. It is within Lake Nakuru National Park and is served by five main rivers: Njoro, Makalia, Enderit Larmudiac and Ngosur (KWS, 2002) (Fig. 1) (Mulama & Ondieki, 2023). River Njoro is a permanent river feeding the lake, and the other four are seasonal rivers. It is an enclosed catchment area as the lake has no outflows. It is at an altitude of 1754 m (5755 ft) above sea level. It has a surface area of about 45 km^2 within latitude −0.3562˚N and longitude 36.1002˚E. The catchment area is in the surrounding vegetation, Mau Forest, Eburu Forest and Dundori forest (Mulama & Ondieki, 2023). The Mau Forest covers an area of 650 km^2. Eburu covers an area of 87.38 km^2 to the South of the lake, while Dundori forest covers an area of 69.56 km^2 to the East of the lake, as observed by Shah (2016).

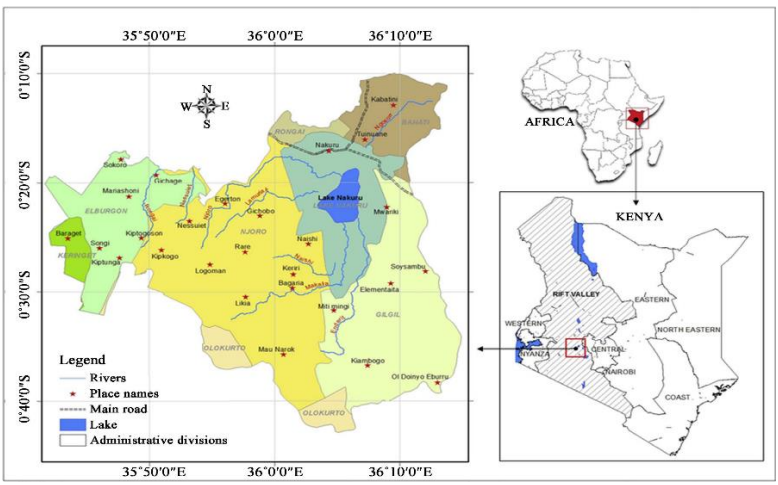


Fig. 1. Geographical location of the study area modified from (Were, et al., 2013)

2.2 Data and Processing Tools

This study was carried out in the Lake Nakuru area in Nakuru County using two methods. In the first method, data were obtained from the USGS website, whereby Landsat 7 and 8 images were used, though images collected after May 31, 2003, had data gaps when the SLC failed; the images had about 78% of their pixels missing. In the second method, shapefiles were used to find out how the lake surface area had changed, this was done by using different methods (Mulama & Ondieki, 2023).

2.3 Data Pre-Processing

Once the images were downloaded from the Earth Explorer website, shapefiles were created for every image in each year. Since the images are already geo-

referenced, NDVI was performed on the images and the different ranges for different classes on the images were saved and thereafter used to calculate statistics for all the classes (Mulama & Ondieki, 2023). The second method was to perform supervised classification, whereby maximum likelihood was used and thereafter the statistics of the area were calculated, which was given in hectares.

2.3.1 Normalized difference vegetation index

It evaluates vegetation by calculating the dissimilarity between Near-Infrared (which vegetation intensely reflects) and Red (which vegetation absorbs).

$$NDVI = (NIR - Red)/(NIR + Red) \quad (1)$$

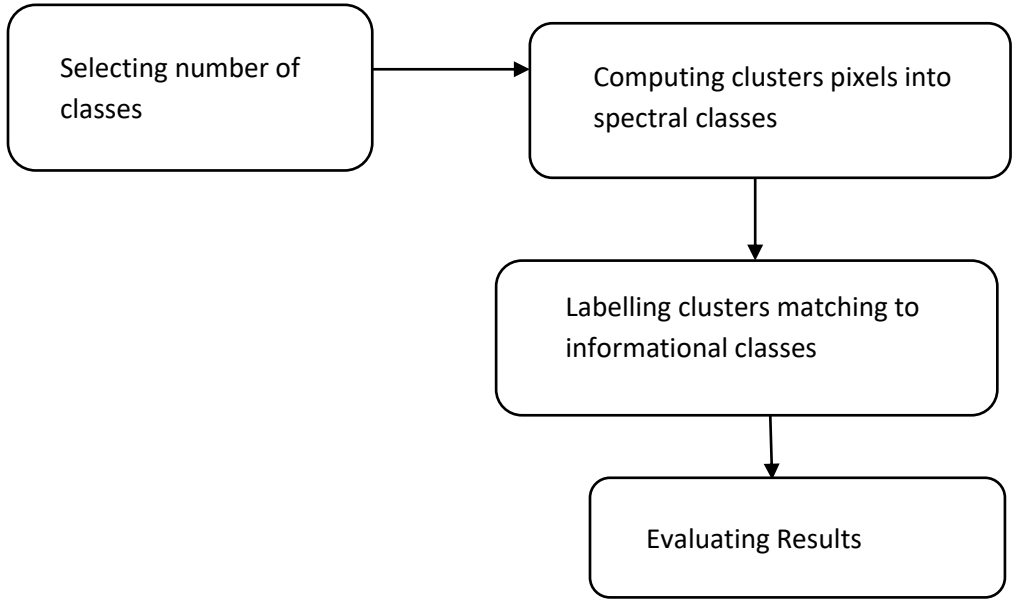
In this case study, NDVI was used to be able to identify the different things in water and be able to distinguish what is making the lake area expand, therefore Density slice was performed in order to use different ranges for water, vegetation and soil (Mulama & Ondieki, 2023). For water, the range is −1 to 0 and Cyan for the color, for vegetation range is 0.3 to 0.9, Green for its color and for soil range is 0.0 to 0.3. Finally, the statistics for the classes were recorded in hectares and recorded.

2.3.2 Unsupervised classification

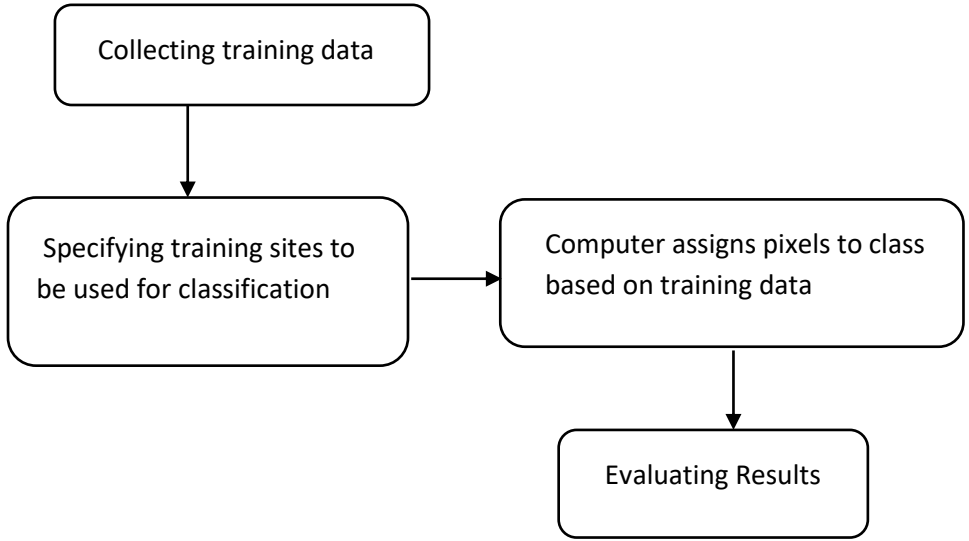
In this classification method, there is no training data provided. Pixels are characterized into clusters based on their resemblances, which can be done using two methods, K-MEANS and ISODATA. Under supervised classification K-MEANS was performed on the images, this was done by setting a number of classes, in this case the number of classes was set to 3, to avoid many classes in the water because the main interest was to assess the spatial change of the lake area not its constituents (Mulama & Ondieki, 2023). After giving an output result, the image was used to get statistics for the lake area in hectares.

2.3.3 Supervised Classification

Training data is present in supervised classification, the user detects different land covers classes and then the computer uses the training data and applies them on the image which results into a thematic map, the computer software determines the different classes by using what is similar to the classes in the training data (Mulama & Ondieki, 2023). In supervised classification both maximum likelihood and spectral angle mapper were performed. First, before performing maximum likelihood, Regions of Interest (ROI) were created from the output image of NDVI, in order to get the different regions, and later the ROI were saved. In maximum likelihood, the saved ROI were used to select the classes, no threshold was used. Statistics were obtained from the output image. The second classification, Spectral Angle Mapper, was used to further classify the image by using the region of interest and later getting the statistics of the region of interest in hectares (Mulama & Ondieki, 2023).



Flowchart 1. Flowchart representation of Unsupervised Classification



Flowchart 2. Flowchart representation of Supervised Classification

2.3.3.1 Maximum likelihood

This method calculates both the variance and covariance of the category spectral response patterns when classifying an unknown pixel. An assumption is made that the statistics for each class in each band are normally distributed, and the probability that a given pixel belongs to a specific class is determined by applying a probability density function for each class derived from its training data.

2.3.3.2 Spectral angle mapper

It is based on spectral classification that uses an n-Dimensional angle to match pixels to training data. It determines the spectral similarity between two spectra by calculating the angle between the spectra and treating them as vectors in a space with dimensionality equal to the number of bands. Smaller angles represent closer matches to the reference spectrum; pixels further away than the specified maximum angle threshold in radians are not classified

$$\cos^{-1}\left[\frac{\sum_{i=1}^{nb} t_i r_i}{\left(\sum_{i=1}^{nb} t_i^2\right)^{1/2}\left(\sum_{i=1}^{nb} r_i^2\right)^{1/2}}\right]$$

α = spectral angle between vectors

nb = number of spectral bands

t = target pixel

r = reference pixel

It takes the arccosine of the dot product between the test spectrums “t” to a reference spectrum “r” with the following equation (Yuhas et al. 1992).

3. RESULTS AND DISCUSSION

In this case study, results on how Lake Nakuru was changing during the past years, were obtained (Mulama & Ondieki, 2023). This was done by finding the area of the lake in hectares for different years using different classification methods, namely, NDVI, Supervised classification, both Maximum likelihood and Spectral Angle Mapper, and lastly unsupervised classification under K-Means. By getting the data from the graphs which were plotted for the different years against the area in hectares, further- more correlation between the area of the lake using the shapefiles and the area of the lake under different classification methods was performed, assuming the area of the lake using the shapefile is more accurate, in order to get the correlation coefficient. With that the results were able to show the spatial change of the lake area (Mulama & Ondieki, 2023).

3.1 Area of Lake Nakuru Using Different Classification Methods

The area of the lake was obtained and recorded in the form of tables, by indicating the years and the area under the different methods of classification performed on the images. Under different classification methods, different areas were found showing how the lake area is changing, both an increase and a decrease of the lake size were seen as it is observed in the Tables (Tables 1-3) (Mulama & Ondieki, 2023).

Table 1. Showing the area of Lake Nakuru in different methods of classification used in different years. Year (1987-2006)

Years	1987	1989	1995	2000	2001	2002	2003	2004	2005	2006
Area of shapefile	3493.44	4019.4	4008.24	4775.51	3626.19	3588.48	3681.72	3825.36	3695.76	3544.47
Area using NDVI	3297.87	3552.57	3287.52	4594.41	2244.87	2045.34	3608.64	3075.84	3005.1	2889.09
Area using K-means	3201.3	3571.56	2966.49	4126.86	3081.33	2665.8	3205.17	2752.74	2057.67	2439.54
Area using M.L.C	3089.61	3358.8	3036.42	4417.2	2504.3	2307.33	3556.44	3000.96	3001.14	2895.39
Area using SAM	3194.1	3525.3	3210.66	4056.84	2325.24	2227.23	3496.23	2595.24	2551.95	2455.11

Table 2. Showing the area of Lake Nakuru in different methods of classification used in different years. Year (2007-2015)

Years	2007	2008	2009	2010	2011	2012	2013	2014	2015
Area of shapefile	3716.1	4060.17	3809.61	3723.57	4206.15	4587.54	5087.07	5688.09	5569.56
Area using NDVI	2913.93	3219.39	3183.03	2784.42	3502.26	3737.79	4200.3	5404.05	5445.36
Area using K-means	1567.71	3149.91	2426.76	1742.04	3362.31	3609.81	4092.66	4010.4	4211.37
Area using M.L.C	2783.7	3081.15	3201.75	2590.83	3505.86	3769.56	4102.2	5076.99	5205.15
Area using SAM	2375.01	2600.01	2964.87	2442.15	3348.72	3600.18	3736.89	5097.24	5126.04

Table 3. Showing the area of Lake Nakuru in different methods of classification used in different years. Year (2016-2019)

Years	2016	2017	2018	2019
Area of shapefile	5495.76	5678.28	6186.51	6249.42
Area using NDVI	5411.16	5538.06	5433.84	5909.58
Area using K-means	5354.37	5470.19	5453.46	4951.44
Area using M.L.C	5195.07	5301.09	5173.2	5645.61
Area using SAM	5072.85	5287.86	5501.52	5847.03

3.2 Analysis

From the tables on the findings of the lake area, using the results, line graphs (Fig. 2) were drawn to make it visible to see the changes in the lake area over the years (Mulama & Ondieki, 2023). From the figure, it is evident that the area was expanding steadily from the year 2010 to 2019.

3.2.1 Correlation between the lake area using its shapefile and the area of the lake in different methods

The When the line graph was plotted to show the change of the lake area, MS Excel was used to obtain correlation. In order to show the accuracy of using the shapefile to obtain the area of the lake in each classification method used. The correlation coefficient is 1, from the graphs below it is seen that they were able to produce a coefficient of 0.82 - 0.93, thus a good correlation (Figs. 3-6) (Mulama & Ondieki, 2023).

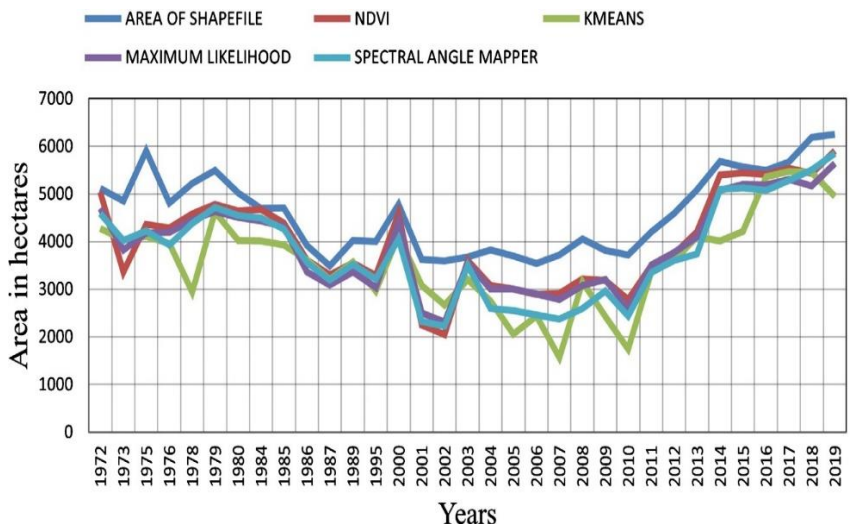


Fig. 2. Variation of the lake area size in hectares using different classification methods over the years

3.3 Discussion

Since different classification methods were used, the change of the lake area is seen to be reducing in the years 1976-1987 and suddenly expanding in the years 2011-2018 (Mulama & Ondieki, 2023). The most appropriate line graph is Area of the shapefile, because it shows clearly how the lake reduced, then expanded, even though over the years.

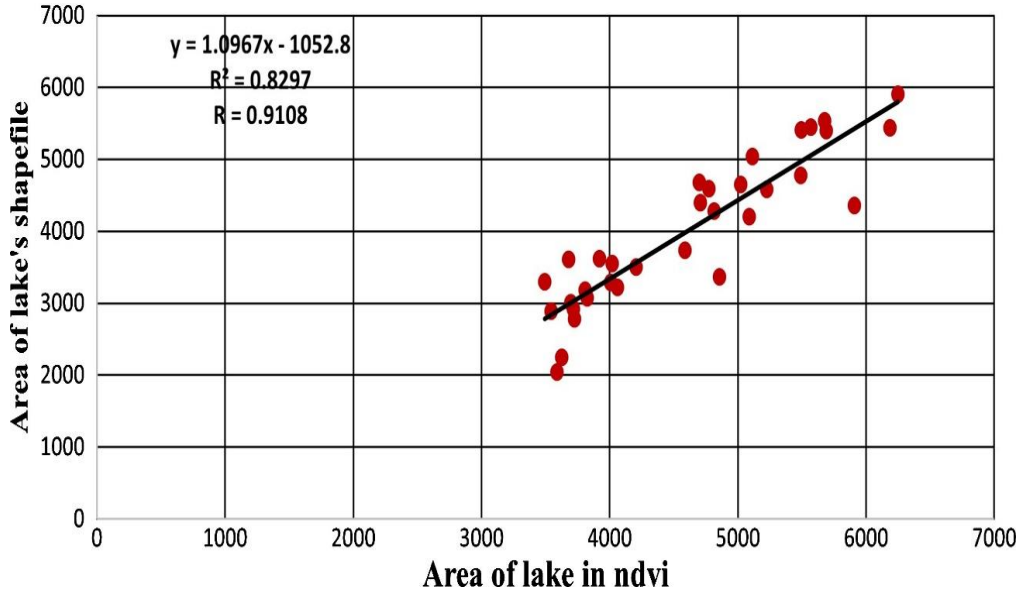


Fig. 3. Correlation of the lake's shapefile and area using NDVI method

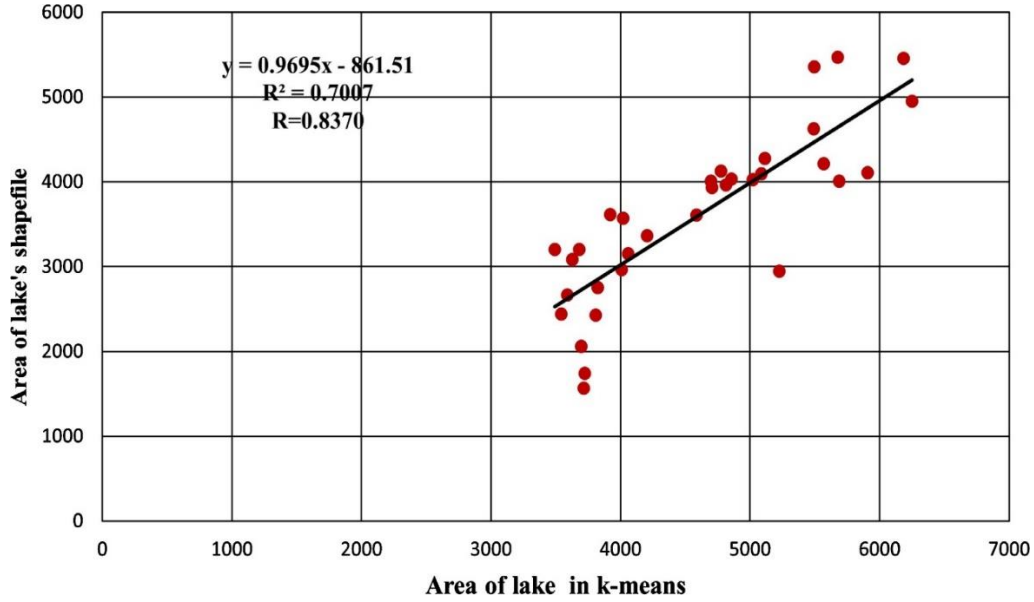


Fig. 4. Correlation of the lake's shapefile and area using K-means method

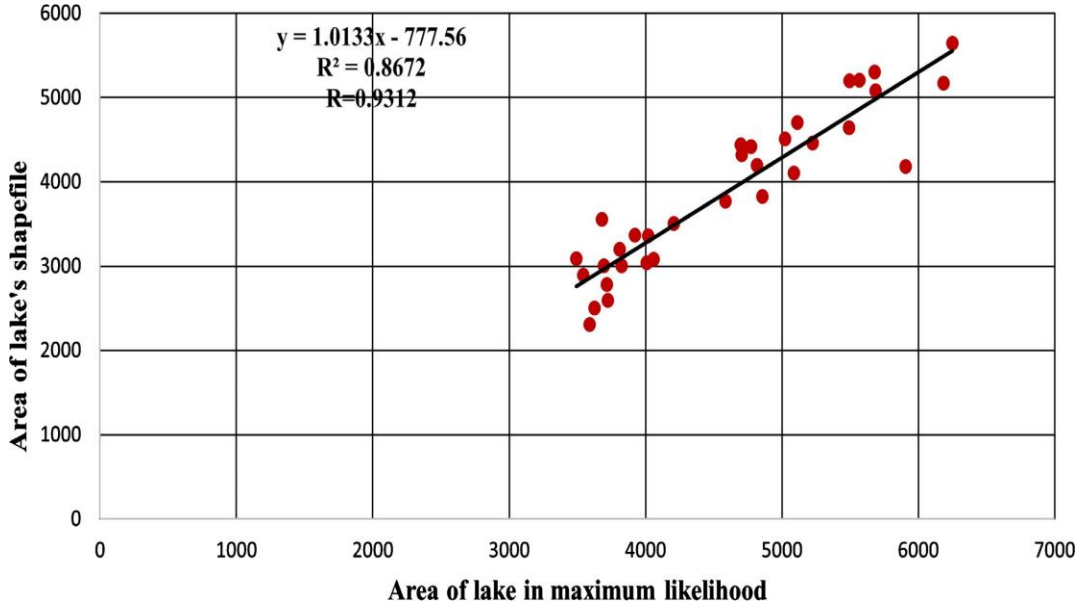


Fig. 5. Correlation of the lake's shapefile and area using maximum likelihood method

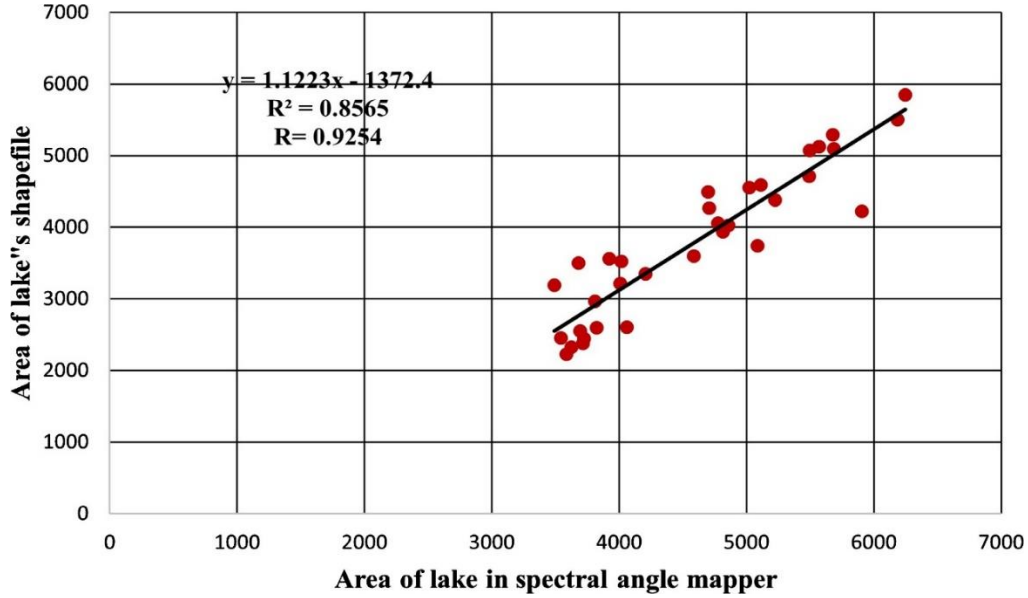


Fig. 6. Correlation of the lake's shapefile and area using spectral angle mapper method

In 2005 and 2011, Landsat 7 its Scan Line Corrector in the ETM+ instrument failed, that is why on the satellite images it showed diagonal lines; it was still able to show the lake's area with a bit of reduction and still provided a good line graph (Mulama & Ondieki, 2023).

Using NDVI, the lake area reduced by about 1343.97 Ha, which is about a 29.2% decrease, then the area expanded with a 19.9% increase and finally, from 2011 to 2018, it expanded again by about 1535.4 Ha, thus 39.35% (Mulama & Ondieki, 2023).

With K-MEANS, from 1979 to 1995, the area reduced by about 1657.71 Ha, that is a 35.84% reduction of the water area. Then increased in 2000 and had a "constant like" drop and rise in 2001 to 2010 (Mulama & Ondieki, 2023). From 2010 to 2019, the lake area then expanded by about 3209.4 Ha, which is a 64.81% increase.

Supervised classification was used and one method that was used is Maximum Likelihood whereby it showed from 1973 to 1984 the lake area abridged by about 610.76 Ha thus a 13.75% decrease. The following two years 1985 and 1987 the area decreased, then rose suddenly from 1995 to 2000 and later dropped till 2010. From 2010 to 2019 the lake area expanded by about 3054.78 Ha showing a 54.10% intensification.

The other supervised classification, that is, Spectral Angle Mapper, showed a smoother graph than the other classification method, from the year 1979 to 1995, the lake area lessened by about 1503.6 Ha, showing a 31.89% decrease. The lake area sustained an increase from 2004 to 2019 by about 3251.79 Ha which is a 55.61% showing a surge (Mulama & Ondieki, 2023).

After getting the area of the lake in hectares and recording the results in a table, the area of the shapefile was considered to be more accurate; therefore, it was used for correlation with the other areas found from using different classification methods. The perfect correlation coefficient is 1; the graphs for correlation gave 0.91 for NDVI, 0.83 for K-MEANS, 0.93 for Maximum Likelihood and 0.92 for Spectral Angle Mapper (Mulama & Ondieki, 2023).

4. CONCLUSION AND RECOMMENDATIONS

By using different classification techniques, the results were able to provide a smooth line graph to show the change in the Lake Nakuru area. Spatial distribution maps of the lake showed different images of the lake in each of the classification used, though the graph of the area of the shapefile is assumed to be more accurate because it's whereby the lake is calculated before any classification is per- formed on the image and is used to correlate the area of the lake with the other methods (Mulama & Ondieki, 2023). Though the cause of the lake's change over the years isn't known. The change was seen using a line graph, and the Lake Nakuru area is expanding from the years 2010 to 2018.

The land use practices of the region have been changing with the built-up area recording an increase of almost 400% over the past 30 years, the changes in these practices make a major contribution to the soil loss of the area as the vegetation and forest are destroyed leading to an increase in the susceptibility of the soil to erosion (Mulama & Ondieki, 2023). The results of the classification also revealed that the area of Lake Nakuru has increased by 50% from the year 1990 to 2018. The modeled factors, despite assumptions being made on the soil erodibility factor and the LS factor, are the results obtained.

To find better results, both data from satellite images and field survey data should be used to produce the best results for the area of Lake Nakuru. The use of satellite data from Landsat images on the Earth Explorer website proved to be useful and successful in this case study and has proved to be appropriate in mapping and monitoring a lake area. Correlation graphs proved that Maximum Likelihood would be the best method to use to classify the image because it gave a coefficient of 0.93 (Mulama & Ondieki, 2023). Since the lake is showing an expansion from the results of the line graphs. It's prudent to monitor it which will help predict the future change of Lake Nakuru.

DATA AVAILABILITY

The data shall be available upon request from the corresponding author.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc.) and text-to-image generators have been used during the writing or editing of this manuscript.

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COMPETING INTERESTS

Authors have declared that no competing interests exist.

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Chapter 6
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Impact of Volcanic Ash on Acidity, Mineral and Nutrient Content of Creeks and Soils Following the Montserrat Soufriere Hills Eruptions

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ABSTRACT

Volcanic activity occurs in hundreds of locations on the planet, extending from the far reaches of Iceland in the Northern hemisphere to the south Pacific 'rim of fire' and beyond. Montserrat's once-deemed near-extinct Soufriere Hills volcano roared to life after many decades of dormancy in 1995 and continued belching its plumes into the atmosphere and its volcanic **(volcano)** ash flow into the surrounding lower elevations for more than a decade, forcing over half of the population to evacuate to safer domains. The volcano finally experienced relaxation of the most aggressive phases of eruptive activity in 2010. Volcanic eruptions typically deliver enormous amounts of gases, minerals and ash residues to the nearby land and streams patenting contaminating the surrounding environment. Because agricultural soils and potable water supplies provide a major source of minerals known to be essential for plant and animal health, it was of interest to determine the pH, mineral, and nutrient composition of the creeks and surroundings following Montserrat's volcanic eruption. The volcano wreaked severe havoc and irreparable damage to the geography and ecosystems, and created a serious immediate and chronic health risk to the remaining inhabitants of the Island. The active pyroclastic ash flow reached temperatures of 1500 °F or more and progressed down the hillsides at speeds often exceeding 80 mph. The extreme temperatures of the ash flow caused instant incineration and virtual destruction of everything in its path, while the gaseous and particulate-laden plume reached altitudes of 7 miles (10.7 km),

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impacting air quality and aircraft navigation in the region. A total of 19 individuals became entrapped in the lava flow and perished quickly as a direct result of the volcanic eruption. Volcanic soils derive from deeper layers of the Earth's surface and are notoriously rich in mineral nutrients representative of those deep mineral deposits. While virtually all combustible objects in the path of the raging ash flow become incinerated within minutes, the aftermath of the eruptions delivers the mineral-rich constituents over a wide swath of the surrounding geography, often reaching miles from the volcanic origin. Soils and consumable water sources are well known to provide a reliable community source of essential minerals in human nutrition, and in their chronic absence, numerous pathologic sequela may develop. To determine the mineral content of surrounding surface soils, samples of surface ash layers and nearby creeks were obtained from 7 regions and 3 free-flowing creeks and subjected to mineral analysis. The surface ash specimens contained abundant amounts of the nutrients Iron, Calcium, Sodium, Magnesium, Sulfur, and non-nutrient Aluminum, intermediate amounts of potassium, Phosphorus, and Manganese, and small amounts of Boron, Copper and Zinc. Creek water runoff originating from higher elevations was consistently acidic with a healthy mean pH of 5.1, and contained smaller proportions of the same minerals, reflective of their relative effective solubility in the acidic water. Thus, the volcanic ash flow while devastating to the environment and public health in the immediate aftermath, provided numerous minerals important to plant and animal nutrition and to human health and health water sources which contributed to the longer-term reinstitution of edible plants and to the recovery of the vibrant lush greenery of the Emerald Isle of the Caribbean.

Keywords: Volcanic ash; soils; minerals; Montserrat.

1. INTRODUCTION

There are hundreds of potentially live and active volcanoes in the Northern hemisphere, including 170 in North America and 21 in the Caribbean, distributed on 11 different island locations. Volcanoes are considered to be live if they are currently active or likely to erupt again, despite the duration since the last eruption. Volcanic eruptions often pose multiple hazards to health and well-being, depending on one’s proximity to the visible and less visible elements that may be contained and released during volcanic events. The immediate volcanic cloud contains both gaseous and fine particulate matter, which are presumed to pose the most grave immediate risk to human and animal exposure. The unseen potentially hazardous gaseous elements typically include hydrogen sulfide, carbon monoxide, hydrogen chloride, sulfur dioxide, hydrogen, and water, which are caustic and may rapidly become dissipated into the airborne environment, including subsequent deposition on convenient surfaces, including plant life and other materials. Volcanic eruptions can pose significant health risks, including respiratory problems, eye and skin irritation, and potential contamination of water and food sources, especially for those with pre-existing conditions. The particulate residues, if left undisturbed, may become injurious to both living and nonliving surfaces upon which the gaseous materials may become disbursed. Much of the plant life may succumb over the next few days and weeks unless it

becomes cleared of the surface debris via natural rainfall, air currents (i.e. wind) or flushing with water, where such becomes a real possibility nearest to homes and gardens. The sulfurous nature of the gases is easily detected in vanishingly small concentrations and often described as a smell of 'rotten eggs 'or 'sewer gas' and this serves as a natural physical deterrent for prolonged human exposure. Because the gaseous constituents rapidly become diluted with surrounding airflows, they effectively become sufficiently diluted and mixed with the surrounding air and eventually dispersed to sublethal concentrations throughout a greater environment. The specific gases released typically include variable concentrations of Carbon dioxide (CO_2), Sulfur dioxide (SO_2), Hydrogen chloride (HCl), Hydrogen sulfide (H_2S), Radon (Rn), Hydrogen fluoride (HF), Sulfuric acid (H_2SO_4), and water vapor, depending on the chemical composition of the underlying magma. Thus, the caustic nature of the escaping gases attests to their dose-related and potentially harmful or lethal health effects on exposed individuals and animal life. The gases become diluted with the atmosphere as they are carried downstream from the eruption, thus, the further distant one is to the escaping gaseous elements, the less immediate risk one would be exposed to, although long-term cumulative residual risks remain unclear. Typical early symptoms include Irritation of eyes or airways, difficulty breathing, vomiting, dizziness, visual disturbances, tremors, and headaches, while longer term effects can include bronchitis and respiratory infections, loss of consciousness, and potentially death especially in individuals who have compromised immune systems or other comorbidities (Pierson et al., 2025; American Geosciences Institute, 2025; Volcano Hazards Program, 2025; Siebert & Simkin, 2005; CDC, 2025; Tulp et al., 2023).

1.1 Plants Derive their Essential Mineral Bionutrients and Moisture to Support their Metabolism from the Soil

Plant nutrition is heavily dependent on the surface soils' mineral and sediment composition, such that it can retain sufficient water and bioavailable nutrients throughout the periods of sunlight and darkness to support the accompanying growth of vegetative species and ground organisms (Tulp et al., 2023). The soil may obtain additional essential nutrients from decaying plants and animal matter that can leach into and contribute to the fertilization of the environment (Ciriminna et al., 2022; Browing et al., 2014; Jones & Gíslason, 2008). The sources of the nutritive content of the surface soils used in agriculture and in natural growths of plant life typically derive from environmental conditions supported by fertilization and exogenous water supplementation to replace consumed nutrients that may have become depleted during repeated cycles of agricultural operations or other seasonal environmental events (Tulp et al., 2023). The micronutrients thus obtained facilitate efficient plant metabolism and endogenous biosynthesis of numerous phytochemicals of nutritional interest in the metabolism of man and animals that may consume the plants as food while contributing to the biological life cycle. Effective crop rotation can often minimize the impact and urgency of nutrient depletion in soils, but artificial replacement of some depleted elements may still be required on a timely schedule (Tulp et al., 2023). Volcanic soils have a well-earned reputation as potentially valuable sources of minerals and essential

micronutrients for agriculture, and that over time may contribute to the development of nutrient rich soils known as andosols that are capable of supporting abundant new plant and aquatic growth for decades during prolonged periods of volcanic recovery Ciriminna et al., 2022; Browing et al., 2014; Jones & Gíslason, 2008). The consistency of the volcanic particulates when blended into the agricultural layers can also assist in the hydration of the soils, thereby acting as a hedge against periods of low rainfall.

1.2 Composition of Volcanic Ash

Most volcanic ash is made of tiny jagged low density rock fragments of varying diameter, most less than 0.2 mm in diameter, plus finer dust particles as small as 0.2 µm in diameter which like the larger particles are virtually insoluble in water but can add to a hydration matrix that is conducive to penetration by natural or added water sources (Jones & Gíslason, 2008; Rose & Durant, 2009; Walker, 1981). The larger, more coarse particles of volcanic ash have the consistency and feel like grains of sand and can help to 'break up' the soil and assist hydration when blended into the agricultural layer (Tulp et al., 2023). The acidic nature of the aqueous runoff provides a healthy mineral-enriched water source for plants and animals indigenous to the area. While the very fine particles are blended into the mixture, they contribute a smooth and powdery consistency and can assist the soils by binding other constituents. Electronics including computers are susceptible to the adverse effects of volcanic ash, as the smallest particulates have the capability of penetrating the most innermost parts of computers, electronics, aircraft engines, machinery and other finely manufactured surfaces where then can form an impermeable surface covering often contributing to instrument or mechanical failure (Walker, 1981; USGS, 2025; Delmelle et al., 2007).

Volcanic eruptions also result in the release of numerous potentially hazardous gasses into the atmosphere, including water, carbon dioxide, hydrogen, sulfur dioxide, hydrogen sulfide, carbon monoxide and hydrogen chloride (Pierson et al., 2025; American Geosciences Institute, 2025; Volcano Hazards Program, 2025; Siebert & Simkin, 2005; CDC, 2025; Tulp et al., 2023; Jones & Gíslason, 2008; Rose & Durant, 2009; Walker, 1981). The sulfur and halogen gases and associated metals are removed from the atmosphere by processes of chemical reactions, including dry and wet deposition on convenient surfaces and materials, and by adsorption onto the surface of the volcanic ash and granular particles. The crystalline-solid structures of the salts that may be formed from these chemical reactions may act more as an insulator than a conductor when in contact with electronic surfaces, including computers and other electronic instrumentation devices, in contrast to the liquid forms, which may act as corrosive ionic conductors. In addition, the released gases and particulates of the plume cloud block the sun's rays, temporarily disrupting ultraviolet radiation, impart a transient environmental cooling effect, and impact factors of bird feeding and nesting activities in addition to migration patterns (Delmelle et al., 2007; Horwell & Baxter, 2006).

1.3 Characteristics and Hazards of Volcanic Eruptions

During an active eruption, the volcanic materials are forced into the atmosphere by the rapidly expanding magma under great pressure and may form a tephra plume extending to several miles high and which may be carried by prevailing air currents many miles from their origin (USGS, 2025) (Tulp et al., 2023). The force of the eruption is such that large boulders 1-2 meters or more in diameter and at high temperatures may be forced upwards and out of the volcano, and are then carried by the resulting ash flow and hurtling down and away from the volcano (Delmelle et al., 2007; Horwell & Baxter, 2006; Fergus, 1996). The intensity of the temperature of the descending matter may ignite flammable objects in their path of destruction for considerable distances from their volcanic origin. The volcanic particles of the tephra typically consist of jagged pieces of rock, minerals, and silicates, sometimes referred to as volcanic glass due to their glass-like crystalline characteristics (Rose & Durant, 2009). The ash particles often take on an acidic, caustic consistency, have a highly abrasive nature and may cause topical skin irritation to exposed skin and mucosal surfaces upon direct physical contact (Tulp et al., 2023). If the volcanic dusts are inhaled the fine lightweight particles, some of which are only the size and diameter of cigarette smoke, can become impacted deep within the alveoli of the lungs and cause severe and sometimes chronic breathing problems (Horwell & Baxter, 2006; Avery, 2004; Forbes et al., 2003; Nadaf et al., 2022). If left untreated, the damage to the lungs may lead to chronic respiratory illness or, even more dire, acute exposures may result in suffocation, the most common primary cause of early and often immediate death following volcanic eruptions (Koreni-Both et al., 2000). Thus, the need for individual preventive measures, including masking and covering of all exposed skin surfaces in addition to relocation to a safe area, is imperative (Tulp et al., 2023). The types of minerals usually present in volcanic ash are dependent on the chemistry of the magma from which it erupted. Considering that the most abundant elements typically found in the silicate-rich magma of volcanos like the one in Montserrat are silicone and oxygen, the various types of magma (and therefore ash) produced during volcanic eruptions are often commonly explained in terms of their silica content, which can range from approximately 45 to 70 percent by weight in any given volcano (Rose & Durant, 2009; Walker, 1981).

1.4 History of the Montserrat Volcano

The Montserrat Soufriere Hills volcano, which once was believed to be virtually extinct, roared back to life with severe explosive eruptions of an unprecedented magnitude after a long dormancy beginning in July 1995 through late 1997 (Tulp et al., 2023). The volcano then began a long series of less violent eruptions which persisted for well over another decade (Tulp et al., 2023; Fergus, 1996). The volcanic eruptions resulted in virtually total destruction of much of the Southern half of the lush green island Nation, including the capital city of Plymouth and numerous surrounding towns and villages. The entire Southern half of the island became covered with hundreds of tons of volcanic ash, sand, and volcanic boulders which often reached several meters or more in areas in

closest proximity to the volcano (Tulp et al., 2023). Prior to the reactivation of the volcano, the vast majority of the island population resided in towns and villages located on the Southern half of the Island due to the more favorable terrain, abundant availability of water from the numerous natural aquifers, and the more abundant lush vegetation and greenery throughout the Southern sector. Most of the island's industry was also located on the Southern half of the island and suffered irreparable harm from the volcanic eruptions. A total of 19 Montserratian lives were lost after having been trapped by the rapidly moving ash, sand flows and poisonous gases (Fergus, 1996).

The hot ash flow encircled, incinerated, and buried scores of trees and vegetation, dozens of vehicles, homes, and other buildings located in harm's way with great and virtually unrelenting force (Tulp et al., 2023). The composition of the ash and sand flow was largely unknown, and the temperature of the rapidly moving ash flows were recorded and exceeded 1500 °C and traveling at rates up to 80 miles per hour or more in areas closest to the volcano, virtually precluding opportunities to escape the oncoming onslaught (Witham et al., 2005). Thus, the intense force and momentum were easily capable of rapidly incinerating much in its direct path and nearby surroundings (Tulp et al., 2023). The land area of the Island reportedly increased from 39.5 to 40 square miles due to the voluminous quantities of ash released into the coastal areas located east and Westerly downstream from the ravaging impact of the volcano during the repeated eruptions (Tulp et al., 2023; Witham et al., 2005). Montserrat is a mountainous island, thereby enabling the ash to flow an easy downhill slope until it reached the sea on both the Easterly and Westerly sides of the Island, leaving only the northern half of the island somewhat protected from direct ash flow and the most severe exposures to the expanding tephra. The ash fall ultimately covered much of the island and reached other nearby islands and surrounding sea surfaces and seabed over 50 miles distant from the erupting volcano (Tulp et al., 2023). The fine dust particle clouds common to erupting volcanos contaminated the airspace and posed a serious threat to aircraft operating in the immediate area should their carburation and air handling functions become compromised, thereby and disrupting air traffic throughout the region and leaving sea travel the only access for escape to safer environs (Jones & Gíslason, 2008).

1.5 Elemental Composition of Volcanic Eruptions

The chemical content of the minerals and particulate matter present in volcanic ash is heavily dependent on the chemistry of the magma from which the volcano erupted. Silicon and Oxygen are the most abundant elements found in most magma, and the various types of magma (and therefore ash) produced during volcanic eruptions are most commonly explained in terms of their percentage of silica content (Tulp et al., 2023). Low energy eruptions containing basalt generally produce a characteristically dark colored ash containing only ~45–55% silica, in association with iron and magnesium, which contribute to the darker coloration. The most explosive eruptions are termed rhyolite eruptions and produce an ash with a silica content that approaches 70%, while other types of ash including andesite and dacite have an intermediate content of silica that

ranges between 55–69% by weight (Siebert & Simkin, 2005) (Tulp et al., 2023). The particles in ash tend to be porous and lighter in weight than most stones due to the high gaseous pressures exerted on them during their formation, and which becomes infused into the rock particulates and larger formations, in somewhat of a 'Swiss Cheese' manner (Avery, 2004; Forbes et al., 2003; Nadaf et al., 2022).

1.6 Global Expansions of Native Populations often Encroach on Previously Undeveloped Areas, Increasing Environmental Stress

The current trends in expansions in the global population have caused the progressive encroachment of urban development into higher risk areas, often in closer proximity to volcanic centers. In so doing, this population growth has increased the potential for human exposure to volcanic ash fall, earthquakes and other severe environmental events (Tulp et al., 2023). The direct health effects of volcanic ash on humans residing in close proximity to such danger zones are usually short-term and mild for persons **(people)** in normal health, as given adequate forewarning, such areas tend to be evacuated to safer areas in advance of the eruptions. However, chronic prolonged exposure, should it occur, potentially poses an increased risk of silicosis among unprotected workers and local inhabitants who may choose to delay relocation. Ash particles of less than 10 µm in diameter can become suspended in the air and are known to be inhalable, therefore, people exposed to ash falls have often experienced immediate respiratory discomfort, difficulty in breathing, eye and skin irritation, and nose and throat symptoms (Tulp et al., 2023). Most of these early symptoms tend to be of short duration and are not considered to pose an immediate significant health risk to those without pre-existing respiratory conditions (Avery, 2004; Forbes et al., 2003; Nadaf et al., 2022).

The more long-term adverse health effects of volcanic ash depend largely on the particle size and chemistry of the exposure, the mineralogical composition of the ash and the reactive chemical coatings adherent to the surface of the ash particles inadvertently inhaled or in contact with exposed surfaces and tissues (Avery, 2004; Forbes et al., 2003). Additional factors related to potential respiratory symptoms are the frequency and duration of exposure, the concentration of ash particles and residues in the air and the respirable ash fraction; the chemical composition of the particles inhaled, and the proportion of ash particles of less than 10 µm diameter (Tulp et al., 2023). The tiniest particles are commonly known as the $PM_{10.}$ or the most minute particles (Nadaf et al., 2022), which, when inhaled due to the small particle size and chemical composition, can impact deep in the respiratory tree and alveolar tissues. In such instances, individuals with pre-existing comorbid conditions, including COPD and others, can experience further compromise of patient outcomes (Forbes et al., 2003). Chronic health effects from volcanic ash fall are indeed possible, as exposure to the free crystalline silica from any source and in any amount is known to be a primary contributory cause of silicosis, where the amount inhaled correlates with the severity of respiratory illness produced (Tulp et al., 2023). A high calcium content of the finely divided silicates may also facilitate a more

profound respiratory outcome as appeared to have occurred in the 'Gulf War syndrome' (El Askan Disease) (Koreni-Both et al., 2000) during the recent Persian Gulf conflict, where unsuspecting participants were inadvertently exposed to the fine particles of the desert sands that were rich in calcium silicates and chemically adherent to mucous membranes. The social context may also be important, as rescue workers and others involved in recovery efforts may be inadvertently over-exposed to unknown particulates and other noxious agents during recovery efforts, and clinical sequela may identify both post-traumatic and pathophysiologic medical symptoms (Tulp et al., 2023).

1.7 The Chemical Composition of the Soufriere Hills Volcanic Materials had not been Previously Examined, and Was Largely Unknown when the Volcano Erupted

The chemical nature and consistency of materials released from volcanic eruptions can take many forms, ranging from large rock and boulder-like projectiles larger than your neighborhood automobiles to ultra-fine powders considerably smaller than a typical grain of sand (Tulp et al., 2023). While much of the recent ash flow from the Soufriere Hills took the form of fine powders and granular sand, the mineralogic composition of the ash which fell over Montserrat and nearby islands and the gaseous exposures at the time were largely unknown, causing much alarm and apprehension among residents. While the Soufriere Hills volcano has now subsided to a constant release of moisture-filled steam from its peak, and is now considered little more than an environmental nuisance, as its peak eruptive activity posed a serious threat to life and limb to all in the vicinity. Montserrat now has several black sand volcanic beaches formed where the ash flow reached the shorelines, and which have now been well washed via local weather and tidal flow during the years following the volcanic eruptions and are somewhat unique for beaches among Caribbean Islands (Tulp et al., 2023). To this end, mixed ash and sand samples were collected from 7 locations on the Island from close in and several miles distant from the volcano and assayed for mineral content via mass spectrometry to determine not only the mineral composition but to also determine the safety of the once volcanic environment should any noxious or otherwise toxic minerals be found present in the efflux. In addition, water samples were collected from 3 distinct sites in a similar fashion for chemical analysis and determination of pH due to the acidic nature of the eruptive materials and the obvious odorous presence of sulfur smells in the air during and for several weeks following the eruptions (Tulp et al., 2023).

2. METHODS

Approximately 100 g samples of the surface layer of ash + sand mixture were aseptically collected in sterile vials from 7 randomly selected sites labelled A through G that were accessible to the investigators via well-established protocols (Forbes et al., 2003). In addition, water samples of approximately 100 ml each were collected in a similar fashion and labeled Plymouth, Olveston and Egret House in St Peter's, corresponding to their area of collection (Tulp et al., 2023).

Care was taken to not permit the solid or liquid samples to contact the skin of the collection personnel so as to avoid possible contamination (Tulp et al., 2023). Mineral analyses were determined via atomic absorption mass spectrometry at the University of Vermont Agricultural Analytic Laboratory, Burlington, Vermont and are expressed as parts per million of the individual minerals, as a percentage for Carbon and Nitrogen, and a pH pm a 0-14 scale for H+ concentration. Microbial content was not determined, but all samples were clear and colorless, and visually clear of obvious gross microbial contamination. The water samples are derived from aquifer runoff from three of the thirteen known aquifers surrounding the volcano. The study was approved by the USAT Institutional Review Committee as an environmental research activity (Tulp et al., 2023).

3. RESULTS

The results of the organic residue and mineral analysis of the ash-enriched soils are depicted in Fig. 1 through 6 below, from those constituents with the greatest concentrations in Fig. 1 to those with the lowest prevalence I Fig. 3 in descending order (Tulp et al., 2023). In Fig. 1, the mean mineral content expressed as ppm of the 7 specimens indicates that the highest mineral concentrations were found for Sodium, Calcium and Iron, all of which are nutritionally essential for sustained plant and animal growth and development (Tulp et al., 2023). For sodium, the range of recorded values was from 1081 to 2789 ppm, for Calcium from 6222 to 9413 ppm and for Iron, 17542 to 35318 ppm thus with the higher concentrations found in closer proximity to the volcanic source.

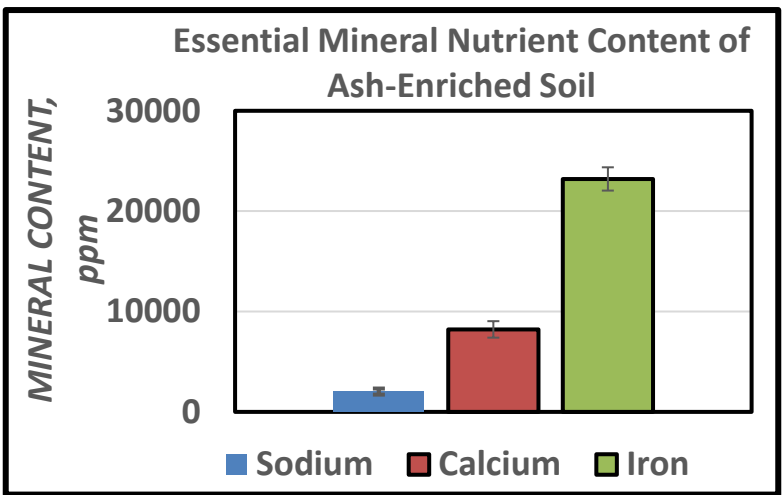


Fig. 1. Essential minerals: Sodium, Calcium and Iron content of volcanic ash

Data are the mean ppm ± 1 SEM, N=7

The results of the analysis for the nutrients Phosphorous, Potassium, Magnesium, Sulfur and Mangnese **(Manganese)** are depicted in Fig. 2, and

indicate that while the average mean concentrations were of lesser magnitude than was found for the Sodium, Calcium and Iron depicted above, the volcanic soils still provided an abundant source of the above mentioned and nutritionally essential minerals. The ranges for Phosphorus were from 255 to 677 ppm, for Potassium 54 (one sample) to 663 ppm, for Magnesium 411 to 1299 (one sample), for Sulfur 1081 to 2789 ppm, and for Manganese 135 to 210 ppm.

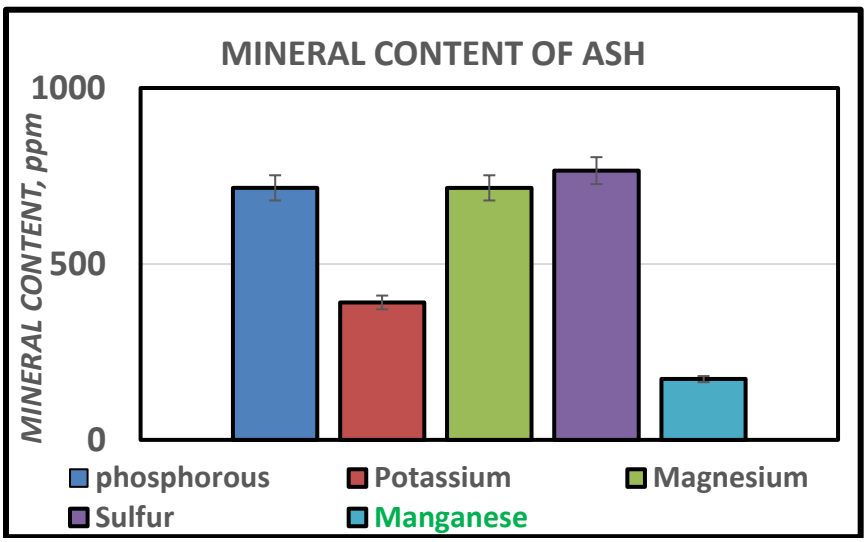


Fig. 2. Nutrient Phosphorus, Potassium, Magnesium, Sulfur and Manganese content of volcanic ash
Data are the mean ppm ± 1 SEM, N=7

The lowest mineral nutrient concentrations were found for Boron, Copper and Zinc, deemed micronutrients for animal nutrition, are depicted in Fig. 3 below (Tulp et al., 2023). All are deemed essential or trace micronutrients for the growth and development of plants and animals. The range of values for Boron was 3 to 30 ppm; for copper, 3 to 14 ppm, and for Zinc, 24 to 44 ppm, respectively.

The pH and mineral analysis of the water drainage are depicted in Fig. 4 through 8, and indicate that while the average water mineral concentrations were lower than the soil concentrations as predicted, there was significant mineral content in the creek waters surveyed (Tulp et al., 2023). In Fig. 4, the pH of three creeks surveyed in close proximity up to approximately 7 miles distant is depicted (Tulp et al., 2023). The results indicate that the water was consistently acidic in pH, with the most acidic in the St Peter's sample, also the most distant from the volcanic origin. Since virtually all of the creeks flow from higher elevations closer to the volcano is was not unexpected that the pH and compositions of the three creeks were similar in their relative acidity and mineral content (Tulp et al., 2023).

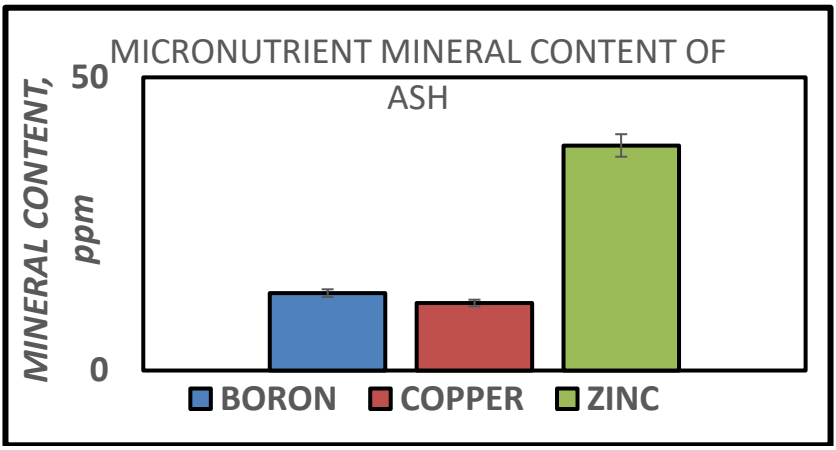


Fig. 3. Boron, Copper and Zinc content of volcanic ash
Data are the mean ppm ± 1 SEM, N=7

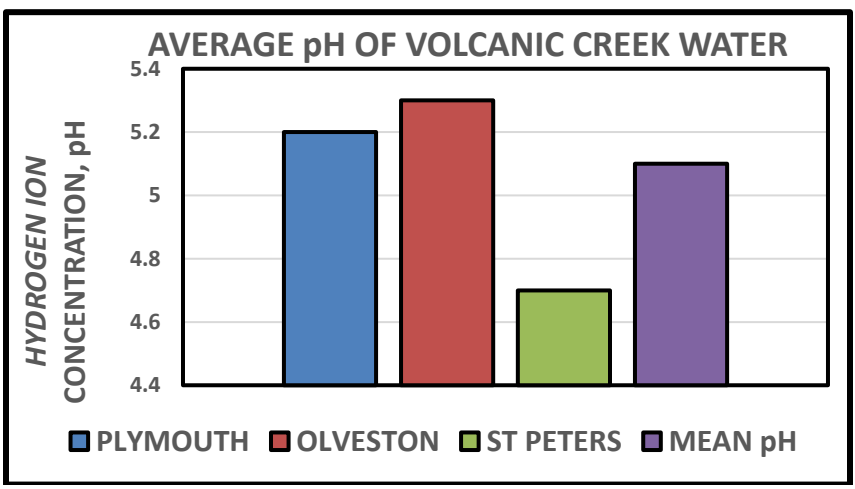


Fig. 4. Creek water pH of free-flowing creeks near Plymouth, Olveston and St Peter's, Montserrat. Each value represents a single specimen, performed in duplicate. The far right bar represents the arithmetic average of the 3 sites

Measures of calcium content in creek waters emanating from higher elevations close to the volcano are depicted in Fig. 5 below and indicate that the greatest calcium concentrations were found in waters draining closest to the volcano and decreased proportionately in samples obtained at the collection sites more distant from the volcanic source (Tulp et al., 2023). The magnitude of the calcium levels would likely have been great enough to cause caking and congestion in household plumbing over time had communities used creek water for household

use. Montserrat has over a dozen aquifers at the higher elevations, therefore enabling Montserratians to enjoy an abundance of some of the highest quality of water in the Caribbean on a year-round basis (Fergus, 1996). Thus, the calcium content of the creek waters depicted in Fig. 6 indicates that the calcium content is greatest in the creek draining closest to the volcano and becomes progressively less concentrated as the creek becomes diluted in addition to potential calcium uptake in supporting soils in the most distant source measured (Tulp et al., 2023). It is of interest that the acidic nature of the creek water is antagonistic to the growth of some infectious microorganisms, thereby contributing to the environmental safety of the open water sources.

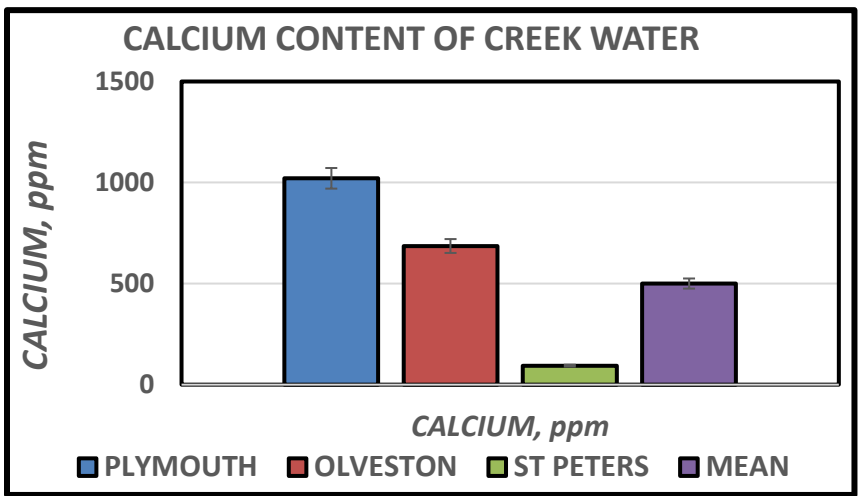


Fig. 5. Calcium content of creek water in Montserrat. Data are individual samples obtained from three locations. The mean concentration as the arithmetic average of the three samples ± 1 SEM, is depicted in the far right bar of the figure

The water concentrations of salts, phosphate, and zinc are depicted in Fig. 5, and are contrasted with the pH values shown in the left panel of the figure (Tulp et al., 2023). The water concentration of the salt and phosphate were similar in the Plymouths and Olveston samples, while the concentrations of dissolved salt and phosphate expressed as ppm were of less magnitude in the St Peters sample, the most distant source from the volcanic origin. In contrast, Zinc concentrations were lower.

The potassium, magnesium, phosphate and aluminum concentrations of the volcanic creek water are depicted in Fig. 7 below, and indicate that the relative concentrations are considerably lower than those of the other minerals depicted in Figs. 5 and 6 above (Tulp et al., 2023). Nonetheless, they are sufficient to deliver those nutrients

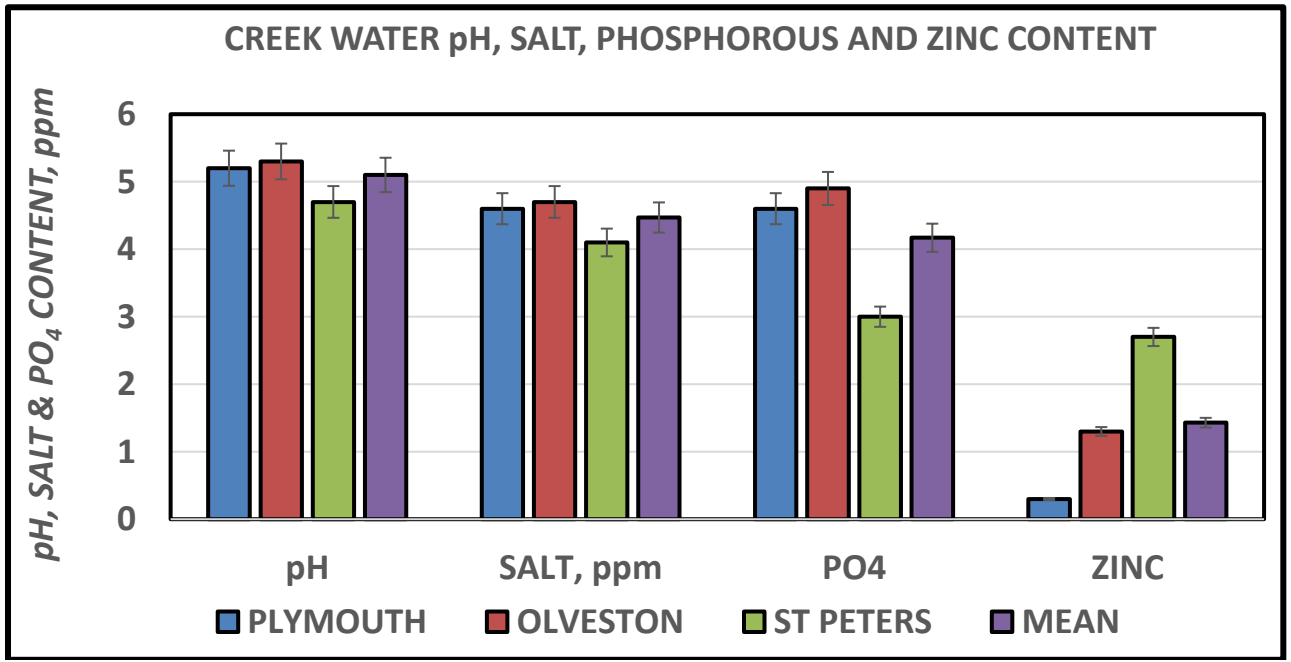


Fig. 6. The pH, Salt, Phosphate and Zinc concentrations of the volcanic creek water. N as stated for Fig. 5 above

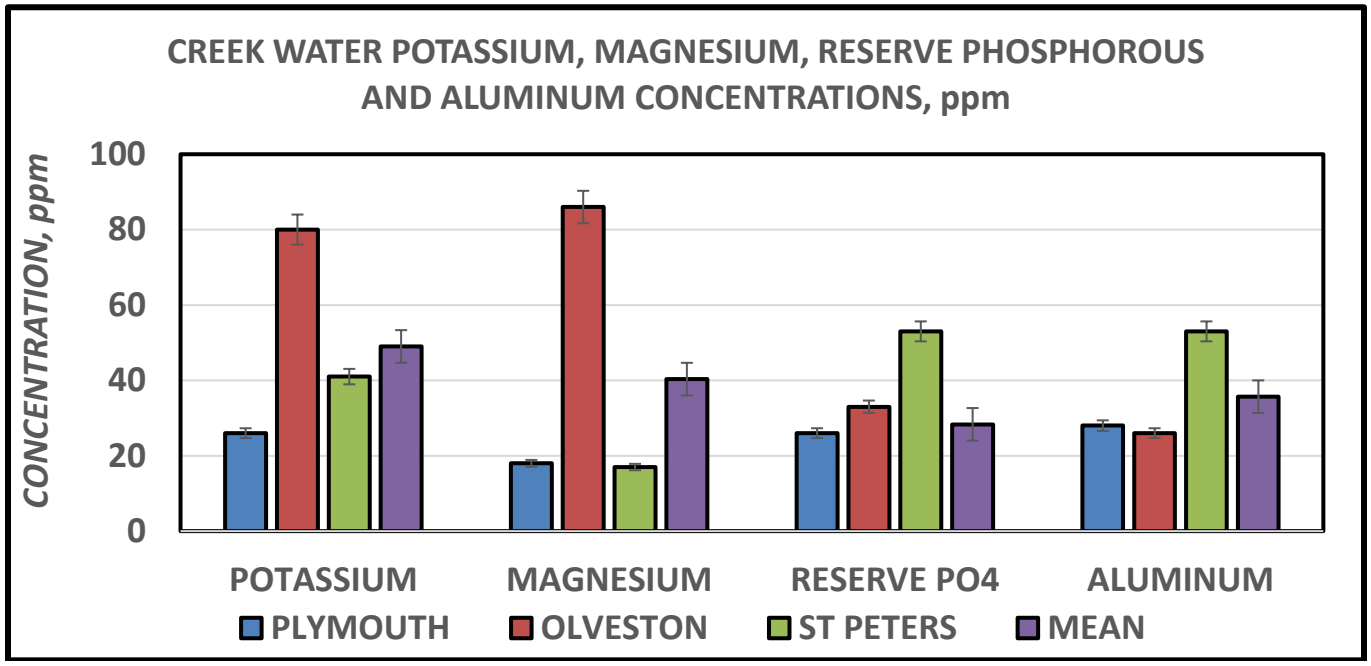


Fig. 7. Potassium, Magnesium, Reserve Phosphate and Aluminum content of creek water in Montserrat. Data are individual samples obtained from three locations. The mean concentration as the arithmetic average of the three samples ± 1 SEM is depicted in the far-right bar of the figure

and additional minerals to surrounding soils as they make their way to the sea, where collectively they can contribute nutrient support of the sea organisms including fish, corals and other live organisms and sea plant species (Tulp et al., 2023).

4. DISCUSSION

Since before the eruption of Vesuvius in 79A.D, hundreds of volcanic eruptions with a volcanic explosivity index (VEI) of 4 or above have occurred in dozens of countries through the globe, incurring severe havoc and destruction often with dire consequences for those individuals unfortunate enough to be in harm's way with no path to escape the fire and venom often released with little warning. The VEI quantifies the relative magnitude and intensity of a volcano expressed in a logarithmic format in a manner similar to the Richter scale used to report the force and intensity of earthquakes (Richter, 1935; Gutenberg & Richter, 1956; Hutton & Boore, 1987). The Montserrat Soufriere Hills volcano has become one of the most closely monitored volcanoes in the world since its eruption began, with the establishment of the Montserrat Volcano Observatory, operated by the British Geological Survey and which takes detailed seismic and frequent air quality measurements and reports on the activity of the volcano to the government and population of Montserrat and made available to interested parties globally.

Volcanic soils have a well-earned and longstanding reputation as being nutritionally beneficial for a wide variety of plant growth, including the propagation of many tropical fruits and other tropical vegetation (Ciriminna et al., 2022; Browing et al., 2014; Jones & Gíslason, 2008). The aqueous runoff from the volcanic soils, combined with prevailing air currents that may transport volcanic dust particles for many miles before they return to earth, and rainfalls that help to bring the airborne particles down, can extend the range of nutrient enrichment of soils distant from the eruptive source. The compositional analysis of the Montserrat ash and sand mixture is fully consistent with this earned reputation. The mineral content of the sand includes the nutritionally essential nutrients Magnesium, Potassium, Manganese, Phosphorus, Calcium, and Zinc are important on both plant and animal life and are especially impressive in their relative abundance. The acidic pH of the ash flow, as indicated by the pH of the water obtained from various free-flowing creeks draining the volcanic areas supports such an observation, enabling the ash to compliment fertilizers when needed, although for most agricultural uses, on island experience has shown that additional fertilizer or lime treatment is seldom needed to restore adequate growth potential to the formerly volcanic-enriched soils. Volcanic eruptions significantly impact air quality by releasing harmful gases and particles, including sulfur dioxide (SO2), particulate matter (PM), and volcanic smog (vog), which can lead to respiratory problems and exacerbate existing health conditions.

While the ash is generally limited in organic matter, averaging less than 1% for Carbon and Nitrogen in the present samples and likely derived from animal life consumed during the lava flow process, the rich mineral content is unique, not

only adding much nutritional benefit to the soil on the island but also contributing to an abundant supply of construction grade sand (Hutton & Boore, 1987). Availability of construction quality sand is often a scarce commodity among Caribbean Islands, where their only other source of sand for construction would be their common sandy beaches that contribute to recreational tourism and to the vibrant tourist economies of the islands (Hutton & Boore, 1987). Thus, the abundant availability of the Montserrat volcanic sand residuals has become a daily export and enables Montserrat and other Island jurisdictions to preserve their otherwise pristine sandy beaches for other commercial, recreational, and environmental endeavors and helps to sustain their economic viability. Many of the Caribbean Islands were formed by volcanic eruptions throughout the region, as evidenced by the geographic features, which typically include mountainous central areas dotted with extinct volcanoes. The Montserrat volcano vents frequently releasing steam clouds, which may linger over the higher elevations and thus is still considered an active volcano.

Wildlife soon re-established residence in the volcanic-impact areas, including wild chickens, birds, goats, wild boars, ground lizards and other animal species endemic to the area (Tulp et al., 2023). In addition, local streams that survived the volcanic eruptions were also observed to have established aquatic life, thereby affirming the absence of substances which may have been toxic to mammalian, reptilian or freshwater aquatic life and enabling those species of animal life to flourish in the affected areas once again. The entry into volcanic areas continues to be restricted to only those individuals with a need to enter. Individual masking was required throughout the island during the active eruptions and among construction workers during the sand excavation and rock crushing activities, where strict safety precautions were implemented. The health of the residents who remained on Island during and after the volcanic eruptions has been closely monitored in the local health facilities including the Glendon Hospital and thankfully to date, no acute or chronic respiratory or other health hazards from the volcano have been reported in the two plus decades since the peak of the volcanic activity occurred.

Volcanic activity is common in many regions of the planet, and while initially immensely destructive to nearby environments and their populations, in the longer term, they may contribute to the regeneration of the environments, continuing for decades after the eruptions have occurred. It is unfortunate that in the past communities have all too often become established in areas in uncomfortably close proximity to earlier volcanic eruptions and therefore proven to be in harm's way should an active volcano spontaneously re-emerge and come back to life as occurred in Montserrat, where it obliterated numerous towns and villages including the capitol City of Plymouth, and which communities have not been reestablished as the memories of 1995 and subsequent years are still fresh in the history and memory of Montserratians. The Island of Montserrat rests on two tectonic plates, the more Southern of which supported the volcanic activity, while the more Northern plate remains stable and largely unaffected by the direct effects of the ash flow activity. Thus, the geography of the Northern half of the island provided an area of safe refuge for those who remained on the

Island during the acute phase of the volcanic eruptions and provided a safe land mass for the redevelopment of Montserrat, in addition to providing ample high-quality construction-grade sand to support the redevelopment activities. The mineral content of the volcanic sand from Montserrat includes naturally occurring cement hardening agents and is highly valued in the construction industry throughout the Caribbean. Thus, the Montserrat Soufriere Hills volcano has played an important role in the establishment and development of Montserrat over the recent centuries and is likely to continue making its beneficial contributions to the economy and well-being of Montserrat and its residents well into the future, whether it be in the valued construction, agriculture, or tourism industries. The sand is safe, the soil is nutritionally fertile, the skies are clear and unpolluted, the beaches and waters are clean and warm, and the environment highly inviting. At least for now, the Soufriere Hills volcano has had its day in the sun for all to see and enjoy and to enable the Emerald Isle of the Caribbean to restore its lush green environment.

5. CONCLUSION

The recent eruptions of the Montserrat Soufriere volcano dating from 1995 and subsequent years, caused vast destruction and incineration of a wide swath of terrain during aftermath of the ongoing eruptions, in addition to the delivery of megatons of volcanic ash and caustic volatile gases to the land and sea masses surrounding the volcano, in addition to gaseous emissions into the atmosphere. The prevailing winds carried the gaseous and finer particulate matter to surrounding islands and sea beds and significantly impacting the airspace of air traffic in Montserrat and surrounding islands during active phases of the volcano. While the immediate impact of the volcanic eruptions was devastating to the populace and environment, the residual effects of the mineral content of the particulate matter served to enhance the mineral content of the soils impacted by the volcanic ash, providing essential minerals for eventual restoration of plant growth. The surface ash specimens obtained contained abundant amounts of the essential nutrients Iron, Calcium, Sodium, Magnesium, Sulfur, and non-nutrient Aluminum, intermediate amounts of potassium, Phosphorus, and Manganese, and small amounts of Boron, Copper and Zinc. Creek water runoff originating from higher elevations was consistently acidic with a healthy mean pH of 5.1, and contained smaller proportions of the same minerals, reflective of their relative effective solubility in the acidic water. Thus, the volcanic ash flow while devastating to the environment and public health in the immediate aftermath, provided numerous minerals important to plant and animal nutrition and to human health and health water sources which contributed to the longer-term reinstitution of both edible and inedible plants and to the recovery of the vibrant lush greenery characteristic of the Emerald Isle of the Caribbean. As an added bonus, the acidic nature of the aqueous runoff was likely instrumental in impeding microbial growth in the creeks surveyed, despite the warm climate of the environment and the presence of potential organic and mineral microbial nutrients, which ultimately drained into the nearby sea, where they were effectively buffered and neutralized, there minimizing potential damage to the seabed and its continued aquatic growth and survival.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

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COMPETING INTERESTS

Authors have declared that no competing interests exist.

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Chapter 7
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Satellite-based Monitoring of Vegetation Cover of Lalbagh Botanical Garden, Bengaluru, Karnataka, India

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ABSTRACT

Lalbagh Botanical Garden is Bengaluru's oldest garden, spread across 240 acres, and is home to a wide array of plants from around the world, a treasure house of biological diversity. Lalbagh has 2,950 spp. of plants and trees. It was started in 1760, was given the status of a Government Botanical Garden in 1856, and since then it has been an internationally renowned center for scientific study of plants and botanical artwork, and also conservation of plants. Lalbagh Botanical Garden, with a high diversity of species, provides a storehouse of genetic resources within the region, and it also provides food and pharmacological resources for a range of species such as birds, mammals, reptiles, and insects. The aim of the study is to give an overall picture of vegetation distribution and canopy coverage in different parts of Lal Bagh through satellite imagery.

In the present study, Quick Bird imagery of 2005 is used for comparative assessment of NDVI and TNDVI indices with supervised classified imagery of Lalbagh Botanical Garden, one of the largest and oldest botanical gardens in Bengaluru. The NDVI and TNDVI transformed imagery of Lalbagh showed variation in their values. The different land use classes of supervised classified imagery showed more resemblances to the NDVI than TNDVI. It was observed that 33% of Lalbagh's 240 acres is covered with a thick canopy comprising mainly large mature trees. Small-statured ornamental specimens and many small trees growing in thickets like groupings constitute a thin canopy, covering 20% area.

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Water constitutes about 13%. The rocky area is present on the North West, comprising about 12%, and the Grass area about 16.6%. The study concluded that in order to achieve environmental sustainability, the city should increase both its percentage of green open space and m2 of urban green open space per capita. More and more efforts should be made in this regard to increase the green spaces in the city by maintaining the old green spaces like botanical gardens, parks, and avenue trees. There is a need to have a clear account of existing green spaces. This study gives various land uses of the entire Lalbagh in a broader picture and a more supplementary study is needed to give tree-wise distribution in the entire Lalbagh.

Keywords: Remote sensing; comparative study; UTM; NDVI.

1. INTRODUCTION

Urban areas span approximately three percent of the Earth's land surface area, and accommodate more than 4.2 billion people, representing 55% of the global population. Within cities, vegetation is present in parks, woodlands, abandoned sites, residential areas, and along streets - collectively referred to as urban forests (Esperon-Rodriguez et al., 2022). Urban forests are defined as individuals or groups of trees that grow within an urban or suburban area. Urban forests can be termed as the green infrastructure of a city, which contributes to the physiological, sociological, and economic well-being of urban societies. The sustainability of any city depends on optimized interaction among its grey infrastructure (e.g. buildings, roads, utilities, etc.), blue infrastructure (e.g., lakes, ponds, rivers, etc.), and green infrastructure (e.g. urban forests) (Bherwani et al., 2024; Dadhich et al., 2023). In this context, urban forestry is the management of trees and other forest resources in the urban community. Ecosystem for the environmental, economic, social, health, and aesthetic benefits that trees provide to society (Peng et al., 2008). Bengaluru's urban forest canopy comprises hundreds and thousands of such trees and they are an important component of the arboreal biomass. From majestic alleles to imposing individuals, many of these trees were planted as a part of original park designs and have achieved specimen status over time. Yet we know very little about them. Urban green areas have considerable ecological benefits that serve to sustain human and environmental health (Myneni, 1995; Che Lam et al., 1999; Witcher & Griffith, 2011; Myeong et al., 2006).

In the current research, Lalbagh, the leading botanical garden, is selected as it is the focal point of the city, linking old, new, and attracting a large number of local, national, and international visitors (Shetty & Somashekar, 2015). At 240 acres, it is the largest Botanical Garden in Bengaluru and the State. Trees in Lalbagh individually and in groups comprise some of the city's most valuable landscape specimens in terms of age, aesthetic, environmental and economic benefits and community impacts. Lalbagh is a forested oasis in a highly urbanized setting in Bengaluru. The present study using RS and GIS helps to give an overall picture of vegetation distribution and canopy coverage in different parts of Lal Bagh through satellite imagery.

The 2005 Quick Bird imagery of Lalbagh is used for studying the various land uses of Lalbagh. The high resolution of imagery has made discretion of various land uses easier and more recognizable with prominent canopy features of individual trees (Shetty & Somashekar, 2015). In the first step, NDVI and TNDVI imagery of Lalbagh was developed. In the second, imagery was subjected to supervised classification with seven land use classes. The NDVI and TNDVI transformed imagery was compared with supervised classification imagery to become acquainted with details of different land uses, with main stress on green cover of Lal Bagh. A similar comparative study was carried out (Lillesand & Kiefer, 2000).

2. MATERIALS AND METHODS

2.1 Data Products

The 2005 Satellite imagery of Quick Bird with 2.4-meter resolution is used for the present study. Analysis is focused on the vegetation distribution in terms of the canopy coverage of trees. Geographical data: Topographical map: Scale: 1:50000; the coordinate system of the baseline data map is applied for geometric correction of satellite imagery. Erdas 9.2 and ArcGIS 9.2 were used for image processing and further analysis (Shetty & Somashekar, 2015; AREA, 2013).

2.2 Methods

Quickbird imagery was utilized as a spatial basis for mapping various land use classes of Lal Bagh (Shetty & Somashekar, 2015).

1. Geometric correction: Image preparation includes geometric rectification to the UTM (WGS 84 datum) coordinate system by 30 ground control points. The geo-corrected image is then used for further processing.
2. The Normalized Difference Vegetation Index (NDVI) and Transformed Normal Difference Vegetation Index (TNDVI) is used to transform multi-spectral data into a single image band, which represents vegetation distribution was computed using a standard algorithm

   NDVI = (NIR – R) / (NIR + R).
   TNDVI= Sqrt ((NIR - red/NIR + red) + 0.5).

3. NDVI-RGB false color composite of vegetation was developed to display and quantify vegetation using Quick Bird.
4. Supervised classification of the imagery was made to differentiate seven land use classes

3. RESULTS AND DISCUSSION

In the present paper the FCC (False Color Composition) of the imagery (Fig. 1) is prepared using 4, 3, 2 bands of the Quick Bird. Quick Bird imagery (2005) of Lal Bagh is used for a comparative study of NDVI and TNDVI with the supervised classified imagery (Shetty & Somashekar, 2015).

The Normalized Difference Vegetation Index (NDVI) is an extensively used index for vegetation studies and is generally used for evaluating the quality of vegetation (Schwab, 2009; Sudha & Ravindranath, 2000; Yang et al., 2008; Gamon et al., 1995; Myeong et al., 2006).

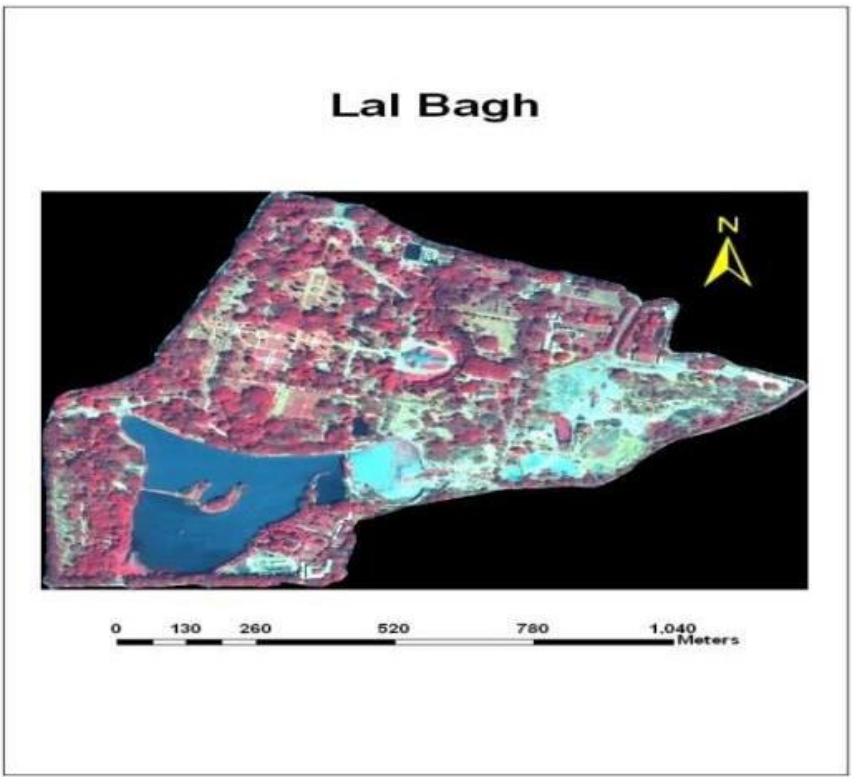


Fig. 1. FCC of Lal Bagh (Shetty & Somashekar, 2015)

The Normalized Difference Vegetation Index (NDVI) is an extensively used index for vegetation studies and is generally used for evaluating the quality of vegetation (Schwab, 2009; Sudha & Ravindranath, 2000; Yang et al., 2008; Gamon et al., 1995; Myeong et al., 2006).

NDVI combined with TNDVI is proven to have an extremely wide range of applications in measuring vegetation cover. The NDVI/TNDVI transformed image is sliced into 7 land use classes for ease of analysis and is given pseudo color (Shetty & Somashekar, 2015). The seven classes were identified as Deep water, Shallow water, Rock and impermeable surface, Soil, Grass, Thin canopy, and thick canopy.

The seven classes of the NDVI transformed image (Fig. 2A) had values ranging from -0.99 (the minimum NDVI value in this dataset) to 0.96 (the maximum NDVI value in this dataset). The majority of old and big trees, which come under the class Thick canopy, showed a mean NDVI value of 0.613, Thin canopy representing small trees and ornamental bushes had values between 0.156 to 0.256, the values vary between - 0.0353 to 0.0414 for bare soil (Shetty & Somashekar, 2015). Rock and impermeable surfaces had values less than -0.04303. Deep and Shallow water classes have the lowest NDVI values as water is a good absorber of near-infrared radiation. The above observation is in conformity with the findings of (Quackenbush et al., 1999), where they appropriately used vegetation indices and image ratios to potentially improve crop classification accuracy.

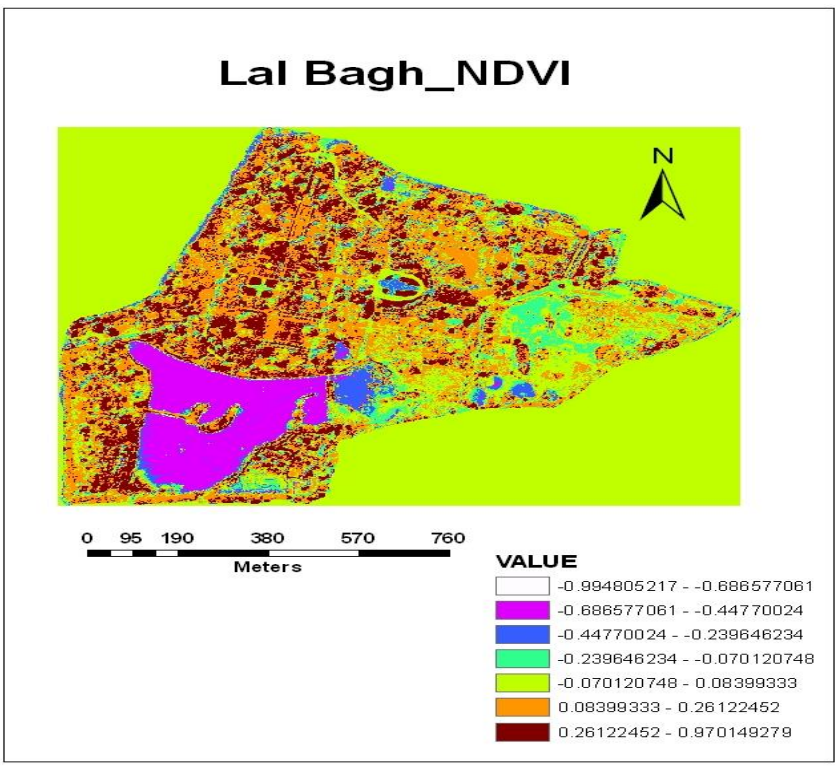


Fig. 2(A)

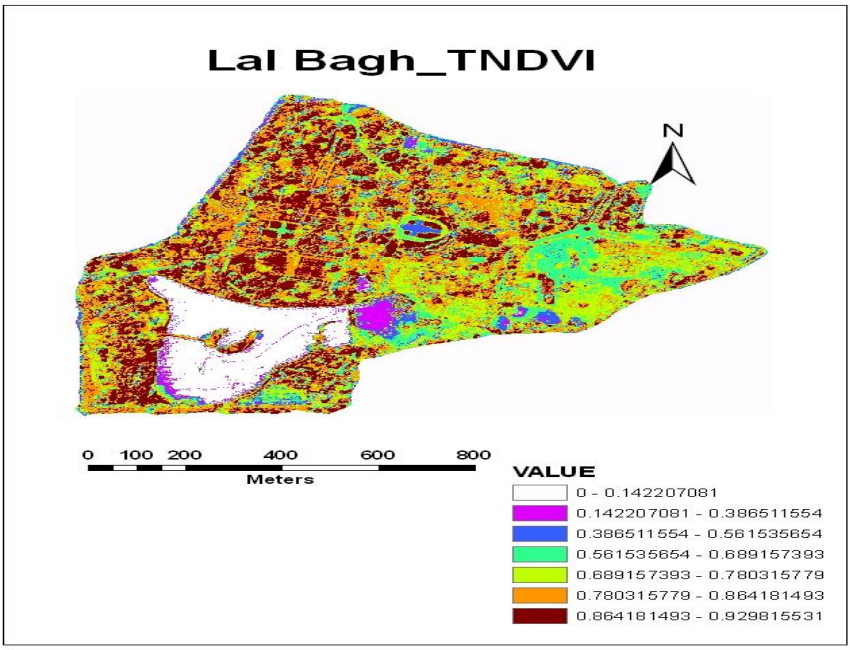


Fig. 2(B)

Fig. 2(A&B). NDVI and TNDVI image of Lal Bagh (Shetty & Somashekar, 2015)

The TNDVI transformed image (Fig. 2B) showed positive values which vary from 0.0363 to 0.926. The thick canopy representing old big trees of the Lalbagh had TNDVI values between 0.897 and 0.926. For thin canopies, the mean value was 0.861. Bare soil had values between 0.722 to 0.694. The values were less than 0.686 for Rock and impermeable surfaces. The lowest TNDVI values are observed for water (Shetty & Somashekar, 2015).

The thick and thin canopy is depicted in a much more elaborate manner in NDVI imagery than in TNDVI. The different land use categories in NDVI imagery showed similarity to supervised classified imagery.

The different land use classes were clearly depicted in supervised imagery (Fig. 3) of Lal Bagh. The Thick canopy class representing old, big specimen trees engrossed an 80-acre area in Lalbagh. The thin canopy comprises 46 acres, and the grass area occupies 41 acres in Lalbagh (Shetty & Somashekar, 2015). The total vegetation area of the Lalbagh is around 70%. The rock and other impermeable surfaces constitute 26.6 acres. The water area is around 30 acres, which is less than the official estimation, as some areas of the lake are covered

with water hyacinth, giving a false impression. Spatial relationships in a botanical garden can examine how close vulnerable plants are to open spaces or high-use visitor areas, how tree canopies change over time, or the density of plantings (Ramachandra & Uttam Kumar, 2009).

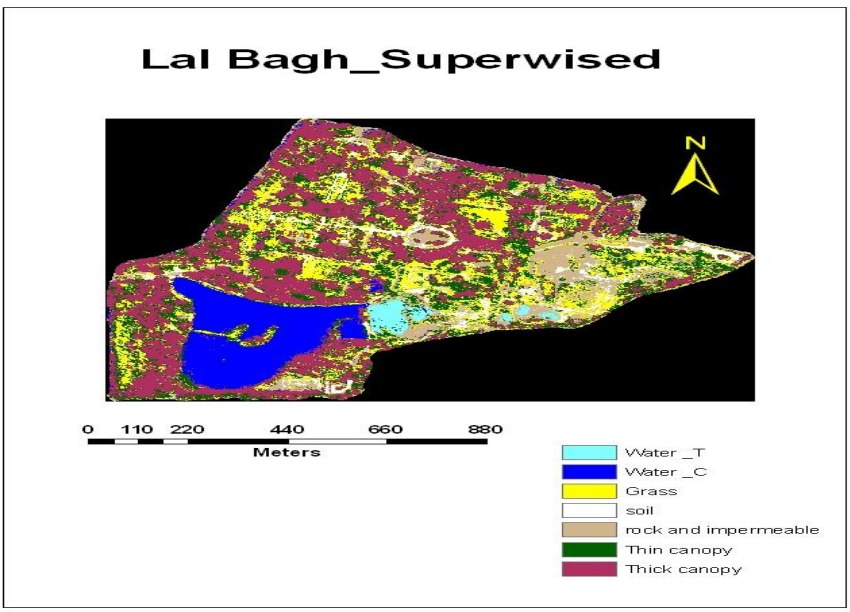


Fig. 3. Supervised Image of Lal Bagh (Shetty & Somashekar, 2015)
Water-Turbid (Water_T)
Water: Clear (Water_C)

Table 1. The Producer's accuracy, user's accuracy, overall accuracy (%), and kappa statistics for different classes of imageries (Shetty & Somashekar, 2015)

Class	Producer's Accuracy	User's Accuracy	Overall Accuracy	Kappa Statistics
D. Water	100	75	98.4	0.7439
S. Water	100	100		1.000
Rock	100	100		1.000
Soil	94.44	100		1.000
Grass	100	100		1.000
Thin canopy	94.74	100		1.000
Thick Canopy	100	96.97		0.9593

The overall accuracy and Kappa coefficient (Table 1) in the current paper show that the pixels, on average, are classified into the correct classes (overall accuracy).

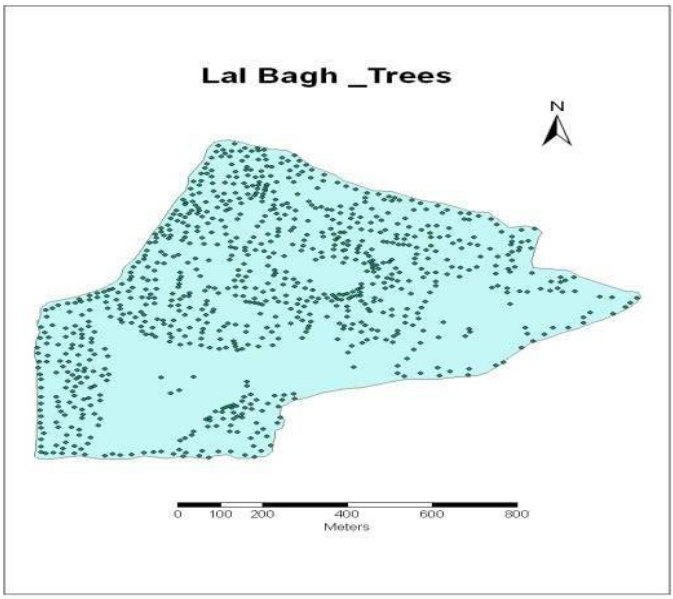


Fig. 4. Tree counting Map of Lal Bagh (Shetty & Somashekar, 2015)

The tree counting is carried out based on canopy cover, showing 3000 trees. Trees were very prominent on the north and south Western part of the garden.
The areas for different land uses of NDVI, TNDVI, and supervised imagery are depicted in the Graph.

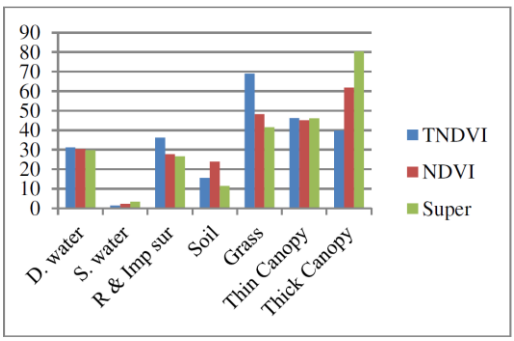


Fig. 5. Graph depicting areas of different classes of NDVI and TNDVI and supervised imagery of Lal bagh (Shetty & Somashekar, 2015)

In the above graph, we could observe similarities in various land use classes between NDVI and supervised imagery, where TNDVI shows delineation with supervised imagery. (Kong, 2005; Nagendra & Gopal, 2009) observed that NDVI helps in mitigating the shadow effect of high-spatial resolution imagery and improves the classification of vegetated areas.

In the imagery, the Rock and impermeable surface class is restricted to the western part of Lalbagh. Lal Bagh Rock is the oldest rock structure on earth, which is supposed to be 3000 million years old. The old big trees which are characteristic specimens of Lalbagh, under a thick canopy, are spread all over the garden. A thin canopy comprising ornamental thickets and small trees is prominent in t h e center spaces of the garden.

Water bodies are depicted more lucidly in NDVI imageries than in TNDVI imageries. Deep and shallow water classes comprising Lalbagh lake and lotus pond, respectively are the focal point of the garden and the main water source and drainage area. It is present in the southwestern part of the garden. The lake is spread across an area of about 40 acres and has a maximum depth of 3.5 meters. Grass class is present as patches throughout the garden. The ornamental and economic plant species, both of exotic and indigenous origin, can be found in Lalbagh, and some of the exotic species are introduced from different parts of the world, like Europe, Australia, and Afghanistan.

4. SUMMARY AND CONCLUSION

To achieve environmental sustainability, the city should increase both its percentage of green open space and m2 of urban green open space per capita. Green spaces of Bengaluru showed more than a 33% decrease since 1992. The increasing population and alarming rate of pollution have led the city to use its sustainable nature, and it has become the 7th most polluted city with a high rate of asthma patients (Shetty & Somashekar, 2015). More and more efforts should be made in this regard to increase the green spaces in the city by maintaining the old green spaces like botanical gardens, parks, and avenue trees. We need to have a clear account of existing green spaces. The insufficient literature (Jain et al., 2011; Ozbakir & Bannari, 2008; Prince & Tucker, 1986) lack of government and public understanding with regard to green spaces and the role played by them in maintaining the sustainability of the city, has resulted in the constant decrease of these vast green spaces.

It is in this direction, an effort has been made to study the vegetation of Lalbagh, the largest botanical garden of Bengaluru, through RS, which provides quick monitoring of large areas with no cost data and gives accurate information to map and analyze vegetation cover (Shetty & Somashekar, 2015). Among the different land use classes thick and thin canopy is depicted in a much more elaborated manner in both supervised and NDVI imagery than in TNDVI. The rock and impermeable surface class is restricted to the western part of Lalbagh.

Lalbagh lake, spread over 40 acres, is depicted in both NDVI and TNDVI transformed imagery. The Thick canopy class representing old, big specimen trees occupied an 80-acre area in Lalbagh. The thin canopy covers 46 acres, and the grass area occupies 41 acres in Lalbagh. The total vegetation area of the Lalbagh is around 70%. Lalbagh supports rich biodiversity with a variety of flora and fauna (Sivasakthivel and Nandini 2014; Fernandes and Ganesh 2020). The rock and other impermeable surfaces constitute 26.6 acres. More endeavor is needed to study the vastness of green spaces and understand its distribution characteristics and the role played by they play in the sustainability of cities.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc.) and text-to-image generators have been used during the writing or editing of this manuscript.

COMPETING INTERESTS

Authors have declared that no competing interests exist.

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Chapter 8
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The Implementation of the Hierarchical Hazard Control Model to Assess the Respiratory Protection of Quarry and Allied Workers in Namibia

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ABSTRACT

The increased risks of adverse respiratory outcomes among quarry and allied workers are affected by high levels of cumulative dust exposure, which is a significant concern in the occupational health sector in developing countries, like Namibia. The study examines respiratory protection practices among quarry and allied workers in Namibia, amidst rising concerns over occupational respiratory disorders due to prolonged exposure to respirable dust. Using the Hierarchical Hazard Control Model, the study assessed the Engineering Controls, Administrative Controls, Educational/Training Programmes, and Medical Surveillance Measures in dust controls for 304 quarry and allied workers in the Erongo, Otjozondjupa, and Kunene regions. This study used self-administered questionnaires to collect data from the quarry and allied workers, between November and December 2022. A Chi-square (χ^2) test was used to determine the association between the level of employee respiratory protection and individual factors significant at p=0.000. Findings showed a statistically significant association between employee respiratory protection and employment status ($\chi 2(1) = 7.592$, $p = 0.000$), job category ($\chi^2(9) = 37.742$, $p = 0.000$), educational level ($\chi^2(3) = 68.517$, $p = 0.000$) and worksite ($\chi^2(8) = 282.178$, $p = 0.001$). A strong positive correlation between worksite and respiratory protection ($r=0.7$, $p<0.01$), negative correlation between worksite protection and education/training programs ($r=0.841$, $p<0.05$) was found. The findings underscore a critical gap in implementing effective engineering controls and training programs, increasing

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the risk of respiratory illnesses among workers. In conclusion, the study revealed that quarry and allied workers could not effectively implement engineering controls to reduce workers' exposure to respirable dust due to the lack of an occupational respiratory health framework, safety regulations and enforcement, and education and training. The study also provided evidence that most allied and quarry workers did not undergo the mandatory medical examination, as a request of limited hazard control measures. Recommendations include a need for an occupational respiratory health framework for quarry and allied workers to ensure occupational respiratory health.
Effective government policy that will enhance occupational safety and health in mining and quarrying is necessary for accident prevention in Namibia.

Keywords: Hierarchical hazard; respiratory diseases; quarry and allied workers; respiratory protection; occupational safety.

1. INTRODUCTION

Quarry mining in Namibia is an important business sector as it employs a significant population as well as contributing to the country's Gross Domestic Product (GDP). The mining and quarrying sector contributes 11.7 percent to the Namibian GDP (Nambinga & Mubita, 2022). Apart from satisfying the ever-growing needs of a rapidly growing population for infrastructural development and urbanization, quarrying activities have helped to provide direct and indirect jobs for many people living in host communities (Ogbonnaya et al., 2016). Quarry mining, charcoal production, and construction are the key industries in which workers are exposed to respirable dust (Szymendera, 2017). In the aforementioned industries, dust particles are generated when sand, gravel, clay, stones, concrete, bricks, and quartz are broken, stirred, or otherwise disturbed through cutting, drilling, mixing, blasting, grinding, or crushing (Draid *et al.,* 2015). Inhaled dust penetrates deep into the lungs, putting the quarry workers at an elevated risk of incurable work-related illness (WRI), particularly respiratory disorders such as lung cancer, pneumoconiosis, asthma, chronic obstructive pulmonary disease, silicosis, and kidney disease (Wadsworth *et al.,* 2019). A high prevalence of silicosis, asthma, and adverse respiratory symptoms like cough, chest pain, and dyspnea has been reported among workers engaged in quarrying. Additionally, quarrying also poses a danger to workers due to injuries caused by rocks falling on the workers, accidents during the use of machinery along with eye problems (Nemer et al., 2020). The prevalence of occupational respiratory hazards and mortality is on the increase in Namibia (Nangolo, 2019). The increased risks of adverse respiratory outcomes among charcoal workers are affected by high levels of cumulative dust exposure (Hamatui *et al.,* 2016). In addition, dust was reported as a common physical hazard at the study sites, with some workers reporting difficulties breathing in Namibia (Nghitanwa & Zungu, 2017). To curb this public respiratory concern, authorities worldwide, including Namibia, have been developing various respiratory safety measures/techniques to help reduce the huge amount of dust inhaled by employees (Shihepo et al., 2024).

Kumar, Gupta, Agarwal, and Singh (2016) acknowledge that Mining and construction industry workers face occupational hazards due to a lack of legal protection and unorganized structures. According to (Geng & Saleh, 2015), effective government policy and safety regulations enhance occupational safety and health in mining and quarrying and are necessary for accident prevention. However, Sanmiquel et al. (2015) argued that despite new laws being introduced to enhance occupational safety, accidents of different nature still occur, including fatal accidents. A Research study conducted by Amukugo & Nangombe (2017) states how a policy framework, such as the Namibian National Health policy framework, outlines the guidelines of actions of the public servants. Namibia lacks an occupational respiratory health framework for quarry and allied workers to ensure occupational respiratory health.

In addition, the Namibian Labour Act makes provision for general protection for workers, but the enforcement of worker protection is inadequate. Thus, the need for training and awareness amongst quarry and allied workers regarding respiratory hazards and the use of PPE which is also lacking in quarry mines in Namibia. The lack of an occupational respiratory health framework, safety regulations and enforcement and education and training may lead to increased injury rates (Long et al., 2015).

Although several national programs are implemented in Namibia to prevent the onset of and reduce occupational respiratory conditions, they appear ineffective. There is a lack of a framework that is directed towards establishing a sustainable and integrated system that accommodates occupational respiratory health. Other complementing frameworks, such as the Namibian National Health Policy Framework, are silent on occupational respiratory conditions. This framework emphasises clinical and public health respiratory conditions more than the private sector, in which most quarries operate (Amukugo & Nangombe, 2017). In line with this, a need to move towards area-specific approaches, such as a framework in occupational respiratory health, as an effective and integrated prevention and control program approach to challenge the high prevalence of occupational respiratory conditions in quarry mines. Therefore, this study sought to explore respiratory protection among Namibian quarry and allied workers, as the study enriches previous studies on dust exposure.

2. MATERIALS AND METHODS

The study used a quantitative cross-sectional design to collect data on respiratory protection among Namibian quarry and allied workers from three regions, namely Erongo, Otjozondjupa and Kunene regions, as shown in Fig. 1 (Shihepo et al., 2024). The sample size was enabled through Taro Yamane's formula, denoted by n=N/(1+Ne^2) n = sample size, while N = population size. Substituting N with 1112 (which is the target population extracted from the total number of quarry and allied workers as indicated in Table 1) at a 95% confidence level and 5% level of significance, the sample size (n) should be more than 286 hence the study adopted 304 as the sample size for quarry and allied workers (Shihepo et al., 2024).

To ensure reliability, the questionnaire was subjected to a pilot test to assess the relevance, sensitivity, and acceptability of the questions. The results validated the questionnaire's consistency and accuracy, determining its effectiveness and allowing for modifications to suit the study's objectives, thus ensuring its validity. A pilot test was conducted in Otavi, with 5% of the study sample being pretested with the questionnaire and the interview guide. The pilot test was carried out on quarry and allied workers from Sargberg Charcoal in August 2022, thus participants with characteristics similar to those selected for the main study. The pilot study enabled the investigator to develop participant approach skills, which was an effective way of recruiting participants for the main study.

This study used self-administered questionnaires to collect data from the quarry and allied workers, between November and December 2022 (Shihepo et al., 2024). Validity was ensured through in-depth scrutiny of the standing literature to classify conceptual dimensions and evaluation of the questionnaire by a series of research specialists from the Namibia University of Science and Technology and research supervisors (Leung, 2015). Research seminars in article writing at NUST/DRIP contributed to the credibility of this study.

The study involved participants who were given the option to decline or withdraw from the research through informed consent, ensuring they understood the importance of making an informed decision before participating in the study. Confidentiality and anonymity were ensured by not revealing the source of data and omitting identifying information. All data was held strictly confidential, as the researcher was the only one who knew the source or origin of the data. Subsequently, the data was kept in a password-secured computer only accessible by the researcher. It was explained to the participants (managerial staff, quarry workers, and allied workers) that they were not liable for punishment, victimisation or mistreatment following their involvement or not in the study. In this study, the participants had no direct benefits from their participation; however, there are indirect benefits from the implementation of recommendations as per the study findings. The researcher obtained ethical clearance and approval to conduct the study from the Namibia University of Science and Technology, with ethical clearance from the Ministry of Health and Social Services, and from the selected participating companies in the study.

Collected data were exported to SPSS version 28, and exploratory data analysis was performed to check for missing data and assess the distribution of numerical variables. Descriptive statistics and precise categorical data were presented as numbers (*n*) and percentages (%) (Shihepo et al., 2024). A Chi-square (χ^2) test was used to determine the association between the level of employee respiratory protection and individual factors significant at p=0.000. A 2-tailed Pearson correlation was done to establish the relationship between selected individual and work-related factors and the level of employee respiratory protection at both 0.01 and 0.05 levels of significance. A linear multiple regression analysis ascertained if selected individual and work-related factors predict the level of employee respiratory protection at 95 CI; p=0.000 (Shihepo et al., 2024).

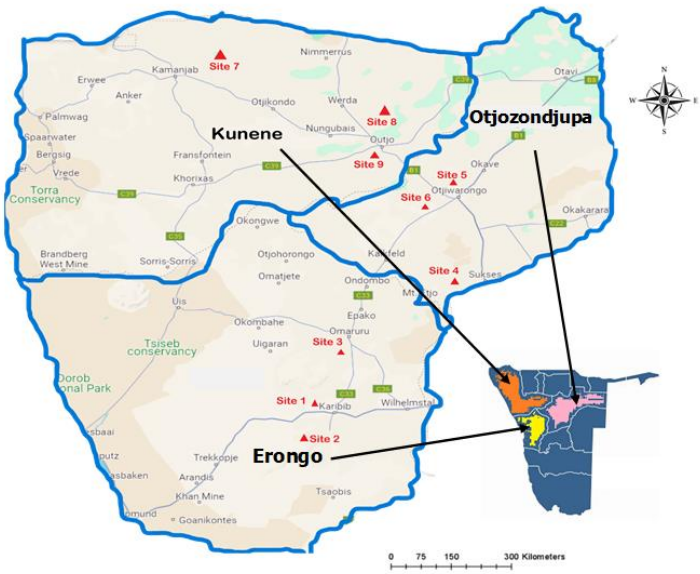


Fig. 1. Research study data collection sites

3. RESULTS

Delineated in Table 1, respondents' demographic characteristics, the most dominant age group is 25-34 years with 53% (n=161), trailed by 35-44 years with 26.3% (n=80) (Shihepo et al., 2024). Participants aged 55-65 years were the least represented with 1% (n=3). There were more males, 74.7% (n=227), than females, 25.3% (n=77), working in quarries and construction sites. Most participants were single, 89.1% (n=271), compared to those who were married, 10.9% (n=33).

The majority of the quarry and allied workers attained a secondary school education level of 53% (n=161), followed by primary education of 31.6% (n=96). Only 6.9% (n=21) reached the tertiary level (Shihepo et al., 2024).

Almost all 92.1% (n=280) quarry and allied workers were fully employed, whereas only 7.9% (n=24) worked on a part-time basis. The bulk of the participants, 27% (n=134), worked under stone blasting, followed by bagging 18.8% and loading 13.2%. Few individuals, 3% of the respondents, worked as builders, and 4.3% worked in stone collection and grinding as well as polishing. The majority of quarry and allied workers, 40.1% (n=122), worked for 1-3 years, followed by those working for 4-7 years, 37.2% (n=113). Only 2.6% (n=8) worked for more than 10 years.

Table 1. Respondents' demographic characteristics

Demographic Variable	Characteristic	Frequency (n)	Percent (%)
Age	18-24 years	40	13.2
	25-34 years	161	53
	35-44 years	80	26.3
	45-54 years	20	6.6
	55-65 years	3	1
	Total	304	100
Gender	Male	227	74.7
	Female	77	25.3
	Total	304	100
Marital Status	Single	271	89.1
	Married	33	10.9
	Total	304	100
Educational Level	None	26	8.6
	Primary level	96	31.6
	Secondary level	161	53.0
	Tertiary level	21	6.9
	Total	304	100.0
Employment Status	Full time	280	92.1
	Part-time	24	7.9
	Total	304	100
Working Experience	Less than a year	34	11.2
	1-3 years	122	40.1
	4-7 years	113	37.2
	7-10 years	27	8.9
	More than 10 years	8	2.6
	Total	304	100

3.1 Dust Exposure at Work

The study pursued the proportion of quarry and allied workers exposed to dust at work, as portrayed by the results in Fig. 2 (Shihepo et al., 2024).

As shown in Fig. 2: Proportion of quarry and allied workers exposed to workplace dust, the majority of quarry and allied workers (91.1%) were exposed to dust at work, while only 8.9% were not (Shihepo et al., 2024).

4. PROTECTION OF QUARRY AND ALLIED WORKERS FROM RESPIRATORY CONDITIONS AND INFECTIONS

The current protection of Namibian quarry and allied workers from respiratory conditions and infections is presented using the hierarchical hazard control model. The hierarchy of controls ranks (5) five levels of actions to reduce or

remove hazards from the most to least effective (Shihepo et al., 2024). Premised on this study's outcomes, three (3) level action controls for reducing respiratory hazards have been identified from the most to least effective as engineering controls, administration controls and use of PPE/C.

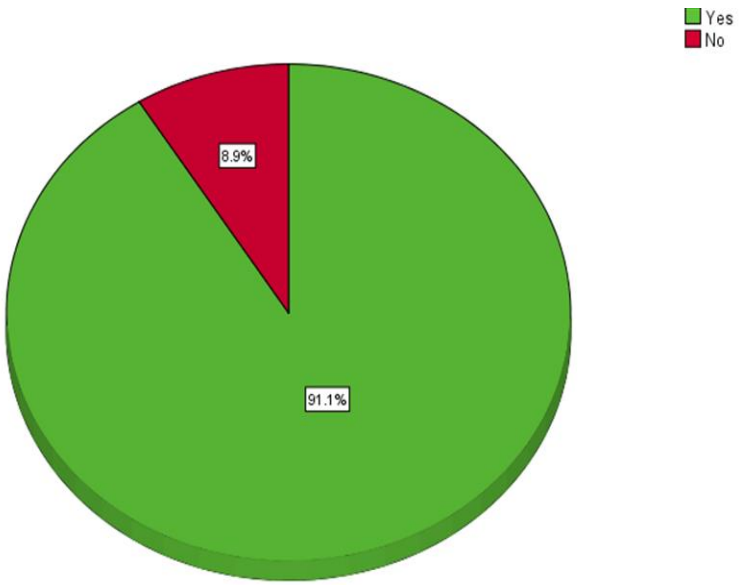


Fig. 2. Proportion of quarry and allied workers exposed to workplace dust

4.1 Engineering Controls

Using water mists and sprays (wet drilling) is a practical engineering control measure to protect employees exposed to dust in quarry mines and construction sites. As shown in Table 2: Engineering controls used in protecting quarry and allied workers below, majority of the participants 87.1% (n=265), indicated that water sprays were not used in reducing the amount of dust emitted in the atmosphere as a result of the mining and construction processes (Shihepo et al., 2024). The application of water mists was not a common practice in protecting quarry and allied workers, as almost all the respondents, 98% (n=298), failed to acknowledge.

4.2 Administrative Controls

The application of administrative controls in protecting quarry and allied workers was revealed in the form of educational/training programs and medical surveillance measures (Shihepo et al., 2024).

Table 2. Engineering controls used in protecting quarry and allied workers

	Yes		No	
	Frequency (n)	**Percentage (%)**	**Frequency (n)**	**Percentage (%)**
Water Sprays (wet drilling)	39	12.9	265	87.1
Water Mists	6	2	298	98

Table 3. Employee educational awareness/training

Question	**Response**	**Frequency(n)**	**Percent (%)**
Are you aware of any health and safety guidelines or company policies	Yes	119	39.1
	No	185	60.9
	Total	304	100
Are you trained on the importance, use and care of PPE/C	Yes	74	24.3
	No	230	75.7
	Total	304	100

4.3 Educational /Training Programmes

Employee education/training is a critical element of any complete protection of employees from workplace hazards, as it imparts information on how to protect themselves and their co-workers. This article assesses health and safety training among quarry and allied workers in Namibia and reveals the following results in Table 3: Employee educational awareness/training (Shihepo et al., 2024).

Less than a quarter of the participants, 24.3% (n=74), were trained on the importance of PPE/C, while the majority, 75.7% (n=230), did not receive such training. Subsequently, most quarry and allied workers, 60.9% (n=185), did not know any guidelines or company policies on PPE/C, while more than a quarter, 39.1% (n=119), were knowledgeable.

4.4 Medical Surveillance Measures

This article assesses the existence of a medical surveillance programme as part of a respiratory protection initiative. All the quarry and allied workers (100%) did not undergo pre-employment medical checkups (Shihepo et al., 2024). Upon employment, only 11.8% (n=36) of the workers acknowledged going for periodic medical examinations compared to 88.2% (n=268) who did not. None of the employees requires medical examinations to ensure the worker can use the PPE without risk to their health.

4.5 Association between Employee Level of Respiratory Protection with Individual and Work-Related Factors

A Chi-square test was performed to ascertain the association between the level of employee respiratory protection and the following individual factors: age, gender, marital status, employment status, working experience, job category, frequency of PPE/C utilisation, and educational level work site.

Table 4. The association between individual and work-related factors and employee respiratory protection

	Value	df	Asymptotic Significance (2-sided)
Age	5.618	3	0.006
Gender	0.053	1	0.818
Marital status	0.338	1	0.561
Employment status	7.592	1	0.000
Working experience	12.275	4	0.015
Job category	37.742	9	0.000
Frequency of PPE/C utilisation	0.708	1	0.004
Educational level	68.517	3	0.000
Worksite	282.178	8	0.000

As shown in Table 4: The association between individual and work-related factors and employee respiratory protection, the Pearson Chi-Square test result portrays a statistically significant association between employee respiratory protection and employment status at $\chi^2(1) = 7.592$, $p = 0.000$; employee respiratory protection and job category at $\chi^2(9) = 37.742$, $p = 0.000$; employee respiratory protection and educational level at $\chi^2(3) = 68.517$, $p = 0.000$; employee respiratory protection and worksite at $\chi^2(8) = 282.178$, $p = 0.000$ (Shihepo et al., 2024). However, there was no statistically significant relationship between employee respiratory protection and age at $\chi^2(3) = 5.618$, $p = 0.006$, employee respiratory protection and gender at $\chi^2(1) = 0.053$, $p = 0.818$, employee respiratory protection and working experience at $\chi^2(4) = 12.275$, $p = 0.015$.

4.6 Correlation of Individual and Work-Related Factors with the Level of Employee Respiratory Protection

A Pearson correlation was performed to establish the relationship between selected individual and work-related factors (work site, age, gender, marital status, religion, level of education, employment status, working experience, job category) and the level of employee respiratory protection as indicated in Table 5: Individual and work-related factors correlation with the level of employee respiratory protection (Shihepo et al., 2024).

Table 5. Individual and work-related factors correlate with the level of employee respiratory protection

Correlation variables	Pearson Correlation
Level of employee respiratory protection and the worksite	.700**
Level of employee respiratory protection and age	.142*
Level of employee respiratory protection and gender	0.013
Level of employee respiratory protection and marital status	-0.033
Level of employee respiratory protection and employment status	0.057
Level of employee respiratory protection and working experience	0.036
Level of employee respiratory protection and job category	.331**
Level of employee respiratory protection and training	-.841*
Level of employee respiratory protection and educational level	-.424**

***Correlation is significant at the 0.01 level (2-tailed). Correlation is significant at the 0.05 level (2-tailed).*

There was a strong positive correlation between employee respiratory protection and worksite (r=0.7) at 99% CI (2-tailed), as well as a weak positive association between employee respiratory protection and age (r=0.142) at 95% CI. The level of employee respiratory protection showed a positive weak correlation with job category (r=0.331) at 99% CI, a strong positive relationship with training (r=0.841) at 95% CI, and a moderately weak association with the level of education (r=0.424) at 99% CI (Shihepo et al., 2024). There was no significant correlation between the level of employee respiratory protection and gender (r=0.013), employment status (r=-0.057), marital status (r=0.033), and working experience (r=0.036).

4.7 Multiple Regression Analysis of the Level of Employee Respiratory Protection and Selected Individual and Work-Related Factors

Selected individual and work factors were assessed in a standard regression analysis to predict the level of employee respiratory protection. Table 6: Model Summary analysis of the level of employee respiratory protection and selected individual and work-related factors provides an overview of the model summary analysis (Shihepo et al., 2024).

Table 6. Model Summary analysis of the level of employee respiratory protection and selected individual and work-related factors

R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	Change Statistics			Sig. F Change
					F Change	df1	df2	
0.72	0.518	0.501	0.285	0.518	31.467	10	293	0.000

Predictors: (Constant), work site, age, gender, marital status, religion, level of education, employment status, working experience, job category.

The prediction model summary was statistically significant, F (10, 293) = 31.467, p = 0.000, as more than half (52%) of the selected individual factors accounted for the level of employee respiratory protection (R^2 = 0.501, Adjusted R^2 = 0.518). There was a medium-high degree of correlation denoted by R=0.72 (Shihepo et al., 2024). These findings imply that selected individual and work-related factors highly influenced the level of employee respiratory protection.

5. DISCUSSION

This study revealed the inability of quarry mines and construction employers to apply feasible respirable dust engineering controls in the form of water mists and sprays. According to Gürcanl, Baradan and Uzun (2015), respirable dust engineering controls are designed to remove or reduce the hazard at the source, meaning suppressing, diluting, or diverting dust generated by mining, construction and associated activities. Respirable dust engineering controls include proper ventilation systems, the use of water sprays, water mist (wetting agents) and control booths or environmental cabs to enclose equipment operators (National Institute for Occupational Safety and Health, 2021). These findings entail that Namibian quarry and allied workers have no engineering controls in place to reduce their exposure to respirable dust (Shihepo et al., 2024). You *et al*. (2019) argue that engineering controls provide more consistent and reliable protection than other interventions because the controls are not dependent on an individual's performance, supervision, or intervention to function as intended.

There is a lack of administrative controls, such as educational/training programmes, among Namibian quarry and allied workers. According to Zhou *et*

al. (2018), administrative controls provide additional guidance and procedures, demonstrating the company's commitment to safety and enhancing the overall safety culture of the workplace. In protecting quarry and allied workers, administrative controls come in the form of educational programmes and medical surveillance. With insufficient educational/training programmes, employees are prone to developing knowledge gaps on respiratory protection, making it difficult for them to cooperate and actively participate in precautionary safety and protection measures. The lacklustre of training entails that Namibian quarry and allied workers are more inclined to occupational-related lung disease risks as well as negative attitudes, inadequate knowledge, and bad practices in respiratory protection equipment (Shihepo et al., 2024). During training, employees interact with trained professionals who can provide instruction and understanding in the correct use of PPEs, which will serve to overcome employee resistance to proper PPE usage (National Institute for Occupational Safety and Health, 2021). Educational and training programmes on safety precautions and behaviours, especially proper use of PPE, are effective methods to increase knowledge and prevent respiratory symptoms and diseases related to quarry, construction, and allied exposures (Alemu *et al.,* 2020).

Moreover, the study results showed that all the quarry and allied workers did not undergo a mandatory initial medical examination (Shihepo et al., 2024). Health surveillance in respiratory protection is an outline of repeated employee medical assessments and getting information, which includes history tracking, regular physical examinations, chest X-rays, and lung function tests (Almberg, Friedman, Rose, Go, & Cohen, 2020). In addition, NIOSH (2019) states that initial examination assists in the early detection of respirable dust-related illnesses and conditions that may make the miner more susceptible to the toxic effects of respirable dust. Thus, without initial examination, Namibian quarry and allied workers are the least protected from respiratory conditions, thereby heightening the risk of becoming impaired without early detection (Chen, Cheng, Xie, & Su, 2022). Without initial examination, early detection and intervention of respiratory illness are slowed, allowing greater disease progression and deteriorating health outcomes. Also pointed out in the findings is the inability of most quarry and allied workers to undergo periodic medical examinations. Deprived of an individual medical baseline (initial medical examinations), valuable in assessing any future health changes and a comparable point (periodic examination), the protection of quarry and allied workers from respirable dust-related illnesses is minimal (Coffman *et al*., 2021; Shihepo et al., 2024).

Furthermore, employees were evaluated medically to be sure about their ability to use a respirator, because using a respirator may create a psychological burden on employees, depending on the type of respirator, the type of job, and the workplace conditions in which the respirator is used (Shihepo et al., 2024). Medical surveillance measures include medical examinations that review an employee's medical and work history, and a physical examination. The medical and work history covers an employee's present and past work exposures, illnesses, and any symptoms indicating respirable dust-related diseases and compromised lung function (Lancet, 2019).

In the present article, most quarry and allied workers were not provided with PPE/C despite it being a mandatory obligation enshrined in the Namibian Labour Act. Without PPE/C protection, from respiratory conditions is jeopardised. Personal protective equipment (PPE) controls are the last line of defence in protecting workers from dust exposure when engineering and administrative controls are not feasible or do not provide adequate protection (Alemu *et al.,* 2020). The study noted that the majority of those provided with PPE/C were given free of charge. These findings are in line with the fact that respirators approved by NIOSH and suitable for their intended purpose must be provided by employers at no cost to mine and construction workers to effectively protect themselves from respirable dust (Alemu *et al.,* 2020). Workers being given respirators for free at work lessen their susceptibility to respirable dust (Shihepo et al., 2024).

The use of PPE/C among quarry and allied workers was inconsistent, as most individuals occasionally used PPE/C. These findings imply that Namibian quarry and allied workers lack adherence to the use of PPE/C; hence, they become susceptible to developing respiratory conditions and related illnesses (Shihepo et al., 2024). According to Ashley and O'Connor (2017), PPE provides direct protection to workers as a supplementary control. Compliance with respiratory protection guidelines in many industries is troublingly low, especially among workers in construction and mining (Dhatrak and Nandi, 2020). Conversely, the use of PPE decreased the intermediate risk of respiratory injury by 44%, the maximum risk by 32%, and increased the chances of no risk of respiratory effects by 24% (Yarpuz-Bozdogan, 2018).

Premised on the findings, the type of PPE/C given to Namibian quarry and allied workers is biased towards protecting the general body (overalls) rather than the respiratory system, which is the priority given the nature of their work. With very few employees using respirators, the degree of protection becomes low, thereby placing an increased risk of respiratory adverse health effects associated with exposure to respirable dust. NIOSH (2019) recommends the use of respirators as an interim measure when engineering and administrative controls are not effective in maintaining worker exposure to respirable dust at or below the proposed PEL (Shihepo et al., 2024). Contrary to the findings of the current article, the use of respirators as an interim measure in maintaining worker exposure to respirable dust at or below the proposed PEL is recommended (NIOSH Health, 2019).

Evidence from the article's findings revealed the use of disposable N95 and surgical masks against respirable dust exposure. Lee *et al.* (2005) contend that respirators have different levels of protection and are used in a variety of conditions; hence, it is important to understand which respirator will protect against specified dust PEL exposures. Disposable N95 respirators protect against larger dust particles, of which quarry mines and construction sites can generate smaller and finer particles (Zgambo, 2015). Conversely, half-facepiece respirators protect from smaller particles of dust, and full-facepiece respirators provide eye and respiratory protection. On the other hand, surgical masks are the

least protective with a filtration efficiency between 25.7-61.5%, and half-facepiece P100 respirators were the most protective with 96.5-98.9% filtration efficiency (Sapbamrer *et al.,* 2021; Shihepo et al., 2024). Therefore, the continued dust exposure by the quarry and allied workers emphasises the importance of filtering facepiece respirators (FFRs) in dusty environments (Lee *et al.,* 2005).

The findings also showed that of the employees supplied with respirators, none did not obtained respirator fit testing. In the total absence of respirator fit testing, it is axiomatic that quarry and allied works in Namibia are not adequately protected against breathing respirable dust and contaminated ambient air (Shihepo et al., 2024). Research supports the use of respirators for lung protection while working with hazardous occupational inhalants; hence, fit testing becomes necessary to ensure maximum respiratory protection (National Center for Environmental Health, 2014). Respirator fit testing is performed to confirm the correct mask size without air leaks to protect individuals from hazardous inhalants (Ashley & O'Connor, 2017). This means that before an employee is required to use any respirator facepiece, the employee must be fit-tested with the same make, model, style, and size of respirator that will be used. As revealed by the study findings, less than half of the quarry and allied workers were not adequately protected by respirators due to excessive facial hair. The use of respirators is commonly affected by fit factors such as facial hair (Shihepo et al., 2024). If a tight seal is not maintained between the facepiece and the employee's face, contaminated air will be drawn into the facepiece and breathed by the employee (Rengasamy et al., 2017). In general, the presence of beards and wide sideburns had a detrimental effect on the performance of the respirators (Jacobsen et al., 2021). This means that Namibian quarry and allied workers with excessive facial hair, including stubble and wide sideburns, which interfere with the seal, cannot expect to obtain as high a degree of respirator performance as clean-shaven individuals (Shihepo et al., 2024). OSHA concludes that poorly fitting facepieces expose workers to contaminants and that the use of an effective fit testing protocol is the best way of determining which respirator facepiece is most appropriate for each employee (Momyer, 2016).

Premised on the regression model, the level of education, age, working experience, and worksite risk perception predict or influence how quarry and allied workers are protected from respiratory infections and conditions (Shihepo et al., 2024). Research published across various occupations shows several individual factors as important determinants of the level of respiratory protection among quarry and allied workers. These factors include personal characteristics such as age, gender, work experience, level of education, knowledge (training) of respiratory protection hazards, and risk perceptions of respiratory protection (Z'gambo, 2015). The educational level of quarry and allied workers greatly influenced respiratory protection. Lower educational levels of primary level and below, as noted among most Namibian quarry and allied workers, entail lower respiratory protection, as some of the employees might fail to comprehend and apply respiratory protection information. A study by Almberg, Friedman, Rose, Go, and Cohen (2020) found that the respondents' level of education significantly

affected respiratory protection, particularly when comparing respondents with and without formal education. However, the difference between individuals with basic education and those with higher education was not statistically significant. Research findings by Jacobsen, Schaumburg, Sigsgaard and Schlünssen (2021) showed that people who did not complete formal education also did not protect themselves from respiratory hazards as they shunned the proper PPEs at work (Shihepo et al., 2024).

Worksite risk perception has shown an influence on quarry and allied workers' respiratory protection. The protection motivation theory states that risk perception and use of personal protective equipment increase when workers have a reason for concern. In most instances, employees are cognisant of protecting themselves from respiratory conditions as a result of having suffered from some respiratory illness. Sehsah*,* El-Gilany, and Ibrahim *(*2020) found that workers who had experienced an accident or incident in the past felt less safe and were more aware of the risks than those who had not experienced an accident or incident, hence, they are more prone to protect themselves. When workers are made aware of the hazards in their workplace, they are more inclined to use PPE to protect themselves from exposure to the hazards. Conversely, if workers are provided with PPE but are not told why or how to use PPE, likely, such PPE will not be utilised. Kumar *et al.* (2016) point out the importance of workers having knowledge of the hazards and risks posed by their work and how they can protect themselves from these hazards (Shihepo et al., 2024).

6. CONCLUSION

The study revealed that quarry and allied workers could not effectively implement engineering controls to reduce workers' exposure to respirable dust due to the lack of an occupational respiratory health framework, safety regulations and enforcement, and education and training. The study provided evidence that most allied and quarry workers did not undergo the mandatory medical examination, as a request of limited hazard control measures. This contradicts the claim made by NIOSH, which argued that construction and mining companies should provide new employees with such examinations. Despite being mandated by the Labour Act of Namibia to provide protective equipment, the study revealed that 91 percent of quarry and allied workers in the study were not given the necessary equipment, and water sprays and misting systems were limited, thus exposing the quarry and allied workers to respiratory dust. The exposure to respiratory dust by quarry and allied workers is aligned with the Hierarchical Hazard Control Model, which states that implementing hazard controls in quarry mines highly prevents respiratory infections and is regarded as effective. Without proper equipment, workers are at risk of experiencing respiratory conditions that are detrimental to their health (Shihepo et al., 2024).

The study found that the type of PPE used by workers in Namibia is biased toward protecting the general body instead of the respiratory system. In addition, the use of PPE in the study was not consistent, and only a few quarry and allied workers used a respirator, which means that their level of protection was low.

This could result in adverse health effects because of exposure to dust. The study revealed that none of the employees had obtained a fit test. Allied and quarry workers in Namibia do not have the required protection against dust and air contamination. The study showed that less than half of the employed individuals had adequate protection due to their facial hair (Shihepo et al., 2024). The study supports the concept of the protection motivation theory, which states that when workers are concerned about their safety, they tend to use more protective equipment, but also workers with inadequate knowledge, training, and education about workplace hazard control measures still risk respiratory dust exposure.

7. RECOMMENDATION

The study recommends the need for an occupational respiratory health framework specifically for quarry and allied workers to ensure control measures are adequately implemented and enforced. Quarry mines should also ensure daily health and safety checks and periodical medical testing for all quarry and allied workers to ensure that medical fit testing for PPE is conducted in dust conditions. Finally, the study also recommends education and training to enhance safety awareness for quarry and allied workers.

8. STUDY IMPLICATIONS

The practicality of the study implies that there is a need for an occupational respiratory health framework, policy enforcement, and education and training to enhance quarry and allied workers' safety. These implementations will adequately help the Namibian Labour Act to make provisions specifically for quarry and allied worker protection.

The study also contributes to the literature as it is aligned with the Protection motivation theory and the Hierarchical hazard control model, by revealing that when workers are concerned about their safety, they tend to use more protective equipment.

Finally, on a policy level, the study shows an urgent need for an occupational respiratory health framework for quarry and allied workers to ensure occupational respiratory health.

Effective government policy that will enhance occupational safety and health in mining and quarrying is necessary for accident prevention in Namibia.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc) and text-to-image generators have been used during the writing or editing of this manuscript.

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COMPETING INTERESTS

Authors have declared that no competing interests exist.

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